

**STUDIES ON VARIABILITY OF *Rhizoctonia bataticola*
(Taub.) Butler CAUSING DRY ROOT ROT OF
CHICKPEA & ITS MANAGEMENT**

Submitted by

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M.Sc. [Agri.]

THESIS

Submitted to

**Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani in partial
fulfilment of the requirements for the degree of**

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IN

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DEPARTMENT OF PLANT PATHOLOGY

COLLEGE OF AGRICULTURE,

VASANTRAO NAIK MARATHWADA KRISHI VIDYAPEETH

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2020

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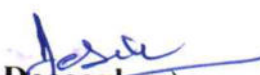
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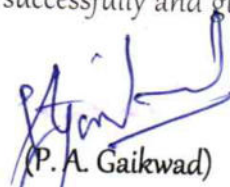
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CONTENTS

Sr. No.	Title	Page No.
I	INTRODUCTION	1-6
II	REVIEW OF LITERATURE	7-92
III	MATERIALS AND METHODS	93-118
IV	RESULTS AND DISCUSSION	119-160
V	SUMMARY AND CONCLUSIONS	161-169
	LITERATURE CITED	I-XXXI
	THESIS ABSTRACT	-
	CURRICULAM VITAE	-
	APPENDIX - I	-

LIST OF TABLES

TABLE NO.	CONTENTS	PAGE NO.
1.	Tahsil-wise chickpea dry root rot incidence, during <i>Rabi</i> , 2016-17 and 2017-18	119-120
2.	District-wise chickpea dry root rot incidence, during <i>Rabi</i> , 2016-17 and 2017-18	122
3.	Variety wise chickpea dry root rot incidence, during <i>Rabi</i> , 2016-17 and 2017-18	122
4.	Agro-climatic zone-wise chickpea dry root rot incidence, during <i>Rabi</i> , 2016-17 and 2017-18	123
5.	Nomenclature of <i>R. bataticola</i> isolates	125
6.	Pathogenic variability among the isolates of <i>R. bataticola</i> (Sick soil method)	127
7.	Pathogenic variability among the isolates of <i>R. bataticola</i> (Paper towel method)	129
8.	Grouping of <i>R. bataticola</i> isolates based on their pathogenic variability, by Sick soil and Paper towel methods	129
9.	Cultural variability among the isolates of <i>R. bataticola</i>	130
10.	Morphological variability among the isolates of <i>R. bataticola</i>	132
11.	RAPD primers and the number of total polymorphic bands produced in 9 isolates of <i>Rhizoctonia bataticola</i> .	135
12.	The similarity matrix representing coefficient of similarity among 9 isolates of <i>R. bataticola</i> based on RAPD marker	136
13.	Effect of various Carbon sources on growth and Sclerotial production of <i>R. bataticola</i>	137
14.	Effect of various nitrogen sources on growth and Sclerotial production of <i>R. bataticola</i>	138
15.	Effect of soil moisture regimes on dry root rot disease incidence in chickpea cv. JG-62 (pot culture)	140
16.	Effect of Different soil types on development of dry root rot incidence of chickpea	142
17.	<i>In vitro</i> efficacy of systemic fungicides against <i>R. bataticola</i>	144

18.	<i>In vitro</i> efficacy of non-systemic/contact fungicides against <i>R. bataticola</i>	146-147
19.	<i>In vitro</i> bioefficacy of bioagents against <i>R. bataticola</i>	151
20.	Efficacy of organic amendments against <i>R. bataticola</i> , causing dry root rot of chickpea	153
21.	Integrated efficacy of fungicides, bioagents, and organic amendments against <i>R. bataticola</i> , causing dry root rot of Chickpea (Pot culture)	156
22.	Reactions of chickpea varieties, cultivars, elite lines against <i>R. bataticola</i> (Pot / polybag culture)	159
23.	Categorization of chickpea varieties, cultivars, elite lines based on reaction against <i>R. bataticola</i> (Pot culture)	160

LIST OF PLATES

PLATE NO.	CONTENTS	PAGE NO.
I	Chickpea crop fields surveyed for recording dry root rot disease incidence	119-120
II	Typical symptoms of dry root rot (<i>R. bataticola</i>) on chickpea crop	124-125
III	Pure culture of <i>R. bataticola</i> test isolates	125-126
IV	Microphotograph of <i>R. bataticola</i> , causing dry root rot of chickpea	125-126
V	Mass multiplication of <i>R. bataticola</i>	127-128
VI	Pathogenicity and Pathogenic variability among the isolates of <i>R. bataticola</i> (Sick soil method)	127-128
VII	Pathogenicity and Pathogenic variability among the isolates of <i>R. bataticola</i> (Paper Towel method)	129-130
VIII	Cultural variability among the test isolates of <i>R. bataticola</i>	130-131
IXA	Morphological variability among the test isolates of <i>R. bataticola</i>	132-133
IXB	Morphological variability among the test isolates of <i>R. bataticola</i>	132-133
XA	Morphological variability among the test isolates of <i>R. bataticola</i>	132-133
XB	Morphological variability among the test isolates of <i>R. bataticola</i>	132-133
XIA	RAPD fingerprint profile of 9 pathogen isolates of <i>R. bataticola</i> by using primer OPA-02, OPA-03: Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of <i>R. bataticola</i> .	135-136
XIB	RAPD fingerprint profile of 9 pathogen isolates of <i>R. bataticola</i> by using primer OPA-07, OPA-09: Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of <i>R. bataticola</i> .	135-136
XIIA	RAPD fingerprint profile of 9 pathogen isolates of <i>R. bataticola</i> by using primer OPA-13: Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of <i>R. bataticola</i> .	135-136
XIII	Effect of various Carbon sources on growth and Sclerotial production of <i>R. bataticola</i>	137-138

XIV	Effect of various nitrogen sources on growth and Sclerotial production of <i>R. bataticola</i>	138-139
XV	Effect of soil moisture regimes on dry root rot disease incidence in chickpea cv. JG-62 (pot culture)	140-141
XVI	Effect of Different soil types on development of dry root rot incidence of chickpea	142-143
XVII	<i>In vitro</i> efficacy of systemic fungicides against <i>R. bataticola</i> (Rb-6 isolate)	144-145
XVIII	<i>In vitro</i> efficacy of non-systemic and combi-fungicides against <i>R. bataticola</i> (Rb-6 isolate)	146-147
XIX	<i>In vitro</i> efficacy of bioagents against <i>R. bataticola</i> (Rb-6 isolate)	151-152
XX	<i>In vitro</i> efficacy of organic amendments against dry root rot of chickpea (Cv. JG-62)	153-154
XXI	Integrated management of chickpea dry root rot (Pot Culture)	156-157
XXII	Screen-house reactions of chickpea varieties, cultivars, elite lines against <i>R. bataticola</i> (Pot culture)	159-160

LIST OF FIGURES

TABLE NO.	CONTENTS	PAGE NO.
1.	Various agro-climatic zones of Marathwada region surveyed for dry root rot of chickpea	96-97
2.	Various districts of Marathwada region surveyed for dry root rot of chickpea	121-122
3.	District-wise incidence of chickpea dry root rot in Marathwada region (<i>Rabi</i> , 2016-17 and 2017-18)	120-121
4.	District-wise pooled mean incidence of chickpea dry root rot in Marathwada region (<i>Rabi</i> , 2016-17 and 2017-18)	121-122
5.	Variety-wise incidence of chickpea dry root rot in Marathwada region (<i>Rabi</i> , 2016-17 and 2017-18)	122-123
6.	Variety-wise pooled mean incidence of chickpea dry root rot in Marathwada region (<i>Rabi</i> , 2016-17 and 2017-18)	122-124
7.	Agro-climatic zone wise incidence of chickpea dry root rot in Marathwada region(<i>Rabi</i> , 2016-17 and 2017-18)	123-123
8.	Agro-climatic zone wise pooled mean incidence of chickpea dry root rot in Marathwada region (<i>Rabi</i> , 2016-17 and 2017-18)	123-124
9.	Cultural variability among the isolates of <i>R. bataticola</i>	130-131
10.	Graphical representation of comaprative analysis of RAPD primer and polymorphic percentage.	135-136
11.	Average similiarity of nine isolates of <i>Rizocotina Bataticola</i> based on UPGMA analysis.	136-137
12.	Dendrogram generated based on RAPD primer data through UPGMA analysis	136-137
13.	Effect of organic and inorganic sources of carbon	137-138
14.	Effect of organic and inorganic sources of Nitrogen	138-139
15.	Effect of soil moisture regimes on the incidence of <i>R. bataticola</i> .	140-141
16.	Effect of Different soil types on development of dry root rot incidence of chickpea	142-143
17.	<i>In vitro</i> bioefficacy of systemic fungicides against <i>R. bataticola</i>	144-145
18.	<i>In vitro</i> bioefficacy of non-systemic and combi-fungicides against <i>R. bataticola</i>	146-147
19.	<i>In vitro</i> bioefficacy of bioagents against <i>R. bataticola</i>	151-152

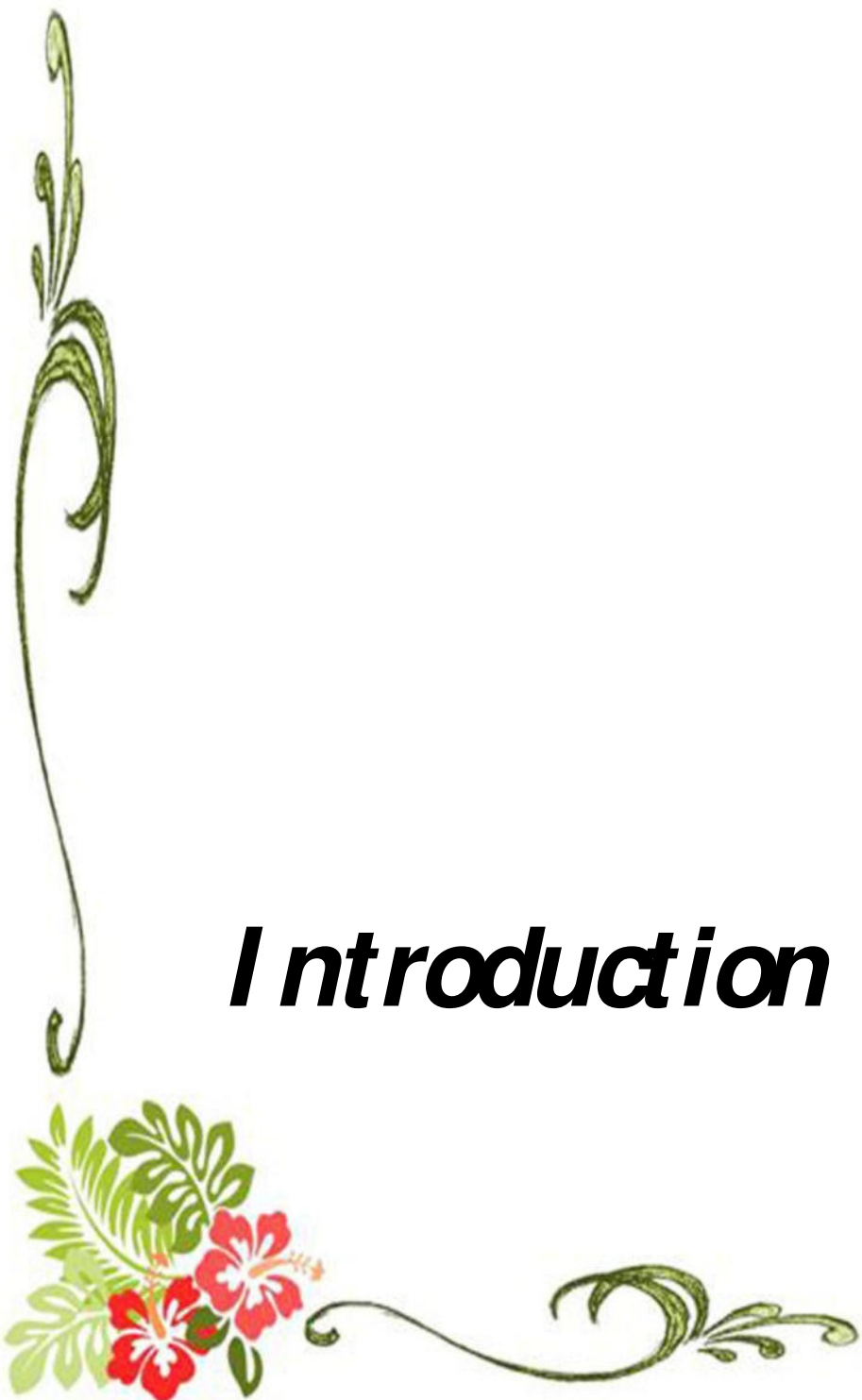
20.	Efficacy of organic amendments against <i>R. bataticola</i> (Pot culture)	153-154
21.	Integrated efficacy of fungicides, bioagents & organic amendments against (Pot Culture) <i>R. bataticola</i> causing dry root rot of chickpea.	156-157

ABBREVIATIONS

%	=	Per cent
/	=	Per
@	=	At the rate of
µg	=	Microgram
µM	=	Micromolar
BOD	=	Biological Oxygen Demand
bp	=	Base pair
C.D.	=	Critical Difference
cm	=	centimeter (s)
CRD	=	Completely Randomized Design
cv.	=	Cultivar
DAS	=	Days after sowing
Dia.	=	Diameter
DNA	=	De-oxy ribonucleic acid
dNTP	=	Deoxyribose Nucleotide Tri-Phosphate
e.g.	=	Exempli gratia (for Example)
EDTA	=	Ethylene Diamine Tetra Acetic Acid
<i>et al.</i>	=	and others
EtBr	=	Ethidium bromide
etc.	=	Etcetera
Fig.	=	Figure (s)
g	=	Gram
ha	=	Hectare (s)
hrs.	=	Hours
<i>i.e.</i>	=	That is
ISR	=	Induced Systemic Resistance
IU	=	International Unit

kb	=	Kilobase pair
kg	=	Kilogram (s)
l	=	litre
m	=	meter (s)
Max.	=	Maximum
mg	=	milligram
MgCl ₂	=	Magnesium chloride
Min.	=	Minimum
ml	=	millilitre
mm	=	millimetre
mM	=	millimolar
NaCl	=	Sodium chloride
ng	=	Nanogram
No.	=	Number (s)
°C	=	degree Celsius
OD	=	Optical density
PCR	=	Polymerase chain reaction
PDA	=	Potato Dextrose Agar
PDB	=	Potato Dextrose Broth
PDC	=	Per cent Disease Control
pH	=	Log H ⁺ ion concentration
pmol	=	picomole
PRESR	=	Pre emergence Seed rot
POESM	=	Post Emergence Seedling Mortality
psi	=	pounds per square inch
q/ha	=	quintal per hectare
RAPD	=	Randomly Amplified Polymorphic DNA
RBD	=	Randomized Block Design

RNA	=	Ribonucleic acid
RNase	=	Ribonuclease
rpm	=	revolutions per minute
SDS	=	Sodium Dodecyl Sulphate
Sec.	=	Seconds
SEm	=	Standard error of mean
sp.	=	Species
ST	=	Seed treatment
T	=	Treatment
t/ha	=	tonnes per hectare
TAE	=	Tris acetate EDTA buffer
Taq	=	Thermus aqaticus
Tris HCL	=	Tris Hydrochloride
U	=	Unit
UV	=	Ultra Violet
V	=	Volts
<i>viz.</i> ,	=	videlicet (namely)
w/v	=	weight/volume
w/w	=	weight/weight



Introduction

CHAPTER - I

INTRODUCTION

India is largely a vegetarian country and pulses are an important part of the daily diet. India is the largest cultivator & consumer of pulses. Maharashtra state is a leading supplier of pulses, by making pulses growing in the State an attractive and remunerative proposition to farmers. This is achievable by improving yield and quality of pulses by use of quality inputs, adopting best management practices. This would ensure increased returns to pulse growers with improved nutritional values to the consumer.

Pulses like Moong, Urad, Tur and Chickpea are grown in Maharashtra. This research work is focused on chickpea. Chickpea (*Cicer arietinum* L.) belongs to family *leguminaceae* also commonly known as Chana, Bengal gram or garbanzo and one of the major pulses cultivated and consumed in India. It is a major and cheap source of protein compared to animal protein.

Chickpea is a *Rabi* crop preferably sown in September-November and harvested in February. Crop matures in around 90-120 days, depending on the variety. Indian local varieties are short duration while kabuli varieties take comparatively longer period to mature. Similarly temperature also plays a vital role, cooler regions like northern India take longer period to mature, compared to relatively warm weather in southern parts of India. Areas having low to moderate rainfall and a mid-cold weather are best suitable for the crop, preferably moderate rainfall of 60-90 cm per annum. It has an indeterminate growth habit, which means that the growth cycle extends as long as moisture is available.

Chickpea seeds are rich source of protein (17 -20%) and form an important part of vegetarian diet plans. Sprouted seeds of Chickpea are eaten as a vegetable or salad. Chickpea is split without its seed coat dried and cooked

into a thick soup called *Dal* or ground into flour and used as a main ingredient for snacks and sweet meat. Remaining plants are used for livestock feed, whole seeds may be milled or directly fed to horses and cattle. Malic acid exudates from the leaves can be used in medicines or used as vinegar.

At present, it is grown in over 50 countries of Asia, Africa, America and Oceania in rain-fed environments (Sharma *et.al.* 2015). In India, chickpea accounts for about 45% of total pulses produced in the country. As compare to other pulses, India is the major producing country for chickpea, contributing for over 75% of total production in the world and the area under its cultivation was 105.73 lakh/ha with production 111.58 lakh/tones and yield was about 1056 Kg/ha (Anonymous, 2019a). The Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Andhra Pradesh, Karnataka, Chhattisgarh, Bihar and Jharkhand are the major chickpea producing States in India contributing more than 95% to the total chickpea production. Madhya Pradesh is the single largest in the country accounting for over 40% of total production while Rajasthan, Maharashtra, Uttar Pradesh and Andhra Pradesh contribute about 14%, 10%, 9% and 7%, respectively. Whereas the area, production and yield of chickpea in Maharashtra were 1847.7 hectare, 1880.9 tonnes and 1018Kg/ha, respectively (Anonymous, 2019b).

The most affecting abiotic constraints to yield of chickpea, include drought, heat, cold and salinity and the major biotic constraints are Ascochyta blight (*Ascochyta rabii*), Fusarium wilt (*Fusarium oxysporum*), Dry root rot (*Rhizoctonia bataticola*), Botrytis grey mould (*Botrytis cinerea*), Collar rot (*Sclerotium rolfsii*), Root-knot nematode (*Meloydogyne incognita* and *M. javanica*), Stunt-virus, Pod borer (*Helicoverpa armigera*) and Cutworm (*Agrotis ipsilon*). Amongst the causal agents of biotic stresses, about 67 fungi, 3 bacteria, 22 viruses and 80 nematodes have been reported on chickpea (Nene *et al.*, 1996).

Among the various soil borne fungal diseases affecting Chickpea productivity, dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler (*Macrophomina phaseolina* (Tassi.) Goid)) is the most severe disease particularly in the central and southern zone of India. It was first reported from India by (Mitra, 1931); later, the disease has been reported from most chickpea-growing areas in India and other countries. It can affect chickpea production, causes considerable yield losses that vary from 5 to 50 % and may cause 100% losses in susceptible cultivars under favorable condition (Pande *et al.*, 2012) and considerable yield losses which may be as high as 50 to 71% (Veena *et al.*, 2014a). The disease is more prevalent during hot temperature of 30 to 35°C and low soil moisture conditions (Taya *et al.*, 1988; Pande *et al.*, 2010).

In chickpea, dry root rot can be mistaken for *Fusarium* wilt as the general symptoms of these two diseases are apparently similar. Also, it is affected by different types of rot such as wet root rot and collar root rot. Hence, an attempt was made to differentiate dry root rot from *Fusarium* wilt and other root rot based on visible symptoms at seedling, foliage and roots. Comparative symptomology of these diseases is provided in following table (Sharma *et.al.* 2015)

Plant stage/part	Dry root rot	Fusarium wilt
Seedling stage	➤ not seen at this stage, unless favourable conditions occurs	➤ Young seedlings killed within 3 weeks of sowing. ➤ Whole seedling collapse and retain their dull-green colour
Flowering to podding stage	➤ Symptoms mostly observed at flowering to pod formation stage ➤ Drooping of petioles and leaflets confined only to the top of infected plant ➤ Stems of affected plants are usually straw	➤ Most commonly observed at vegetative to flowering stage ➤ Affected plant show typical wilting i.e. drooping of petioles, rachis and leaflets. ➤ Lower leaves are chlorotic but most of the other leaves droop while green ➤ Stem retain green colour

	coloured, sometime brown	
Root	➤ Tap root is generally dark and rotten and most of the lateral and finer roots are dried ➤ Micro-sclerotia can be seen underneath bark ➤ Entire root system is rotten ➤ Affected plant when uprooted breaks easily	➤ Roots generally healthy and shows no sign of rotting ➤ No microsclerotia formed ➤ Xylem vessels show brown to black discoloration when split vertically ➤ Difficult to uproot easily

Plant stage/part	Wet root rot	Collar root rot
Seedling stage	➤ Most often seen at the seedling stage (up to 6 weeks after sowing) in soils with relatively high moisture content ➤ Affected seedlings gradually turn yellow and petioles and leaflets droop. ➤ Seedlings do not usually collapse.	➤ Mostly seen at the seedling stage (up to 6 weeks after sowing), particularly if the soil is wet. ➤ Affected seedlings turn yellow. Young seedlings may collapse, but older seedlings may dry without collapsing. ➤ Leaves do not droop. ➤ When uprooted, the seedlings show rotting at the collar region and downwards. ➤ The rotten portion is covered with whitish mycelial strands
Root	➤ A distinct dark brown lesion appears above the collar region on the main stem and can extend to lower branches in older plants. ➤ The stem and root below the lesion show rotting, frequently with pinkish	➤ A white mycelial coating can be seen on the tap root of completely dried seedlings, even several days after death. ➤ If affected seedlings are uprooted from moist soil ➤ in the earlier stages of infection, rapeseed-like

	mycelial growth. ➤ Sclerotia are not usually seen.	sclerotia (1 mm in diameter), attached to mycelial strands around the collar are seen
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The disease was previously known as “*Rhizoctonia* wilt” in chickpea, later it was named as “Dry root rot”. The dry root rot disease of chickpea generally appears around flowering and podding time in the form of scattered dried plants. The seedlings can also get infected. The susceptibility of the plant to the disease increases with age. Drooping of petioles and leaflets is confined to those at the very top of the plant. Sometimes when the rest of the plant is dry, the topmost leaves are chlorotic. The leaves and stems of affected plants are usually straw colored, but in some cases the lower leaves and stems are brown. The lower portion of the tap root usually remains in the soil when plants are uprooted. The tap root is dark, shows signs of rotting, and is devoid of most of its lateral and finer roots. A grayish mycelium can sometimes be seen on tap root. Dark, minute sclerotial bodies can be seen on the roots exposed or inside of the bark or whenever split open at collar region vertically.

Dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid] is a soil and seed borne necrotrophic fungal pathogen that has a global distribution, which can infect more than 284 plant species throughout the world including monocot and dicots (Farr *et al.*, 1995). This species morphology is composed of hyphae and sclerotia (hyphal propagules). The pathogen has a predominant sclerotial stage and less frequently, a pycnidial stage in its life cycle. The sclerotia are viable for more than 12 months in the soil. It survives between crop seasons on infected plant debris. These sclerotia serve as a primary means of survival. The sclerotia float freely on soil surface when field is flooded for irrigation and become primary inoculum for emerging seedlings.

**Taxonomic Position of *Rhizoctonia bataticola*
(Gupta and Sharma, 2015)**

Division	: Basidiomycota
Sub Division	: Agaricomycotina
Class	: Agaricomycetes
Order	: Cantharellales
Family	: <i>Ceratobasidiaceae</i>
Genus	: <i>Rhizoctonia</i>
Species	: <i>bataticola</i>

Successful survival and adaptation of *Rhizoctonia bataticola* to various geographic environments has been confirmed by several workers demonstrating morphological (Mayek-Pérez *et al.*, 2001), pathogenic (Mayek-Pérez *et al.*, 2001; Su *et al.*, 2001) and genetic diversity (Vandemark *et al.*, 2000; Mayek-Pérez *et al.*, 2001; Su *et al.*, 2001; Almeida *et al.*, 2003; Jana *et al.*, 2003; Aboshosha *et al.*, 2007) in various crops. Techniques such as RAPD markers are useful for measuring genetic relationships and variations within and among populations of *Rhizoctonia bataticola* in different countries like Australia (Fuhlbohms, 1997), Brazil (Almeida *et al.*, 2003) and India (Jana *et al.*, 2003). Pathogenic variability in the target production area is a prerequisite for identifying genotypes with a stable resistance to the variable pathogen populations.

Environmental conditions like temperature, soil moisture and pH play an important role in the viability and growth of *Rhizoctonia bataticola* (Khan, 2007). *Rhizoctonia bataticola* is able to produce microsclerotia under relatively low water conditions while viability of microsclerotia is drastically reduced at high water potentials (Olaya and Abawi 1996). Temperature and soil moisture are the two important weather parameters influencing the dry root rot infection, colonization and development in chickpea. Understanding conditions of temperature and soil moisture will help in standardization of dry root rot resistance screening techniques which will help breeders in developing breeding strategy for dry root rot resistance over wider geographical areas.

R. bataticola is soil / seed borne pathogen in many crops which can survive for longer duration, about 2 to 15 years, on plant residues and / or in soil (Ratul and Singh, 2018) for longer duration. Due to its enormous inoculums potential, it seems practically difficult to manage this pathogen / disease with chemicals alone or with any single practice. Abundant and / or frequent use of chemicals alone to control such soil borne plant pathogens is impractical and uneconomical. Therefore, it is imperative to explore and exploit alternative means by integrating chemicals to manage effectively and economically, the root rot causing plant pathogens, such as *R. bataticola*.

The dry root rot of chickpea is more challenging in the abruptly changing climatic conditions where growing crop is predisposed to high temperature and moisture stress. *R. bataticola* is mainly a soil-inhabiting pathogen, many environmental and soil factors are responsible for its growth and development of disease. Systematic research related to the biology, ecology and epidemiology of dry root rot in chickpea is a need of time. Research is needed to improve the identification and characterization of variability within its epidemiological and pathological niches. Limited literature available on host plant resistance for dry root rot indicated lack of resistant sources for this disease (Sharma *et al.*, 2015).

Keeping all the said factors in view along with economic importance of chickpea and yield losses caused by dry root rot causing fungi in chickpea, the present studies were planned and conducted, during *Rabi*, 2016-17 and 2017-18 seasons, at Department of Plant Pathology, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani, with following specific objectives.

OBJECTIVES:

1. To survey chickpea fields for recording dry root rot incidence in Marathwada region.
2. To isolate, identify and prove pathogenicity of the selected isolates of dry root rot of chickpea
3. To study cultural, morphological, pathogenic and molecular variability of the *Rhizoctonia bataticola* isolates.
4. To study the effect of organic and inorganic sources of carbon and nitrogen on growth of *Rhizoctonia bataticola*.
5. To study the effect of soil moisture regimes and soil types on the incidence of *Rhizoctonia bataticola* (Pot culture)
6. Disease management strategies
 - 6.1 To evaluate *in-vitro* the bioefficacy of fungicides and bioagents against *Rhizoctonia bataticola*.
 - 6.2 To evaluate the bioefficacy of organic amendments against *Rhizoctonia bataticola*.
 - 6.3 To integrate the efficient fungicides, biocontrol agents and organic amendments for management of dry root rot (Pot culture).
7. To evaluate chickpea varieties, cultivars and germplasm against dry root rot.

***Review of
Literature***

CHAPTER – II

REVIEW OF LITERATURE

The available literature pertaining to the present investigation is reviewed under the following heads.

- 2.1. Occurrence, distribution and losses
- 2.2. Symptomatology
- 2.3. Isolation, identification and Pathogenicity
of *R. bataticola* (*M. phaseolina*)
- 2.4. Variability among the isolates of *R. bataticola* (*M. phaseolina*)
 - 2.4.1. Cultural variability
 - 2.4.1.1. Variability in radial growth
 - 2.4.1.2. Variability in colony texture
 - 2.4.1.3. Variability in colony pigmentation
 - 2.4.2. Morphological variability
 - 2.4.2.1. Variability in hyphae
 - 2.4.2.2. Variability in sclerotia
 - 2.4.3. Pathogenic virulence / variability
 - 2.4.4. Molecular variability
- 2.5. Organic and inorganic carbon and nitrogen sources
- 2.6. Soil moisture regimes and soil types
- 2.7. Disease management strategies
 - 2.7.1. Efficacy of fungicides
 - 2.7.2. Efficacy of bioagents
 - 2.7.3. Efficacy of organic amendments
- 2.8. Integrated disease management
- 2.9. Screening of chickpea cultivars / varieties against *R. bataticola*

The genus *Rhizoctonia* was first established by De Candolle (1815). Members of the genus *Rhizoctonia* D.C. are considered as a complex mixture of filamentous fungi, possessing in common a non-spored imperfect

state, usually referred to as the *Rhizoctonia* anamorph (Gonzalez *et al.*, 2006) of which several species have been recorded (Saksena and Vaasteraja, 1961; Yamamoto, 1962; Hussain and Mckeen, 1963). Among them, *Rhizoctonia bataticola*, an important species causes seed rot, root rot, charcoal rot and ashy blight in most of the cereals, pulses, oilseeds, fibre crops, vegetables, aromatic and medicinal plants, plantation and horticultural crops (Uppal, 1934; Likhite and Kulkarni, 1934; Young, 1949). The fungus *R. bataticola* is known to infect a wide range of host plants belonging to monocots and dicots with a high degree of aggressiveness, which are economically important.

2.1. Occurrence, distribution and losses

Grover and Sakhuja (1981) reported that *Rhizoctonia bataticola* [*Macrophomina phaseolina* (Tassi.) Goid] caused a severe leaf blight of *Vigna radiata*.

Ahmad and Mohammad (1986) reported dry root rot of chickpea (*Cicer arietinum* L.) incited by *Rhizoctonia bataticola* (Taub.) Butler it caused considerable reduction in yield. The incidence of dry root rot has been reported as high as 24.7 per cent in Madhya Pradesh.

Kaushik *et al.*, (1988) reported seed-borne nature of *Rhizoctonia bataticola*, causing leaf blight of Mung bean. Seeds of twenty cultivars of *Vigna radiata* showed 2.2 to 15.7% infection by *Rhizoctonia bataticola*.

Raut and Ingale (1989) reported that the ubiquitous sclerotial fungus *Rhizoctonia bataticola* (Taub.) Butler in its pycnidial state is known as *Macrophomina phaseolina* (Tassi.) Goid and caused complex disease syndrome like charcoal rot of stem, root rot, seedling blight, foliage blight, tuber decay, dry rot, fruit rot, pod and seed rot in several crops.

Prajapati *et al.*, (2003) conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh and Uttaranchal and reported that the

incidence of dry root rot of chickpea caused by (*Macrophomina phaseolina*) at 20 locations in Uttar Pradesh and Uttaranchal.

Kohire *et al.*, (2006) carried out a survey in 2004-2006 and reported that wilt; dry root rot and stunt were associated with infected chickpea in Marathawada region of Maharashtra, disease incidence varied from 1.6 to 16.5 %. Maximum average incidence was observed in Parbhani (18.5%).

Gurha and Trivedi (2008) reported that *F.oxyspora ciceri* and *R. bataticola* caused wilting and dry root rot diseases respectively in all districts of Himachal Pradesh.

Rana *et al.*, (2009) conducted survey on pea diseases and identification of fungi associated with wilt and root rot complex in Himachal Pradesh. The disease were prevalent in all pea growing areas of state during 2006-2008. The incidence of wilt and root rot was 20 per cent in temperate wet, 15 per cent in sub temperate, 12 per cent in temperate dry and 11 per cent in subtropical regions.

Manjunatha *et al.*, (2011) conducted a survey during rabi season of 2008 for recording the dry root rot incidence in chickpea and revealed that the mean maximum incidence noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18%) in Bidar. The overall incidence of dry root rot ranged from 1 to 19 per cent across the districts. The higher disease incidence in Gulbarga is due to variation in soil temperature (35-37°C) and moisture stress conditions during rabi season which ultimately led to more production of sclerotia of *Rhizoctonia* sp. on chick pea plants roots.

Moradia and Khandar (2011) surveyed a field during 2002-2003 at Main oilseeds Research station, Junagadh Agricultural university, Junagadh to study the loss of yield of groundnut due to dry root caused by *Macrophomina phaseolina* and found that maximum plant mortality (root rot) of 29.3 per cent due to *Macrophomina phaseolina* with highest yield loss of 435kg/ha. in Keshod tehsil of Junagadh district of Saurashtra region.

Rhiaz *et al.*, (2011) conducted survey in Jammu and Kashmir, Dry root rot disease caused by *Rhizoctonia bataticola* was observed on variety Radhey in Jammu Kashmir.

Khan *et al.*, (2012) surveyed during 2010-11 for the occurrence and severity of chickpea dry root rot in Jammu and Kashmir and recorded that both the divisions is concerned highest (40.00 per cent) disease was observed in village Shangus and lowest (4.11 per cent) in village Naina because of same chickpea crop sow year after year in same field where no irrigation is possible due that disease perpetuates through the debris in the field. They also recorded that the intensity of the disease incidence high in the month of February and March during late flowering and podding stage.

Ghosh *et al.*, (2013) conducted a survey in 2010-2011 rabi cropping season and obtained the information on the distribution and incidence of chickpea diseases in respect to soil type, cultivar used, seed treatment in central and southern parts of India. Local cultivars predominated in most farmers' fields (25% - 48%). 63% of the farmers were practices seed treatment with fungicide. Dry root rot and collar rot diseases were found at all of the sites and incidence ranged from 8.9% - 10.3% and 7.1% - 10.5% respectively irrespective of cultivar type and locations. Incidence of wilt and black root rot disease ranged from 9.7% - 13.8% and 6.6% - 7.4% respectively. Black root rot disease was found in Andhra Pradesh and Karnataka states only and indicated that dry root rot and collar rot is currently highly distributed in all surveyed chickpea growing areas of central and southern parts of India.

Veena *et al.*, (2014a) noted that dry root rot by *Rhizoctonia bataticola* (Taub.) Butler cause considerable yield losses in chickpea which may be as high as 50 to 71 per cent.

Shrinivas P. (2016) conducted a survey of dry root rot disease caused by *Rhizoctonia bataticola* (Taub.)Butler [pycnidial stage: *Macrophomia phaseolina* (Tassi) Goid] in January, 2014 rabi cropping season in different

chickpea growing locations of central (Madhya Pradesh and Maharashtra) and southern (Andhra Pradesh, Telangana and Karnataka) India and observed that the maximum dry root rot incidence was in Telangana (18.28 %) and the least in Maharashtra (5.38%) due to diversified weather conditions and variations in sowing dates in different states, different crop growth stages *i.e.* from seedling (20 days old) to podding stage. Disease observed in all stages and occurrence observed irrespective of cropping system, soil types and cultivars. Mostly the disease incidence was low in the irrigated fields compared to rainfed fields.

Kadam M.A. (2016) conducted a roving survey in Marathwada region to record the disease incidence of dry root rot of chickpea and observed that highest disease incidence in Latur district (23 %) and lowest in Aurangabad district (10.20 %).

2.2. Symptomatology

Dastur (1935) reported that only mature plants were affected, showing bronzing of the leaves on one or more of the lower branches. The colour of leaves later changed to yellow and then brown. The affected branches and leaf stalks of diseased plants were stiff and turned upwards and leaflets stand more or less vertically and shed prematurely. The terminal part of the tap root was black or brown and shriveled.

Dhingra and Sinclair (1978) described that *Rhizoctonia bataticola* (Taub) Butler (Pycnidial stage: *M. phaseolina*) caused various diseases *viz.*, root rot, seedling blight, charcoal rot, stalk rot, wilt, stem blight, seedling decay, leaf blight etc.

Nene *et al.*, (1978) reported that *R. bataticola* in chickpea caused continuous black discoloration of pith and xylem vessels of roots and basal shoots, as one of the diagnostic symptoms.

Casas and Diaz (1985) described the symptoms induced by *Macrophomina phaseolina* in chickpea, as dry collar and root rots and affected

plants developed dark brown to black roots and dried up without showing any flaccidity.

Trapero and Jimenez (1985) described the symptoms of dry collar and root rot symptoms found affected during disease survey. Disease occurred at seed development stage. Stems and leaves dried up and became straw colored and roots and collars became dry and brittle. Sclerotia of *M. phaseolina* occurred under the epidermis as well as in inner tissues of the lower stem, which appeared grayish. Affected plants produced pods that were either empty or contained small, poor quality seed.

Vishwadhar and Sarbhoy (1989) reported that *Rhizoctonia bataticola* (Taub.) Butler (Pycnidial stage - *Macrophomina phaseolina* (Tassi.) Goid) causes seed / seeding rot, root rot, ashy stem blight and charcoal rot of soybean in India.

Singh *et al.* (1990) reported that roots inoculated with *Rhizoctonia bataticola* (*M. phaseolina*) caused disintegration of cortical tissues with mycelial and sclerotial bodies plugging the xylem vessels.

Haware *et al.*, (1990) reported that the disease development was influenced by dry soil condition especially at flowering stage. The plant suddenly dries in the field. The leaves and stem of affected plants usually become straw colored. The tap root is dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissue.

Abawi and Pastor (1990) reported that *Macrophomina phaseolina*, caused dark lesions on the epicotyls and hypocotyls of the beans seedlings and death followed by obstruction of xylem vessel resulted in wilting. In adult plants, the pathogen induced red to brown lesions on roots and stems and produced dark mycelia and black microsclerotia. Affected stem showed longitudinal dark lesions, plants defoliated and wilted.

Nene *et al.*, (1991) described that the dried plants due to disease generally appeared during flowering and podding time. These were scattered in the field. The seedlings can also get infected. Drooping of petioles and leaflets was confined to those at top of the plant. Leaves and stem of affected plant usually straw coloured but in some cases, the lower portion of the stem was brown. When the plant was uprooted, lower portion of the tap root remains in the soil. It was devoid of lateral and finer roots. Dark minute sclerotial bodies can be seen on the roots exposed inside the wood.

Beniwal *et al.*, (1992) reported sudden drying of the whole plant was the most conspicuous symptom of this disease. The tap root of infected plant become dark showed signs of rotting, and is devoid of most of the lateral and finer roots. The dead root was brittle and its tip was easily broken.

Pande *et al.*, (2004) noted that dry root rot (*R. bataticola*) of chickpea, generally appeared during late flowering and podding stages and the infected plants dries completely. Root system of the diseased plants showed extensive rotting, lateral roots gets brittle and minute sclerotial bodies appeared in the pith cavity and on the outer surface of the tap root.

Jana *et al.*, (2005) reported that *Macrophomina phaseolina*, a soil borne pathogen, infected soybean plants through roots, moved up to collar region of the stems, caused discoloration or charcoal colour at collar region and foliar chlorosis.

Meena *et al.*, (2006a) reported that charcoal rot (*Macrophomina phaseolina*) of maize favored by high soil temperature (30 to 42⁰ C) and low soil moisture. The pathogen over-winters as sclerotia in soil and upon germination penetrate roots and lower stems, during growing season. Produce numerous, minute and black sclerotia, particularly on the vascular bundles and outside of rind of the stalk.

Das *et al.*, (2007) reported that charcoal rot (*M. phaseolina*) infected sorghum plant roots and lower stem showed water-soaked lesions that

slowly turned brown or black and pith of the infected stalk disintegrated across several nodes.

Monga *et al.*, (2007) reported root rot (*R. bataticola* and *R. solani*) as one of the most serious soil borne diseases of cotton, particularly in the irrigated northern cotton zone. The disease appeared in June and characterized by sudden and complete wilting of the plants. The affected plants can easily be pulled out of ground and the bark of affected roots developed cracks giving yellowish appearance.

Monga and Sheo Raj (2008) reported the symptoms of dry root rot (*R. bataticola*) in cotton as: complete sudden wilting of plants and every leaf from top to bottom drooped down in such plants. In the beginning, a few plants randomly wilted, which later get converted into circular diseased patches. The pathogen actually infects much earlier to the cotton crop, but wilting symptoms manifested at later stage of crop. The affected plants can easily be pulled out of the ground and root bark broken into shreds, giving yellowish appearance.

Weem *et al.*, (2011) reported first sunflower charcoal rot (*M. phaseolina*) from Illinois and described its symptoms as: silvery lesions causing girdling of lower stems and premature death of sunflower plants. When lower stem and roots were split open, the pith tissue compressed into layers. Black micro-sclerotia produced on outside of the lower stem tissue and on stem vascular tissues.

Khan *et al.*, (2012) noted the symptoms of the dry root rot of chickpea disease were yellowing of the leaves within a day or such leaves drop and plant showed completely dried symptoms within a week after the appearance of the first symptoms. If the plants were pulled out from the soil and examined the basal stem and main root system of diseased plant showed extensive rooting with most of the lateral root destroyed. The tissues weakened

and break off easily. In advance cases the sclerotial bodies may scatted in the pith cavity and on the outer surface of the tap root.

Shrama and Pande (2013) noted the symptoms of DRR affected chickpea plants that the plants are usually straw coloured, in some cases the lower leaves and stems show a brown colour. The tap root is black, rotten and devoid of most of the lateral and fine roots. The dead root is quite brittle and shows shredding of the bark. Dark minute sclerotial bodies can be seen on the roots exposed or inside the wood.

Sharma *et al.*, (2015) described that the DRR symptoms are most commonly observed in chickpea during post-flowering stage which include drooping and chlorosis of petioles and leaflets, initially confined to top leaves of the plant. Leaves and stems of affected plants are usually straw coloured and in some cases, the lower leaves and stems are brown. The tap root turns black with signs of rotting and is devoid of most of the lateral and finer roots. The dead roots are quite brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark or when split open at the collar region vertically.

2.3. Isolation, identification and pathogenicity of *R. bataticola* (*M. phaseolina*)

Halsted (1890) recognized *Rhizoctonia bataticola* (Taub.) Butler as a plant pathogen for the first time.

Taubenhaus (1913) gave the name of the genus as *Sclerotium* because of absence of spores and the species named as *bataticola* because it was pathogenic to *Ipomea batatus* (L.) Lam.

Briton Jones (1925) transferred the fungus to the genus *Rhizoctonia* based on the identification of cultures by Butler (1918).

Ashby (1927) accepted *Macrophomina* and rejected the binomial *Macrophomina phaseoli* and proposed a new binomial *M. phaseolina* as the pycnidial stage of *R. bataticola* on the basis that *M. phaseoli* was the earliest applicable binomial. He was not aware that Tassi had earlier described *Macrophomina phaseolina*.

Haigh (1930) suggested that *R. bataticola* be used for sclerotia isolate and pycnidial strains should be called *M. phaseolina*.

Goidanich (1947) examined the original material of Tassi and compared it with *Macrophomina phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinensis*, *Dothiorella cajani* and *D. phaseoli*. He confirmed all of them were identical. He corrected the mistake made by Ashby. According to the International Code of Botanical Nomenclature, the binomial *Macrophomia phaseolina* was the valid name for the pycnidial stage of *R bataticola*.

Reichert and Hellinger (1947) reported that mycelia width varied from approximately 2-11µm and distance between two consecutive septa measured 46µm. However, the most important character regarding taxonomy and classification were the production size and composition of microsclerotia.

Short *et al.*, (1978) described that during the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies called microsclerotia. The microsclerotia were black and variable in size from 50–150 µm depending on the available nutrients of the substrate on which the propagules were produced.

Surichandraselvan and Seetharaman (2002) reported that most virulent isolate of *M. phaseolina* showed significant faster growth (41.3mm) on PDA even 24 hrs after inoculation, while the least virulent exhibited slowest growth (19.7mm).

Katariya and Gaur (2007) isolated nine isolates of *R. bataticola* and tested pathogenicity of these isolates both in sterilized and unsterilized soil in Chickpea variety C-325.

Sharma *et al.*, (2012a) isolated the 50 isolates of *R. bataticola* collected during different years (2004 to 2010) from different chickpea growing regions of India. They reported that all the isolates proved to be pathogenic on chickpea cultivar BG 212. There was significant variation in virulence of the pathogen, but there was no clear pattern of distribution linked to the virulence of the isolates.

Sharma and Pande, (2013) isolated a pathogenic isolate of *R. bataticola*, from naturally infected chickpea plant at ICRISAT, Patancheru and purified using mono-sclerotia and maintained on PDA slants at 5°C in refrigerator.

Veena *et al.*, (2014a) isolated a virulent isolate of *Rhizoctonia bataticola* from infected chickpea plants by using tissue segment method and identified the pathogen based on its mycelial and sclerotial characters and pathogenicity test proved by soil inoculation method.

Veena *et al.*, (2016) isolated the pathogen *Rhizoctonia bataticola* from dry root rot infected chickpea plants by using tissue segment method and purified by single hyphal tip method and maintained on PDA.

2.4. Variability among the isolates of *R. bataticola* (*M. phaseolina*)

2.4.1. Cultural variability

2.4.1.1. Variability in radial growth

Dhingra and Sinclair (1972) collected nine isolates of *M. phaseolina* from various parts of United States reported variation in growth rate and colony characters. Further, the growth rate was correlated with pathogenicity.

Chohan and Kaur (1976) studied cultural variability among four isolates of *R. bataticola*, RB-1 (from sunflower roots) RB-2 and RB-3 (sunflower stems) and RB-4 (stem of mash, *Phaseolus mungo*). Mean colony diameter of the four isolates viz., RB-1, RB-2, RB-3, and RB-4, was 9.31, 7.67, 8.50 and 8.57 cm, respectively. Hyphal diameter of RB-4 was minimum (1.15 μ m) and RB-1 maximum (1.94 μ m). All the four isolates formed mostly black sclerotia; there was no difference in the manner, or behaviour pattern of sclerotial formation and in colour (black).

Raut and Ingle (1989) studied cultural and sclerotial variations among the isolates of *R. bataticola* isolated from different cultivated crops belonging to 7 families and reported variations in their growth rate, sclerotia production, pycnidia and size of pycnidiospores. On PDA, sesamum, pigeonpea, soybean and cluster bean isolates (*R. bataticola*) showed excellent mycelial growth (between 90 and 80 mm); isolates from cotton, cowpea, chickpea, sunflower, safflower and chilli showed moderate growth (between 80 and 51 mm); and green gram, black gram, okra and citrus isolates showed poor growth (from 31 to 50 mm). Comparatively cotton and cowpea isolates produced more sclerotia and their size varied with the host. The sclerotia of cotton isolates were larger (115.06 μ m) and of soybean isolates smaller (56.33 μ m).

Lokesha and Benagi (2007) reported among four isolates of *M. phaseolina* in pigeon pea maximum growth was recorded in ICRISAT isolate which was significantly superior over other isolates (82.7 mm) followed by Bidar isolate (78.35 mm).

Chowdary and Govindaiah, (2007) reported that the radial growth was found maximum after 24 hours of incubation on PDA (40.3 mm) followed by potato sucrose agar (39.0 mm) and it completely covered the PDA plate by 48 hours of incubation.

Aghakhani and Dubey, (2009a) collected twenty three isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their cultural characters. After 48 hours of incubation, highest radial growth in Rb1 (Bangalore) followed Rb4 (Faridkot) was observed. Rb3 (New Delhi) and Rb5 (Faridkot) were next best with statistically similar colony diameters (83.5 mm). Least and similar colony diameter of 51.3 mm was recorded in Rb7 and Rb10 (Kanpur) isolates.

Charul and Biwas (2009) studied morphological variability among eight isolates of *Rhizoctonia bataticola*, causing leaf spot/blight of pigeonpea. They reported that T₁, T₂, T₃, and T₄ isolates as fast growing with colony diameter in the range of 8.0 to 8.7 cm. While, T₅, T₆ and T₈ isolate were slightly slow growing, with colony diameter between 7.1 to 7.5 cm. Mycelium of the T₁, T₂, T₃ and T₄ isolates was partially fluffy and dark brown coloured; while, of the isolates T₅, T₈, T₆ and T₇ isolates was fluffy and dark brown.

Manjunatha and Naik, (2011) collected thirty isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea from three major chickpea growing regions were highly variable in their cultural characters. Among these, twelve isolates were fast in growth, another twelve were moderate and the remaining six isolates were slow in growth.

Garg and Kumar (2012) reported that *Macrophomina phaseolina* differed widely in morphological and cultural characteristics. The mycelium was hyaline and light to dark brown, profusely branched and constricted at septa. Sclerotia were oval to spherical, dark brown to black and measured 62.5 µm to 100 µm in diameter in culture.

Raja Mohan and Balabaskar (2012) studied morpho-cultural characters of 20 isolates of *M. phaseolina*, causing dry root rot of groundnut. They reported that all of the isolates produced cottony white, light gray / gray /

black gray and black coloured mycelium on PDA medium. The isolates MP1, MP11, MP13 and MP18 produced significantly maximum (90 mm) mycelial growth, while it was minimum (75.00 mm) in MP5. The biggest (105.10 μm) and the smallest (85.70 μm) sclerotia were produced by MP11 and MP5 isolate, respectively.

Sharma *et al.*, (2012a) isolated 94 isolates of *R. bataticola*, causing dry root rot of chickpea and collected from different chickpea growing regions of India. Most of the isolates produced aerial mycelium and the growth rate varied between 0.97-2.2 cm/day. Colony colour varied from light black to black and light gray to gray, colony shape varied from round to irregular. The reverse side colony colour varied from light black to black. The sclerotia produced within 36 to 48 hrs. and their intensity varied from moderate to very high. Shape of the sclerotia varied from oblong, ellipsoidal, irregular and round.

Sundravadana *et al.*, (2012) explored morpho-cultural variability among *R. bataticola* isolates (roots, leaf and seeds) of various pulses, causing dry root rot. They reported on PDA maximum colony growth of redgram isolates (74.44-76.71 mm) followed by cowpea isolates (64.72-70.76 mm) blackgram isolates (58.78-71.77 mm) greengram isolates (58.74-62.44 mm) and soybean isolates (54.63-58.19 mm). Among the isolates, the sclerotia size ranged from 43.6 μm to 89.9 μm . The blackgram root isolate produced large sized (89.9 μm) sclerotia, followed by redgram root isolate (86.85 μm). The sclerotia of the shoot and root isolates of soybean were the smaller (43.6 and 43.8 μm , respectively). Shape of the sclerotia was categorized into two groups viz., round and irregularly oblong. Blackgram and redgram shoot sclerotia were round, while rest were irregularly oblong.

Iqbal and Mukhtar (2014) collected 65 isolates of charcoal rot caused by *M. phaseolina* in mung bean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan were studied for cultural variability. Significant differences were noticed among 65 isolates with sixteen isolate rated as fast

growing, 11 as slow growing and the rest as medium growing. Maximum colony diameter of 87.17 and 86.67 mm was observed in isolate MP- 7 (Dera Ghazi Khan) and MP-26 (Layyah) proved to be the fast growing, while isolates MP-8, MP-29 and MP-30 showed minimum radial growth and rated as slow growing.

Meena *et al.*, (2015) studied morpho-cultural variability among 6 isolates of *R. bataticola*, causing charcoal rot of peanut. They reported that three isolates (I₁, I₄, and I₅) showed fast growth (>80mm) and rest three (I₂, I₃, and I₆) showed slightly slow growth (<75mm) on PDA. The mycelium was partially fluffy (I₁ and I₂) to fluffy (I₃, I₄, I₅ and I₆). The colony colour was light brown (I₁ and I₅) dark brown (I₂, I₃, I₆) and light green (I₂, I₃, I₆).

2.4.1.2. Variability in colony texture

Dhingra and Sinclair (1978) noticed that fluffy growth in stem isolates, partially fluffy growth in root isolates and appressed growth in all other plant part isolates of *Macrophomina phaseolina* on soybean.

Devi and Singh, (1998a) collected thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India and observed MP-1 isolate had cottony mycelial growth while MP-2 isolate showed 5 µm appressed mycelial growth.

Devi and Singh (1998b) studied cultural variations among 36 isolates of *M. phaseolina* (= *R. bataticola*) isolated from roots, pods and seeds of diseased urdbean plants, collected from various locations in India and categorized them as MP-1, MP-2 and MP-3 isolates. MP-1 isolates produced cottony mycelium with pale gray colonies, MP-2 isolates produced appressed mycelium growth with brown colonies and MP-3 isolates produced smokey gray profuse mycelium. Sclerotia were irregular shaped and big sized (400 x 280 µm) in MP-1 and MP-2 and round in MP-3 isolate. Pycnidia size was maximum in MP-3 (192 µm) and medium (186 µm) in MP-1 and MP-2

isolates, with globose shepe of the pycnidia. The pycniospores measured 19.5x6.2 μm and 17.6x5.9 μm in the isolates MP-2 and 3, respectively.

Fernandez *et al.*, (2006) analyzed the diversity patterns of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans based on presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth.

Meena *et al.*, (2006a) studied morpho-cultural variability among seven isolates *Macrophomina phaseolina*, causing charcoal rot of maize, collected from different agro-ecological zones of India. Based on colony colour, two each isolates were grayish white, blackish gray and dark gray, and one isolate was whitish. On the basis of mycelial growth and colony texture, the Coimbatore isolate produced thick fluffy, cottony mycelial growth; lesser cottony growth in Albhavi, Bangalore and Udaipur isolates; while it was less fluffy in Dehli and Ludhiana isolates. Based on number of sclerotia per microscopic field and the size of sclerotia, large sized sclerotia (95.7 μm) with maximum number (52.00) were produced in Hyderabad isolate, followed by the isolates of Dehli (86.4 μm and 50.3) Ludhiana (77.8 μm and 49.6) Bangalore (48.6 μm and 76.5) Udaipur and Arabhavi (45.0 μm and 76.7-78.8) and least in Coimbatore isolate (66.9 μm and 44.0).

Aboshosha *et al.*, (2007) isolated 33 *M. phaseolina* isolates from diseased sunflower crop grown (2003, 2004) in various region of El-Behera and characterized them for pathogenicity, pathogenic variability and colony phenotype. Colony phenotype of the test isolates was also variable. Colonies were mostly grey, dark green, black or brown on PDA; whereas, on Czapek dox agar colonies were mostly black, brown, dark green or white. The colony growth pattern was either dense, light dense or sparse dense on different media tested.

Charul and Biwas (2009) studied morphological variability among eight isolates of *Rhizoctonia bataticola*, causing leaf spot/blight of pigeon pea. They reported that T₁, T₂, T₃, and T₄ isolates as fast growing with colony diameter in the range of 8.0 to 8.7 cm. While, T₅, T₆ and T₈ isolate were slightly slow growing, with colony diameter between 7.1 to 7.5 cm. Mycelium of the T₁, T₂, T₃ and T₄ isolates was partially fluffy and dark brown coloured; while, of the isolates T₅, T₈, T₆ and T₇ isolates was fluffy and dark brown.

Aghakhani and Dubey, (2009) collected twenty three isolates of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) causing dry root rot of chickpea from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chhattisgarh) of India were highly variable in colony texture. The majority of nineteen isolates produced suppressed mycelium whereas 4 isolates in which three from Kanpur and one from Rohtak produced aerial mycelium.

Edraki and Banihashemi (2010) studied phenotypic diversity among 60 isolates of *Macrophomina phaseolina*, causing charcoal rot in various crops hosts, collected from different parts of Iran. Based on morphological features and growth pattern on PDA, the isolates were grouped into four phenotypes such as fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth. Fluffy isolates with few sclerotia produced smaller sclerotia and maximum growth rate. Sclerotial dimensions varied from 80-220 µm (isolate fluffy with few sclerotia) 110-167 µm (isolate fluffy with abundant sclerotia) 114-180 µm (isolate partially fluffy) and 87-220 µm (isolate appressed growth).

Manjunatha and Naik (2011) collected thirty isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea from three major chickpea growing regions were highly variable in their colony texture. These isolates were categorized into three groups. The first group comprised of all isolates from Bidar (except RB7) exhibited light brown with fluffy growth and

formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur, Gulbarga (except RG3 and RG7) and an isolate from Bidar (RB7) exhibited brownish black with flat colony and sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown colour sclerotia formed at the center of the culture.

Sundravadana *et al.*, (2012) categorized variability among the eleven isolates of *R. bataticola* from different pulse crops (redgram, greengram, cowpea, soybean, blackgram) into three groups viz., linear, fluffy and linear at the end with fluffy growth at the centre. All isolates were more aggressive on the original host from which it was isolated, which was shown by the variability in pathogenic characters.

Sharma *et al.*, (2012a) studied morpho-cultural variability among 94 isolates of *R. bataticola* (*M. phaseolina*) causing dry root rot of chickpea. Average mycelia radial growth of the 94 isolates showed significant differences. Colony colour varied from light black to black and light grey to grey, colony varied from radial to irregular. Most of the isolates produce too low aerial mycelium. Reverse side colony colour ranged from light black to black but most of the isolates showed black colour. Most of the isolates grew very fast and covered the plate within 96 hrs., while other isolates like RB 21, RB31, RB7, RB84 and RB87 grew slowly and showed suppressed growth. Most of the isolates produced the sclerotia within 36-48 hrs. of incubation and their density varied from very less to high and was moderate in majority of isolates. Shape of the sclerotia varied from oblong, ellipsoidal, irregular and round.

Sayyad *et al.*, (2015a) studied cultural characteristics of *Macrophomina phaseolina* isolated from chickpea, on six culture media. They reported on PDA maximum mycelial growth and excellent sclerotial production, abundant mycelium, aerial as well as submerged wooly; colour

initially white later changed to dark brown after 5 days. Sclerotia appeared after 5 days of inoculation.

2.4.1.3. Variability in colony pigmentation

Devi and Singh, (1998a) collected thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India were studied for colony colour. MP-1 isolate had pale grey colonies while brown in MP-2.

Fernandez *et al.*, (2006) The diversity patterns of 30 isolates of *M. phaseolina* of beans 15 from Mexico and 15 from other countries were analyzed based on morphological characteristics such as colour of colony (grey and black) and observed 22 isolates were grey, 4 isolates were white and 4 isolates were black in colour.

Aboshosha *et al.*, (2007) observed colonies of *Macrophomina* isolates of sunflower charcoal rot were mostly grey, dark grey, black and brown for fourteen stem isolates in frequencies of 4, 3, 5 and 1 on PDA and 4, 5, 2 and 2 on Czapek-Dox Agar while white colony was observed in only one isolate on both media.

2.4.2. Morphological variability

2.4.2.1. Variability in hyphae

Devi and Singh (1998a) collected thirty six isolates of *M. phaseolina* of mung bean from different region of North, South, North East and Central India were studied for morphological variability. Typical right-angled branching of mycelium was found in isolate MP-1 and it was acute to right angled in isolates MP-2 and MP-3.

Ratnoo *et al.*, (1997) studied cultural, morphological and pathogenic variations among the isolates characters. Studied two isolates, from which it was found that M1 isolate showed dark gray mycelium, fluffy growth,

hyphae thin, sclerotia formation abundant, blackish globose to round and measure 49 - 76 μm in diameter. M2 isolate was gray cottony and fast growth of mycelium, hyphae thin dark brown, round and measured 45-53 μm in diameter.

Suryawanshi *et al.*, (2008) isolated the Pathogen from T9, PU 30, TPU 4, UK 17, UPU97-10 of black gram plants from stem and leaves. The mycelium was superficial, hyaline to brown, septate and tree like in form.

Gupta *et al.*, (2012) observed variation in size of hyphal width of 40 *Rhizoctonia bataticola* isolates from chickpea. 18 highly virulent isolates were classified under very thin (5.2-6.5 μm) 17 highly virulent and 2 virulent isolates under thin (6.6 -7.9 μm) and only three highly virulent isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 μm) thick hyphal width.

Sangeetha and Jahagirdar (2013b) studied variability among the pathogens causing root rot/wilt of soybean viz., *Sclerotium rolfsii*, *Rhizoctonia bataticola* and *Fusarium* sp. The association of these pathogens involving two or three varied from region to region. On the basis of morphological characters, the pathogens were identified as *Sclerotium rolfsii*, which produced mustard sized sclerotial bodies, *R. bataticola* produced brown to black, right angle branched mycelia with septa and *Fusarium* sp. produced three kinds of spores viz., microconidia, macroconidia and chlamydospores.

Sharma *et al.*, (2015) reported that *R. bataticola* causing dry root rot of Chickpea grew better on PDA and produced brown to gray coloured mycelium, which turned darker with age. The young hyphae thin, hyaline, aseptate and dichotomously branched, which latter produced black sclerotia. The sclerotia formed were black, smooth, spherical or oblong to irregular shaped.

2.4.2.2. Variability in sclerotia

Hildebrand *et al.*, (1945) observed the sclerotia were black, smooth varying from spherical to oblong to irregular shaped. The sclerotial shape of any isolate varied with substrate on which fungus was grown.

Ghosh and Sen (1973) reported that sclerotial diameter of *Rhizoctonia bataticola* in gram varied from 98.1-119.2 μm .

Raut and Ingle (1989) studied cultural and sclerotial variations among the isolates of *R. bataticola* isolated from different cultivated crops belonging to 7 families and reported variations in their growth rate, sclerotia production, pycnidia and size of pycnidiospores. On PDA, sesamum, pigeonpea, soybean and cluster bean isolates (*R. bataticola*) showed excellent mycelial growth (between 90 and 80 mm); isolates from cotton, cowpea, chickpea, sunflower, safflower and chilli showed moderate growth (between 80 and 51 mm); and green gram, black gram, okra and citrus isolates showed poor growth (from 31 to 50 mm). Comparatively cotton and cowpea isolates produced more sclerotia and their size varied with the host. The sclerotia of cotton isolates were larger (115.06 μm) and of soybean isolates smaller (56.33 μm).

Ratnoo *et al.*, (1997) studied cultural, morphological and pathogenic variations among the isolates characters. Studied two isolates, from which it was found that M1 isolate showed dark gray mycelium, fluffy growth, hyphae thin, sclerotia formation abundant, blackish globose to round and measure 49 - 76 μm in diameter. M2 isolate was gray cottony and fast growth of mycelium, hyphae thin dark brown, round and measured 45-53 μm in diameter.

Devi and Singh (1998) Morphological variability among thirty six isolates of *M. phaseolina* from mung bean collected from different region of North, South, North East and Central India were studied for by and observed bigger sclerotia in isolate MP-2 (400 x 280 μm).

Shekhar *et al.*, (2006) collected Seven isolates of *M. phaseolina* on maize collected from Ludhiana, Hyderabad, Delhi, Arabhavi, Udaipur, Bangalore and Coimbatore, observed Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia 9 mm⁻¹ disc) of bigger size (95.7 µm) compared to other isolates. On the basis of sclerotial morphology, isolates were divided into oblong shape having a regular edges and round with regular edges. On the basis of disease expression, Hyderabad isolate was more virulent compared to other isolates.

Fernandez *et al.*, (2006) analyzed the diversity patterns of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans based on morphological characteristics such as shape (round, ovoid and irregular) and production of microsclerotia. Microsclerotia were higher in most Mexican isolates than in non-Mexican isolates. However, no significant differences between Mexican and non-Mexican isolates were found in length, width and length/width ratios of microsclerotia.

Meena *et al.*, (2006a) studied morpho-cultural variability among seven isolates *Macrophomina phaseolina*, causing charcoal rot of maize, collected from different agro-ecological zones of India. Based on number of sclerotia per microscopic field and the size of sclerotia, large sized sclerotia (95.7 µm) with maximum number (52.00) were produced in Hyderabad isolate, followed by the isolates of Delhi (86.4 µm and 50.3) Ludhiana (77.8 µm and 49.6) Bangalore (48.6 µm and 76.5) Udaipur and Arabhavi (45.0 µm and 76.7-78.8) and least in Coimbatore isolate (66.9 µm and 44.0).

Monga *et al.*, (2007) studied morphological characters of 23 *Rhizoctonia bataticola* isolates and reported variations in colony colour which ranged from creamish to brown mycelium. Sclerotial production was abundant their size was minute to large. Sclerotia were singly large and circular in the isolates RS-DWD, RJ-3-6 and RS-CMB. In *R. bataticola*, whitish mycelial growth on the surface of black mycelial mats was observed and based on pigmentation; the isolates were grouped into four.

Suryawanshi *et al.*, (2008) isolated *Macrophomina phaseolina* from T9, PU 30, TPU 4, UK 17, UPU97-10 cultivars of black gram from stem and leaves. The typically dark black to brown colony observed on PDA medium. The mycelium was superficial, hyaline to brown, septate and tree like in form. The sclerotia produced were round, oblong and irregular in shape.

Aghakhani and Dubey (2009a) collected twenty three isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their morphological characters. The sclerotia formed in different isolates were dark brown to black and size varied from 40 - 600 μm . The isolates from Bangalore (Rb1) and Faridkot (Rb 5) produced the largest size sclerotia ranged from 200 – 600 μm and 100 – 400 μm respectively. Dumka (Rb19) isolate produced small sized sclerotia which ranged from 43.9- 81 μm . They observed most virulent isolate (Rb1 from Bangalore, Karnataka) was fast growing and produced largest sclerotia. The isolates were highly variable in virulence.

Edraki and Banihashemi (2010) studied phenotypic diversity among 60 isolates of *Macrophomina phaseolina*, causing charcoal rot in various crops hosts, collected from different parts of Iran. Based on morphological features and growth pattern on PDA, the isolates were grouped into four phenotypes such as fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth. Fluffy isolates with few sclerotia produced smaller sclerotia and maximum growth rate. Sclerotial dimensions varied from 80-220 μm (isolate fluffy with few sclerotia) 110-167 μm (isolate fluffy with abundant sclerotia) 114-180 μm (isolate partially fluffy) and 87-220 μm (isolate appressed growth).

Linhai *et al.*, (2011) studied Variations in the 35 isolates of *M. phaseolina* collected from the major sesame producing regions in central China comprising of Hubei, Henan, Anhui, Jiangxi province. Sclerotia quantity and

size had high variations. Pingyu and Xiaogan produced no sclerotia after 24 h of culture and sclerotium size of 7 days old culture varied between 0.10 mm (Wuxue) and 0.16 mm (Jiayu) with average of 0.13 mm.

Manjunatha and Naik (2011) collected thirty isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions variable in their characters were categorized into three groups. The first group comprised of all isolates from Bidar except RB7 and exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur and Gulbarga except RG3 and RG7 and also an isolate from Bidar RB7 and showed brownish black with flat sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown coloured sclerotia formed at the center of the culture.

Mahdizadeh *et al.*, (2011) recovered fifty two *Macrophomina phaseolina* isolates from 24 host plant species and studied for average size and time taken for microsclerotia formation. Isolates from Khorasan and Isfahan kashan had the largest microsclerotia while isolates Yazd and Kerman had the smallest microsclerotia.

Sharma *et al.*, (2012a) isolated 94 isolates of *R. bataticola*, causing dry root rot of chickpea and collected from different chickpea growing reasons of India. Most of the isolates produced arial mycelium and the growth rate varied between 0.97-2.2 cm/day. Colony colour varied from light black to black and light gray to gray, colony shape varied from round to irregular. The reverse side colony colour varied from light black to black. The sclerotia produced within 36 to 48 hrs. and their intensity varied from moderate to very high. Shape of the sclerotia varied from oblong, ellipsoidal, irregular and round.

Garg and Kumar (2012) reported that *Macrophomina phaseolina* differed widely in morphological and cultural characteristics. Sclerotia were oval to spherical, dark brown to black and measured 62.5 µm to 100 µm in diameter in culture.

Gupta *et al.*, (2012) studied forty isolates of *Rhizoctonia bataticola* from chickpea and observed sclerotial sizes varied from 72.7- 117.2 x 57.1-106.5 µm. Based on sclerotial size, the isolates were divided into three groups' viz. minute size (72.7 - 87.5 x 57.1-73.0 µm) medium size (87.6 - 103.2 x 73.5 - 90.0 µm) and bold (103.3 -117.2 x 90.1-106.5 µm) size of sclerotia groups. Bold size of sclerotia was observed in six highly virulent isolates belong to Madhya Pradesh and Maharashtra. Rest of the thirty two highly virulent and two virulent isolates placed in minute and medium size of sclerotial groups. The isolate from Betul, Madhya Pradesh produced maximum number of sclerotia (22.3 sclerotia per microscopic field) and isolate from Dhule, Maharashtra exhibited minimum number of sclerotia (8.1 sclerotia per microscopic field). The shape of sclerotia ranged from oblong (12 isolates) to round (28 isolates).

Kataria *et al.*, (2012) studied cultural variability among the isolates of *R. bataticola*, causing dry root rot of chickpea and reported that there was no direct relationship between cultural characters and virulence of the isolates. The isolate BKN with scanty and suppressed growth was highly pathogenic. Isolates of SGN, NHR, DUR, HAU, DEL and HNG with varied colony growth were moderately pathogenic and the rest of two isolates CHR and UPR with different mycelium characters were less pathogenic. Sclerotia were produced at 40 to 48h after inoculation on Petri plates in most of the isolates. Sclerotia were oval to spherical shaped, ablong, ellipsoid and round, black to dark brown coloured, intensity varied from moderate to high in majority of the isolates and size ranged between 62.4-76.8 µm.

Raja Mohan and Balabaskar (2012) studied morpho-cultural characters of 20 isolates of *M. phaseolina*, causing dry root rot of groundnut.

They reported that all of the isolates produced cottony white, light gray / gray / black gray and black coloured mycelium on PDA medium. The isolates MP1, MP11, MP13 and MP18 produced significantly maximum (90 mm) mycelial growth, while it was minimum (75.00 mm) in MP5. The biggest (105.10 μm) and the smallest (85.70 μm) sclerotia were produced by MP11 and MP5 isolate, respectively.

Sundravadana *et al.*, (2012) reported that eleven isolates of *R. bataticola* from different pulse crops viz., redgram, greengram, cowpea, soybean, blackgram showed variability in sclerotial intensity, shape and intensity of pigment synthesis. All isolates were more aggressive on the original host from which it was isolated than on other hosts.

Varma and Pathe (2013) isolated 22 isolates of *R. bataticola*, causing soybean charcoal rot and studied their morphological and cultural variability. They reported that 22 isolates (*R. bataticola*) do not differed in the type of mycelial growth and growth rate on Potato dextrose agar medium. Morphologically, the test isolates revealed variability in size of sclerotia (72.1-99.9 x 57.1-106.5 μm) and number of sclerotia (9.2-22.2) based on which the isolates were categorized into three groups *i.e.* small, medium and large sized. They concluded that the isolates with fluffy growth, small size sclerotia, dark black colony and maximum sclerotia production were highly virulent, but no positive correlation, with virulence were confirmed.

Prasad *et al.*, (2014) collected isolates of *M. phaseolina* from semi arid regions of Maharashtra, Andhra Pradesh and Karnataka states of India and studied variability in shape, length, width and length/width in morphological characters of microsclerotia and observed that AP 21 isolate varied in length/width ratio (483.3/200.0 μm) while no significant differences were observed in others.

Singh *et al.*, (2014) studied morpho-cultural variability of 15 isolates of *Rhizoctonia solani*, infecting maize and rice, from trans-genetic

plains of India. The isolates from maize were fast growing and produced compact dense/fluffy, pale white to yellowish colony on PDA; produced scattered to regular smooth surfaced sparse sclerotia. The rice isolates produced light weight, sub-central to scattered and pitted sclerotia; medium sized, compact sparse colony growth and light brown to dark brown colony.

Iqbal and Mukhtar (2014) collected 65 specimens of charcoal rot caused by *M. phaseolina* in mungbean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan. They classified nine isolates under large sized, 26 under small sized and remaining 30 isolates as medium sized sclerotia. Maximum sclerotial size was observed in isolates MP-20 and MP-3 showing 29.00 and 27.33 μm diameter respectively, while the isolates MP-39 and MP-28 were found to be the smallest in size. The average sclerotial size of isolates ranged from 17.00 to 29.00 μm . Sharma *et al.*, (2015) reported that *R. bataticola* causing dry root rot of Chickpea grew better on PDA and produced brown to gray coloured mycelium, which turned darker with age. The young hyphae thin, hyaline, aseptate and dichotomously branched, which latter produced black sclerotia. The sclerotia formed were black, smooth, spherical or oblong to irregular shaped.

Gade *et al.*, (2018) studied morphological variability of 40 isolates of *R. bataticola*, causing dry root rot of soybean. Maximum sclerotial size was reported in isolate Rb-33 (120.11 μm) while, the isolate Rb-40 produced smallest size sclerotia (42.03 μm). Average sclerotial sizes of the isolates ranged from 42.03-120.11 μm . Large sized sclerotia ($>80 \mu\text{m}$) were produced by 16 isolates, 8 isolates had medium sized sclerotia (60-80 μm) and rest 16 isolates had small sized sclerotia ($<60 \mu\text{m}$).

2.4.3. Pathogenic virulence / variability

Jayanti and Bhatt (1993) sown pre-germinated seed of 21 chickpea varieties in soil inoculated with *R. bataticola* which caused seed rot within 24h in NEC874 and EG234. Bold 2375, BG209, JG62, JG315, ICC

3357 and JG1133 developed necrotic lesion in 3-5 cells deep at hypocotyl region within 7 days. RSG-44, AGC677, NEC41, GL269, JG74, ICC8983 and ICC 5003 developed only superficial necrosis along the hypocotyl region. BGM 416, BG 416, ICC 1376 and ICC 113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases.

Suriachandraselven and Seetharaman (2002) isolated 25 isolates of *M. phaseolina*, causing charcoal rot of sunflower from sunflower growing areas of Tamilnadu. They reported that all the 25 isolates were varied in respect of growth rate and virulence. Based on growth rate, the isolates were grouped as fast growing (41.3mm/day) and slow growing (19.7mm/day) on PDA. On the basis of pathogenicity test all the test isolates were pathogenic to the sunflower which caused about 38.30 to 88.30 per cent disease incidence.

Pande *et al.*, (2004) screened 29 chickpea germplasm lines and 10 cultivars against dry root rot caused by *Rhizoctonia bataticola* in chickpea using paper towel method developed by Nene *et al.* (1981). Germplasm line ICC 14395, cultivar ICCV 2 and advanced breeding line (ICCX830203-BH-BH-11H) were found resistant to dry root rot, 22 moderately resistant, 19 susceptible and 3 highly susceptible were observed in the remaining lines. The disease severity in the two susceptible line BG 212 and ICC 12267 used as control was rated highest. The identified genotypes were used as source of resistance to dry root rot.

Monga *et al.*, (2004) studied pathogenic variability among 15 isolates of *R. bataticola* causing root rot of cotton and reported that the test isolates caused seedling mortality in the range of 31.3-91.6 per cent. However, it was significantly highest (range 65.6-91.6 %) with the isolates RJ-9, PB-2, HR-25, RJ-6, RJ-16 and HR-18; moderate (42.7 %) with the isolates HR-15 and RJ-23 and minimum (19.8-31.3 %) in the isolates RJ-19, RJ-22, RJ-24, RJ-28 and HR-17.

Meena *et al.*, (2006a) isolated seven isolates of *Macrophomina phaseolina*, causing charcoal rot of maize, collected from different agro-ecological zones of India. On the basis of disease expression the most virulent was Hyderabad isolate with maximum disease rating score (8.8) followed by Dehli isolates (7.0) whereas, Coimbatore isolate was least virulent (2.3-5.2).

Aboshosha *et al.*, (2007) isolated 33 *M. phaseolina* isolates from diseased sunflower crop grown (2003, 2004) in various region of El-Behera and characterized them for pathogenicity, pathogenic variability and colony phenotype. All of the 33 isolates were pathogenic to both sunflower cultivars, H11-008 and H11-009, in the pathogenicity test. Almost 35.8% isolates were highly pathogenic (> 75% aggressiveness) 48.4% were low pathogenic ($\leq$ 25% aggressiveness) and 15.8% were intermediate pathogenic (< 75% to > 25% aggressiveness).

Monga *et al.*, (2007) studied pathogenic and morphological variability among 23 isolates of *R. bataticola* and 25 isolates of *R. solani* obtained from cotton root rot affected plants, collected from various locations in North India. In pathogenicity, *R. solani* five isolates (HR-7, RJ-1, HR-3-10, RJ-3-6 and RS-CMB) caused 100% mortality of cotton seedlings, 12 isolates caused mortality between 75.3 to 91.6% and it was least with the isolate HR-4 (16.6%). In *R. bataticola* highest seedling mortality was with the isolate RJ-28 (83.3%) followed by HR-3-6 (75.0%) and rest of the isolates showed minimum mortality and did not differed.

Jayalakshmi *et al.*, (2008) conducted Pot culture experiment for screening 12 promising entries of chickpea (JG-11, GBS 964, GBS 963, GBC 2, GBM 6, GBM 2, GBM 10, BGD 103, GCP 107, GCP 101, A1, ICCV 10, L-550 + 1 susceptible check JG-62) against dry root rot during rabi season, 2004-05 and 2005-06. Out of 12 genotypes tested against root rot disease, 4 resistant, 2 moderately susceptible, 3 susceptible and 3 highly susceptible to dry root rot disease were observed.

Aghakhani and Dubey (2009a) studied pathogenic variability among 23 isolates of *R. bataticola*, causing dry root rot of chickpea, by blotter paper and sick soil method and by sowing highly susceptible chickpea cultivars. They reported all the test isolates as highly variable in virulence, tested by both the methods. In sick soil method, root rot incidence in chickpea Cv. Pusa-212 was very high (31.7-100%) whereas, it was minimum in Pusa-1103. Among the isolates, Rb1 was highly virulent with maximum mean root rot incidence (76.4%) whereas, Rb5 and Rb11 were less virulent. Similarly, in blotter paper method, Cv. Pusa-212 and DCP-17 were susceptible to the all isolates.

Charul and Biwas (2009) studied pathogenic variability among 8 isolates of *R. bataticola* causing leaf spot / blight of Pigeonpea. On the basis of disease severity expressed on susceptible Pigeonpea Cv. Prabhat, the isolates T₁, T₃, T₆, T₇ and T₈ were highly virulent with disease severity > 50.00 %; whereas the isolate T₂, T₄ and T₅ were moderately virulent with < 50.00 % disease severity.

Rayatpanah *et al.*, (2012) studied pathogenic variability among 24 isolates of *M. phaseolina* (charcoal rot) isolated from soybean and sunflower plants grown in northern Iran. Pathogenicity was attempted by seed inoculation method. None of the isolate was pathogenic on corn, while all of the isolates infected soybean and sunflower seedlings and were more virulent on soybean than sunflower. The disease index on sunflower and soybean varied between 19 - 24 and 27 - 30 per cent, respectively.

Gupta *et al.*, (2012) Collected and screened forty isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea from four major chickpea growing states viz., Madhya Pradesh, Chhattisgarh, Gujarat and Maharashtra of central India for virulence of the isolates on susceptible chickpea variety BG 212 by blotter paper technique. The highly virulent isolates from Madhya Pradesh, Chhattisgarh, and Maharashtra were fast

growing isolates and exhibited largest sclerotia. The sclerotial size and number of sclerotia of *R. bataticola* isolates had positive correlation with virulence.

Sharma *et al.*, (2012a) collected and evaluated fifty isolates of *Rhizoctonia bataticola* on chickpea from different agro climatic zones in India for the degree variability in pathogenicity test. All the isolates proved pathogenic on chickpea cultivar BG 212. The dry root rot disease severity ranged from 4 to 9 (1-9 severity rating scale). The effect of isolates on severity was significant. The pathogenic and non pathogenic isolates were not concentrated in any one particular state/region.

Sharma *et al.*, (2012b) performed pathogenicity test for 94 isolates of *Rhizoctonia bataticola* in laboratory on a susceptible cultivar BG 212 of chickpea by paper towel technique. All 94 isolates were highly pathogenic with disease severity ranging from 7 to 9 rating (1 to 9 scale) except two isolates [Jodhpur (RB 14) and Delhi isolate (RB 16)] with a disease severity of 4.

Sundravadana *et al.*, (2012) studied pathogenic variability among *R. bataticola* isolates causing dry root rot of various pulses (soybean, cowpea, green gram, red gram and black gram) from Tamilnadu. The seedling mortality at 15 DAS was maximum with root isolates (6.60-20.33 %) followed by shoot isolates (5.3-16.10 %) and seed isolates (3.20-13.28 %). The root rot incidence at 60 DAS was with shoot isolates (48.00 to 90.60 %) followed by root isolates (53.10 to 88.58 %) and seed isolates (38.00 to 58.00 %).

Lekhraj *et al.*, (2012) conducted pathogenic variability of *Macrophomina phaseolina* isolates collected on infected chickpea from 9 various parts of Rajasthan, Hissar and New Delhi. Pathogenicity of these isolates tested on 40 day old chickpea cv. C-235 showed there was no direct relationship between cultural characters and virulence of the isolates. The isolate BKN had scanty, suppressed growth was highly pathogenic. Isolates SGN, NHR, DUR, HAU, DEL and HNG having different types of colony

growth were moderately pathogenic and the isolates CHR and UPR with different mycelium characters were less pathogenic. Similarly, there was no apparent correlation between the virulence and growth rate of the fungal colonies.

Iqbal and Mukhtar (2014) studied pathogenic variability among 65 isolates of *M. phaseolina*, causing charcoal rot of mungbean, from different agro-ecological regions of Punjab and Khyber Pakhtunkhwa provinces of Pakistan. They reported that based on differential reactions of the test isolates on three mungbean cultivars (NM-92, NM-51 and AEM-96) the isolates exhibited varied pathogenic reactions. In Cv. NM-92 six isolates showed highly virulent reaction with mean disease severity score of 7.3, five isolates were least virulent with severity disease score between 2.3 to 3.7. Similarly on Cv. NM-51, nine isolates showed virulent reactions with disease severity score from 7.0 to 7.7; while, six isolates were least pathogenic with severity score from 2.0 to 3.0 and rest of the isolates were intermediate. On Cv. AEM-96, 17 isolates were highly virulent with disease severity score above 7.0; while six isolates were least virulent with disease severity score 3.0 and rest of the isolates were moderately virulent. Based on cluster analysis, the 65 isolates were categorized into five clusters. Highly virulent isolates (08) belonged to cluster five, least virulent isolates (10) belonged to clusters 2 and 3 and rest of the isolates with moderate virulence belonged to the cluster 1 and 4.

Mahmoud and Kamal (2014) isolated five isolates from *R. solani*, causing root rot of soybean and studied their pathogenic variability test. All tested isolates were able to infect soybean plants causing root rot with different degrees of virulence. Isolates R5 showed the highest pathogenic capability and gave high significantly root rot rating. There was a great diversity among all the tested isolates studied, and was in the range of 36 to 100%.

Gade *et al.*, (2018) studied pathogenic variability among 40 isolates of *R. bataticola*, causing root rot of soybean and reported all isolates as pathogenic to susceptible soybean Cv. TAMS-38. Among different isolates

tested, two *viz.*, Rb-3 and Rb-17 were highly pathogenic with 71-100 per cent mortality due to root rot; whereas eight isolates *viz.*, Rb-10, Rb-13, Rb-18, Rb-22, Rb-29, Rb-34, Rb-39 and Rb-40 were strongly pathogenic with 51-70 per cent mortality due to root rot, 24 isolates were moderately pathogenic with 21 to 50 per cent mortality due to root rot and rest six isolates were weakly pathogenic with 1-20 % mortality due to root rot.

2.4.4. Molecular variability

Vandemark *et al.*, (2000) studied genetic relationship among the isolates of *Macrophomina phaseolina*, by employing AFLP technique and using 15 selective primer pairs. Relationship matrices generated with each group of AFLP primers were highly correlated ($r^2 > 0.92$). Isolates were very diverse and could not clearly be grouped based on the geographic locations from which they were obtained. Relationships were determined using both the genetic similarity method of Nei and Li and the 'genetic distance' method of Skroch.

Almeida *et al.*, (2003) studied genetic diversity among 55 isolates of *M. phaseolina*, isolated from soybean roots collected from different regions of Brazil, by RAPD analysis. UPGMA cluster analysis revealed three divergent groups with an average similarity coefficient of 99 per cent, 92 per cent and 88 per cent. The three groups corresponded to 5.45, 59.95 and 34.6 per cent, respectively of all isolates used. A single plant had three different haplotypes, while 10.9 per cent of the analyzed plants had two different haplotypes. In another study the genetic similarity was evaluated among *M. phaseolina* isolates from different hosts *viz.*, soybean, sorghum (*Sorghum bicolor*) sunflower (*Helianthus annuus*) cowpea (*Vigna unguiculata*) corn (*Zea mays*) and wheat (*Triticum aestivum*)] as well as two soil samples from native areas. They reported that more divergent isolates originated from areas with a single crop. Isolates from areas with crop rotation were less divergent, showing high similarity values. Amplification of ITS regions using primers ITS1 and ITS4,

produced only one DNA fragment of 620 bp. None of the isolates were differentiated through PCR-RFLP.

Jana *et al.*, (2003) showed single RAPD primer A13 could be used to identify and discriminate several isolates of *M. phaseolina* and *Fusarium* sp. obtained from 20 hosts including soybean, cotton, chickpea and safflower.

Jana *et al.*, (2005) used Universal rice primers to cluster 40 isolates of *M. phaseolina* from soybean, cotton and chickpea.

Purkayastha *et al.*, (2006) demonstrated genetic variations among 59 isolates of *Macrophomina phaseolina*, isolated from cluster bean as well as other hosts, using RAPD assay and PCR-RFLP. They reported that RAPD assay clearly distinguished the isolates on the basis of chlorate phenotype and host origin. Isolates from a single host were generally similar to each other but differed distinctly from those from other hosts. Chlorate-sensitive isolates were distinct from chlorate-resistant isolates, within a given host. Also, for the first time high degree of polymorphism in restriction patterns of the ITS region, including part of 25S rDNA had been reported in the charcoal rot fungus by them.

Reyes Franco *et al.*, (2006) studied genetic variability among 96 isolates of *Macrophomina phaseolina* from Mexico and other countries, by AFLPs and using four combinations of oligonucleotides. The analysis revealed 418 amplified products, of which 7.2 % were monomorphic and 92.8 % were polymorphic. Cluster analysis of the isolates showed substantial genetic diversity among the test isolates. Two major clusters were first group (A) included 90 isolates from all countries with genetic dissimilarity from 3 % to 11 %, while the second group (B) included six isolates from Argentina, Australia, USA (two isolates) Japan and Brazil.

Meena *et al.*, (2006b) studied genetic variability among seven isolates of *Macrophomina phaseolina*, incitant of maize charcoal rot, RAPD

techniques using 100 primers (OPA-E). They reported that UPGMA cluster analysis of 706 loci revealed three main clusters. Similarity matrix and Jaccard's similarity coefficient indicated maximum genetic variations between the isolates of Arabhavi and Coimbatore (70.8 %) followed by Ludhiana and Coimbatore (69.5 %). The most closely related isolates were Hyderabad and Delhi with similarity coefficient of 65.5 per cent, followed by Udaipur and Bangalore isolates (62.9 %).

Aboshosha *et al.*, (2007) studied molecular variability among 33 *M. phaseolina* isolates, affecting sunflower from Egypt, by RAPD-PCR analysis and using eight random oligonucleotide primers. They revealed DNA fingerprints 2 - 9 bands ranging from 350bp to 3kbp. No variations revealed for number of bands between isolates with the primers OPS-16 or OPA-2, while more variations revealed with the primers OPA-1, OPA-4, OPA-5, OPA-8, OPA-13 and OPR-14. However, dendrogram analysis revealed two clusters, one of high pathogenic isolates and other of low pathogenic isolates.

Monga *et al.*, (2004) studied genetic variability among 15 isolates of *Rhizoctonia bataticola*, causing root rot of cotton, by RAPD technique and 18 primers. The similarity coefficient among the *R. bataticola* isolates ranged from 0.27 to 1.00% with an average of 0.63% among all the isolates. The UPGMA cluster analysis revealed four sub groups (a, b, c and d). The largest sub group c consisted of seven isolates with 0.85-1.0 genetic similarity coefficient.

Shekhar *et al.*, (2006) analyzed *M. phaseolina* isolates of maize from Ludhiana, Hyderabad, Delhi, Arabhavi, Udaipur, Bangalore and Coimbatore through RAPD marker for genetic diversity. The UPGMA cluster analysis for 706 loci score permitted for identification of three main clusters. Similarity matrix and Jaccard's similarity coefficient between the isolates indicated that the maximum genetic variation was among isolates of Arabhavi and Coimbatore with 70.8% followed by Ludhiana and Coimbatore with 69.5%. The most closely related isolates were Hyderabad and Delhi with an

affinity per centage of 65.5% followed by Udaipur and Bangalore isolates with 62.9% similarity.

Monga *et al.*, (2007) studied genetic variability among twenty five isolates of *Rhizoctonia bataticola*, causing cotton root rot, by using RAPD markers. All the fifteen primers used generated scorable polymorphic bands for the isolates and classified the isolates into two broad groups. The group-1 consisted of isolate no.16, with genetic similarity 0.59% and was pathogenic (83.3 %). Group-2 was further clustered into five sub-groups. The largest sub-groups were II b and II c which consisted of 9 isolates each, with 0.79-1.0% genetic similarity coefficient. The sub-groups IIa, IId and IIe consisted of two isolates each, with similarity coefficient in the range of 0.63-1.0.

Rajkumar and Kuruvinashetti (2007) studied genetic diversity among 10 isolates of *M. phaseolina*, causing charcoal rot (stalk rot) of *Rabi* sorghum, by PCR-RAPD markers. From a set of 40 random primers tested, amplicon profiles of 15 were reproducible. A total of 149 amplicon levels, with an average of 9.9 bands per primer were available for analysis, of which 148 were polymorphic (99.3%). UPGMA cluster analysis indicated very low genetic similarity coefficient index in the range of 0.14 to 0.72. A high level of genetic heterogeneity of 0.28 was recorded among the isolates.

Aghakhani and Dubey (2009b) studied genetic diversity among 27 isolates (23 from chickpea and 4 from other host crops) of *R. bataticola* representing 11 different states of India, by random amplified polymorphic DNA (RAPD) internal transcribed spacer restriction fragment length polymorphism (ITS-RFLP) and ITS sequencing. The isolates showed virulence and UPGMA cluster analysis, grouped the isolates into six categories at 40% genetic similarity, indicating high level of diversity. RAPD markers *viz*; OPN 4, OPN 12, and OPN 20 clearly distinguished majority of the isolates into area specific groups. The ITS I, 5.8rDNA and ITS II regions of 11 isolates representing different RAPD groups were amplified with primers ITS 1 and ITS 4 and digested with seven restriction enzymes. The restriction enzymes

DraI, MboI, RsaI, and AluI were found suitable for differentiating the isolates into five categories by showing isolate specific ITS-RFLP patterns.

Manjunatha (2009) studied thirty isolates of *Rhizoctonia bataticola* from chickpea for molecular variability by using three primers OPO-10, OPO-12 and OPN-12. There were four isolates in Group-I, 16 isolates in Group-II and 10 isolates in Group-III. However, in Group-II, three isolates (RG-18, RG-19 and RG-20) exhibited 100 per cent similarity. RB-29, RB-25 and RG-16 isolates from Bidar and Gulbarga respectively showed genetically diversity among thirty isolates.

Bayraktar and Dolar (2009) observed intra and inter specific polymorphism among fungal pathogens that cause root rot on chickpea by using 30 RAPD primers. UPGMA cluster analysis of RAPD datasets using Dice's coefficient differentiated all fungal isolates from each other and revealed considerable genetic variability between the isolates.

Sundravadana *et al.*, (2011) studied molecular variability among 11 isolates of *R. bataticola*, causing root rot of pulse crops from Tamilnadu by RAPD-PCR analysis. The black gram root isolates showed a maximum genetic similarity of 73 % with soybean shoot isolate. Red gram shoot isolate showed 61% genetic similarity with greengram isolates. The findings from this study confirm the variability in *R. bataticola* isolates from pulses, according to their pathological as well as genetic characters.

Kumar and Gaur (2011) collected and studied fourteen isolates of *Macrophomina phaseolina* from different groundnut growing regions in India for their virulence and genetic diversity using PCR-RAPD markers. A total of 85 amplicon levels were obtained with 7 primers with an average of 12.1 bands per primer were available for analysis, of which 83 were polymorphic (97.74%). It was possible to discriminate all the isolates with any of the 7 primers employed. The UPGMA clustering of data indicated the isolates shared varied levels of genetic similarity within a range of 0.069 to 0.65 similarity

coefficient index. Isolates Bikaner, Churu, Delhi, Hissar which were clustered in one group happened to be more virulent reflecting correlation of similarity with pathogenicity to some extent but there was practically no direct correlation between the virulence and RAPD genetic markers of all the isolates.

Pancheshwar *et al.*, (2012) studied genetic variability among 21 isolates of *R. bataticola*, causing charcoal rot of soybean, by RAPD-PCR analysis. Randomly selected 8 decamer primers amplified 64 RAPD marker loci. Out of these 64 bands, 29 bands (45.3%) were polymorphic. Similarity coefficient values ranged from 0.69-0.98, which indicated narrow genetic variability among the test isolates.

Rayatpanah *et al.*, (2012) studied RAPD-PCR amplification profiles, by using six random OPA primers (kitA) and showed polymorphisms among *M. phaseolina* isolates. The primer OPA-13 (5'CAGCACCCAC3') amplified genomic DNA of all isolates of *M. phaseolina* producing 4-12 bands of sizes between 0.25 and 2.3 kb. UPGMA analysis classified the isolates into nine major groups with 64 % similarity. The isolates from same location showed closer genetic relatedness and the isolates from the areas with crop rotation were less diversion.

Sharma *et al.*, (2012a) studied genetic variability among 94 isolates of *R. bataticola*, causing dry root rot of chickpea, collected from different agro-climatic regions of India, by amplified fragment length polymorphism (AFLP) analysis. Five AFLP primer combinations provided a total of 121 fragments, which were polymorphic with an average polymorphism of 0.213. The dendrogram based analysis showed diversity among maximum number of *R. bataticola* isolates, irrespective of their geographical origin. Both morphological and molecular data clearly indicated diversity among the test isolates of *R. bataticola*, in India.

Ganeshamoorthi and Dubey (2013) tested 10 random amplified polymorphic DNA (RAPD) primers to assess molecular diversity of 50 *R.*

solani isolates, causing wet root rot of chickpea and reported a great diversity in the range of 52 to 93 per cent. The isolates were highly variable in aggressiveness and caused upto 100 per cent wet root rot incidence in chickpea. RAPD analysis revealed considerable variability among the test isolates. The number of amplified DNA fragments varied, and the primers produced reproducible bands with band size ranging from 0.25 - 3 kb. The primers OPA18 and R28 produced maximum number of bands whereas; the primers A03 and A08 produced minimum number of bands. OPN20 also gave 100 per cent polymorphism. RAPD primers produced 84 polymorphic and one monomorphic band. The similarity values of RAPD profiles ranged from 0.52 - 0.93. The dendrogram showed similarity coefficient from 52 - 93 per cent, reflecting a wide range of molecular variability among the isolates. Based on UPGMA analysis, 50 isolates were classified into six major groups at 55 per cent of similarity coefficient.

Mishra *et al.*, (2015) studied genetic variability among 22 isolate of did molecular characterization of *Rhizoctonia solani* (Kühn) isolates of three hosts, *viz.* rice, maize and green gram, by RAPD analysis. The phylogenetic analysis based on RAPD revealed three clusters comprising two major clusters of maize and rice, and one minor cluster with four outliers of the isolates. The two green gram isolates, PRS21 and PRS22, were clustered with maize isolates.

Belkar and Gade (2016) studied molecular variability among 20 isolates of *Rhizoctonia bataticola*, causing root rot of soybean, collected from various states (Maharashtra, MP, Karnataka, AP, Rajasthan) by using RAPD markers / primers. The primers OPF-01, OPF-13, OPF-17 and OPF-20 exhibited cent per cent polymorphism among the isolates. Based on Jaccards coefficient, the genetic similarity coefficient was in between 0.63-0.99 and on this basis the isolates Rb-23 and Rb-15 were found closely related. UPGMA dendrogram generated four major clusters. The cluster I comprised of the isolates Rb-40, Rb-26 and Rb-34; cluster II : Rb-39, Rb-37, Rb-10, Rb-14, Rb-

15, Rb-13, Rb-5 and Rb-23; cluster III : Rb-24, Rb-25, Rb-22, Rb-3, Rb-17 and Rb-19 and cluster four only two isolates Rb-29 and Rb-30.

2.5. Organic and inorganic carbon and nitrogen sources

2.5.1. Effect of organic and inorganic carbon sources

Shanmugam and Govindaswamy (1973) tested 12 different carbon sources for the growth of *Macrophomina phaseolina* and reported that the highest growth was obtained in Dextrose followed by Maltose and D-Fructose.

Hooda and grover (1983) reported that glucose and sucrose increased the virulence as well as mycelial growth of *Rhizoctonia bataticola* infecting mung bean (*Vigna radiata*).

Diazfranco (1984) studied the effect of carbon sources on growth and sclerotial formation of *Macrophomina phaseolina* causing black rot of bean. He observed that dextrose, fructose, sucrose and galactose were the best for mycelia growth, while sucrose and dextrose were excellent for sclerotial production.

Mehta and Gupta (1992) observed that sucrose was the best carbon source for growth of *Rhizoctonia bataticola* infecting cowpea.

Lakhpale *et al.*, (1995) studied the effect of different carbon sources on growth and sclerotial formation of paddy isolates of *R.solani*. They reported that, out of six carbon sources (starch, manitol, galactose, lactose, sorbitol, and dextrose) dextrose was the best source of carbon which promoted good growth and abundant sclerotial formation than lactose and galactose.

Sharma and Tripathi (2002) studied the effect of 5 carbon sources on sclerotial formation in urd bean isolate of *R.solani*. Dextrose as carbon source resulted in the greatest mycelia growth (90mm) at 144hrs. after inoculation, followed by mannitol (82.0mm) and sorbitol (80.0mm). Radial

growth did not significantly vary among the treatments at 168hrs.after inoculation.Mycelial growth was inhibited in starch (70mm) galactose (75.3mm) and lactose(76.6mm) at 144hrs. after inoculation.Dextrose resulted in the highest number of sclerotia per petri plate (60) followed by mannitol (44) and sorbitol (39).

Priya santosh (2006) studied the effect of nine different carbon sources on growth and sporulation of *Macrophomina phaseolina* isolated from patchouli leaves and reported that sucrose supported maximum (308mg) vegetative growth of the *Macrophomina phaseolina* followed by D-fructose and starch.Maltose and glucose supported good growth and both were at par.Moderated growth was recorded in mannitol, lactose, sorbitol and citric acid. Mannitol and lactose and sorbitol and citric acid were at par. Growth was very poor in control (26mg). The excellent sclerotial formation was recorded on sucrose and D-fructose and good on glucose, maltose and starch.Mannitol and sorbitol supported moderate sclerotial formation and poor on lactose and citric acid.No sclerotial formation was observed in control.

Sharma and Vishnupriya (2006) studied *in vitro* effect of various carbon sources on the growth and sporulation of *Macrophomina phaseolina* isolated from *Coccinia indica* . they reported that among the 21 carbon sources, i.e. D-xylose, L(+)-arabinose, L(+)-rhamnose, D-glucose, D-fructose, D-galactose, L-sorbose, D(+)-mannose, sucrose, lactose, maltose, raffinose, starch, dextrin, glycerol, dulcitol, malic acid, citric acid, tartaric acid, fumaric acid and control (without carbon) sucrose was found to be best source and supported good sporulation of the fungus. No growth was recorded in the control. Other sources of carbon induced good, moderate or poor growth and various degrees of sporulation of the test fungus.

Salunkhe *et al.*, (2009) studied the effect of nutritional and physiological factors on growth and sclerotial formation of *Rhizoctonia bataticola* isolates and reported that among six carbon sources, highest growth was obtained in dextrose followed by sucrose, mannitol, starch, maltose and

cellulose. Sclerotial production in cellulose was maximum followed by dextrose, sucrose, maltose, starch while, it was minimum in mannitol.

Jha and Sharma (2009) studied the growth and sporulation morphology of seven *Rhizoctonia bataticola* isolates on 25 carbon-nitrogen combinations and noticed that no difference in linear growth was observed after 72 hrs, but linear growth rate difference was noticed after 24 and 48 hrs in all the isolates. Higher carbon amount resulted more aerial growth, large and good sclerotia formation as compared to lower carbon amount. Smaller and poor sclerotia formation was observed at higher nitrogen (KNO_3) amount when amount of carbon was limited.

Kumar and Gaur (2010) studied *in vitro* effect of 14 carbon sources on growth sporulation of *Macrophomina phaseolina* causing groundnut root rot and reported that maximum mycelia growth of isolates AJM (55.67mm) BKN (84.33mm) DO (63.33mm) DLI (88.00mm) HMH (73.67mm) JJN (63.67mm) NHR (77.15mm) SRDR (70.33mm) SWM (62.67mm) and UDZ (49.33mm) was observed on sucrose as compared to dextrose, maltose and glycerol. While, mycelial growth of isolates AWR (59.00mm) CUR (82.67mm) and>NNL (79.00mm) was comparatively more on dextrose as compared to sucrose, maltose and glycerol. The growth of isolates AJM, AWR, BKN, CUR, DO, DLI, HSR, HMH, JJN,>NNL, NHR, SRDR, SWM and UDZ was less on glycerol as compared to other carbon sources. The growth of isolate BKN, CUR and HSR was comparatively good on glycerol as compared to other isolates. With maltose as a carbon source BKN, HSR,>NNL produced good growth. Thus, growth variability on different carbon sources distinguished fourteen isolates.

Thombre and Kohire (2018) reported that Sucrose (90.00 mm) Fructose and Dextrose (83.50 mm) among carbon and Potassium nitrate (82.14 mm) as well as Sodium nitrate (78.33 mm) among nitrogen sources yielded maximum growth and sporulation of the test pathogen *Macrophomina phaseolina* causing mungbean blight.

2.5.2. Effect of organic and inorganic nitrogen sources

Shanmugam and Govindaswamy (1973) found that Asparagine was significantly superior to other nitrogen sources for maximum growth of pathogen *in vitro*. This was followed by glutamine, peptone and potassium nitrate which were at par.

Lakhpale *et al.*, (1995) studied the effect of different nitrogen sources on growth and sclerotial formation of paddy isolates of *R.solani* and reported that, among nitrogen sources tested, asparagines was best for growth and sclerotial formation of *R.solani* isolates followed by sodium nitrate, potassium nitrate and ammonium nitrate.

Sharma and Tripathi (2002) studied the effect of 13 nitrogen sources on sclerotial formation in urd bean isolate of *R.solani*. Among the nitrogen sources, the greatest radial growth (63.3mm) was obtained with asparagines after 120 and 144hrs. of inoculation . However, after 168hrs. of inoculation, the greataest radial growth (90mm) was observed in petri plates containing asparagines, potassium nitrate, urea and potassium nitrate. At 168 hrs. after inoculation mycelia growth was recorded for asparagines (72) urea(70) potassium nitrate (68) sodium nitrite (66) and ammonium chloride (65). Ammonium chloride was the optimum nitrogen source for mycelia growth and sclerotial formation in urd bean isolate of *R. solani*.

Sharma and Vishnupriya (2006) studied *in vitro* effect of various nitrogen sources on the growth and sporulation of *Macrophomina phaseolina* isolated from *Coccinia indica*. They reported that among the 21 nitrogen sources tested, i.e. potassium nitrate, sodium nitrate, calcium nitrate, ammonium nitrate, ammonium chloride, ammonium sulfate, ammonium phosphate DL-alanine, glycine, L-serine, DL-valine, DL-aspartic acid, L-glutamic acid, L-histidine, DL-methionine, DL-tryptophan, lysine, thio-urea ,urea and control (without nitrogen) maximum growth and fair sporulation was supported by DL- valine and no growth was recorded in the control. Other

sources of nitrogen induced good, moderate or poor growth and various degrees of sporulation of the test fungus.

Salunkhe *et al.*, (2009) studied the effect of nutritional and physiological factors on growth and sclerotial formation of *Rhizoctonia bataticola* isolates and reported that Sodium nitrate proved better nitrogen source for growth and sclerotial formation than ammonium nitrate.

Jha and Sharma (2009) studied the growth and sporulation morphology of seven *Rhizoctonia bataticola* isolates on 25 carbon-nitrogen combinations and noticed that no difference in linear growth was observed after 72 hrs, but linear growth rate difference was noticed after 24 and 48 hrs in all the isolates. Higher carbon amount resulted more aerial growth, large and good sclerotia formation as compared to lower carbon amount. Smaller and poor sclerotia formation was observed at higher nitrogen (KNO_3) amount when amount of carbon was limited.

Kumar and Gaur (2010) studied *in vitro* effect of 4 nitrogen sources on growth sporulation of *Macrophomina phaseolina* causing groundnut root rot and reported that isolates BKN, CUR, DLI, HSR, HMH, NNL, NHR and SRDR were having significantly better growth on all the four nitrogen sources viz., sodium nitrate, ammonium nitrate, asparagines and leucine as compared to control. Isolate HSR was having mycelia growth *i.e.* 88.67, 77.00, 66.67 and 58.33mm on sodium nitrate, ammonium nitrate, asparagines, leucine respectively. Isolate BKN had growth similar to isolate HSR on ammonium nitrate (77.00mm) and leucine (58.33mm). The isolates AJM, AWR, BKN, CUR, DO, DLI, HSR, HMH, JJN, NNL, NHR, SRDR, SWM and UDZ had more growth on sodium nitrate as compared to ammonium nitrate, asparagines and leucine. All the tested nitrogen sources supported the mycelia growth of isolates BKN, DLI, HSR, NNL, NHR, SRDR and SWM. Minimum mycelia growth of all the isolates of *Macrophomina phaseolina* was observed highest (73.33mm) in sodium nitrate as nitrogen source.

2.6. Soil moisture regimes and soil types

Ghaffar and Erwin (1969) reported that when cotton plants grown in soil temperatures of 20 - 40⁰ C were subjected to soil water stress, the severity of *Macrophomina phaseolina* was much greater in moisture stress condition than those provided with sufficient soil water.

Singh and Mehrotra (1982) reported that Dry root rot was observed in gram when exposed to temperature more than 30⁰ C. The disease was aggravated by dry soil conditions especially at flowering stage leading plants to sudden drying.

Chandrasekaran and Shanmugam (1984) studied the effect of soil type on wilt incidence on chickpea root rot (*Fusarium solani*) and reported highest disease incidence in sandy loam soil (96.30 %) followed by black soil (53.80 %) and red soil (35.00 %).

Taya *et al.*, (1988) observed the DRR incidence of chickpea was maximum (86.6 %) in sandy soils as compared with clay soils (61.6%) in sick pots at greenhouse.

Husain and Ghaffar, (1995) evaluated that colonization of *M. phaseolina* on chickpea roots was high when subjected to moisture stress conditions compared to unstressed plants. Soils artificially inoculated produced poor growth of plants both in low and high moisture compared to natural soil. Root colonization was greater at 40-50% moisture holding capacity as compared to 10-20% soil moisture holding capacity.

Ratnoo *et al.*, (1997) reported the infection and development of ashy grey stem blight in cowpea caused by *M. phaseolina* was most favoured by higher temperature 25-40°C and low in flooded soil compared to dry soil of 40-60% soil moisture.

Sagar and Sugha (1998) reported highest pea wilt incidence in silty loam soil (85.70 %) sandy loam soil (84.70 %) and comparatively minimum wilt incidence in clay soil (78.50 %).

Rajkrishan *et al.*, (1999) studied the Role of edaphic factors on the incidence of dry root rot of sesame caused by *Rhizoctonia bataticola*. Among various soil types, sandy soil supported maximum disease incidence (78.33%) as compared with clay soil (51.56%). More disease incidence of 48.33% was observed when pots were irrigated after the interval of seven days as compared to pots receiving irrigation every day.

Chauvan *et al.*, (2000) reported that sandy soil favoured maximum stem rot incidence (63.47%) in cauliflower followed by sandy loam soil (54.19%) and least in clay soil (38.89%).

Wokocha (2000) observed the disease incidence of dry root rot of soybean was significantly low when the soil moisture was 60 - 70% but it was 5.0% at 10 - 20% soil moisture.

Patel and Anahosur (2001) conducted an experiment with chickpea cultivars ICC 4951 (susceptible) and Bheema (resistant) in earthen pots filled up with artificially infected soil with *Macrophomina phaseolina* at four levels of soil moisture viz., 25, 50, 75 and 100%. They observed *M. phaseolina* infection was maximum at 25% and minimum at 100% soil moisture. ICC 4951 developed higher disease than Bheema at 25% soil moisture. Both cultivars showed maximum disease incidence at the lowest soil moisture level compared to high moisture.

Sharma and Pande (2013) studied the unraveling effects of temperature and soil moisture stress response on development of dry root rot [*Rhizoctonia bataticola* (Taub.)] Butler in chickpea and reported that out of five temperature regimes (15⁰ C, 20⁰ C, 25⁰ C, 30⁰ C and 35⁰ C) and four moisture levels (40%, 60%, 80%, and 100%) a combination of high

temperature (35⁰ C) and soil moisture content levels less than or equal to 60 % predisposes chickpea to DRR.

Mallaiah and Krishna Rao (2016) conducted a survey on the incidence of dry root rot of greengram caused by *Macrophomina phaseolina* and also reported the influence of soil type on dry root rot incidence. They observed that high incidence of dry root rot was in sandy loam (11.1%) followed by sandy clay loam (8.1%) and least in clay soils (4.3%).

2.7. Disease management strategies

2.7.1. Efficacy of fungicides

Singh and Chohan (1981) reported that Benlate @ 2.5 ppm, 5.0 ppm, 10.0 ppm and 20.0 ppm inhibited mycelial growth of *M. phaseolina* up to 85.0, 95.0, 97.8 and 100.0 per cent, respectively.

Raut and Somani (1987) evaluated the efficacy of various fungicides as seed treatment against *R. bataticola*, causing root rot of chickpea and reported that seed treatment with Carbendazim 50% @ 1g/kg Bavistin gave more disease incidence *i.e.* 19.45%, followed by Thiram 18.50% and Rovral 16.24% disease incidence. But Deconil and Captan gave least disease incidence 15.25% and 15.18% respectively.

Chauhan (1988) evaluated *in vivo* the efficacy of various fungicides as seed treatment to control seedling disease (*R. bataticola*) of cotton. They reported that seed treatment with Carbendazim @ 2g / kg gave least disease incidence in H 777 (8.6%) and in G 27 (9.2%) followed by Quintozene @ 5.0% (9.4 and 9.8%) Benomyl @ 2.0% (11.7 and 12.6%) Carboxin @ 2.5% (15 and 16.2%) Captaf @ 4.0% (16 and 16.9%) Thiobendazole @ 3.0% (16.9 and 17.3%) Thiophanate methyl @ 1.5% (17.8 and 18.6%) Captafol @ 3.0% (19.6 and 21.0%) Quintozene + Captaf @ 2.5 + 2% (22.4 and 23.3%) Thiram @ 3% (22.7 and 24.3%) and MEMC @ 3% (26.6 and 27.3%).

Srivastava and Tripathi (1998) evaluated *in vivo* the fungicides viz., PCNB (brassicol) TMTD (Thiram) Carboxin (vitavax) and Carbendazim (bavistin) in combination (mixed in equal proportion) @ 2.5g / kg seed, for management of seedling disease complex of sugarbeet, caused by *R. solani*, *R. bataticola*, *S. rolfsii* and *P. aphanidermatum*. They reported maximum seedling emergence with Bavistin + Thiram seed treatment (50.7%) followed by Vitavax + Thiram (50.1%) Vitavax + PCNB (49.9%) and Bavistin + PCNB (48.6%). Significantly minimum pre- emergence mortality was reported with the fungicides viz., Bavistin + Thiram (7.7%) followed by Vitavax + Thiram (9.0%) Vitavax +PCNB (9.3%) Bavistin + PCNB (12.4%) and PCNB + Thiram (16.5%). Also all these seed dressing fungicides were reported to reduce post-emergence mortality.

Ebenezar and Wesely (2000) evaluated *in vitro* six fungicides on against *M. phaseolina*, causing root rot disease in green gram. Carbendazim (0.1%) completely inhibited mycelial growth of the pathogen, followed by Copperoxy chloride (0.25%) Thiram (0.2%) Thiophanate methyl (0.1%) Captan (0.2%) and Methoxy ethyl chloride (0.05%) with mycelial growth inhibition of 95.89, 91.11, 89.67, 78.56 and 77.44 per cent respectively.

Prashanti *et al.*, (2000) evaluated *in vitro* systemic fungicides viz., Carbendazim 50 % WP, Metalaxyl MZ 35 % WS and Propiconazole 25 % EC each @ 250, 500 and 1000 micro g/ml and non systemic fungicides viz., Chlorothalonil 75 % WP, Copper oxychloride 50 % WP, Captan 50 % WP and Iprodione 50 % EC each @ 1000, 2000 and 3000 micro g/ml against *R. bataticola*, causing root rot safflower. Carbendazim 50 % WP and Propiconazole 25 % EC showed 100 % inhibition of mycelial growth, at all concentrations. Among the non systemic fungicides, Chlorothalonil @ 3000 ppm was the most effective (82.45 %). In pot culture study, healthy safflower cv. A-1 seeds were treated with Carbendazim (25 DS, 50 WP and 50 WP) Metalaxyl 8 % + Mancozeb 64 % (72 % WP) Captan (50 % WP) Chlorothalonil (75 % WP) (each at 2g/kg seed) and were sown in soil-filled pots inoculated

with *R. bataticola* to determine the effect of seed dressing fungicides on disease incidence. Although all the treatments were superior over the control (35.41% plant survival) subeej 25 DS was the most effective with 87.91% plant survival.

Lambhate *et al.*, (2002a) evaluated *in vitro* the efficacy of different fungicides viz., Bavistin (Carbendazim 50% WP) Blitox- 50 (Copperoxy chloride 50 % WP) Captan (50 % WP) Ridomil (Metalaxyl 8 % + Mancozeb 64 % (72 % WP)) Thiram (75 % WP) and Topsin-M (70 % WP) each @ 0.1, 0.2, and 0.3% against *M. Phaseolina*, causing root of cotton. They reported that the fungicides viz., Bavistin, Ridomil, and Topsin-M resulted with 100% mycelial growth inhibition, followed by Captan (77.00%) and Thiram (83.00%) but Blitox-50 was less effective.

Prajapati *et al.*, (2002) reported that Carbendazim 50 % WP, Carbendazim 50 % WP + Thiram 75 % WP, Carboxin 75 % WP and Thiophanate methyl 50 % WP completely inhibited the growth of *R. bataticola* and were found to be most effective.

Dubey and Kumar (2003) evaluated *in vitro* the efficacy of Azaridirachtin, Mancozeb 75 % WP and Carbendazim 50 % WP against mycelial the growth and sclerotial survival of *M. phaseolina*, causing charcoal rot of soybean. They reported that Azaridichtin (@ 25%) inhibited mycelial growth of the test pathogen by 84.20 per cent; whereas, Mancozeb and Carbendazim caused inhibition of 86.30 and 86.40 per cent, respectively. At 300 ppm and above, growth of the test pathogen was completely inhibited by Azadirachtin and Carbendazim; whereas, Mancozeb inhibited growth by 87.30 per cent. The sclerotia lost their viability after 96 hours of treatment with Azadirachtin, Mancozeb and Carbendazim, as against 83 per cent sclerotial germination in untreated control.

Malathi and Doraisamy (2003) evaluated *in vitro* three fungicides viz., Carbendazim 50 % WP, Captan 50 % WP and Thiram 75 % WP (each @

5,10, 50, and 100 µg /ml) for their inhibitory effect on growth and sclerotial production of *M. phaseolina*. They reported that Carbendazim @ 100 µg/ml gave 100 per cent mycelial growth inhibition, followed by Carbendazim @ 50, 10 and 5 µg /ml (85.2, 75.9 and 72.2 per cent, respectively). Thiram @ 100 and 50µg/ml (70.3 and 66.7%) Captan @ 100µg/ml (65.9%) and it was least with Captan (@ 5µg/ml, and Thiram @ 5µg/ml).

Prajapati *et al.*, (2004) evaluated the fungicides viz., Carbendazim (25 DS) Thiram (80 WP) Carboxin (75 WP) Topsin-M (50 WP) Mancozeb (75 WP) Captan (50 WP) and Thiobendazole (60 WP) as seed treatment against *R.bataticola* causing dry root rot of chickpea. They reported that significant increase in seed germination was obtained with Carbendazim (73.1%); followed by Carboxin (71.4%) Topsin-M (69.4%) Thiram (66.9%) and Mancozeb (65.3%). However, highest reduction in root rot incidence was reported with Carbendazim + Thiram and Carbendazim alone (each, 97.5%); followed by Topsin-M (61.5%) Thiram (51.9%) Mancozeb (47.8%) and Captan (32.8%).

Suryawanshi *et al.*, (2005) evaluated *in vitro* seven fungicides against *Macrophomina phaseolina*, causing root rot / wilt of blackgram and reported that Mancozeb 75 % WP (@ 0.2 %) significantly inhibited mycelial growth and sclerotial production of the test pathogen followed by Carbendazim 50 % WP (0.1%) Tricyclazole 75 % WP (0.1%) and Iprodione 25% + Carbendazim 25% (0.2%).

Sandipan *et al.*, (2007) reported that Carbendazim significantly inhibited *M. phaseolina* at 100 ppm while non- systemic fungicide MEMC and Mancozeb completely inhibited the growth of test fungus at all the concentrations.

Kaushal (2008) reported that seed treatment with Thiram 75 % WP and Carbendazim 50 % WP at 2g/kg was effective in control of chickpea dry root rot.

Khan and Ganopadhyay (2008) evaluated *in vitro* the efficacy of four fungicides and two strains of *P. fluorescens* against *M. phaseolina* causing root rot of chickpea. They reported that Carbendazim 50 % WP (@ 500 ppm) gave significantly least mycelial dry weight (72mg) followed by Captan 50 % WP (86 mg) Chlorothalonil 75 % WP (106mg) and Carboxin 98 % TC (110mg). In pot culture, seed treatment with Carbendazim (@2g/kg seed) exhibited least root rot incidence (36.7%) with highest disease control (62.10%) followed by Captan @ 3g /kg seed (4.04%, 58.60%) Chlorothalonil @2g/kg seed (48.80%, 53.70% and Carboxin @ 3g/kg seed (43.3,55.25%) with root rot incidence and its control, respectively.

Khonde *et al.*, (2008a) evaluated *in vitro* the fungicides viz., Carbendazim 50 % WP (0.1%) Thiram 75 % WP (0.3%) Mancozeb 75 % WP (0.25%) Thiophanate methyl 70 % WP (0.1%) and Penconazole 50 % EC (0.1%) against *R. bataticola*, causing root rot of soybean. They reported that the Carbendazim + Thiram (0.1 + 0.2 %) Penconazole (0.1%) and Thiophanate-M (0.1%) as were most effective with cent per cent (100%) inhibition of mycelial growth, followed by Thiram @ 0.3% (97.67%).

Monga and Sheo Raj (2008) evaluated *in vitro* 12 fungicides (Carbendazim 50 % WP, Carboxin 98 % TC, MEMC 3% WS, Captafol, Mancozeb 75 % WP, Copperoxy Chloride 50 % WP, Captan 50 % WP, Thiophanate-M 70 % WP, Sulphur 90 % WP, Dodine, Kitazin and Celest) against *R. solani* and *R. bataticola*, causing root rot of cotton, at four concentrations (50, 100, 500 and 1000 ppm). They reported Carbendazim, MEMC and thiophanate- M as highly toxic, with cent per cent (100%) mycelial growth inhibition @ 50 ppm. While, Mancozeb, Captan and celest showed complete inhibition of *R. solani* at 50 ppm, whereas, Captan at 100 ppm and celest and Mancozeb 1000 ppm showed complete inhibition of *R. bataticola*.

Jaiman *et al.*, (2009) evaluated seed dressing fungicides viz., Carbendazim (@2 g/kg) Tebuconazole (2 g/kg) Thiophanate methyl (2 g/kg) Captan (2.5 g/kg) Mancozeb (2.5 g/kg) and Thiram (2.5 g/kg) against *M.*

Phaseolina, causing root rot in Cluster bean. They reported that fungicide Carbendazim recorded significantly minimum pre-emergence seed rot (5.0%) and post-emergence seedling rot (5.0%) with least root rot incidence (10%). This was followed by the fungicides Thiophanate methyl (6.0, 6.0 and 13.0%) Thiram (10.0, 11.0 and 15.0%) Captan (11.0, 14.0 and 18.0%) Mancozeb (18.0, 19.0 and 23.0%). All the fungicides tested recorded significant increase in seed yield over untreated control.

Ammajamma *et al.*, (2009) screened systemic and non- systemic fungicides against *R. bataticola*, among systemic fungicides Hexaconazole 5 % EC and Metalaxyl 35 % WS have given cent per cent inhibition. Thiram was found most effective among the contact fungicides and Thiram + Carboxin was also found effective.

Rawal and Singh (2010) evaluated the efficacy of chemical seed protectants viz., Metalaxyl (0.3%) Carbendazim (0.1%) Captan (0.2%) Mancozeb (0.2%) Thiram (0.2%) Carboxin (Vitavax 0.1%) against five different pathogenic fungi of Mungbean viz., *Rhizoctonia bataticola*, *Aspergillus niger*, *Fusarium oxysporum*, *F. moniliforme* and *F. solani*. They reported that Mancozeb (0.2%) obtained seed germination of 96.7% and seedling mortality of 1.0%; followed by Carboxin 0.1% (92.0 and 2.0%) Carbendazim 0.1% (90.7 and 3.7%) Captan 0.2% (83.3 and 3.3%) Metalaxyl 0.3% (76.2 and 13.3%) and Thiram 0.2% (75.3 and 13.7%) over control.

Moradia (2011) evaluated *in vitro* the efficacy of non-systemic fungicides each @ 500, 1000, 2000 and 3000 ppm against *M. phaseolina* and reported Copperoxy chloride @ 500 ppm as most effective with per cent mycelium mycelial growth inhibition (100.00%) followed by Sodium metabisulphide @ 2000 ppm (100.00%) and Boric acid (95.80%).

Kumar *et al.*, (2011) evaluated *in vivo* various fungicides against *R. bataticola*, causing root rot of Jatropha. Under *in vitro* evaluation, 100 % mean mycelia growth inhibition was reported with Carbendazim 50 % WP and

Carboxin 75 % WP, followed by Kitazin 48 % EC (96.4 %) Copperoxy chloride 50 % WP (82.3 %) Captan 50 % WP (81.5 %) and Thiram 75 % WP (73.4 %). Under field evaluation, Carbendazim@ 2 g / each seed treatment + soil drenching resulted with significantly least pre- (16.67 %) and post- (10.07 %) emergence mortality and highest disease control (61.9 %) followed by Carboxin ST + SD (20.83 %, 13.25 % and 51.5 %) Carbendazim SD (22.91 %, 16.24 % and 44.2 %) and Carboxin SD (27.08 %, 19.95 % and 33.0 %).

Magar *et al.*, (2011) evaluated *in vitro* seven fungicides viz., Carbendazim 50% WP, Copperoxy chloride 50 % WP, Difenconazole 25 % EC, Propiconazole 25 % EC, Triadimefon 25 % WP, Mancozeb 75 % WP and Benomyl 50 % WP against *Macrophomina phaseolina*, causing leaf blight of greengram. They reported that the fungicides Carbendazim @ 0.1%, Propiconazole @ 0.1% and Mancozeb @ 0.25%, caused cent per cent mycelial growth inhibition (100%). Triadimefon, Difenconazole and Benomyl (each @ 0.1%) resulted with 75.93, 79.63 and 81.46 per cent mycelial growth inhibition, respectively.

Daniel *et al.*, (2012) evaluated *in vitro* nine fungicides each at two different concentrations against *M. phaseolina* causing root rot of coleus. They reported that the fungicides tested each @ 0.1 % concentration resulted with maximum mycelia growth inhibition Carbendazim 50 % WP (82.00 %) Propiconazole 25 % EC (83.88 %) Captan 50% WP (82.00 %) Thiophanate methyl 70 % WP (81.66) and Copperoxy chloride 50 % WP @ 0.25 % (87.25 %). In pot culture, soil drenching with Carbendazim @ 0.1 % resulted with highest disease control (72.65 %) followed by Copperoxy chloride @ 0.25 % (65.65 %) and Propiconazole @ 0.1 % (63.11 %).

Khan *et al.*, (2012) evaluated *in vitro* eight fungicides viz., Indofil M-45 (75 % WP) Bavistin 50 % WP, Companion, Copperoxy chloride 50 % WP, Benlate 50 % WP, Indofil Z-78 (75 % WP) Ridomil 72 % WP, Sulphur 90 % WP (each @ 0.2 %) against *R. bataticola*, causing dry root rot of chickpea. They reported that Indofil M-45, Bavistin, Companion, Copperoxy chloride

and Benelate resulted with cent per cent mycelial growth inhibition (100 %) of the test pathogen while, Indofil Z-78, Ridomil and Sulphur were least effective with 21.17 to 38.47 per cent mycelial growth inhibition.

Ikkar *et al.*, (2013a) evaluated *in vitro* nine fungicides (each @ 500, 1000, and 1500 ppm) against *M. Phaseolina*, causing root rot of mungbean. They reported Carbendazim 50 % WP, Metalaxyl 35 % WS, Carbendazim + Thiram 75 % WP and Carbendazim + Mancozeb 50 % WP caused highest mean mycelial growth inhibition (each 94.94%) followed by Thiram (93.50%) and Hexaconazole (93.43%). Rest of the fungicides tested resulted with mean growth inhibition in the range of 25.16 to 78.41%.

Sangeetha and Jahagirdar (2013b) evaluated *in vitro* the fungicides against soybean root rot / wilt causing pathogens viz., *Rhizoctonia bataticola*, *Sclerotium rolfsii* and *Fusarium* spp. and reported Mancozeb 75 % WP, Carbendazim 50 % WP, Thiophanate methyl 70 % WP, Hexaconazole 5 % EC, Propiconazole 25 % EC, Carbendazim 25 % + Mancozeb 50 % WP and Carboxin 37.5 % + Thiram 37.5 % WP as most effective in inhibiting mycelial growth of all three pathogens. In *R. bataticola*, among the non- systemic fungicides evaluated, maximum inhibition was observed with Mancozeb, followed by Captan (85.00%) at 0.3% concentration. Complete inhibition was observed with Thiophanate methyl and Carbendazim at all concentrations. Followed by, Propiconazole (94.92%) and Hexaconazole (91.60%). Among different combinations assessed, maximum inhibition (96.41%) was observed in Carbendazim + Mancozeb which was at par with Carboxin + Thiram (94.53%).

Veena *et al.*, (2014a) tested the efficacy of two non systemic fungicides (Copper oxychloride and Captan) two systemic fungicides (Hexaconazole and Tebuconazole) and one antifungal antibiotic validamycin each @ 1000, 1500, 2000 and 2500 concentrations were tested against *Rhizoctonia bataticola*, incident of dry root rot of chickpea under in vitro conditions. The fungicides Copperoxy-chloride, Captan, Hexaconazole and

Tebuconazole were found to be highly effective (100 %) in inhibiting the mycelial growth of highly virulent pathogen at all the concentrations tested.

Deshmukh *et al.*, (2014) evaluated *in vitro* 10 fungicides (each @ 500, 1000, 2000ppm) against *M. Phaseolina*, causing charcoal rot of *Rabi* sorghum. They reported cent per cent average mycelial growth inhibition with Carbendazim (100.00%) followed by Carbendazim 12% + Mancozeb 63% (92.91%) Metalaxyl 8% + Mancozeb 64% (90.62%) Mancozeb (84.74%) Thiram (82.35%) and Thiophanate methyl (80.74%).

Ingole *et al.*, (2014) evaluated *in vitro* some fungicides against *Macrophomina phaseolina*, causing root rot of cotton. They reported that non-systemic fungicide, Zineb 75 % WP @ 500 ppm caused per cent inhibition of mycelial growth and sclerotial formation, followed by Copper hydroxide 77 % WP (53.63 %). Systemic fungicides, Carbendazim 50 % WP proved best with highest mean mycelial growth inhibition (97.81 %) followed by Thiophanate methyl 70 % WP (84.57%) and Tebuconazole 25.9 % EC (80.50%).

Pawar *et al.*, (2014) evaluated *in vitro* eight fungicides viz., Carbendazim 50 % WP, Mancozeb 75 % WP, Thiram 75 % WP, Copperoxy chloride 50 % WP, Pyraclostrobin 20 % SG, Picoxystrobin and Azoxystrobin 25 % SC on the growth and sporulation of *Rhizoctonia bataticola*. They reported that Pyraclostrobin, Picoxystrobin, Azoxystrobin were highly effective in reducing mycelial growth and sclerotial formation of *R. bataticola*. Fungicides viz., Carboxin, Copperoxy chloride, Mancozeb, Thiram, Carbendazim were also effective in reducing the sclerotial formation of *R. bataticola*.

Suryawanshi *et al.*, (2014a) evaluated *in vitro* (each @ 500, 1000 and 2000 each) fungicides against *Macrophomina phaseolina*, causing charcoal rot of sunflower. They reported average mycelial growth inhibition with the test fungicides was ranged from 59.39 % (Copper oxy-chloride 50 % WP) to 94.45% (Carbendazim + Mancozeb 75 % WP). However, highest average

mycelial growth inhibition was with Carbendazim + Mancozeb (94.45%) followed by Carbendazim (93.36%) and Metalyxyl + Mancozeb (90.62%).

Chaudhary *et al.*, (2017) evaluated *in vitro* ten fungicides at various concentrations against *M. Phaseolina*, causing dry root rot of soybean. They reported that Carbendazim 50 % WP @ 250, 500, 1000 ppm, Mancozeb 75 % WP @ 1500, 2000, 2500 ppm, Carbendazim 12 % + Mancozeb 63 % WP @ 1500, 2000, 2500 ppm resulted with cent per cent mycelia growth inhibition (100 %). These were followed by Propiconazole 25 % EC @ 1000 ppm (92.64 %) Trifloxystrobin 25 % + Tebaconazole 50 % WP @ 1000 ppm (82.56 %) and Chlorothalonil 75 % WP @ 2500 ppm (79.07 %).

Maruti *et al.*, (2017b) tested *in vitro* efficacy of six systemic (@ 500, 1000 and 1500 ppm) seven contact and six combi fungicides (@ 1000, 2000 and 3000 ppm) against dry root rot (*R. Bataticola*) of Pigeonpea. Among contact fungicides tested, Ziram 27 % SC recorded 100 per cent mean mycelia growth inhibition followed by Mancozeb 75 % WP (99.34 %) Thiram 80 % WP (98.91 %) Chlorothalonil 75 % WP (89.28 %) and Captan 50 % WP (70.94 %). Among systemic fungicides, highest mean mycelia growth inhibition was with Tebuconazole 5 % EC (100 %) followed by Propiconazole 25 % EC, Hexaconazole 5 % EC, Difenconazole 25 % EC, Carbendazim 50 % WP (99.00-99.60 %) and Thiophanate methyl 70 % WP (96.74 %). Combi-fungicides Carbendazim 12% + Mancozeb 63% WP, Trifloxystrobin 25% + Tebuconazole 50% EC and Carboxin 37.5% + Thiram 37.5% WP resulted with 100 % mean mycelia growth inhibition, followed by Zineb 68 % + Hexaconazole 4 % Wp (99.29) Captan 50 % + Hexaconazole 5 % WP (99.34) and Tricyclazole 18 % + Mancozeb 62 % WP (97.28 %).

Parmar *et al.*, (2017) tested *in vitro* various six systemic (each @ 100, 250, 500 ppm) and non-systemic (each @ 2000, 2500 and 3000 ppm) against *Macrophomina phaseolina*, Castor root rot. Among contact (non-systemic) fungicides, maximum mean mycelial growth inhibition was reported with Propineb 70 % WP and Mancozeb 75 % WP (99.97 %) followed by

Captan 50 % WP (85.63 %). Thiram 75 % WP (62.00 %) Copper hydroxide 77 % WP (60.77 %) and Copperoxy chloride 50 % WP (59.75 %). At higher concentration (3000 ppm) all contact fungicide Carbendazim 50 % WP resulted with maximum mean mycelia growth inhibition (81.96 %) followed by Difenconazole 25 % EC (60.52 %) Thiophanate methyl 70 % WP (56.71 %) and Tebuconazole 2 % DS (54.51 %). Sclerotial formation was totally absent in all systemic fungicides tested @ 250 and 500 ppm.

Ravichandran & Hegde (2017) reported that among combi products evaluated, Carbendazim 25% + Mancozeb 50% (Sprint), Carboxin 37.5% + Thiram 37.5% (Vitavax power 75% WP) and Carbendazim 12% + Mancozeb 63% (Saaf) were found to be most effective with complete inhibition of mycelial growth of *R. bataticola* at all the concentrations tested. Among the contact fungicides tested, Chlorothalonil and Mancozeb at 0.2% were effective in inhibiting cent per cent inhibition. Among the four systemic fungicides evaluated against *R. bataticola*, Carbendazim, Difenconazole and Tebuconazole were best with cent per cent inhibition of mycelial growth at all concentrations tested.

Sharma and Kumari (2017) evaluated *in vitro* the efficacy of fungicides viz., Carbendazim 50 % WP, Thiram 75 % WP, Carboxin 75 % WP, Phenylpyrrole 50 % EC, Thiophanate Methyl 70 % WP and Chlorothalonil 75 % WP as seed treatment and soil drenching under both pot and field conditions to manage dry root rot of chickpea. In pot experiment, seed treatment with Carbendazim @ 1 g/kg seed was found most effective with lowest disease incidence of 16.66 and 13.33 per cent in 2014-15 and 2015-16, respectively. Followed by Thiophanate methyl and Thiram with disease incidence of 19.99 and 23.33 in 2014-15; and 18.33 and 21.66 in 2015-16. Carboxin, Phenylpyrrole and Chlorothalonil were comparatively less effective in reducing the disease incidence, but significantly superior over check.

Pawar *et al.*, (2018) conducted an experiment of efficacy of fungicides, bioagents and organic amendments to manage root rot in green

gram and reported that, *in vitro* evaluation of fungicide showed cent per cent growth inhibition (100%) radial mycelial growth of *Rhizoctonia bataticola* in all fungicide treatments. In respect of bioagent, *T. viride* showed (86.02%) highest growth inhibition against the pathogen.

Uikey *et al.*, (2018) reported that fungicides were used at different concentration viz., Carbendazim @ 0.1% and Carboxin + Thirum @ 0.3% were efficient in completely inhibiting the growth of *Rhizoctonia bataticola* (100%) *in vitro* and also biocontrol agent *Trichoderma viride* was found antagonistic and the *in vitro* mycelia inhibition of *Rhizoctonia bataticola* was 50.9% which causes dry root rot in chickpea

2.7.2. Efficacy of bio-agents

Selvarajan and Jeyrajan (1996) worked on inhibition of chickpea root rot pathogens, *F. solani* and *M. phaseolina* by antagonists and reported that *T. harzianum* induced maximum inhibition zone against *M. phaseolina*.

Jakhar *et al.*, (1997) conducted the experiment in screenhouse by using two antagonists and reported that both the antagonists viz. *Trichoderma viride* and *T. harzianum* were found effective in controlling root rot of cotton (*Rhizoctonia* spp.) however In this study *T. harzianum* was found superior to *T. viride*.

Raguchander *et al.*, (1997) reported that seed pelleting with *T. viride* isolates significantly reduced the dry root rot (*M. phaseolina*) incidence in mungbean. Among the isolates, *T. viride*-III supported higher plant growth, better native *Rhizobium* nodulation and high grain yield. Sclerotial number and root rot incidence were greatly reduced in seed pelleting with *T. viride*. Maximum seed germination (89.51%) was in seed pelleting with *T. viride* III, followed by *T. viride* I.

Ushamalini *et al.*, (1997) reported that the test antagonists (*T. harzianum*, *T. viride*, *T. hamatum*, *T. pseudokoningii*, *T. koningii*, *Bacillus*

subtilis and *Pseudomonas fluorescens*) significantly mycelial inhibited growth of *M. phaseolina*, causing charcoal rot of cowpea.

Pant and Mukhopadhyay (2001) attempted integrated management of seed and seedling rot complex of soybean. They reported the test that antagonists exhibited varied antagonistic potential against *R. solani*, *S. rolfsii*, *M. phaseolina* and *Fusarium* spp. *G. virens* attained maximum growth against test pathogens in comparison to other three isolates of *T. harzianum*. Among *Trichoderma* isolates, *T. harzianum*-3 proved to be superior on account of its faster growth attained against test pathogens. *G. virens* completely overgrew (100 % overgrowth) colonies of *M. phaseolina* and *Fusarium* sp. after 108 hrs, while it took longer time against *R. solani* (120 hrs) and *S. rolfsii* (144 hrs). *T. harzianum* took 120 hrs against *M. phaseolina* and *Fusarium* spp., 144 hrs against *R. solani* and 160 hrs against *S. rolfsii* for complete colonization.

Lambhate *et al.*, (2002b) reported that the antagonists *Trichoderma viride*, *T. harzianum* and *Aspergillus niger* showed inhibitory effects on the growth of *M. phaseolina* to the extent of 87.60, 77.50, 73.90 per cent, respectively.

Singh *et al.*, (2003) the efficacy of *Trichoderma harzianum*. *T. Viride*, *T. hamatum*, *gliocladiumvire*, *Pseudomonas fluorescens* and *Bacillus subtilis* in controlling *Rhizoctonia bataticola* causing dry root rot in chickpea was determined *in vitro* and in field experiment conducted in Kanpur, Uttar Pradesh, India 2001-02. *T. harzianum* recorded the highest control of the pathogen both *in vitro* and *in vivo*.

Malathi and Doraisamy (2003) evaluated efficacy of eight biocontrol agents viz., *T. viride* (isolate 4) *T. viride* (isolate 6) *T. harzianum* (isolate 1) *T. harzianum* (isolate 5) *T. hamatum*, *T. longibrachialum*, *T. koningii*, *T. pseudokoningii* against *M. phaseolina* causing dry root rot of groundnut. They reported that the maximum inhibition zone and per cent

inhibition of pathogen growth obtained by *T. harzianum* (isolate 5) was 6 mm and 66.7 per cent, followed by *T. koningii* (5mm and 64.4%) *T. longibrachialum* and *T. viride* (isolate 6) (4 mm and 61.1%) *T. harzianum* (isolate1) (5 mm and 60.0%) *T. pseudokoningii* (3 mm and 46.7%) *T. hamatum* (3 mm and 44.4%) and *T. viride* (isolate 4) (4 mm and 32.2%) over control.

Ananthan *et al.*, (2003) studied the antagonistic effect of four *Trichoderma* sp. in dual culture technique. All the *Trichoderma* spp. significantly inhibited the growth of *M. phaseolina* to an extent of 16.11% as compared to control.

Kaswate *et al.*, (2003) evaluated *in vitro* the biocontrol agents against eight isolates of *Rhizoctonia bataticola*. They reported that *Trichoderma viride* to be completely suppressed mycelium growth of the eight isolates and *Pseudomonas fluorescens* as least effective. The antagonist had also showed antagonism against *R. bataticola* isolates of pigeonpea, chickpea, greengram, clusterbean, field pea, cotton, okra and safflower.

Patil *et al.*, (2004) evaluated *in vitro* the antagonists *Trichoderma* spp. and *P. fluorescens* against *Rhizoctonia bataticola* and *Sclerotium rolfsii*, causing seed root stem root rot of sunflower. They reported that *T. harzianum* and *T. viride* caused maximum mycelial growth inhibition of 89.61 and 88.01 per cent of *R. bataticola* respectively, followed by *P. fluorescens* (65.37%).

Suriachandraselvan *et al.*, (2004) tested *Trichoderma viride* and *Trichoderma harzianaum* as potential antagonists to *Macrophomina phaseolina*. These two antagonists inhibited the mycelial growth and reduced the number, size and germination of sclerotia of *M. phaseolina*. Both *T. viride* and *T. harzianaum* produced non-volatile metabolism in the culture filtrates, which showed the maximum fungitoxic effect against the pathogen. Further they conducted pot culture experiment to manage the charcoal rot of sunflower. Seed treatment with talc based formulation (4g/kg-1) of *T. viride* or *T.*

harzianaum significantly reduced the disease incidence (22-27 %) compared to untreated control (67 %).

Gaur *et al.*, (2005) conducted multi-locational field studies (1999-2000 to 2002-2003) to find out best application technique of *T. harzianum* to control dry root rot of chickpea. They reported that dry root rot of chickpea can be managed either with the soil application of 10 to 15 days pre incubated *T. harzianum* (TG-1) @ 10 kg / ha in 200 kg of FYM or by seed treatment with talc based formulation of *T. harzianum* (2×10^6 c.f.u. /g) @ 10g/kg seed, in combination with soil application of ZnSO_4 @ 25 kg/ha. Soil treatment of *Trichoderma* was found significantly superior over seed treatment.

Chattannavar *et al.*, (2006) evaluated *in vitro* 12 isolates of *Trichoderma* and two isolates of *G. virens* against soil borne pathogens (*F. solani*, *R. solani*, *R. bataticola* and *S. rolfsii*) of cotton. They reported that *T. harzianum* (Dharwad) caused maximum mycelial growth inhibition (51.77%) of *R. bataticola*, followed by *T. koningii* TNAU isolates (49.31%) and *T. harzianum* TNAU isolate (45.52%).

Lokesha and Benagi (2007) evaluated *in vitro* the efficacy of biocontrol agents against *M. phaseolina*, causing dry root rot of pigeonpea and reported that both *T. virens* (PDBC TVS-2) and *P. fluorescens* (PDBC Pf1) significantly inhibited mycelial growth of *M. phaseolina* by 78.22 per cent and 76.66 per cent, respectively.

Patil *et al.*, (2007) evaluated *in vitro* biocontrol agents to manage dry root rot (*R. bataticola*) of groundnut. They reported that seed treatment with *Trichoderma* (USA) resulted with least root rot incidence (30.18%) followed by *Bacillus* sp-1 (3.37%) and *Bacillus* sp-2 (7.18%) VAM (5.85%) and soil application of *Trichoderma* (TNAU) with 19.51%, compared to untreated control (53.08).

Pan and Bhagat (2008) studied *in vitro* antagonistic potential of *Trichoderma* and *Gliocladium* spp. against *M. phaseolina* and reported that all

the bioagents tested caused significant inhibition of test the pathogen. However, highest growth inhibition was reported with *T. viride*-13 (55.90%) followed by *T. viride*-10 (54.20%) *T. viride*-4 (52.40%) *T. viride*-1 (49.50%) and *G. virens* (45.50%).

Singh *et al.*, (2007) evaluated *in vitro* the fungal antagonists against *M.phaseolina* causing root rot of mung bean. They reported that all the bioagents (*Aspergillus* spp., *G. virens*, *T. harzianum*, *T. viride*) significantly inhibited mycelial growth of the test pathogen and reduced the root rot incidence under green house conditions. However, *T. harzianum* was reported most effective followed by *G. virens* and *T. viride*. *Aspergillus* spp. was reported less effective against the test pathogen.

Khan *et al.*, (2008) evaluated the fungal antagonist viz., *T. viride*, *T. harzianum*, *T. hamatum*, *Aspergillus niger*, and *A. flavus* against *R. bataticola* and reported that all antagonists significantly inhibited radial growth of *R. bataticola*. However, it was significantly highest with *T viride* (96.39%) followed by *A. niger*, *T. harzianum*, and *A. flavus* were with mycelial growth inhibition of 84.81, 77.59 and 76.11 per cent, respectively.

Khan and Ganopadhyay (2008) evaluated *in vitro* the efficacy of *P. fluorescens* (strains PFBC-25 and PFBC-26) against *M. phaseolina*, causing dry root rot of chickpea by paper disc, streaking and seed treatment (@4,6,8 g/kg seeds). They reported maximum mycelial growth inhibition with both the strains of *P. fluorescens* (77.7%) by streaking method, followed by PFBC-25 strain (71.9%) and PFBC-26 strain (70.1%) by paper disc method. Seed treatment with PFBC-25 and PFBC-26 strains (@ 8g/kg seed) exhibited minimum root rot incidence of 36.7 and 34.4 per cent, respectively and corresponding disease control of 62.1 and 64.4 per cent, respectively.

Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot.

Khonde *et al.*, (2008a) tested the five antagonists' of soybean and observed that *Trichoderma viride* (96.39%) was superior in inhibiting the growth of *R. bataticola*.

Jaiman *et al.*, (2009) evaluated (pot culture) five bioagents as seed treatment (each @ 4g / kg seed) against root rot (*M. phaseolina*) of cluster bean and reported significantly least pooled mean disease incidence with *T. viride* (19%) followed by *T. harzianum* (22%) *G. virens* (26%) *B. substilis* (30%) and *P. fluorescens* (31%).

Choudhary *et al.*, (2010) evaluated *in vitro* efficacy of the bioagents viz., *T. harzianum*, *A. versicolor*, *T. viride*, *B. firmus* against *M. phaseolina*, causing dry root rot of mungbean. They reported *T. viride* as superior in inhibiting mycelial the growth (64.81%) of the pathogen, followed by *T. harzianum* (61.85%) *A. versicolor* (61.11%) and *Bacillus firmus* (39.63%).

Gaur and Sharma (2010) evaluated *in vitro* and *in vivo* potential biocontrol agents viz., *T. harzianum*, *T. viride*, *T. virens*, *T. koningii*, and *Chaetomium globosum* etc. against *R. bataticola* causing dry root rot of cotton. They reported significantly highest mycelial growth inhibition with *T. viride*-1 (76.53%) followed by *T. harzianum* Th-1 (72.78%) and *G. virens* (72.36%). In pot culture study, seed treatment and soil application of *T. viride* and *T. harzianum* resulted with minimum root rot incidence of 35.78 and 38.98 per cent, respectively and corresponding disease reduction of 56.28 and 52.53 per cent, respectively.

Lodha and Mawar (2010) conducted field experiment (2003 and 2004) to assess the efficacy of native biocontrol agents against dry root rot incidence (*M. phaseolina*) in cowpea, guar and sesamum. They reported that FYM amended soil application of *T. harzianum*, *A. versicolor* and *P. oxalicum* @ 500 g per plot of 3x4 m² and *B. firmus* was coated to seeds @5 ml / kg seed, significantly reduced the dry root rot incidence in test host crops and increased

the yield, over untreated control. However in cowpea, significantly least pooled mean dry root rot incidence was recorded with *A. versicolor* (3.65%) followed by *P. oxalicum* (7.00%) *B. firmus* (8.65%) and *T. harzianum* (9.3%). In guar *B. firmus* gave least mean root rot incidence (10.95%) followed by *T. harzianum* (11.75%) and *P. oxalicum* (12.25%). Whereas, in sesamum, *B. firmus* was most effective with minimum mean root rot incidence (5.10%) followed by *A. versicolor* (5.85%) *P. oxalicum* (6.85%) and *T. harzianum* (7.05%).

Doley and Jite (2012) evaluated *in vitro* the efficacy of *T. viride* against *M. phaseolina* by inhibiting the mycelial radial growth of *M. phaseolina* by 71.42%. The results showed variable mycelial growth rate for all fungal isolates which was determined after 6 days of incubation.

Khan *et al.*, (2012) evaluated *in vitro* the biocontrol agents viz., *T. viride*, *T. harzianum* and *P. fluorescens* against *R. bataticola*, causing dry root rot of chickpea and reported that all the antagonists significantly inhibited mycelial growth of the test pathogen. However, it was significantly highest with *T. viride* (81.96 %) followed by *T. harzianum* (56.26 %) and *P. fluorescens* (29.15 %).

Hegde *et al.*, (2013) evaluated *in vitro* the efficacy of various bioagents against *R. bataticola*, causing root rot of Jatropha. They reported maximum mycelial growth inhibition with *T. virens* (57.78 %) which was at par with *T. harzianum* (57.23 %) followed by *T. viride* (55.83 %) and *T. koningii* (54.45 %) and *P. fluorescens* (30.11 %).

Ikhar *et al.*, (2013a) evaluated *in vitro* the efficacy of seven bioagents against *M. phaseolina*, causing root rot of mungbean. They reported *Trichoderma harzianum*, as most effective with significantly highest mycelial growth inhibition (61.01%) followed by *Bacillus subtilis* (55.08%) and *T. hamatum* (51.12%) and rest of the bioagents tested recorded mean mycelial growth inhibition in the range of 32.58 to 42.52 %.

Kumar *et al.*, (2013) evaluated *in vitro* and *in vivo* four bio-agents viz., *T. harzianum*, *T. viride*, *T. atroviride* and *Bacillus subtilis* against eight isolates of *M. phaseolina*, causing dry root rot of moth bean. They reported that maximum mycelial growth inhibition of the test isolates resulted with *T. harzianum* (61.1 to 70.1%) followed by *T. viride*. Inhibition of the mycelium growth of *M. phaseolina* isolates by *T. harzianum*, *T. viride* (58.6 to 66.6%) *T. atroviride* (52.0 to 63.1%) and *B. subtilis* (45.8 to 54.8%). In a greenhouse study, these antagonists recorded reduction in root rot incidence by 69.0, 65.0, 62.5 and 58.3 per cent, respectively, with *T. viride*, *T. harzianum*, *T. atroviride* and *B. subtilis*.

Thombre *et al.*, (2013) evaluated *in vitro* seven bioagents against *M. Phaseolina*, causing blight of mungbean. They reported *T. harzianum* as most effective with significantly highest mycelia growth inhibition (77.59%) followed by *A. niger* (68.17%) *T. viride* (65.46%) *T. hamatum* (63.26%) *P. fluorescens* (60.90%) *T. longibrachiatum* (60.10%) *B. subtilis* (51.37%) and *T. koningii* (49.55%); whereas, *G. virens* was least effective (31.54%).

Solunkhe and Deshpande (2013) evaluated *in vitro* the efficacy of bioagents against *M. phaseolina* causing root rot of safflower and reported that *T. viride* resulted with expressed significantly highest (81.54%) mycelia growth inhibition, followed by *T. harzianum*, *T. hamatum*, *P. fluorescens*.

Ashwini *et al.*, (2014) evaluated *in vitro* the efficacy of the antagonists viz., *Trichoderma viride* and *Pseudomonas fluorescens* against *M. phaseolina* of black gram and reported that *P. fluorescens* resulted with maximum mycelial growth inhibition (62.41%) followed by *T. viride* (44.94%).

Navgire and Wadhawe (2014) evaluated (pot culture) the bioagents against *M. phaseolina*, causing root rot of safflower and reported that test bioagents significantly reduced seedling mortality, over untreated control. Significantly least seedling mortality was recorded with *T. hamatum*

(2.08 %) *T. viride* (2.41 %) followed *T. harzianum* (2.16 %) *T. viride* (2.41 %) and *P. fluorescens* (2.74 %).

Shahare (2014) evaluated *in vitro* the bioagents against *Macrophomina phaseolina*, causing stem rot of sesame and reported maximum mycelial growth inhibition with *T. harzianum* (87.5%) followed by *T. viride* (80.00%) *Aspergillus flavus* (60%); while, it was lowest with *A. niger* (25%) and *Penicillium* spp. (10%).

Suryawanshi *et al.*, (2014a) evaluated *in vitro* the efficacy of bioagents against *M. phaseolina*, causing charcoal rot of sunflower and reported *T. viride* as most effective with significantly highest mycelial growth inhibition (77.30%) followed by *T. harzianum* (75.30%) and *T. longibrachiatum* (70.53%). Rest of the bioagents tested recorded mycelia inhibition in the range of 56.19 (*B. Subtilis*) to (*T. hamatum*) per cent.

Wadhawe and Navgire (2014) evaluated *in vitro* the bioagents against *M. phaseolina* causing root rot of safflower and reported highest mycelial growth inhibition with *Trichoderma hamatum* (79.28%) followed by *T. harzianum* (75.15%) *T. viride* (70.15%) *T. lingorum* (68.36%) and *P. fluorescens* (65.16%).

Veena *et al.*, (2014b) isolated ten *Trichoderma* spp. from chickpea rhizosphere and root endophytic region by using serial dilution technique and purified by single hyphal tip method. They reported that out of the ten isolates tested against *Rhizoctonia bataticola*, *Trichoderma* isolate-7 showed highest inhibition per centage (83.33). In compatability tests with commonly used fungicides, *Trichoderma* isolate-7 showed highest compatability with validamycin (72.22%) followed by copper oxychloride (66.66%).

Gautam *et al.*, (2015) studied the effect of *Trichoderma* sp viz. *Trichoderma viride* and *T. harzianum* on germination and growth of Mungbean and against its fungal pathogen viz., *Fusarium oxysporium* and *Rhizoctonia*

bataticola. The inhibition in colony of *F. oxysporum* was found 30.66% by *T. viride* and in the *R. bataticola* was 40%. The *T. harzianum* showed inhibition in *F. oxysporum* 42.54% and in *R. bataticola* 48.14% after 3 days of inoculation. The antagonists viz., *T. harzianum* and *T. viride* were also found to affect the growth response of mungbean (*Vigna radiata* L.). The germination of mungbean seeds was found maximum (80%) in *T. harzianum* and *T. viride* (57.77%) in comparison to control (23.33%).

Karthikeyan *et al.*, (2015) evaluated *in vitro* (dual culture and pot culture) the efficacy of *Trichoderma* spp. against *M. phaseolina*, causing root / charcoal rot of black gram. They reported that *T. viride* caused highest mycelial growth inhibition (77.77%) followed by *T. harzianum* (75.55%) *T. reesii* (72.22%) *T. hamatum* (68.88%) and *T. pseudokoningii* (48.88%). In pot culture, maximum reduction of root rot incidence was with *T. viride* seed treatment (59.54%) and its soil application (55.16%) followed by *T. harzianum* (49.71 and 28.60%) *T. hamatum* (32.13%) and *T. reesii* (38.00%).

Wagh *et al.*, (2015) evaluated *in vitro* 22 isolates of *P. fluorescens* and 15 isolates of *Trichoderma* spp. against *R. bataticola*, causing dry root rot of chickpea. They reported among the isolates of *Trichoderma* spp., T40 (Bemetara) isolate resulted with higher mycelial growth inhibition (67.65%) followed by the isolates T36 Dhamdha (64.80%) and T14 Khadsiya (64.07%). Average mortality of chickpea in sick pots after 30 days of challenge inoculation was minimum (20.83%) with T40 (Bemetara) and T23 (Bilaspur) isolates. Among the isolates of *P. fluorescens*, maximum growth inhibition was with P33 (Lohara) isolate (63.21%) followed by P40 (Bemetra) isolate (62.35%) and P36 (dhamda) isolate (61.91%). Average mortality of chickpea in sick pots after 30 days of challenge inoculation with RB4 (Kawardha) was minimum with P3 (Rajnandgaon) followed by P8 (Raigarh) and P40 (Bemetra) isolates.

Arya *et al.*, (2017) evaluated *in vitro* and *in vivo* the efficacy of various biocontrol agents against *Macrophomina phaseolina*, causing dry root

rot of groundnut. Under *in vitro* evaluation, maximum mycelial growth inhibition was with *T. viride* (74.72 %) followed by *T. harzianum* (71.54 %) *T. viride* (62.55 %) *P. fluorescens* (48.32 %) and *B. subtilis* (47.51 %). Under field evaluation, maximum disease control was with *T. viride* + *T. harzianum* ST each @ 4 g/kg + their soil application each @ 5 kg/ha (77.30 %) followed by *T. viride* + *P. fluorescens* ST each @ 4 g/kg + their soil application each @ 5 kg/ha (72.35 %) *T. harzianum* + *P. fluorescens* ST each @ 4 g/kg + their soil application each @ 5 kg/ha (65.91 %) *T. viride* ST @ 8 g/kg + its soil application @ 10 kg/ha (59.36 %) and *T. harzianum* ST @ 8 g/kg + its soil application @ 10 kg/ha (55.54 %).

Maruti *et al.*, (2017b) evaluated *in vitro* the efficacy of bio-agents against *R. Bataticola*, causing dry root rot of Pigeonpea and reported that *T. viride* resulted with maximum mycelia growth inhibition (77.20 %) followed by *T. virens* (75.76 %) *T. harzianum* (73.91 %) and *P. fluorescens* (38.12 %).

2.7.3. Efficacy of organic amendments

Waghe and Lanjewar (1981) evaluated (pot culture) different oil cakes as soil amendments to manage of root and stem rot (*M. phaseolina*) diseases of Sesamum in pot culture. They reported that all the oil cakes tested (@ 15 qt / ha) resulted with maximum seedling emergence (90-98 %) as against 84 per cent in untreated control. Significantly highest reduction in seedling mortality was recorded with groundnut cake (77.55%) followed by Karanj cake (75.51%) Mauwa and Sesamum cakes (each 69.38%) linseed cake (66.67%) Kokam cake (60.00%) and Neem seed cake (35.35%).

Singh *et al.*, (1990) studied the effect of soil amendments with inorganic and organic sources of nitrogen on the incidence of root rot (*M. phaseolina*) of sesame. They reported that Urea + FYM recorded minimum root rot incidence (10.00 %) and maximum seed yield (87.86 g/plot) followed by mustard cake, FYM and Neem seed cake.

Muthusamy and Mariappan (1991) studied antifungal effect of the extracts (@ 1,2 and 3 %) of neem cake, groundnut cake, coconut cake and ginger against germination, disintegration and number of sclerotia of *M. phaseolina*, causing charcoal rot of soybean. They reported that sclerotial germination was significantly inhibited with the test oil cake extracts. However, it was maximum with groundnut cake extract (40.00 to 42.00 %) followed by ginger extract (6.0 to 8.8 %) and Neem seed cake extract. Significantly highest disintegration of sclerotia was recorded with neem cake extract (76.00 to 95.00 %) followed by ginger cake extract (81.00 to 92.00 %) and groundnut cake extract (35.00 to 41.00 %).

Hundekar *et al.*, (1998) tested (pot culture) organic amendments and reported that neem, cotton, groundnut and safflower cakes reduced the inoculum level of *Macrophomina phaseolina* and *Fusarium moniliforme* in the soil and reduced the incidence of stalk rot of sorghum. Except neem cake, the population of beneficial bacteria, fungi and actinomycetes were higher in amended soil than non-amended soil.

Rathore (2000) evaluated the efficacy of organic amendments viz., Neem leaf mould (@ 0.12%) Tumba and mustard oilseed cake (@ 0.04 %) and FYM (@ 0.20%) against *M. phaseolina*, causing seed rot and seedling blight in mothbean. He reported that all the treatments significantly increased the plant emergence and plant growth parameters and also reduced pre and post-emergence plant mortalities. Mustard cake and FYM recorded highest seedling emergence (89.50 and 87.00%) lowest pre-emergence mortality (10.50 and 13.00%) and post-emergence mortality (23.60 and 24.80%) respectively.

Aiyanathan and Salalrajan (2008) evaluated (pot culture) efficacy of organic soil amendments against *R. solani*, causing root rot of fenugreek. They reported reduced root rot incidence 4.8, 5.7 and 6.0 per cent with neem cake, FYM and poultry manure, respectively as against 25.6 per cent in untreated control. These amendments were on par with reduced root rot incidence of 81.2, 77.7 and 76.5 per cent, respectively.

Sudha and Prabhu (2008) evaluated (pot culture) the efficacy of organic soil amendments against *M. Phaseolina*, causing sunflower charcoal rot. They reported that FYM (@ 12.5 t/ha) and neem cake (@ 250 kg/ha) significantly reduced charcoal rot incidence by 13.30 and 15.00 per cent, respectively, as against 63.30 per cent in untreated control. The rhizosphere population of bacteria and fungi increased in amended soils, compared to non-amended soils and also increased the seed germination, seedling growth and seedling vigour index of sunflower.

Jaiman *et al.*, (2009) field evaluated (2001 and 2002) the soil amendments viz., neem cake, mustard cake, wood ash, gypsum, FYM and poultry manure against *M. phaseolina*, causing root rot of cluster bean and reported that the test amendments significantly reduced pre- and post-emergence mortality incidence with increased grain yield. Lowest pooled mean root rot incidence was with neem cake (18%) followed by mustard cake (20%) FYM (25%) and Poultry manure (27%).

Mehta *et al.*, (2010) evaluated (pot culture) seven organic amendments (each 20g/kg soil) against *Sclerotinia sclerotiorum*, causing white stem rot of mustard. They reported minimum mortality with mustard cake (4.10 %) followed by sesame cake (4.80%) seed powder of *S. cuminii* (7.80%) FYM (23.40%) and Poultry manure (24.10%). Minimum disease incidence was recorded with poultry manure (29.90%) followed by neem seed powder (30.00%) and cotton cake (31.70%). Whereas, it was maximum with FYM (41.70%) and mustard cake (45.80%).

Dhingani *et al.*, (2013) evaluated *in vitro* efficacy of aqueous extract of FYM, neem seed cake, castor seed cake and mustard cake against *M. phaseolina*, causing root rot of chickpea. They reported significantly maximum mycelial growth inhibition with the extracts of neem cake @ 20% (59.40%) and @ 10% (53.69%) FYM @ 20% (42.56%) FYM @ 10% (29.5%) castor cake @ 20% (19.78%) and mustard cake @ 20% (15.19%).

Ikhar *et al.*, (2013a) evaluated *in vitro* efficacy of organic amendments aqueous extracts (each @ 10 and 20%) against *Macrophomina phaseolina*, causing root rot of mungbean. They reported aqueous extract of neem seed cake as most effective with significantly highest mean mycelial growth inhibition (79.56%) followed mustard cake extract (62.25%) and karanj cake extract (57.32%). Rest of the organic amendments extracts tested recorded mean mycelial growth inhibition in the range of 12.40 to 50.08 per cent.

Suryawanshi *et al.*, (2015) evaluated (pot culture) efficacy of 11 organic amendments to manage charcoal rot (*M. phaseolina*) disease of sunflower. They reported neem seed cake soil application (preventive and curative) @ 50g/kg potting mixture as most effective with significantly highest seed germination (85.00, 71.67%) least pre -emergence (15.00, 28.33%) post-emergence (22.66, 34.92%) and average (18.83, 31.63%) mortality, significantly highest reduction in pre-emergence (78.57, 59.52%) post-emergence (70.03, 55.62%) and average (74.30, 57.57%) mortality, respectively in preventive and curative methods. The second and third best amendments reported were cotton seed cake and groundnut cake, followed by the amendments *viz.*, Sunflower cake, Soybean cake and Poultry manure. However, FYM and compost were least effective.

Pandhare *et al.*, (2015) evaluated 10 organic amendments for the control of charcoal rot disease (*M. phaseolina*) of *Rabi* sorghum. They reported Neem seed cake as most effective with significantly highest seed germination (83.33%) lowest average mortality (20.37%) and highest average mortality reduction (73.15%). This was followed by the amendments *viz.*, Compost + Neem seed cake (73.33%, 29.11% and 61.36%) Cotton seed cake (73.33%, 30.40% and 59.57%) Compost + Cotton seed cake (66.67%, 36.50% and 51.56%) and Groundnut cake (60.00%, 42.10%, 43.99%) of seed germination, average mortality and average mortality reduction, respectively. Whereas, FYM was found least effective with significantly least seed germination (43.33%) maximum average mortality (57.50%) and its minimum reduction.

2.8. Integrated disease management

Asghari and Mayee (1991) reported that application of *T. harzianum* and soil drenching with 0.2 per cent Carbendazim reduced the stem and pod rot of groundnut by 44-60 per cent and increased the pod yield by 17-47 per cent.

Kaur and Mukhopadhyay (1992) reported that chickpea wilt complex can be effectively controlled by *T. harzianum* and its integration with fungicides. Soil application of *T. harzianum* at various dosages gave 53.50-85.70 per cent disease control, under in glass house conditions. Under field, integration of *T. harzianum* with different fungicides significantly reduced the wilt incidence and increased the crop yield. Seed treatment with Vitavax-200 and Ziram resulted in 29.90 per cent disease control, which was further increased to 63.30 per cent when integrated with soil application of *T. harzianum*.

Vyas (1994) reported that simultaneous application of *T. viride* or *T. harzianum* with Carbendazim to soybean seeds reduced dry root rot (*R. bataticola*) incidence and increased the seed yield of soybean.

Rajeswari (1997) reported that dry root rot (*M. phaseolina*) of mungbean effectively controlled by integration of *Trichoderma harzianum* soil application (5g/kg) + seed treatment (10^8 cfu/ml) and Carbendazim (0.02%) significantly reduced dry root rot incidence (95.3 %) over soil treatment (91.5%) and seed treatment alone.

Kumar and Singh (2000) conducted experiments with different fungicides as a seed treatment. They found that Carbendazim 50 % WP (2.0 g/kg seed) and Thiophanate-methyl 70 % WP (1.0 g/kg seed) were effective in eliminating the pathogen *M. phaseolina* from infected seeds of soybean. Seed treatment with Captan at 3 g/kg or Thiram at 3 g/kg or Thiram + Carbendazim (2:1) at 3 g/kg or Thiram + Carboxin at 2 g/kg.

Kumar and Dubey (2001) reported that seeds treatment with Captan 50 % WP @ 1g/kg + *T. harzianum* (10⁶ spores/ml) @ 10 kg resulted with maximum seed germination, minimum collar rot least disease incidence and maximum green pod yield of pea and which was statistically on par with seed treatment with *T. harzianum* alone, Thiram + *T. harzianum*, Carboxin + *Gliocladium virens* and *G. virens* alone.

Pant and Mukhopadhyay (2001) attempted integrated management of seed and seedling rot complex of soybean with bio-agents and fungicides. They reported that integration of *G. virens* and *T. harzianum* with Vitavax (0.1 %) as most effective, which improved seedling emergence, plant stand and yield.

Mani and Hepziba (2003) evaluated *in vitro* the biocontrol agents *Trichoderma viride* and *T. harzianum* and oilseed cake as soil organic amendment to manage sunflower root rot (*Macrophomina phaseolina*). They reported that *T. viride* (ST) @ 4g/kg + neem seed cake (SA) @ 250 kg/ha resulted with least mean root rot incidence (19.18%) followed by *T. harzianum* (ST) @ 4g/kg + FYM (SA) @ 12.5 t/ha (23.82%) *T. harzianum* (ST) @ 4g/kg + Neem seed cake (SA) @ 250kg/ha (29.68%) and alone neem seed cake (SA) @ 250 kg/ha (30.79%) as against maximum disease incidence in untreated control (53.33%).

Patil *et al.*, (2003) evaluated *in vivo* the biocontrol agents and fungicides to manage cotton root rot disease (*R. bataticola* and *R. solani*) for three years (1998 to 2001). They reported significantly highest pooled mean disease control by *T. viride* seed treatment @ 5g/kg seeds (57.14%) followed by *T. harzianum* (42.86%) *P. fluorescens* (39.43%) Thiram 75% WP and Captan 50% WP (each 32.57%) and Carbendazim 50% WP (28.57%) seed treatment.

Gaur *et al.*, (2005b) conducted green house and multi-locational field studies to evolve the best application technique of *Trichoderma*

harzianum for controlling dry root rot (*R. bataticola*) of deshi cotton (*Gossypium arboreum*). They reported that soil application of 10-15 days pre incubated *T. harzianum* (TG-1) @ 10kg/ha in 200 kg FYM proved significantly superior to alone application of bioagent. Wheat bran based formulation of *Trichoderma* gave better control of disease than talc based formulation. Similarly soil application of *Trichoderma* was found significantly superior over seed treatment.

Vijay-Mohan *et al.*, (2006) reported in fungicidal trails on management of dry root rot of chickpea caused by *R. bataticola* carbendozin (0.2%) and Etaconazole (0.1%) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 and 18.2 per cent highest grain yield of 192 and 18.9 q/ha respectively, during *rabi* 2001-2002 and 2002-2003 crop seasons. The above treatment recorded 57.2 and 52.4% higher yield over control with per rupee return of 12.78 and 11.80 respectively.

Christopher *et al.*, (2007) integrated antagonist *T. viride* and organic amendments by seed and soil application dry root rot (*M. phaseolina*) of urdbean. They reported that application of *T. virens* (seed, soil) plus soil application of FYM as most effective. Application of *T. virens* (seed, soil on 0 DAS and 30 DAS) alone resulted with minimum root rot incidence (6.07%) and maximum disease control (84.55%) followed by *T. virens* (seed and soil on 0 DAS) plus FYM with 7.17 per cent root rot incidence and 81.75 per cent disease control.

Khonde *et al.*, (2008b) tested the efficacy of bio-agents and fungicides as seed treatment and soil application under pot and field conditions for management of root rot (*R. bataticola*) of soybean. In pot experiment, they reported seed treatment with Thiram 75 % WP + Carbendazim 50 % WP @ 2+1g/kg seed effective with maximum control of root rot (50.43%) followed by Thiram @ 3g/kg seed (45.38 %) Thiram + *T. viride* @ 1.5 + 4g/kg seed (41.29 %) with maximum seed germination. In field experiment, seed treatment with

Carbendazim + Thiram @ 1+2 g/kg seed was effective in reducing root rot disease incidence (74.64%) followed by seed treatment with Thiram @ 3g/kg seed (72.81 %) Carbendazim @ 1g/kg seed (66.65 %) and soil application of *T. viride* + *T. harzianum* @ 1+1kg/ha (66.00 %) which also gave maximum grain yield (1808 kg/ha).

Vanitha and Alice (2008) conducted field experiment for three years at (RRS, Paiyur) to manage root rot disease of horse gram with *Trichoderma viride* and organic amendments. They reported that seed treatment of Carbendazim 50% WP @ 1g/kg resulted with lowest disease incidence (8.3%) highest seed yield (1504 kg/ha) and maximum cost benefit ratio (1:3). Soil application of vermiculite @ 2 tonns/ha plus *T. viride* seed treatment was also effective with minimum disease incidence (11%) maximum seed yield (1335kg/ha) and cost benefit ratio (1:2.5).

Rajani and Parakhia (2009) conducted the field trial to manage root rot (*M. phaseolina*) of castor with bioagents and organic amendments. They reported that soil application of neem cake @ 500 kg/ha and *T. harzianum* @ 5kg/ha resulted with least disease incidence (14%) maximum castor seed yield (218 kg/ha) and highest ICBR (1:7.91). Soil application of neem or mustard cake @ 500 kg/ha along with talc based preparation of *T. harzianum* (10^6 cfu/g) @5kg/ha applied just before sowing in furrow, can effectively control root rot disease.

Khodke and Raut (2010) evaluated *Trichoderma viride* and fungicides (Carbendazim 50 % WP, Thiram 75 % WP) as seed treatment and soil application for the management of root rot/collar rot (*M. phaseolina* / *S. rolfsii*) of soybean. They reported that all the treatments significantly increased the seed germination and reduced pre- and post-emergence mortality. However, maximum seed germination was recorded by seed treatment with Thiram 75% WP + Carbendazim 50% WP+ *T. viride* (@ 3+1+4 g/kg) followed by Thiram + Carbendazim ST (81.15%) and seed treatment with *T. viride* + its soil application (77.31%). Significantly least total mortality was recorded by seed

treatment with Thiram + Carbendazim + *T. viride* (4.34%) followed by Thiram + Carbendazim ST (4.86%) *T. viride* ST + its soil application (5.72%) and *T. viride* (6.86%).

Kumar *et al.*, (2011) integrated *in vitro* the fungicides, bioagents and phyto extract to manage root rot (*R. bataticola*) of Jatropha. They reported significantly least pre- (17.78 %) and post- (8.12 %) emergence mortality and highest disease control (67.31 %) with Carbendazim ST @ 2 g/kg seed + NSKE 20 % ST, followed by NSKE 20 % ST + *T. harzianum* (24.45 %, 11.86 % and 54.18 %) Carbendazim ST + *T. harzianum* ST (28.89 %, 15.45 % and 44.04 %) and Carbendazim ST (35.35 %, 24.07 % and 24.76 %).

Moradia and Khandar (2011) evaluated fungicides and bioagents against *Macrophomina phaseolina* causing root rot of groundnut. They reported highest disease control (69.40%) and lowest disease incidence (11.10%) with of Vitavax ST + *T. viride* SA + Neem seed cake, followed by Raxil ST + *T. viride* SA + Neem seed cake and *T. viride* + Neem seed cake SA.

Nagamani *et al.*, (2011) investigated the effective management strategy for dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler and reported that seed treatment with Carbendazim @ 2g/kg of seed + seed treatment with *T. viride* @4g/kg of seed+ soil application of FYM fortified with *T.viride* recorded was the best treatment to decrease the per cent mortality of root rot caused by *R. bataticola*.

Rajpurohit (2011) integrated the bio-control agents and fungicides to manage stem / root rot (*M. phaseolina*) of sesame, under field conditions. He reported significantly least mean root rot disease incidence (4.40%) and highest disease control (74.27%) with the treatment *T. viride* ST (@ 4 g/kg seed) + *P. fluorescens* ST (4 g/kg seed) + soil application (SA) of *T. viride* (2.5 kg/ha) + soil application of *P. fluorescens* (2.5 k/ha). This was followed by the treatment *T. viride* ST + *T. viride* SA + *P. fluorescens* SA (5.2%) *T. viride* ST + *P. fluorescens* ST + *T. viride* SA (6.30%) *T. viride* ST +

T. viride SA (6.50%) *T. viride* ST + *P. fluorescens* ST + *P. fluorescens* SA (7.30%) and Carbendazim ST (@ 0.1%) + Thiram ST @ 0.2% (9.40%) of root rot incidence, as against 17.10 per cent incidence in untreated control.

Kumari *et al.*, (2012) conducted an experiment to determine the compatibility of biocontrol agents and herbal oils with effective fungicides manage root rot (*M. phaseolina*) of mung bean. They reported that Vermicompost (10 %) + Carbendazim 50 %WP (0.1 %) as most effective treatment with significantly least pre and post-emergence mortality (5.74 and 5.04 %) followed by Vermicompost (10 %) + Carbendazim 50 % WP (0.1 %) + *T. harzianum* (4%) with 5.76 and 5.04 per cent pre and post-emergence mortality, respectively.

Lahre *et al.*, (2012) conducted field experiment to evaluate the efficacy of *Trichoderma* (@ 4 g/kg seeds) and three organic amendments (Neem seed cake, Mustard cake and Karanj cake) as soil application alone and in combination with *Trichoderma* against *Sclerotium rolfsii*, causing collar rot of chickpea. They reported under natural conditions, maximum seed germination (93.05%) and minimum total mortality (8.32%) with *Trichoderma* ST (@ 4 g/kg seed) + Neem seed cake soil application (@ 500 kg/ha) followed by *Trichoderma* ST (@ 4 gm/kg seed) + Mustard cake soil application (85.00 and 9.55%) *Trichoderma* ST + Karanj cake soil application (80.86 and 11.14%) and *Trichoderma* ST (@ 4 gm/kg seed) alone (72.99 and 14.02%) of seed germination and total mortality, respectively. Under sick soil condition, maximum seed germination (96.29%) and total mortality (18.51%) were recorded with *Trichoderma* ST + Neem seed cake soil application, followed by *Trichoderma* ST + Mustard cake soil application (92.59% and 25.92%) and *Trichoderma* ST + Karanj cake soil application (92.59 and 25.92%).

Ikhar *et al.*, (2013b) evaluated (pot culture) the fungicides, bio-agents and oilseed cakes for integrated management of root rot (*M. Phaseolina*) of mungbean and reported that all the treatments significantly reduced pre-

emergence seed rot (PESR) and post- emergence seedling mortality (PESM) over untreated control. Significantly least average mortality (5.49 %) was recorded with Carbendazim 50 % WP seed treatment @ 1g/kg, followed by seed treatment with Metalaxyl M-Z 35 WS @ 2g/kg (9.49 %) and soil drenching with garlic bulb crude extract (10.16 %). Rest of the treatments found effective with minimum average mortality were *T. harzianum* (14.49 %) *T. hamatum* (16.49 %) Neem leaf extract (16.49 %) Neem seed cake (31.16 %) and Mustard cake (35.33 %). Significantly highest reduction in average mortality was with Carbendazim (88.07 %) followed by Metalaxyl (79.69 %) garlic bulb extracts (78.10 %) *T. harzianum* (68.75 %) *T. hamatum* and Neem leaf extract (each 64.60%) and Neem seed cake (33.19 %) and Mustard cake (24.65 %).

Mane *et al.*, (2013) attempted management of root and collar rot of soybean with bio-agents and fungicides. They reported maximum reduction in collar root rot incidence of 52.81 and 44.10 % with soil application of *Trichoderma* @ 3 kg / 50 kg FYM /ha and its seed treatment @ 4 g/kg + soil application @ 2 kg / ha, respectively. Maximum seed yield was obtained with seed treatment of Thiram + Carbendazim + *Trichoderma* @ 3 + 1 + 4 g/kg (14.75 q/ha.) followed by seed treatment with *Trichoderma* @ 4g + its soil application @ 2kg/ha (14.33 q/ha) which was at par with soil application of *Trichoderma* @ 3 kg/ha (13.94 q /ha). Seed treatment and soil application 4g + 2 kg/ FYM of *Trichoderma* and seed treatment with Thiram + Carbendazim + *Trichoderma* (3 + 1 + 4 kg) was most effective with maximum disease reduction. Seed treatment with Thiram + Carbendazim + *Trichoderma* (3+1+4 g/kg) also recorded, higher seed yield and maximum ICBR (1:31) followed by seed treatment with *Trichoderma* @ 4 g/kg + 50 kg FYM/ha.

Manjunatha *et al.*, (2013) evaluated *in vitro* the efficacy of bio-control agents viz., *T. viride* and *P. fluorescens* and fungicides viz., Carbendazim 50% WP and Hexaconazole 5 % EC for management of dry root rot (*M. phaseolina*) of chickpea. *T. viride* seed treatment (@4 g/kg) followed

by soil application of FYM enriched with *T. viride* (@ 4 kg/plot) resulted with highest seed germination (97.60 %) and lowest root rot incidence (2.67 %) followed by *P. fluorescens* seed treatment + soil application through FYM @4 kg/plot with 96.27 % seed germination and 2.93 % root rot incidence, Carbendazim seed treatment (@ 2.0 g/kg seed) with 94.67 % seed germination and 4.09 % root rot incidence, *T. viride* seed treatment (@ 4 g/kg seed) with 94.23 % seed germination and 3.15 % root rot incidence *P. fluorescens* seed treatment (@ 5 g/kg seed) with 93.63 % seed germination and 3.38 % root rot incidence and Hexaconazole seed soaking (@ 2 ml/kg seed) with 88.27 % seed germination and 4.32 % root rot incidence.

Suryawanshi *et al.*, (2014b) conducted experiment (pot culture) for integrated management of sunflower charcoal rot (*M. Phaseolina*) disease and reported that all the treatments significantly improved seed germination and significantly reduced the pre-, post-emergence and average mortalities, over untreated control. However, the treatment Carbendazim (ST @ 1g/kg seed) + *T. viride* (ST @ 10 g/kg soil) + NSC (SA @ 50 g/kg soil) was most effective with significantly highest seed germination (96.67%) and significantly highest reduction in average mortality (92.61%) respectively with seed germination of 93.34 and 90.00 per cent and average mortality reduction of 89.95 and 84.80 per cent. Rest of the treatments was also found effective with seed germination in the range of 80.00 to 86.67 per cent and average mortality reduction of 66.75 to 82.12 per cent.

Pandhare *et al.*, (2015) integrated various fungicides, bioagents, and organic amendments for management of charcoal rot (*M. Phaseolina*) of sorghum. They reported that Carbendazim 50 % WP (ST @ 1g/kg seed) + *T. viride* (ST @ 10g/kg seed) + NSC (SA @ 50g/kg soil) resulted with significantly highest seed germination (96.67 %) significantly least average mortality (5.00 %) and highest reduction in average mortality (92.61 %) followed by Carbendazim 50 % WP (ST @ 1g/kg) + *T. viride* (ST @ 10g/kg) and Carbendazim + Neem leaf extract (@10 %) with seed germination,

respectively of 93.34 and 90.00 per cent, average mortality, respectively of 6.85 and 10.37 per cent and average mortality reduction, respectively of 89.95 and 84.80 per cent.

Sayyad *et al.*, (2015b) integrated various fungicides, bioagents and organic soil amendments to manage dry root rot (*M. phaseolina*) of chickpea. They reported maximum seed germination with *T. viride*, ST (92.0%) followed by Carbendazim 50% WP (90.0%) Neem seed cake soil application (85.6%) *T. harzianum* (84.9%) Thiram 75% WP @ (84.6%) *T. hamatum* (84.5%) Groundnut cake soil application (82.3%) Cotton cake soil application (81.6%) and Captan 50% WP (81.3%). Least disease incidence was with Carbendazim (14.0%) followed by Neem seed cake (14.7%) Cotton cake (21.2%) Groundnut cake (24.5%) Thiram (26.3%) and Captan (28.2%) *T. viride* (31.1%) *T. harzianum* (35.3%) and *T. hamatum* (43.3%).

Viswanathan *et al.*, (2015) conducted field experiment (Rabi-2010 and 2011) at ARS, Nandyal (Andhra Pradesh) to assess the efficacy of seed treatment with chemicals, bio-control agents and organic amendments against dry root rot of chickpea. Pooled mean data revealed that the treatment T₄ i.e. Carboxin 37.5% + Thiram 37.5% WP seed treatment @ 2.5 g/kg seed, soil application of *Trichoderma* @ 2.5 kg/ha and organic amendments (neem cake /castor cake) @500 kg/ha resulted with significantly least dry root rot incidence (4.4%) as against 23.4 per cent incidence in untreated control. In rest of the treatments, the disease incidence ranged from 6.5 to 12.5per cent. Maximum plant stand was recorded in treatment T₄, with pod yield of 2157 kg/ha, which was significantly superior over other treatments.

Gupta (2016) conducted field experiment during *Kharif* 2014 and 2015 to manage charcoal rot disease (*M. phaseolina*) of sesame. They reported that treatment *P. fluorescens* + SA of *P. fluorescens* 2.5 kg/ha enriched in 100 kg FYM + Neem cake @ 250 kg/ha and (*T. viride* + *P. fluorescens* SA of *T. viride* + SA of *P. fluorescens* enriched in 100 kg FYM + Neem cake @ 250

kg/ha) resulted with minimum disease incidence of 15.0 and 15.3 per cent respectively, as against 38 per cent incidence in untreated control.

Dhingani and Solanky (2016) evaluated the efficacy of various Bioagents viz. *Tricoderma viride* and *T. harzianum*, Oil cakes viz. neem cake and castor cake, FYM and Carbendazim through seed treatment and soil application against *Macrophomina phaseolina* (Tassi.) Goid causing root rot disease of Chickpea. They reported that soil application of *T. harzianum* @ 5 kg in 500 kg neem cake/ha in furrow 5 days prior to sowing resulted in higher seed germination (74.90%). In case of disease incidence and yield of crop, soil application of *T. harzianum* @ 5 kg in 500 kg FYM /ha in furrow 5 days prior to sowing has lowest disease incidence (11.81%) and gave highest yield (1533 kg/ha).

Ingle *et al.*, (2017) evaluated effectiveness of organic amendments, beneficial microbes and chemical fungicides tested under natural field conditions against root rot of soybean incited by *Rhizoctonia bataticola*. Results revealed that seed treatment with Carboxin + Thiram (2.0 g/kg seed) + soil drench with Carboxin + Thiram (0.1%) at disease inception and Pre- soil application of FYM enriched with *T. harzianum* and *P. fluorescens* 2.5 kg and 5 kg respectively + neem cake 500 kg/ha + ST with *T. harzianum* (4g/kg seed) and *P. fluorescens* (10g/kg seed) stood best by recording lowest (6.17 and 8.15%) root rot with 74.48 and 66.28% reduction in disease respectively.

Deepa *et al.*, (2018) tested *in vivo* efficacy of bio-agents and fungicides as seed treatment under field condition for management of dry root rot (*R. bataticola*) of soybean. Among the treatments employed, the treatment containing seed treatment with Mancozeb 50% + Carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of Mancozeb 50% + Carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (T3) recorded significantly lowest incidence of dry root rot (7.70%) when compared to all other treatments except (T7) seed treatment with *T. harzianum* @ 5g/kg seed + *P. fluorescence* @ 5 g/kg seed followed by soil drenching of Mancozeb 50% + Carbendazim

25% WS @ 3 g/l water to infected and surrounding plants (9.57%). However, untreated control treatment recorded the highest incidence of dry root rot (33.45%).

2.9. Screening of chickpea cultivars / varieties against *R. bataticola*

Krishnamohan *et al.*, (1981) tested 20 chickpea forms screened under field conditions and artificially in the green house, using a 1-6 score (1- highly resistant) BG205 and BG206, although moderately susceptible in the green house, were, highly resistant under field conditions against *R. bataticola*.

Singh and Mehrotra (1982) reported resistant to *R. bataticola* was shown by the cultivars BG-203, G-543 and Hare Chhole when grown in infested soil.

Nene *et al.*, (1989) evaluated 225 chickpea genotypes at the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) Patancheru and the Punjab Agricultural University, Ludhiana for the broad based resistance to *Rhizoctonia bataticola*, *R. solani* and *F. solani* at 24 locations in 11 countries during 5 years period (1978-82). They reported that at each of the 24 locations, few to several lines were found resistant to wilt and/or root rots. However, the genotypes ICC-2862, ICC-9023, ICC-9032, ICC-10803, ICC-11550 and ICC-11551 were resistant to wilt and root rots at 11 to 12 locations of the 18 locations tested.

Singh *et al.*, (1989) screened 312 cultivars of Sesamum obtained from Oilseed Research Station, Virudhachalam (Tamilnadu) against root rot caused by *M. phaseolina* under natural condition. They reported that AK-77, ES-4, IS-175, KIS-430, N66-135, SO 013, ES-004, SI-72, SI-1056, SI-57, SI-3319, SI-3169, SI-1740, SI-1153, SI-1875, SI-1249, SI-935, SI-260, SI-2173, SI-1543, SI-189, IS-285, SI-1088, TMV1, SI-820, M-3-1, RJS-133, IS-140 and T10 were resistant cultivars followed by 158 moderately resistant and 124 susceptible and highly susceptible cultivars.

Reddy *et al.*, (1990) tested resistant to wilt and different root rots of chickpea and found that the mortality of variety JG.-62 (100%) Avrodhi and ICC-48 was (20%).

Baker and Ahmed (1991) tested the resistance of 90 genotypes of chickpea in field infested with wilt, dry root rot pathogens and found that ICC12263 was most resistant.

Jayanti and Bhatt (1993) pre-germinated seed of 21 chickpea varieties were sown in contaminated soil. *R. bataticola* caused seed rot within 24h in NEC874 and EG234, Bold 2375, BG209, JG62, JG315, ICC 3357 and JG1133 developed necrotic lesion 3 to 5 cells deep on the hypocotyls region within 7 days, RSG-44, AGC677, NEC41, GL269, JG74, ICC8983 and ICC 5003, developed only superficial necrosis along the hypocotyle region. BGM416, BG416, ICC1376 and ICC113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases.

Jayalakshmi *et al.*, (2008) investigated the different sources of resistance to dry root rot of chick pea caused by *Rhizoctonia bataticola* and conducted a pot culture experiment for screening of chickpea breeders promising entries (12 + 1 susceptible check JG-62) during rabi season of 2004-2005 and 2005-2006. They reported that out of 12 genotypes tested against root rot disease 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot disease.

Pande *et al.*, (2004) screened 29 chickpea germplasm lines and 10 cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* using paper towel technique and in addition also evaluated, 8 advanced lines that were identified to have field resistance (<20% plants infected) either to wilt, dry root rot and collar rot in multiple disease sick plot at ICRISAT to confirm their resistance to dry root rot. They reported that of 47 lines tested none were immune to dry root rot. One germplasm accession (ICC14395) a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BH-BH-11H)

were resistant to dry root rot. Of the remaining lines, 22 were moderately resistant, 19 susceptible and 3 highly susceptible.

Mishra *et al.*, (2005) have tested 470 germplasm lines and found KG-86 KWR-4, KWR-108 and KWR-277 as a resistant genotype.

Dinakaran *et al.*, (2005) screened 1300 sesame germplasms maintained at Regional Research Station, Vriddhachalam against root rot caused by *M. phaseolina* under field conditions respectively for four seasons in kharif and rabi/summer during 1995 and 1996. They reported that three accessions viz., ORM7, ORM14 and ORM17 showing resistant reaction over seasons were further evaluated with other three susceptible checks viz., TMV3, CO1 and VRI1 at fortnightly intervals from June 1997 to May 1998. The accessions recorded a mean disease incidence of 5.0, 5.2 and 4.2 per cent, respectively as against 30.1, 30.8 and 38.7 per cent respectively in the aforesaid checks. Screening of these accessions under artificially inoculated pot culture conditions during rabi, 1999-2000 exhibited resistant reaction consistently recording 10.0, 9.1 and 8.3 per cent incidence of root rot as against 66.7, 70.0 and 91.7 per cent in checks.

Lokesha and Benagi (2007) screened pigeonpea genotypes against *M. phaseolina* causal agent for dry root rot disease with the sick soil prepared from sand corn meal medium and applied to sterilized soil. Thus by sick plot technique they found that out of 24 pigeonpea genotypes two genotypes viz., PT-221, ICPL-90097 had shown resistant reaction and DEPS-9, Guallali local, GS-1, ICPL-89049, PHY-K-2, TAT-9621 and V-50 recorded moderately resistant reaction.

Kataria and Gaur (2007) evaluated 23 chickpea genotypes at Rajasthan Agricultural University, Bikaner against dry root rot caused by *Rhizoctonia bataticola*. They reported that BG-203 and BG-209 were resistant recorded disease incidence 12.29% and 18.44 per cent, respectively, followed by BG-377 (28.78%) BG-1053 (21.15%) GMS-469 (26.07%) GNG-469

(28.08%) RSG-44 (23.36%) RSG-888 (30.99%) moderately resistant, BG-256 (41.15%) BNG-416 (30.99%) BG-1103 (48.92%) BG-1105 (43.08%) C-550 (52.78%) CSJD-884 (37.22%) GMS-663 (35.22%) GMS-1292 (32.22%) GNG-663 (39.15%) HK-98-155 (456.85%) ICC-4951 (45.00%) RSG-897 (55.08%) RSG-963 (50.85%) RSG-974 (41.00%) moderately susceptible and C-235 (90.00%) highly susceptible against the test pathogen.

Chaturvedi and Dua (2009) have reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot.

Khan *et al.*, (2013) investigated the different resistant germplasm lines of chickpea against dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler and screened sixty germplasm lines in pot and reported that out of 60 germplasm lines screened only 9 namely, KGD-1189, KGD-1201, KGD-1209, KGD-1215, KGD-1217, KGD-1220, KGD-1221, KGD-1248 and KGD-1289, were found resistant. 10 lines namely KWR-7, KWR-26, KWR-28, KWR-50, KGR-18, KGR-48, KGR-159, KKG-103, KKG-111 and KGD-1201 were found moderately resistant. Seven lines namely KWR-54, KWR-55, KWR-60, KWR-63, KWR-77, KWR-78, KLWR-79, were found moderately susceptible. 17 lines namely KGD-1238, KGD-1239, C-304, KWR-12, KWR-14, KWR-15, KWR-16, KWR-17, KWR-18, KWR-22, KWR-61, KWR-65, KWR-70, KWR-71, KWR-1111, KWR-1211 were found susceptible and rest 17 lines were found highly susceptible against the disease.

Patidar *et al.*, (2018) evaluated 126 chickpea genotypes to find out the resistance reaction against dry root rot caused by *R. bataticola* under dry root rot sick micro-plots during rabi season of 2014- 2015 and 2015-2016. Out of 126 genotypes tested against root rot disease 22 (Phule G 0405, GL 10023, RSG 888, BAUG 15, DBGV 102, GJG 1209, GJG 1202, NBeG 453, JG 41, Phule G 08108, BGD 1082, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, GJG 1207, NBeG 439, HK 4, NBeG 460, HK 10-124, NBeG 399, BG 3059) were found resistant, 43 were moderately

resistant, 37 were moderately susceptible, 7 were susceptible and 17 were highly susceptible to dry root rot disease.

***Material and
Methods***

CHAPTER – III

MATERIALS AND METHODS

During the present investigation on chickpea dry root rot disease caused by *Rhizoctonia bataticola* (Taub.) Butler, various experiments (*In vitro* and *In vivo*) were planned and conducted at the Department of Plant Pathology. The details of the materials used and methods applied for various experiments are described here in the following paragraphs.

3.1. Materials

3.1.1. Experimental site

All the laboratory (plate and Pot/Polybag culture) experiments were conducted during 2016-17 to 2017-18 at the Department of Plant Pathology, College of Agriculture, VNMKV, Parbhani. Molecular analysis study of the test isolates of *R. bataticola* was carried out by using services provided by m/s Queincher Biotech Pvt. Ltd., Indapur, Dist. Pune.

3.1.2. Disease samples

Chickpea plants exhibiting typical symptoms of dry root rot (*R. bataticola*) disease were collected in the paper bags during the survey of chickpea crop fields undertaken during *Rabi*, 2016-17 and 2017-18, brought to the laboratory and subjected to the tissue isolation on Potato Dextrose Agar medium.

3.1.3. Culture media

Potato dextrose agar (PDA) was used as basal culture medium for isolation, purification and maintenance of pure cultures of *R. bataticola* isolates and biocontrol agents. Sand: maize (1:3) medium was used for mass multiplication of the pathogen. For studying morpho-cultural characteristics of *R. bataticola*, PDA was used as basal culture medium.

3.1.4. Seeds

Seeds of chickpea varieties / cultivars and germplasm lines were obtained from Agricultural Research Station, Badnapur, VNMKV, Parbhani, and used for various Pot/Polybag culture experiments.

3.1.5. Chemicals

Standard chemicals and reagents, required for various experiments were obtained from the Department of Plant Pathology and Department of SSAC, College of Agriculture, Parbhani.

3.1.5.1. Chemicals for DNA isolation of *R. bataticola* isolates

The chemicals such as: TE buffers (pH 8): 100 mM TrisHCl and 1.0 mM disodium EDTA; DNA extraction buffers : 100 mM Tris-HCl (pH 8.0), 20 mM disodium EDTA, 0.5 M NaCl and 1% SDS, 7.5 M Ammonium acetate, 99% ethanol, 70% ethanol, Phenol : Chloroform : Isoamyl alcohol (25 : 24 : 1), RNase, Agarose gel, (6X) gel loading dye, Electrophoresis buffer, Ethidium bromide and Sterile water, etc. were availed for isolation of DNA of *R. bataticola* isolates (Appendix II).

3.1.5.2. Chemicals for RAPD analysis of *R. bataticola* isolates

10X PCR buffer, 25 mM MgCl₂, 25 mM dNTPs *Taq* DNA Polymerase enzyme, 100 µM Operon primer, sterile water, etc. required for RAPD analysis of *R. bataticola* isolates (Appendix II).

3.1.6. Fungicides

Following 12 fungicides (systemic, contact and combi-products) were used for *In vitro* and *In vivo* experiments conducted during present studies.

Sr. No.	Common / Technical Name	Trade Name	AI and Formulation	Manufacture / Marketed
Systemic fungicides				
1	Carbendazim	Bavistin	50 % WP	BASF India Ltd., Mumbai
2	Metalaxyl	Apron	35 SD	Syngenta, India Ltd., Mumbai
3	Thiophanate methyl	Roko	70 % WP	Biostadt India Ltd., Mumbai
4	Tebuconazole	Folicure	29.9 % EC	Bayer India Ltd., Mumbai
5	Hexaconazole	Contaf	5 % EC	Rallis India Ltd., Mumbai
6	Azoxystrobin	Amistar	23 SC	Syngenta India Ltd., Pune
Contact and Combi-fungicides				
7	Copper oxy-chloride	Blitox	50 % WP	Rallis India Ltd., Mumbai
8	Carbendazim 12% + Mancozeb 63 %	SAAF	75 % WP	UPL, Mumbai
9	Corboxin 37.5 % + Thiram 37.5 %	Vitavax power	75 % WP	Dhanuka Agritech. Ltd., New Delhi
10	Mancozeb	Dithane M-45	75 % WP	Indofil Industries Ltd, Mumbai
11	Captan	Captaf	50 % WP	Rallis India Ltd., Mumbai
12	Cymoxanil 8 % + Mancozeb 64 %	Curzate	72 % WP	Dupont

3.1.7. Equipments and instruments

The laboratory equipments *viz.*, Autoclave, Hot air oven, Laminar airflow Cabinet, BOD Incubator, Refrigerator, Binocular Research Microscope, Electronic balance, pH meter, Spectrophotometer etc. available at the Department of Plant Pathology, College of Agriculture, Parbhani were utilized, as and when required. Distillation unit, Electronic balance, Vortexer, Microwave oven, Variable volume micropipette, Water bath, Deep freezer (-20⁰ C), Horizontal gel electrophoresis apparatus, Gel documentation system, Thermocycler and Gel dryer were used.

3.1.8. Glass-wares

The common glass-wares (Borosil and J-sil) viz., Petri-dishes, test tubes, conical flasks, volumetric flasks, measuring cylinder, glass rods, beakers, funnel, pipette, Autoclavable storage bottles etc. available at the Department of Plant Pathology, College of Agriculture, Parbhani were used.

3.1.9. Biocontrol agents

Pure cultures and talc based formulations of biocontrol agents viz., *Trichoderma asperellum*, *T. harzianum*, *T. virens*, *T. hamatum*, *T. longibrachiatum*, *T. koningii*, *Pseudomonas fluorescens*, and *Bacillus subtilis* were obtained from the Department of Plant Pathology, College of Agriculture, VNMKV, Parbhani; maintained and multiplied on appropriate culture media and used for present studies

3.1.10. Miscellaneous

Black coloured nursery Polybags (20 x 30 cm.), plant protection appliances, inoculation needle, forceps, blotter paper, paper bags, polythene bags, spirit lamp, mercuric chloride, labels, scales, nylon rope, sand, soil, FYM, screen house etc. were used during the course of present investigation.

3.2. Methods

3.2.1. Survey

A field survey of chickpea dry root rot disease, covering three Agro-climatic zones viz., Scarcity Zone (SZ), Assured Rainfall Zone (ARZ) and Moderate Rainfall Zone (MRZ) of Marathwada region of the Maharashtra state (Fig. 1) was undertaken during the months of December to February, 2016-17 and 2017-18 to assess occurrence and distribution of chickpea dry root rot disease. For the purpose, on the farmer's field, 1x1m² area, was randomly selected and name of variety and the per cent dry root rot

disease incidence information was collected. The percent dry root rot disease incidence was calculated by using following formula.

$$\text{Per cent Disease Incidence: } \frac{\text{No. of plants infected}}{\text{Total no. of plants observed}} \times 100$$

3.2.2. Symptomatology

3.2.2.1. Visual examination

On the basis of visual observation of plant the typical symptoms of chickpea dry root rot were recorded and noted.

3.2.2.2. Microscopic examination

Fresh, diseased chickpea plants showing typical dry root rot symptoms collected during survey were brought to the laboratory and washed in running tap water to remove extraneous material. Free hand sections of the diseased plant specimens were cut with sharp blade and also scrapped the fungus growth and dispensed in distilled water. Plants tissue sections and fungal suspension were separately put in a drop of Lactophenol cotton blue stain on separate clean glass slides, covered with cover slip and mounted and observed under compound microscope (make: Labomed Vision 2000). Also, temporary mount of pure culture of *R. bataticola* in a drop of Lactophenol cotton blue stain on glass slide were prepared and examined under microscope.

3.2.3. Sterilization / disinfection

The glasswares were sterilized in Hot air oven at 180°C for 60 to 90 min. Both solid and liquid media were sterilized at 121°C and 15 lbs pressure for 20 min. Plant tissues were surface sterilized using 0.1 per cent mercuric chloride (HgCl₂) solution for 2-3 min. and then washed in three sequential changes with sterile distilled water. The isolations / inoculations were performed under aseptic conditions of laminar air flow cabinet (make:

ACS, Bangalore). The working bench and working hands were also sterilized with 70 per cent ethanol / denaturated spirit. The miner metallic tools (blades, forceps, inoculation needles cork borer etc.) were sterilized by direct heating on flame of the spirit lamp.

3.2.4. Isolation of *R. bataticola* isolates

Naturally diseased chickpea plants showing typical symptoms of dry root rot in standing chickpea crop fields, collected during survey of Marathwada region was brought to the laboratory, washed thoroughly with sterile distilled water, blot dried, cut with sharp sterilized blade into small bits (5mm) and subjected to tissue isolation (Tuite, 1969) on PDA. These bits were sterilized with 0.1 per cent aqueous solution of Mercuric chloride (HgCl_2) for two minutes, washed by giving three successive changes with sterile distilled water in glass Petri plates to remove traces of mercuric chloride and blot dried. These were inoculated separately (location-wise) and aseptically on autoclaved and cooled PDA medium in sterilized glass Petri plates under aseptic conditions of Laminar-air-flow cabinet (make: ACS, Bangalore) and were inoculated in BOD incubator (make: MAC, Delhi) at $28 \pm 2^\circ\text{C}$ temperature. Within 2-3 days of incubation, blackish mycelia mat was developed and within next 7-8 days, microsclerotia were initiated in the plates. Applying hyphal tip and/or single spore/sclerotial isolation technique, the test pathogen was isolated aseptically on PDA medium, sub-cultured and the pure cultures of the test isolates obtained were maintained separately on PDA slant test tubes in refrigerator for further studies. After a week of incubation, the pure culture developed was again transferred aseptically by hyphal tip technique on PDA slant test tubes and incubated at $28 \pm 2^\circ\text{C}$. Applying same procedure, a total of 9 isolates of *R. bataticola* were isolated and their pure cultures were maintained in refrigerator, for further studies. Details of the codes / nomenclature assigned to the 9 test isolates of *R. bataticola* are as given below:



Fig. 1. Various agro-climatic zones of Marathwada region surveyed for dry root rot of chickpe

Sr. No.	Isolate code	Districts	Tahsil	Agro-climatic zone
1	Rb-1	Aurangabad	Vaijapur	SZ
3	Rb-2	Beed	Kaij	ARZ
4	Rb-3	Hingoli	Kalamnoori	MRZ
5	Rb-4	Jalna	Mantha	ARZ
6	Rb-5	Latur	Latur	ARZ
7	Rb-6	Nanded	Loha	ARZ
8	Rb-7	Parbhani	Parbhani	ARZ
9	Rb-8		Gangakhed	ARZ
10	Rb-9	Osmanabad	Osmanabad	ARZ

3.2.5. Mass multiplication of *R. bataticola*

Sand : maize medium was used for mass multiplication of *R. bataticola* test isolates. Sand: maize medium (3 part partially broken maize grains + 1part sand + distilled water to moisten the medium) was prepared, filled into polypropylene bags (9×12cm) and autoclaved at 20 lbs pressure for 30 min, for two consecutive days. After cooling at room temperature, autoclaved sand: maize medium in the bags was inoculated separately with 8-10 mycelial discs (5 mm dia.) obtained from a week old pure culture of *R. bataticola* test isolates and incubated at room temperature, for two weeks. Mass multiplied inoculums of sixteen test isolates of *R. bataticola* were used separately for making sick soil / potting mixture of various polybag / pot culture experiments.

3.2.6. Pathogenicity test

Pathogenicity of *R. bataticola* test isolates was attempted by employing sick soil method. For the purpose, autoclaved and cooled potting mixture of soil : sand : FYM (2:1:1) was filled into black coloured nursery polybags (20 x 30 cm), disinfected with 5 per cent copper sulphate solution. The test isolates multiplied on sand: maize medium was inoculated (@ 50g/kg potting mixture separately in these bags, mixed thoroughly in top 5-6 cm layer,

watered lightly and maintained in screen house for two weeks, so as to proliferate the test pathogen and make the potting mixture sick with *R. bataticola*.

Surface sterilized (0.1% HgCl₂) healthy seeds of susceptible chickpea cv. JG-62 were sown (10 seeds / bag) in these bags, watered lightly and maintained in the screen house. Three bags per test isolate were sown and maintained. The observations on seedling mortality were recorded at two weeks after sowing and based on per cent seedling mortality, pathogenic / non-pathogenic potential of the test isolates was determined.

The test pathogen isolates were re-isolated aseptically on PDA plates, from artificially dry root rot diseased chickpea seedlings (pathogenicity test), compared their cultural and morphological characteristics with the original culture of *R. bataticola* isolates isolated from naturally dry root rot diseased chickpea plants to fulfill Koch's postulates.

3.2.7. Identification of the pathogen

On the basis of symptoms expressed (both on naturally and artificially diseased) on chickpea plants, pathogenicity test, cultural and morphological characteristics and microscopic characteristics, the test pathogen was identified and further confirmed by comparing the description of *R. bataticola* of (Barnett and Hunter, 1972).

3.2.8. Variability among *R. bataticola* isolates

3.2.8.1. Pathogenic variability

Pathogenic variability of 9 isolates of *R. bataticola* was assessed by employing two methods viz., sick soil method and Paper towel technique, the details of which are as given below:

3.2.8.1.1. Sick soil method

Pathogenic variability of 9 test isolates of *R. bataticola* was attempted by employing sick soil method. For the purpose, autoclaved and cooled potting mixture of soil : sand : FYM (2:1:1) was filled into black coloured nursery polybags (20 x 30 cm), disinfected with 5 per cent copper sulphate solution. The test isolates multiplied on sand: maize medium was inoculated (@ 50 g / kg potting mixture) separately in these bags, mixed thoroughly in top 5-6 cm layer, watered lightly and maintained in screen house for two weeks, so as to proliferate the test pathogen and make the potting mixture sick with *R. bataticola*.

Surface sterilized (0.1 % HgCl₂) healthy seeds of chickpea cv. JG-62 were sown (10 seeds / bag) in these bags, watered lightly and maintained in the screen house. Three bags per test isolate were sown and maintained. Black coloured nursery polybags filled with autoclaved potting mixture without any inoculum and sown with surface sterilized healthy seeds of chickpea cv. JG-62 served as untreated control.

Observations on pre-emergence seed rot (PRESR) and post-emergence seedling mortality (POESM) were recorded, respectively at 7-8 days and 15 and 30 days after sowing and total mortality was computed. Per cent PRESR, POESM and total mortality were calculated by applying following formulae:

$$\text{PRESR (\%)} = \frac{\text{No. of Seeds un-germinated}}{\text{Total no. of Seeds sown}} \times 100$$

$$\text{POESM (\%)} = \frac{\text{No. of Seedlings died}}{\text{Total no. of Seedlings}} \times 100$$

$$\text{Total mortality (\%)} = \text{PRESR} + \text{POESM}$$

3.2.8.1.2. Paper Towel Technique

Pathogenicity test for all the 9 isolates was performed in laboratory on a susceptible genotype JG-62 by paper towel technique (Nene *et al.*, 1981). Inoculum was prepared from the seven days old culture of *Rhizoctonia bataticola* grown on 100ml potato dextrose broth medium (PDB). The culture of each isolate was grinded in a blender by adding 50ml of sterile distilled water to each fungal mat. Seven day old seedlings of JG-62 grown in sterilized sand were uprooted, washed under running water and were inoculated by dipping in the inoculum of *Rhizoctonia bataticola* for 2 min. Seedlings inoculated with sterile deionized water served as control. Five inoculated seedlings were placed in paper towel with the shoot left outside, folded, moistened and placed in trays. Trays were transferred to incubators maintained at $35 \pm 1^{\circ}\text{C}$ with 12h photoperiod and regularly moistened with sterile deionized water for seven days.

The inoculated seedlings were observed daily for symptom expression and the data on incubation period was recorded as and when the symptom was observed. At seven days after inoculation, the data on seedling mortality were recorded and finally cumulative average per cent seedling mortality was computed.

Based on virulence / aggressiveness, the test isolates of *R. bataticola* were grouped into following five categories (Pawar, 2010).

Sr. No.	% Mortality	Category	Sr. No.	% Mortality	Category
1	0 %	Avirulent	4	51-70 %	Strongly virulent
2	1-20 %	Weakly virulent	5	>71 %	Highly virulent
3	21-50 %	Moderately virulent	-	-	-

3.2.8.2. Cultural variability

Nine test isolates of *R. bataticola* (Taub.) Butler, were aseptically inoculated separately on autoclaved and cooled PDA plates and incubated at $28 \pm 2^{\circ}$ C. For each test isolate, a triplicate set of PDA plates were maintained. Observations on cultural characteristics viz., colony diameter, colony colour, growth rate, colony texture, colony colour (reverse) and margin were recorded, at seven days of incubation.

3.2.8.3. Morphological variability

For morphological characteristics, temporary mount of a weak old pure culture of the test isolates were prepared separately in a drop of Lactophenol cotton blue stain, on the clean glass slide, covered with glass cover slip and mounted under research microscope. The hyphal width (μm) was recorded, by employing micrometry under five random microscopic fields (10X) for each test isolate and calculated their average hyphal width (Aneja, 2001). From the pure culture PDA plates (two weeks old) of the test isolates, observations on microsclerotia characters viz., shape, colour, size (μm) and number / microscope field (Stereobinocular microscope) were recorded. Observations on number of microsclerotia produced by the test isolates were recorded under five random fields of stereobinocular microscope and averaged the data. The test isolates were categorized on the basis of microsclerotia production (density) / microscopic field, as detailed below (Das, 1988).

Sr. No.	Grade	Description	Av. No of sclerotia / microscopic field
1	-	No sclerotial formation	-
2	+	Poor sclerotial formation	10-20
3	++	Fair sclerotial formation	21-30
4	+++	Good sclerotial formation	31-50
5	++++	Excellent sclerotial formation	>51

3.2.8.4. Molecular variability

Molecular variability among 9 isolates of *R. bataticola* was analyzed by RAPD molecular markers. Standard protocols were used for the isolation of DNA and RAPD analysis. Services provided by m/s Quencher biotech. Pvt. Ltd. were used for purpose of same.

3.2.8.4.1. Extraction of genomic DNA

The extraction of fungal genomic DNA from 9 isolates (Table B) of *Rhizoctonia bataticola* was carried out by following a protocol described by Lee *et. al.*, (1988), Wu *et. al.*, (2001) with some modifications. The liquid culture of each isolate was raised in conical flask. PDB (Potato Dextrose Broth) 100ml was inoculated with 5mm bit of culture disc cut from edge of 5 days old culture of each isolate grown in petridish. The inoculated broth was incubated at $28 \pm 2^{\circ}\text{C}$ for 7 days. The mycelia mat was filtered through Whatman no. 1 filter paper and dried at room temperature. The dry mycelium was transferred to sterile mortar and pestle and ground with quartz sand powder or glass wool. Sufficient extraction buffer was added to the mortar so that quartz sand powder / mycelia mixture become saturated.

A mixture of buffer saturated phenol/ chloroform/ iso-amyl alcohol (25:24:1) was added per 0.5 gm of starting tissue and the solution was mixed thoroughly. The mixture was then transferred into several eppendorf tubes and centrifuged at 16000 xg for 5 min at room temperature to pellet tissue debris and the glass wool pellet to the bottom of tube. The aqueous phase was transferred to a new tube. The DNA in each tube was precipitated with 0.6 volume of isopropanol by incubating the mixture at room temperature for 10 min. The DNA was recovered by centrifugation.

The pellet was rinsed with 75 % ethanol, air dried briefly and resuspended in 200 μl TE buffer containing RNase A at 20 $\mu\text{g/ml}$. The extracted

DNA was resolved on 0.8 % Agarose gel. The quantification was done by spectrophotometer and stored at -20⁰C until further use.

3.2.8.4.2. Quantification of DNA

Spectrophotometer was used for quantitative and qualitative analysis of the DNA of the test isolates. Five µl of DNA sample was added in Cuvette carrying 0.995 µl of sterile H₂O and absorbance was measured at 280 nm wavelength. Similarly, the purity of DNA was checked by measuring the ratio of OD at A260/A280 nm. The quantification of DNA was calculated by using following formula.

$$\text{DNA } (\mu\text{g}/\mu\text{l}) = \frac{\text{OD at 260 nm} \times \text{dilution factor} \times 50}{1000}$$

3.2.8.4.3. RAPD analysis of *R. bataticola* isolates

The PCR protocol for RAPD reaction was optimized with various PCR components and thermal cycler programme as detailed below.

Sr. No.	PCR Components	Required Concentration	Volume / reaction
1	PCR Buffer (10X)	1X	2.5 µl
2	MgCl ₂ (25 mM)	1.5 mM	1.5 µl
3	dNTP mix. (10 mM)	200 µM	2.0 µl
4	Primer (10 µM)	0.4 µM	1.0 µl
5	<i>Taq</i> DNA Polymerase (3 U/µl)	1.25 U	0.5 µl
6	Template DNA	30 ng	1.0 µl
7	Nuclease free water		16.5 µl
	Total	-	25.0 µl

Random primers *viz.*, OPA and OPB (Operon Technologies, USA) were used for RAPD analysis of nine fungal pathogen of *Rhizoctonia bataticola* species. Master mixture (24.0 µl) containing all the above reactants, except template DNA were dispensed in autoclaved PCR tubes (0.2 ml). Genomic DNA of each pathogen of *Rhizoctonia bataticola* species was added to the individual tubes containing the master mixture. The contents of each tube were mixed by tapping with fingers followed by a brief spin to collect contents

at the bottom of the tube. The tubes were placed in Thermocycler and subjected to PCR according to the protocol given below.

3.2.8.4.4. Standardized PCR protocols for amplification of DNA

A PCR master mix in sterile distilled water with all of the above mentioned compounds in required quantities were prepared and amplifications were done through Thermal Cycler, using following PCR conditions.

Sr. No.	Steps	Temperature		Time
1	Initial Denaturation	94 ⁰ C		4 min
2	Denaturation	94 ⁰ C	39 cycles	1 min
3	Annealing	35 ⁰ C		1 min
4	Primer Extension	72 ⁰ C		1.5 min
5	Final Extension	72 ⁰ C		10 min
6	Final hold	4 ⁰ C		Forever

The amplified RAPD PCR product was separated on 1.2 % agarose gel, stained with ethidium bromide and visualized under Gel Documentation System. The polymorphism was detected by comparing RAPD fingerprinting pattern of all *Rhizoctonia bataticola* strains.

3.2.8.4.5. Data scoring and analysis

The RAPD PCR amplicons were scored as present (1) or absent (0). Data matrices were generated and used to plot dendrogram by using Jaccard's similarity coefficient by using software NTSYS-pc.2

3.2.9. Effect of organic and inorganic sources of carbon and nitrogen

3.2.9.1 Organic and inorganic sources of carbon

Experimental details :

Design : Completely Randomised Design (CRD)

Replications : Three

Treatments : Seven

Treatments details :

Tr. No	Treatments	Tr. No	Treatments
T ₁	Dextrose	T ₄	Starch

T ₂	Sucrose	T ₅	Cellulose
T ₃	Mannitol	T ₆	Maltose
T ₇	Control		

Effect of different carbon sources on growth and sporulation of (*R. bataticola*), carbon sources viz., Dextrose, Sucrose, Mannitol, Starch, Cellulose, Maltose and C- Control were used. For the assessment of suitable carbon source, Richards medium without carbon (50 g/lit) was used as basal stock medium. Sucrose when used @ 5.25 g provides 21.052 g of carbon 100/ml. On this basis for each test carbon source, 200 ml Richards agar medium was prepared in 250 ml capacity glass conical flasks and sterilized in autoclave at 15 lbs/inch² pressure for 20 min. Autoclaved and cooled Richards agar medium amended separately with various carbon sources was poured (@ 20 ml/plate) in sterilized glass Petri plates (90 mm dia.) and allowed to solidify at room temperature. On solidification of the medium, plates (three plates / carbon source / replication) were inoculated by placing in the centre 5 mm mycelial disc of actively growing 7 days old pure culture of *R. bataticola* (Rb-6isolate). Each carbon source was replicated thrice. Carbon control (Richard's medium without any carbon source) was also maintained. Plates were incubated at room temperature ($27 \pm 2^{\circ}$ C). Observations on colony diameter and sporulation were recorded after at a week and two weeks of incubations, respectively.

3.2.9.1 Organic and inorganic sources of nitrogen

Experimental details :

Design : Completely Randomised Design (CRD)

Replications : Three

Treatments : seven

Treatments details :

Tr. No	Treatments	Tr. No	Treatments
T ₁	Calcium nitrate	T ₄	Mg nitrate
T ₂	Sodium nitrate	T ₅	Potassium nitrate
T ₃	Ammonium nitrate	T ₆	Peptone
T ₇	Control		

Effect of different nitrogen sources on growth and sporulation of (*R. bataticola*), nitrogen sources viz., Calcium nitrate, Sodium nitrate, Ammonium nitrate, Mg nitrate, Potassium nitrate, Peptone and C- Control were used. For the assessment of suitable nitrogen source, Richards medium without nitrogen (10 g / lit) was used as basal stock medium. As the nitrogen provides 0.872 of nitrogen through 1.386 g/100 ml. For each test nitrogen source, 200 ml of Richards agar medium was prepared and poured in glass Conical flasks (250 ml cap.) and sterilized in autoclave at 15 lbs/inch² pressure for 20 min. Autoclaved and cooled medium was poured (@ 20 ml/plate) in sterilized glass Petri plates (90 mm dia.) and allowed to solidify at room temperature. On solidification of the medium, plates (three plates/nitrogen source/replication) were inoculated by placing in the centre 5 mm mycelial disc of actively growing 7 days old pure culture of *R. bataticola* (Rb-6isolate). Each nitrogen source was replicated thrice. Nitrogen control (with at any source of nitrogen) was also maintained. Plates were incubated at room temperature (27 ± 2 °C). Observations on colony diameter and sporulation were recorded after ten days of incubation. The number of sclerotia was observed microscopically and graded as below. (Tandel and Sabalpara, 2011)

Sr. No.	Score	Grade	Number (Sclerotia/microscopic field)at 100x
1	++++	Excellent	>50
2	+++	Good	30-50
3	++	Fair	21-30
4	+	poor	10-20
5	-	No sporulation	-

3.2.10. Effect of soil moisture regimes and soil types

3.2.10.1. Effect of soil moisture regimes

Experimental details :

Design : Completely Randomised Design (CRD)

Replications : Three

Treatments : Seven (Soil moisture regimes)

Treatments details :

Tr. No	Treatments (Moisture level)	Tr. No	Treatments (Moisture level)
T ₁	35 %	T ₂	45 %
T ₃	55 %	T ₄	65 %
T ₅	75 %	T ₆	85 %
T ₇	100%		

Effect of soil moisture levels (%) on dry root rot disease incidence was studied in pot / polybag culture experiment, using sick soil (*R. bataticola*) under screen house conditions. Black coloured nursery polybags (size: 20 x 30 cm.) filled with autoclaved potting mixture of soil : sand (3:1) were inoculated (@ 50 g / kg mixture) with the test pathogen (Rb-6 isolate) mass multiplied (sand: maize medium) culture, watered adequately and incubated for ten days in the screen house to proliferate the pathogen and make the soil / potting mixture sick. Surface sterilized (0.1% HgCl₂) healthy seeds of chickpea JG-62 were sown (10 seed/pot), watered to maintain the soil moisture levels as per treatments. For each treatment, three pots / replication were maintained and all the treatments were replicated thrice. The seedlings were allowed to grow for 10 days under normal conditions. Different levels of soil moisture were maintained 10 days after sowing. The soil moisture per cent was determined using the gravimetric method on an oven dry basis as per the procedure given by Sunkaram (1969). The method includes saturation of soil sample followed by removal of available soil moisture by oven drying (100 - 110°C), until the weight remains constant. After removing from oven, samples were cooled slowly to room temperature and weighed again. The difference in weight was amount of moisture in the soil. The available per cent soil moisture in the soil was calculated by the following formula.

$$\text{Per cent soil moisture (\%)} = \frac{\text{Saturated soil weight} - \text{Oven dry soil weight}}{\text{Oven dry soil weight}} \times 100$$

The levels of per cent soil moisture was adjusted by maintaining the constant weight by regular weighing and replacing the moisture deficit in each Pot/Polybag by watering.

Observations on disease incidence were recorded at 15, 30 and 45 days after sowing. For scoring the disease incidence, diseased and total plants were counted from each replication and percentage of plants infected in each treatment was calculated by following formula.

$$\text{Per cent Disease Incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants examined}} \times 100$$

3.2.10.2. Effect of soil types

Experimental details :

Design : Completely Randomised Design (CRD)

Replications : Three

Treatments : five (Soil type)

Treatments details :

Tr. No	Treatments	Tr. No	Treatments
T ₁	Black soil	T ₄	Sandy soil
T ₂	Red soil	T ₅	Sandy loam
T ₃	Clay		

Effect of five soil types *i.e.* Black soil, Red soil, Clay soil, Sandy soil and Sandy loam soil was studied on the development of dry root rot incidence under controlled environment. The experiment was conducted on susceptible cultivar JG-62 for dry root rot. Each treatment was replicate three times and each replication consists of two pots (5 seeds/pot). Each sterilized soils were infested with *R. bataticola* (Rb-6) (at 50 g/kg soil) multiplied on sand-maize medium. The pathogen-infested soil was filled in the 6-inch pots (2.0 kg/pot). Sowing in the pots were done four days after pathogen infestation. Ten surface sterilized seeds (soaked in 0.1% HgCl₂ for 1 minute followed by 2

- 3 times washing with sterilized water) of the chickpea dry root rot susceptible cultivar JG-62 were planted in each sick pot.

Observations on disease incidence were recorded at 15, 30 and 45 days after sowing. For scoring the disease incidence, diseased and total plants were counted from each replication and percentage of plants infected in each treatment was calculated by following formula.

$$\text{Per cent Disease Incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants examined}} \times 100$$

3.2.11. Disease management strategies

Based on pathogenicity test and pathogenic variability studies, out of 9 test isolates of *R. bataticola*, the most virulent isolate Rb-6 (Loha, Nanded) was promoted for further *In vitro* studies.

3.2.12.1. *In vitro* evaluation of fungicides

Efficacy of six systemic (each @ 500 and 1000 ppm), six contact and combi-fungicides (each @ 2000 and 2500 ppm) were evaluated *In vitro* against *R. bataticola* by Poisoned food technique (Nene and Thapliyal, 1993), using PDA as basal culture medium. Based on active ingredient, requisite quantity of the test fungicides was calculated, mixed separately and thoroughly with autoclaved and cooled (45° C) PDA medium in sterile glass conical flasks (250 ml. cap.) to obtain desired concentrations. This PDA medium amended separately with the test fungicides was then poured (20 ml / plate) aseptically in sterile glass Petri plates (90 mm dia.) and allowed to solidify at room temperature. For each of the test fungicide and its desired concentrations, three plates / treatment / replication were maintained. After solidification of the PDA medium, all the plates were inoculated aseptically by placing in the centre a 5 mm pure culture disc, obtained from actively growing 7 days old pure culture of *R. bataticola* (Rb-6 isolate) and incubated in an inverted position at 28 ± 2°

C. Petri plates filled with plain PDA (without any fungicide) and inoculated with the pure culture disc of *R. bataticola* (Rb-6 isolate) were maintained as untreated control.

Tr. No	(Systemic)Treatments
T ₁	Carbendazim 50WP
T ₂	Metalaxyl (Apron 35 SD)
T ₃	Thiophanate methyl (Topsin M 70WP)
T ₄	Tebuconazole 29.9%EC
T ₅	Hexaconazole (Contaf 5 EC)
T ₆	Azoxystrobin (Amistar)
T ₇	Control (untreated)

Experimental details :

Design : C.R.D. (Completely Randomized Design)

Replications : Three

Treatments : Seven

Treatment details:

Experimental details :

Design : Completely Randomized Design (CRD)

Replications : Three

Treatments : seven

Treatment details:

Tr. No	(Contact and Combi-fungicides)Treatments
T ₁	Copper oxy chloride 50WP
T ₂	Carbendazim 12% + Mancozeb 63% (SAFF 75WP)
T ₃	Vitavax power (Carboxin 37.55+ Thiram 37.5%)
T ₄	Mancozeb (Dithane M-45 WP)
T ₅	Captan 50WP
T ₆	Curzet (Cymoxanil 8% + Mancozeb 64%)
T ₇	Control (untreated)

Observations on radial mycelial growth / colony diameter were recorded at an interval of 24 hrs., continued till untreated control plates were fully covered with mycelial growth of the test pathogen and averaged finally.

Per cent inhibition of the test pathogen with the test fungicides, over untreated control was calculated by applying following formula (Vincent, 1927).

$$\text{Per cent inhibition} = \frac{C - T}{C} \times 100$$

Where,

C = growth of the test fungus in untreated control plate

T = growth of the test fungus in treated plate

3.2.12.2. *In vitro* evaluation of bioagents

A total of nine biocontrol agents were evaluated *In vitro* against *R. bataticola* (Rb-6 isolate), applying dual culture technique (Dennis and Webster, 1971). Seven days old cultures of the test bioagents and test pathogen grown on PDA and / or NA were used for the study. Two culture discs (each 5 mm dia.), one each of the test pathogen and test bioagent were cut out with sterilized cork borer and placed at equidistance, exactly opposite to each other on autoclaved and cooled solid surface of PDA medium in Petri plates, three plates per treatment per replication were maintained and incubated at $28 \pm 2^\circ \text{C}$. PDA plates inoculated alone with pure culture disc (5 mm) of the test pathogen were maintained as untreated control.

Experimental details :

Design : C.R.D. (Completely Randomized Design)

Replications : Three

Treatments : Nine

Treatment details :

Tr. No.	Treatments	Tr. No.	Treatments
T ₁	<i>Trichoderma asperellum</i>	T ₆	<i>T. longibrachiatum</i>
T ₂	<i>T. harzianum</i>	T ₇	<i>Pseudomonas fluorescens</i>
T ₃	<i>T. virens</i>	T ₈	<i>Bacillus subtilis</i>
T ₄	<i>T. koningii</i>	T ₉	Control (untreated)
T ₅	<i>T. hamatum</i>		

Observations on linear mycelial growth of the test pathogen and test bioagent were recorded at an interval of 24 hours, continued till untreated control plates were fully covered with mycelial growth of the test pathogen and averaged finally. Per cent inhibition of the test pathogen with the test bioagent, over untreated control was calculated by applying following formula (Arora and Upadhyay, 1978).

$$\text{Per cent Growth Inhibition} = \frac{\text{Colony growth in Control plate} - \text{Colony growth in Intersecting plate}}{\text{Colony growth in control plate}} \times 100$$

3.2.12.3. Evaluation of organic amendments (pot / polybags culture)

A total of 11 organic amendments were evaluated against *R. bataticola*, by sick soil method in pot / polybag culture, under screen house conditions. Black coloured nursery polybags (size: 20 x 30 cm.) filled with autoclaved potting mixture of soil : sand (3:1) were inoculated (@ 50 g / kg mixture) with the test pathogen (Rb-6isolate) mass multiplied (sand: maize medium) culture, watered lightly and incubated in screen house for two weeks, so as to proliferate the pathogen and make the potting mixture sick. After two weeks of incubation, these bags were amended separately with the coarse powdered test amendments (each @ 50 g/kg potting mixture), mixed thoroughly, watered lightly and maintained in screen house. After 72 hrs. of amendments application, surface sterilized (0.1 % HgCl₂) healthy seeds of chickpea cv. JG-62 were sown (10 seeds / bag), watered lightly and maintained in the screen house. The black coloured nursery polybags containing sick soil (without any amendment) and sown with surface sterilized healthy seeds of chickpea JG-62 were maintained as untreated control.

Experimental details :

Design : C.R.D. (Completely Randomized Design)
Replications : Three
Treatments : Eleven

Treatment details :

Tr. No	Treatments	Tr. No	Treatments
T ₁	FYM/compost	T ₇	Neemseed cake
T ₂	Vermicompost	T ₈	Karanj cake
T ₃	Poultry manure	T ₉	Safflower cake
T ₄	Groundnut cake	T ₁₀	Mustard cake
T ₅	Cotton seed cake	T ₁₁	Control (untreated)
T ₆	Castor seed cake	-	-

Observations on pre-emergence seed rot (PRESR) and post-emergence seedling mortality (POESM) were recorded, respectively at 7-8 days and 15 and 30 days after sowing and total mortality was computed. Per cent PRESR, POESM and total mortality were calculated by applying the formulae as detailed under sub-head 3.2.8.1.1.

Further, per cent reduction in total mortality with the treatments, over untreated control (sick soil alone) was calculated by following formula

$$\% \text{ Disease control} = \frac{C - T}{C} \times 100$$

Where,

C = Total mortality in untreated control

T = Total mortality in treatment

3.2.13. Integrated disease management (polybag culture)

Those fungicides, bioagents and organic amendments found effective against *R. bataticola* during present *In vitro* and pot / polybag culture studies were selected and used (alone and in-combination) for integrated management of dry root rot of chickpea (polybag culture), by applying ‘sick soil’ method, as detailed under sub-head 3.2.8.1.1.

The test fungicides and talc based formulations of the test bioagents were applied (alone and in-combination) as pre-sowing seed treatment to the healthy seeds of chickpea JG-62 and sown (10 seeds / bag) in the polybags (20 x 30 cm.) containing *R. bataticola* (Rb-6isolate) sick soil /

potting mixture. The powdered test oil cakes were applied (@ 50 g/kg soil or potting mixture) in these polybags, mixed thoroughly, watered lightly and kept in screen house. After 72 hrs, these bags were seeded with surface sterilized healthy seeds (10 seeds / bag) of chickpea JG-62. For each treatment, two bags / replications were maintained. The polybags sown (10 seeds / bag) with surface sterilized healthy seed of chickpea JG-62, containing *R. bataticola* (Rb-6 isolate) were maintained as untreated control.

Experimental details:

Design : CRD

Replications : Three

Treatments : Twelve

Treatment details:

Tr. No.	Treatments	Dosages (g/kg seed or potting mixture)	
		Seed	Soil
T ₁	Carboxin 37.5 % + Thiram 37.5 % (75 % WP) (ST)	3.0 g	-
T ₂	Carbendazim 50 % WP (ST)	1.0 g	-
T ₃	Carbendazim 12 % + Mancozeb 63 % (75 % WP) (ST)	2.5 g	
T ₄	<i>T. asperellum</i> (5x10 ⁷ cfu/g carrier) (ST)	10 g	-
T ₅	<i>T. harzianum</i> (5x10 ⁷ cfu/g carrier) (ST)	10 g	
T ₆	<i>T. hamatum</i> (5x10 ⁷ cfu/g carrier) (ST)	10 g	-
T ₇	T ₄ + T ₅ (ST)	10 g + 10 g	-
T ₈	T ₂ + T ₄ (ST)	1.0 g + 10 g	
T ₉	T ₂ + T ₄ (ST) + FYM	1.0 g + 10 g	50 g
T ₁₀	T ₂ + T ₄ (ST) + NSC (SA)	1.0 g + 10 g	50 g
T ₁₁	T ₁ + T ₄ (ST) + NSC (SA)	3.0 g + 10 g	50 g
T ₁₂	Control (untreated)	Sick soil	

ST: Seed Treatment, SA: Soil Application

Observations on pre-emergence seed rot (PRESR) and post-emergence seedling mortality (POESM) were recorded, respectively at 7-8 days and 15 and 30 days after sowing and total mortality was computed. Per cent PRESR, POESM and total mortality were calculated by applying the formulae as detailed under sub-head 3.2.8.1.1.

Further, per cent reduction in total mortality with the treatments, over untreated control (sick soil alone) was calculated by formula as detailed under sub-head 3.2.12.3.

3.2.14. Screening of chickpea varieties, cultivars, germplasm lines and elite lines against *R. bataticola*

3.2.14.1. Pot / polybag culture

Screening of chickpea entries was attempted by employing sick soil method. For the purpose, autoclaved and cooled potting mixture of soil : sand : FYM (2:1:1) was filled into black coloured nursery polybags (20 x 30 cm), disinfected with 5 per cent copper sulphate solution. The test isolate Rb-6 was multiplied on sand: maize medium and inoculated (@ 50 g / kg potting mixture), separately in these bags, mixed thoroughly in top 5-6 cm layer, watered lightly and maintained in screen house for two weeks, so as to proliferate the test pathogen and make the potting mixture sick with *R. bataticola*.

Surface sterilized (0.1 % HgCl₂) healthy seeds of chickpea JG-62 were sown (10 seeds / bag) in these bags, watered lightly and maintained in the screen house. Two bags per test chickpea entry were sown and maintained.

Observations on pre-emergence seed rot (PRESR) and post-emergence seedling mortality (POESM) were recorded, respectively at 7-8 days and 15 and 30 days after sowing and total mortality was computed. Per cent PRESR, POESM and total mortality were calculated by applying the formulae as detailed under sub-head 3.2.8.1.1.

Based on per cent total mortality incidence, the test chickpea entries evaluated under pot/polybag culture was categorized as described below, by applying following disease rating scale (Nene *et al.*, 1981).

Disease incidence (%)	Disease reaction
1.0-10.0 %	Resistant
10.1-20.0%	Moderately resistant
20.1-30.0%	Moderately susceptible
30.1-50.0%	Susceptible
Above 50 %	Highly susceptible

3.2.15. Statistical analysis

The data recorded in all the experiments (*In vitro* and *In vivo*) was statistically analyzed (Panse and Sukhatme, 1978) using VNMAU-STAT statistical programme, at Central Computer Laboratory, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani.



Result and Discussion

CHAPTER IV

RESULTS AND DISCUSSION

Dry root rot of chickpea is caused by *Rhizoctonia bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid], is a soil and seed borne necrotrophic fungal pathogen which distributed globally and it can infect more than 284 plant species throughout the world. (Farr *et al.*, 1995). This disease influenced by climatic factors, particularly temperature and soil moisture stress, which play an important role in development of this diseases and thereby increases the severity of the disease. In many crops this pathogen survives for longer duration (2-15 years) on plant residues and / or in soil. It can cause significant yield losses, resulting the growers in grate economical loss.

Considering economic importance of this pathogen and the disease, present studies were undertaken on chickpea dry root rot [*Rhizoctonia bataticola* (Taub.) Butler] during 2016-17 to 2017-18 on the particulars such as., survey, symptomatology, isolation, pathogenicity test, variability (pathogenic, cultural, morphological and molecular) of *R. bataticola* isolates, efficacy of carbon and nitrogen sources, effect of soil moisture regimes and soil types, disease management strategies (*in vitro*), and varietal screening. The results obtained on all these studies are being elucidated and discussed under following sub-heads.

4.1. Occurrence and distribution of chickpea dry root rot

A survey of chickpea field was conducted to record dry root rot incidence during the *Rabi* seasons of 2016-17 and 2017-18 (PLATE I). Survey was conducted in 179 and 232 chickpea crop fields of 08 districts, distributed under three agro-climatic zones *viz.*, Scarcity zone (06), Assured Rainfall zone (07) and Moderate rainfall zone (08) of Marathwada region of the Maharashtra state (Fig. 1) respectively during 2016-17 and 2017-18.

The results obtained on dry root rot incidence, tahsil-wise (Table 1), district-wise (Table 2 and Fig. 2, 3 & 4), chickpea variety-wise (Table 3 and Fig. 5 & 6) and agro-climatic zone-wise (Table 4 and Fig. 7 & 8) are being interpreted herein under following sub-heads.

4.1.1. Tahsil-wise incidence of dry root rot of chickpea

The results (Table 1) indicated a widespread incidence of dry root rot, during both the years of survey and it was ranged from 13.93 (Nanded) to 33.32 (Vaijapur) and 16.67 (Ahmadpur) to 38.56 (Ashti) per cent, during *Rabi*, 2016-17 and 2017-18 seasons, respectively. However, maximum dry root rot incidence was observed in Vaijapur tahsil (33.32 and 34.18 %), respectively during *Rabi*, 2016-17 and 2017-18 seasons, followed by Jalna tahsil (32.72 and 35.61 %) and Kalamb tahsil (30.24 and 35.21 %). Whereas in rest of the tahsils dry root rot incidence was in the range of 14.24 (Ardhapur) to 30.12 (Manvat) per cent and 16.97 (Gevrai) to 31.49 (Manvat) per cent, during *Rabi*, 2016-17 and 2017-18 seasons, respectively. However, minimum dry root rot disease incidence was observed in Nanded tahsil (13.93 and 21.33 %), during *Rabi*, 2016-17 and 2017-18 seasons, respectively, followed by Ardhapur tahsil (14.24 and 17.62 %) and Ahmadpur tahsil (15.1 and 16.67 %).

Table 1. Tahsil-wise incidence of dry root rot of chickpea, during *Rabi*, 2016-17 and 2017-18.

Sr. No.	Districts	Tahsils	2016-17		2017-18	
			No. of Fields	Av. Disease Incidence (%)	No. of Fields	Av. Disease Incidence (%)
Scarcity zone						
1	Aurangabad	Vaijapur	2	33.32	3	34.18
		Gangapur	2	24.99	4	26.19
2	Beed	Ashti	1	25.53	3	38.56
		Patoda	2	23.33	2	32.33
		Shirur (Kasar)	2	25.67	3	26.21
3	Osmanabad	Bhoom	2	16.23	3	25.97
		Paranda	3	20.55	2	30.44
Overall Average / Total:			14	24.23	20	30.55
Assured rainfall zone						
1	Aurangabad	Paithan	3	19.32	4	21.55
		Aurangabad	3	21.69	6	18.2
		Soegaon	2	15.45	3	26.81
		Sillod	2	21.37	3	30.33

2	Jalna	Jalna	4	32.72	6	35.61
		Badnapur	4	18.26	5	23.88
		Partur	6	27.34	8	31.66
		Mantha	5	29.1	9	32.52
3	Beed	Gevrai	3	14.98	3	16.97
		Beed	5	19.65	5	21.2
		Majalgaon	4	18.03	4	24.83
		Ambajogai	4	18.99	4	22.23
		Parali (Vaijnath)	4	24.99	5	30.66
4	Osmanabad	Osmanabad	1	26.19	2	30.47
		Kalamb	4	30.24	5	35.21
		Tuljapur	1	15.52	1	18.62
		Umarga	2	22.45	3	26.72
5	Latur	Latur	5	24.11	9	29.33
		Chakur	3	17.96	5	22.87
		Renapur	2	18.65	3	18.44
		Ahmadpur	2	15.1	2	16.67
		Ausa	2	22.98	1	25.11
6	Parbhani	Parbhani	10	22.45	12	28.5
		Jintur	6	22.1	6	31.63
		Gangakhed	4	17.46	4	23.31
		Manwath	5	30.12	6	31.49
		Sailu	5	29.24	5	35.52
		Purna	3	19.65	4	23.23
		Pathri	5	28.44	4	34.65
		Sonpeth	2	25.71	2	28.82
7	Nanded	Loha	3	18.42	4	21.76
		Mukhed	1	20.22	1	27.55
		Naigaon	2	18.81	3	24.74
		Kandhar	2	19.33	3	25.63
	Hingoli	Sengaon	5	29.12	7	35.57
Overall Average / Total:			124	22.17	157	26.63
Moderate rainfall zone						
1	Nanded	Nanded	5	13.93	9	21.33
		Ardhapur	3	14.24	4	17.62
		Mudkhed	2	17.42	3	20.88
		Bhokar	3	15.5	2	17.67
2	Hingoli	Basamat	6	16.93	8	21.97
		Hingoli	9	15.54	11	20.65
		Aundha	8	19.33	10	22.87
		Kalanuri	5	16.56	8	19.4
Overall Average / Total:			41	16.18	55	20.29

4.1.2. District-wise incidence of dry root rot of chickpea

Results (Table 2, Fig. 2, 3 & 4) showed that maximum dry root rot incidence was recorded in Jalna district which was 26.85 % and 30.91%, respectively, during *Rabi* 2016-17 and 2017-18, with maximum pooled incidence of 28.88 %. This was followed by the districts of viz., Parbhani (24.39 %, 29.64 % and 27.01%), Aurangabad (22.69 %, 26.21 % and 24.45 %), Osmanabad (21.86 %, 27.9% and 24.88 %), , Beed (21.39 %, 26.62 % and 24.00 %), Latur (19.76 %, 22.48 % and 21.12 %), Hingoli (19.49 %, 24.09%



Fig. 2. Various districts of Marathwada region surveyed for dry root rot of chickpea

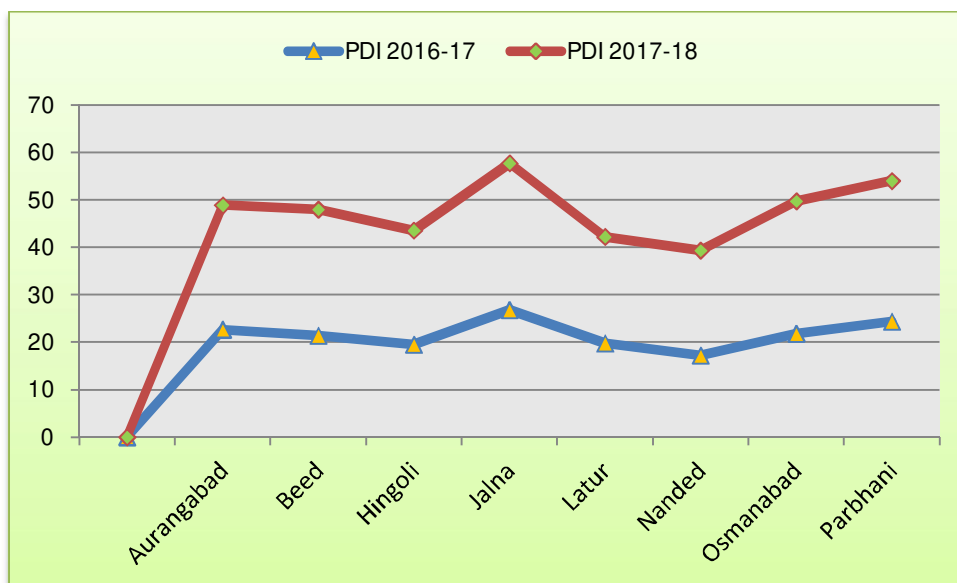


Fig. 3. District-wise incidence of chickpea dry root rot in Marathwada region (Rabi, 2016-17 and 2017-18)

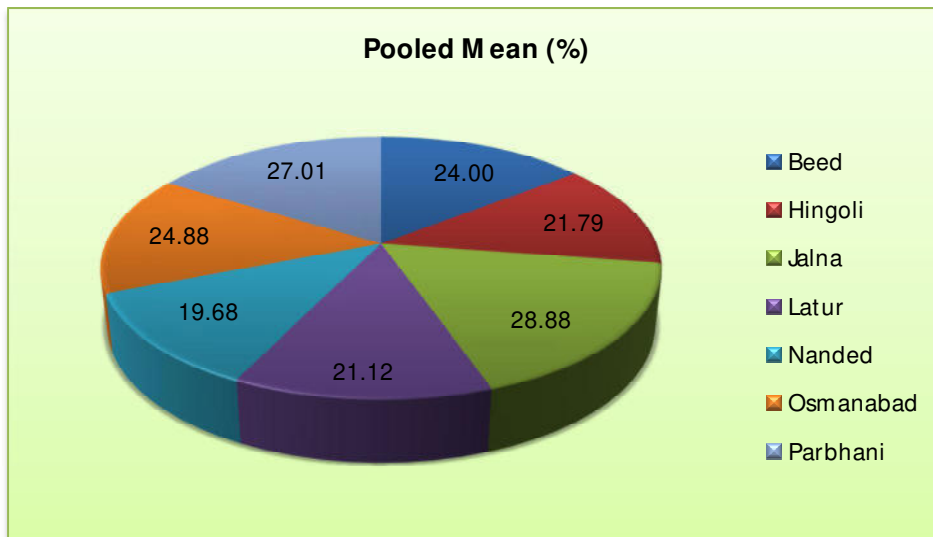


Fig. 4. District-wise pooled mean incidence of chickpea dry root rot in Marathwada region (Rabi, 2016-17 and 2017-18)

and 21.79 %) and Nanded (17.23 %, 22.14 % and 19.68 %). Overall average dry root rot incidence was maximum during *Rabi*, 2017-18 which was (26.24%) and it was comparatively minimum (21.7 %) during *Rabi*, 2016-17.

Table 2. District-wise chickpea dry root rot incidence, during *Rabi*, 2016-17 and 2017-18

Sr. No.	Districts	No of locations		Av. Disease Incidence (%)		Pooled Mean Incidence (%)
		2016-17	2017-18	2016-17	2017-18	
1	Aurangabad	14	23	22.69	26.21	24.45
2	Beed	25	29	21.39	26.62	24.00
3	Hingoli	33	44	19.49	24.09	21.79
4	Jalna	19	28	26.85	30.91	28.88
5	Latur	14	20	19.76	22.48	21.12
6	Nanded	21	29	17.23	22.14	19.68
7	Osmanabad	13	16	21.86	27.9	24.88
8	Parbhani	40	43	24.39	29.64	27.01
	Average / Total	179	232	21.7	26.24	---

4.1.3. Variety-wise incidence of dry root rot of chickpea

Result (Table 3, Fig. 5 & 6) indicated that among these different chickpea varieties cultivated by the farmers of the Marathwada region surveyed during *Rabi* 2016-17 and 2017-18, maximum dry root rot incidence was observed in Vijay 30.56 % and 35.24 %, respectively during *Rabi*, 2016-17 and 2017-18, with pooled mean maximum disease incidence 32.90 %. This was followed by the varieties viz., Digvijay (27.89 %, 30.84 % and 29.37 %), Local (Deshi) (24.21 %, 27.64 % and 25.93 %) and Vishal (20.12 %, 25.86 % and 22.99 %). Whereas with respect to the remaining varieties, the root rot incidence was ranged from 14.33 % to 20.12 % during *Rabi*, 2016-17 and 18.78 % to 25.86 % during *Rabi*, 2017-18, with pooled mean incidence in the range of 16.56 % to 22.99 %.

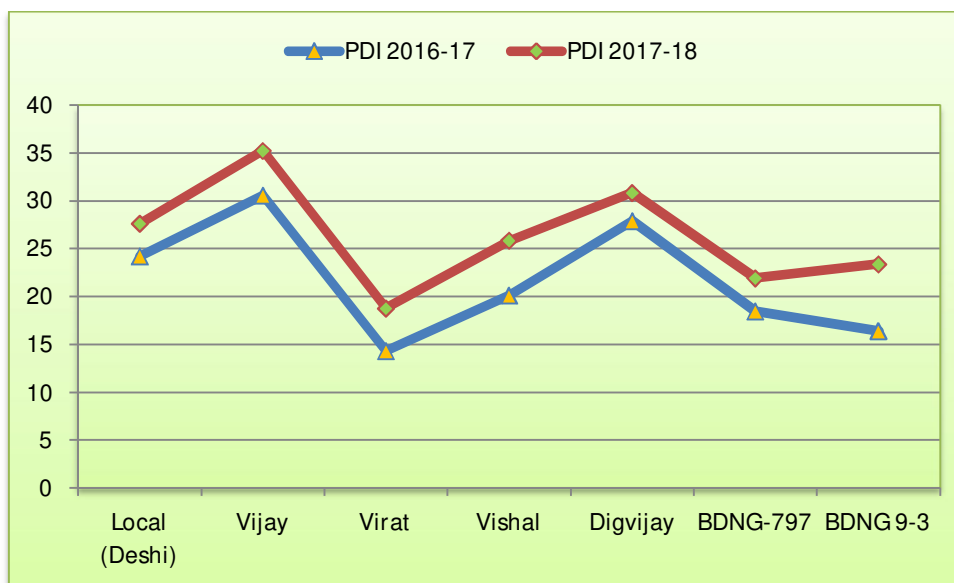


Fig. 5. Variety-wise incidence of chickpea dry root rot in Marathwada region (Rabi, 2016-17 and 2017-18)

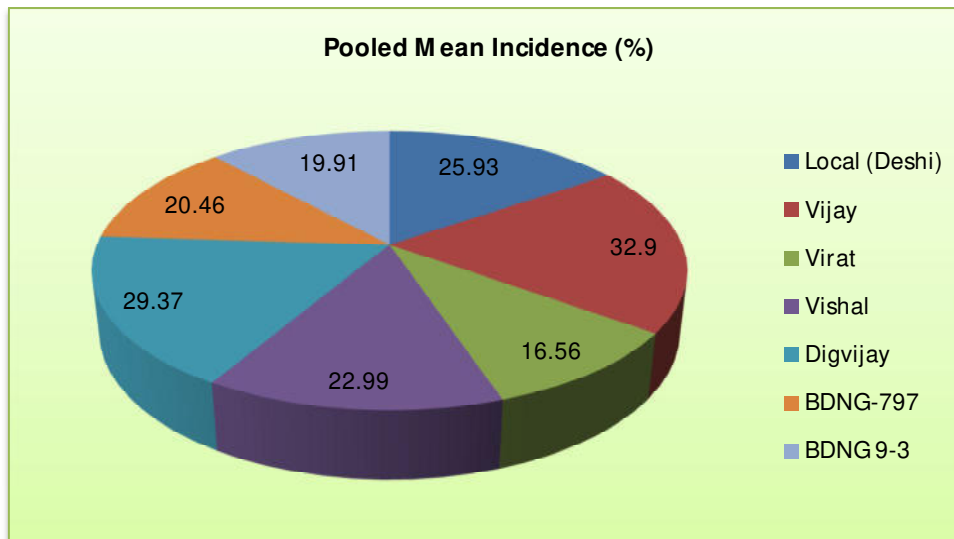


Fig. 6. Variety-wise pooled mean incidence of chickpea dry root rot in Marathwada region (Rabi, 2016-17 and 2017-18)

Table 3. Variety wise incidences of dry root rot of chickpea, during *Rabi*, 2016-17 and 2017-18

Sr. No.	Chickpea Variety	No. of locations		Av. Incidence (%)		Pooled Mean Incidence (%)
		2016-17	2017-18	2016-17	2017-18	
1	Local (Deshi)	34	41	24.21	27.64	25.93
2	Vijay	38	47	30.56	35.24	32.90
3	Virat	25	32	14.33	18.78	16.56
4	Vishal	16	27	20.12	25.86	22.99
5	Digvijay	24	29	27.89	30.84	29.37
6	BDNG-797	26	33	18.48	21.93	20.46
7	BDNG 9-3	16	23	16.42	23.39	19.91
	Overall Average / Total	179	232	21.7	26.24	----

4.1.4. Agro-climatic zone-wise incidence of dry root rot of chickpea, during *Rabi*, 2016-17 and 2017-18

Results (Table 4, Fig. 7 & 8) showed that out of three agro-climatic zones surveyed such as, Scarcity zone (SZ), Assured rainfall zone (ARZ) and Moderate rainfall zone (MRZ), maximum dry root rot incidence was observed which was 24.23 per cent during *Rabi* 2016-17 and 30.55 per cent, during *Rabi* 2017-18, with maximum pooled mean disease incidence of 27.39 per cent in Scarcity zone, followed by Assured rainfall zone (22.17 %, 26.63 % and 24.4 %) and Moderate rainfall zone (16.18 %, 20.29 % and 18.23 %). Whereas, maximum average root rot incidence was observed during *Rabi*, 2017-18 which was 25.82 per cent as compared with the incidence during *Rabi*, 2016-17 which was 20.86 per cent.

PLATE I



Chickpea crop fields surveyed for recording dry root rot disease incidence

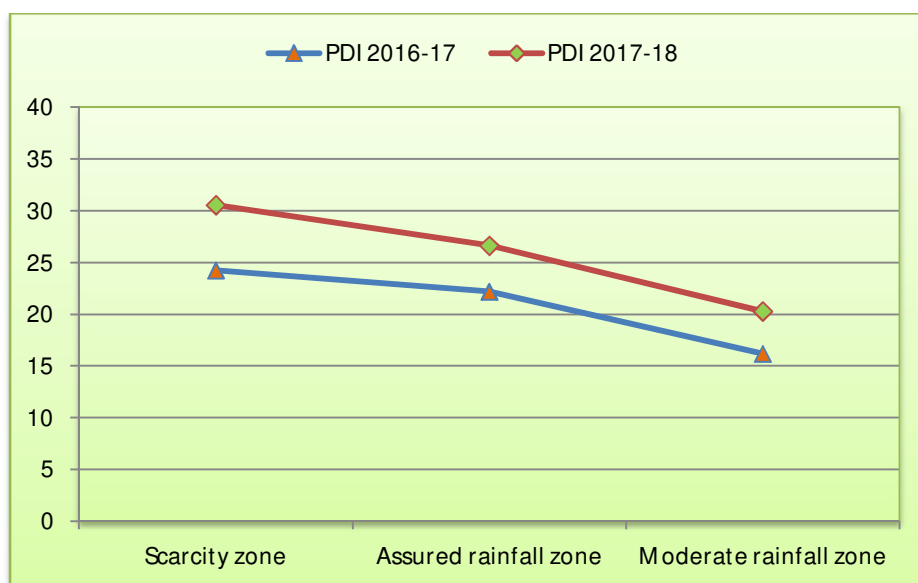


Fig. 7. Agro-climatic zone wise incidence of chickpea dry root rot in Marathwada region (*Rabi*, 2016-17 and 2017-18)

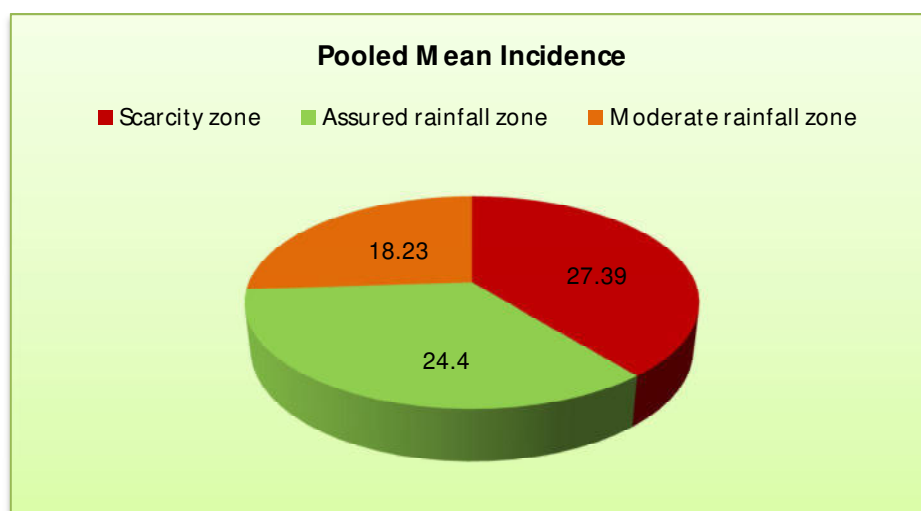


Fig. 8. Agro-climatic zone wise pooled mean incidence of chickpea dry root rot in Marathwada region (*Rabi*, 2016-17 and 2017-18)

Table 4. Agro-climatic zone-wise incidences of dry root rot of chickpea, during *Rabi*, 2016-17 and 2017-18

Sr. No	Agro-climatic Zone	No of locations		Av. Incidence (%)		Pooled Mean Incidence (%)
		2016-17	2017-18	2016-17	2017-18	
1	Scarcity zone	14	20	24.23	30.55	27.39
2	Assured rainfall zone	124	157	22.17	26.63	24.4
3	Moderate rainfall zone	41	55	16.18	20.29	18.23
	Overall Average/ Total	179	232	20.86	25.82	23.34

4.2. Symptomatology of dry root rot of chickpea

During survey, survey was conducted at different growth stages of chickpea *i.e.* from seedling stage to (20days old) to podding stage. At seedling stage, the symptoms were observed such as, infected seedlings become stunted one. When uprooted tap root was found without any lateral and finer roots and it becomes black in color. Whereas in the vegetative to flowering stage, symptoms on affected plants were found as bronzing of the leaves on one or more of the lower branches. In advanced disease symptoms leaves became yellow to brown in affected plants. The terminal part of the tap root and lateral roots became brown to black in color and shriveled one. The tap root was also observed without any lateral roots. A grayish mycelium was also seen sometimes on the tap root. In case of podding and harvesting stage, the disease was scattered in the field as dried plants. It was also observed that, the susceptibility of the disease was increased with the age of plant. Sometimes the apical leaves on the infected plants appeared chlorotic, when the rest of the plant was seen dried. The pods on affected plants were developed poorly and the number of pods per plant was also found to be less. Plants showing such type of different symptoms at different stages of the crop growth (PLATE II) were collected and kept in paper bags for conducting further studies.

Similar symptoms of the dry root rot of chickpea caused by *R. bataticola* and many other crops were reported by several workers. Dastur

(1935) reported yellowing of leaves later on converted in brown colour. The affected branches and leaflets found to be stiff and turned upwards stand vertically and shed prematurely. Nene *et al.* (1991) founded infection in seedling stage also. When the infected plants uprooted, the lower portion of tap root was remain in the soil and it was devoid of most of lateral roots. Khan *et al.* (2012) reported that dry root rot (*R. bataticola*) induced yellowing and drooping of infected plants of chickpea, the diseased plants easily pulled out from the soil, extensive rotting of the basal stem, main and lateral roots. In advanced stage, sclerotial bodies appeared inside the pith and on the outer surface of the tap root and Sharma *et al.* (2015) reported that dry root rot (*R. bataticola*) in chickpea induced drooping and chlorosis of petioles and leaflets, leaves and stems of affected plants turned straw coloured, tap root turned black, rotted and without lateral roots, such roots showed shredding of the bark. Dark, minute sclerotial bodies developed on roots and collar region of the affected plants.

4.3. Isolation of the pathogen

By using tissue isolation technique, the test fungus was isolated from roots and stems of chickpea plants, naturally infected with the dry root rot, on autoclaved and cooled Potato Dextrose Agar plates. After 2-3 days of incubation, black mycelial mat on PDA plates was developed and after 7-8 days of incubation, microsclerotia were developed (PLATE III). Applying same procedure, nine isolates of *R. bataticola* were isolated (Table 5) and identified as *R. bataticola* based on morphological and cultural characters using the descriptions given by C.M.I. (1970). These isolates were labeled as: Rb-1, Rb-2, Rb-3, Rb-4, Rb-5, Rb-6, Rb-7, Rb-8 and Rb-9,. The cultures of these isolates were purified by hyphal tip / single spore isolation technique and their pure cultures were maintained on PDA slant tubes, separately and mass multiplication of the test isolates were worked out by using Sand : Maize and PDB medium for further *in vitro* studies (PLATE V).

Table 5: *R. bataticola* isolates

Sr. No.	Isolate code	Districts	Tahsil	Agro-climatic zone
1	Rb-1	Aurangabad	Vaijapur	SZ
2	Rb-2	Beed	Kaij	ARZ
3	Rb-3	Hingoli	Kalamnoori	MRZ
4	Rb-4	Jalna	Mantha	ARZ
5	Rb-5	Latur	Latur	ARZ
6	Rb-6	Nanded	Loha	ARZ
7	Rb-7	Parbhani	Parbhani	ARZ
	Rb-8		Gangakhed	ARZ
8	Rb-9	Osmanabad	Osmanabad	ARZ

All-inclusive, the test pathogen was developed initially, a white mycelium, which was later on turned dark brown to black in color. Production of aerial mycelium was also observed in some isolates. Branching was occurred mostly at right angle to parent hyphae, but branching at acute angles was also observed in some isolates. The microsclerotia were produced within 7-8 days after incubation, which were dark brown to black in color, having round, ovoid to irregular shape. Observations under Research microscope (Labomed Vision 2000, at 400X), observed as dark black coloured septate mycelium and numerous black coloured microsclerotia (PLATE IV).

Such findings on isolations and characteristics of the *Rhizoctonia bataticola* on Potato Dextrose Agar medium were in accordance with the earlier reports of several workers, who also isolated and characterized *Rhizoctonia bataticola* from dry root rot diseased samples of various host plants (Prameela Devi and Singh, 1998a; Aghakhani and Dubey, 2009 a & b; Edraki and Banihashemi, 2010; Sundravadana *et al.*, 2012; Sharma *et al.*, 2015; Shrinivas P. 2016; Gade *et al.*, 2018).

4.4. Pathogenicity test

Pathogenicity of the test isolate of *R. bataticola* was proved by using sick soil method in polybag / pot culture, under screen house conditions, by sowing the seeds of susceptible chickpea cultivar JG-62 (PLATE VI). The test isolates produced the symptoms such as pre-emergence seed rot, post-

PLATE II



(a) Dry root rot infected seedlings (b) Dry root rot infected plant (c) Field symptoms

Typical symptoms of dry root rot (*R. bataticola*) on chickpea crop

emergence seedling mortality, yellowing and drooping of the leaves, sclerotial production on stems and roots, rotting and shredding of roots and stems etc. Further, this test also elucidated a constant association of *R. bataticola* with the dry root rot disease of chickpea.

On reisolation, from artificially dry root rot diseased chickpea plants specimens *R. bataticola* typical colony growth on PDA plates were observed consistently. Also, their cultural and morphological characteristics were exactly similar with *R. bataticola* pure cultures of the isolates obtained from naturally dry root rot diseased chickpea plants specimens. Thus, by applying Koch's postulates, pathogenicity of the test isolate of *R. bataticola* was conclusively proved.

Similarly, pathogenicity of *R. bataticola*, causing dry root rot/root rot was worked out by sick soil method and proved successfully the pathogenic association of *R. bataticola* / *M. phaseolina*, in various oilseed crops, pulses cereals, vegetables etc, by several workers (Kataria and Gaur 2007; Sharma *et al.*, 2012a; Veena *et al.*, 2014a; Veena *et al.*, 2016; Gade *et al.*, 2018).

4.5. Identification of the pathogen

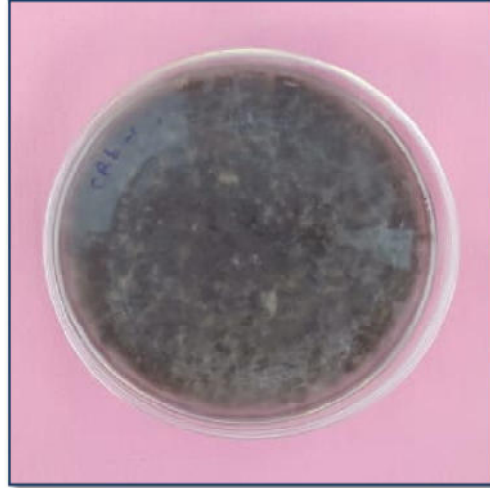
On the basis of the typical symptomatology of naturally and or artificially dry root rot diseased chickpea plants, morphological, cultural characteristics, microscopic observations and pathogenicity test, the test pathogen was identified as *R. bataticola*, which is the cause of dry root rot of chickpea as described by Butler (1918) and Sneh *et. al.*, (1991).

4.6. Variability among isolates of *R. bataticola*

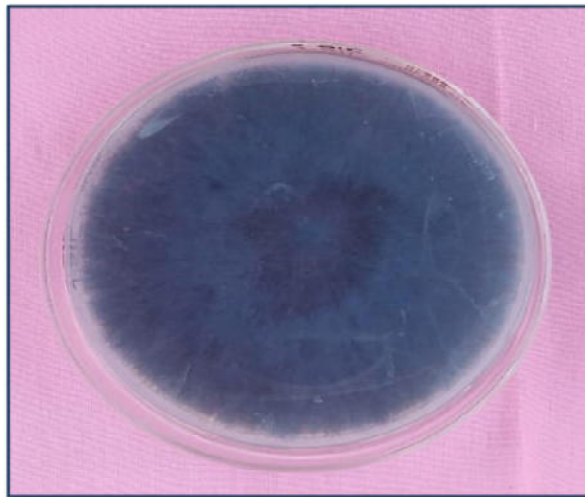
4.6.1. Pathogenic variability

Pathogenic variability of 9 isolates of *R. bataticola* was determined by using two methods viz., sick soil method and paper towel method and the results obtained of this study are being elucidated under following sub-heads.

PLATE III



Pure culture of *R. bataticola*



Reverse Colony Pigmentation of *R. bataticola*

Pure culture & Reverse Colony Pigmentation of *R. bataticola*

4.6.1.1. Sick soil method

The results (Table 6 and PLATE VI) indicated that all of the 9 test isolates of *R. bataticola* as pathogenic and caused pre-emergence seed rot and post-emergence seedling mortality in chickpea (JG-62).

Table 6. Pathogenic variability among the isolates of *R. bataticola* (Sick soil method)

Sr. No.	Isolates	Av. PRESR* (%)	Av. POESM* (%)		Total Mortality (%)	Days to initiation of symptoms
			30 DAS	45 DAS		
1	Rb ₁	18.33	24.39	41.28	84.00	25
2	Rb ₂	13.33	27.39	51.28	92.00	18
3	Rb ₃	17.33	24.39	39.45	81.67	26
4	Rb ₄	17.67	25.48	35.18	78.33	29
5	Rb ₅	15.00	27.18	45.82	88.00	22
6	Rb ₆	6.67	24.14	69.19	100.00	15
7	Rb ₇	10.00	26.20	49.80	86.00	23
8	Rb ₈	11.67	25.34	52.99	90.00	20
9	Rb ₉	16.67	25.32	53.01	95.00	17
10	Control	0.00	0.00	0.00	0.00	-

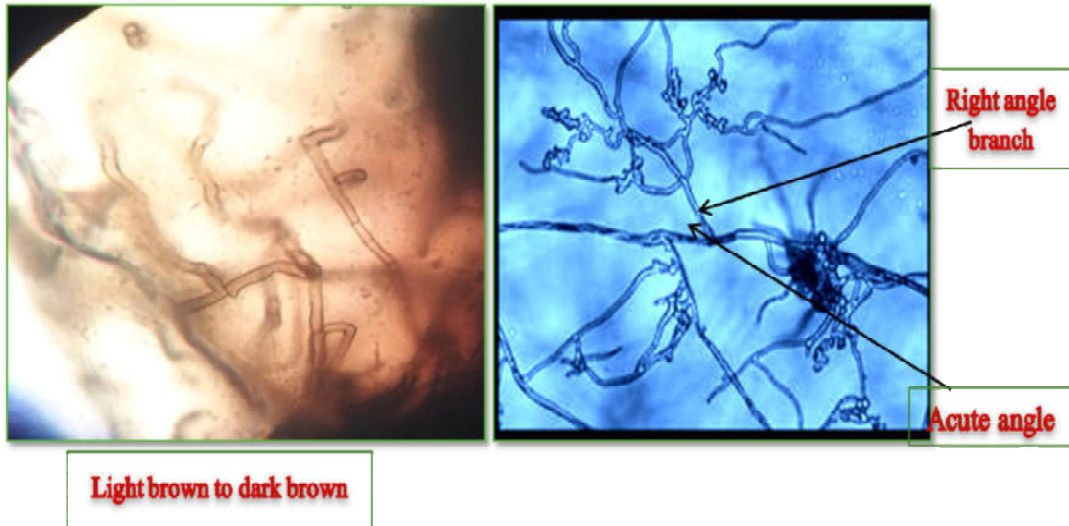
*Based on total no of seed sown

PRESR- Pre-emergence seed rot, POESM- Post-emergence seedling mortality.

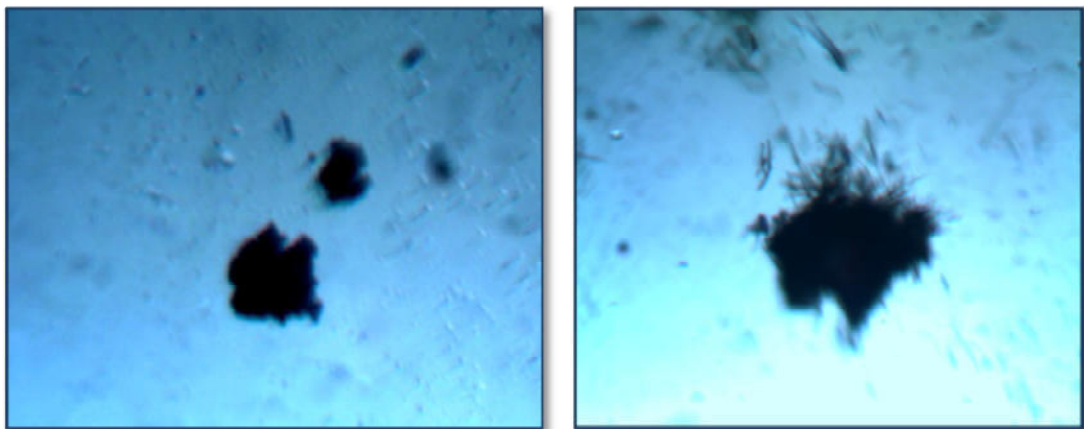
Days to initiation of dry root rot symptoms in chickpea was ranged from 15 DAS (Rb 6) to 29 DAS (Rb 4). Early symptoms was observed in isolate Rb 6 i.e. within 15 days, followed by isolate Rb 9 (17 DAS), Rb 2 (18 DAS) and in Rb 8 within 20 DAS. However, remaining isolates was in the range of 22 DAS to 29 DAS.

Average pre-emergence seed rot (PRESR) was observed with the test isolates was ranged from 6.67 per cent (Rb-6) to 18.33 per cent (Rb-1), as against 0.00 per cent in untreated control. However, it was observed maximum with Rb-1 (18.33 %), followed by Rb-4 (17.67 %), Rb-3 (17.33 %), Rb-9 (16.67 %) and Rb-5 (15.00 %). In rest of the test isolates caused PRESR in the range of 6.67 to 13.33 per cent.

PLATE IV



Hyphal branching



Sclerotial development

Microphotographs *R. bataticola*

Similarly, In case of post-emergence seedling mortality (POESM) which was recorded at 15 and 30 DAS was ranged from 24.14 to 27.39 per cent and 35.18 to 69.19 per cent, respectively. At 15 DAS, Rb-2 induced highest POESM (27.39 %), followed by Rb-5 (27.18%) and Rb-7 (26.20 %). Whereas in remaining isolates, POESM was observed in the range of 24.14 to 25.48 per cent, at 15 DAS. At 30 DAS, the isolate Rb-6 produced highest POESM (69.19 %), followed by Rb-9 (53.01 %), Rb-8 (52.99 %) and Rb-2 (51.28 %). In rest of the isolates produced POESM in the range of 39.45 to 49.80 per cent, at 30 DAS.

Total mortality (PRESR + POESM) produced by the test isolates was ranged from 78.33 to 100.00 per cent. However, it was cent per cent (100 %) in the isolate Rb-6, followed by Rb-9 (95.00 %), Rb-2 (92.00 %), and Rb-8 (90.00 %). In rest of the isolates induced total mortality in the range of 78.33 to 88.00 per cent. Thus, all of the 9 test isolates of *R. bataticola* were found to be highly pathogenic to chickpea.

4.6.1.2. Paper Towel method

The results (Table 7 and PLATE VII) elucidated, all of the 9 test isolates of *R. bataticola* was found highly pathogenic to chickpea, which on 7th day of seedling placed on paper towel , caused average seedling mortality in the range of 48.00 to 93.33 per cent, in chickpea (JG-62) as against no seedling mortality observed in control (untreated). However, it was considerably highest with the isolate Rb-6 (90.00 %), followed by Rb-9 (86.67 %) and Rb-2 (80.00 %). In rest of the test isolates average seedling mortality recorded in the range of 46.67 to 68.88 per cent.

PLATE V



Mass multiplication of *R. bataticola*

Table 7. Pathogenic variability among the isolates of *R. bataticola* (Paper towel method)

Sr. No.	Isolates	Av. Seedling Mortality* (%) on 7 th Day	Level of virulence
1	Rb ₁	66.67	Strongly virulent
2	Rb ₂	80.00	Highly virulent
3	Rb ₃	60.00	Strongly virulent
4	Rb ₄	46.66	Moderately virulent
5	Rb ₅	67.78	Strongly virulent
6	Rb ₆	90.00	Highly virulent
7	Rb ₇	63.33	Strongly virulent
8	Rb ₈	68.88	Strongly virulent
9	Rb ₉	86.67	Highly virulent
10	Control	0.00	----

*Based on the total no. of chickpea seedlings placed in paper towel

Based upon the level of virulence (Pawar, 2010), among 9 test isolates, three isolates viz., Rb-6, Rb-9 and Rb-2 were grouped as highly virulent (Av. seedling mortality >71 %), five isolates as strongly virulent (Av. seedling mortality 51-70 %) and one isolate as moderately virulent (Av. seedling mortality 21-50 %). Thus, also by the paper towel method all of the 9 test isolates of *R. bataticola* were found pathogenic to chickpea.

On the basis of pathogenic variability study carried out by two methods i.e. sick soil and paper towel methods, all of the 9 isolates of *R. bataticola* were divided into various categories (Table 8), according to the rating scale given by Pawar (2010).

Table 8. Grouping of *R. bataticola* isolates based on their pathogenic variability, by Sick soil and Paper towel methods

Sr. No	Category / % Mortality	Sick soil method	Paper towel method
		No. of Isolates and Isolates Codes	No. of Isolates and Isolates Codes
1	Avirulent (0)	None	None
2	Weakly virulent (1-20 %)	None	None
3	Moderately virulent (21-50 %)	None	One: Rb-4
4	Strongly virulent (51-70 %)	None	Five: Rb-1, Rb-3, Rb-5, Rb-7, Rb-8
5	Highly virulent (>71 %)	Nine: Rb-1, Rb-2, Rb-3, Rb-4, Rb-5, Rb-6, Rb-7, Rb-8, Rb-9	Three: Rb-2, Rb-6, Rb-9

PLATE VI



Pathogenicity test on cv. JG 62

**Pathogenicity and Pathogenic variability among the isolates
of *R. bataticola* (Sick soil method)**

Thus, from the comparative evaluation (Table 8) on pathogenic variability study by conducting sick soil and paper towel methods, indicated sick soil method as most efficient method as compared to paper towel method. By using both methods, the isolate Rb-6 was found to be as highly virulent to chickpea. Therefore, the isolate Rb-6 (Loha, Nanded district) was selected and used for further *in vitro* (plate and pot culture) studies.

Similarly, several workers have found both methods as most suitable in establishing the disease caused by *Rhizoctonia bataticola* (Katariya *et al.*, 2007; Veena *et al.*, 2014; Kadam M.A., 2016).

4.6.2. Cultural variability

Results (Table 9, PLATE VIII and Fig. 9) showed that all of the 9 test isolates of *R. bataticola*, expressed very inadequate variability in respect of their cultural characteristics. After seven days of incubation, the colony diameter (mm) of the test isolates were ranged from 79.00 to 90.00 mm and the growth rate was slow, medium and fast.

Table 9. Cultural variability among the isolates of *R. bataticola*

Sr. No.	Isolates	Col. Dia.* (mm) /	Growth rate	Colour	Colony texture	Elevation	Margin
1	Rb ₁	84.00	(++++)	Charcoal black	Appressed	Raised	Smooth
2	Rb ₂	88.50	(++++)	Blackish Grey	Fluffy	Raised	Wavy
3	Rb ₃	81.00	(++++)	Grayish White	Velvety	Flat	Smooth
4	Rb ₄	79.00	(+++)	Blackish Grey	Fluffy	Raised	Smooth
5	Rb ₅	85.00	(++++)	Grayish White	Velvety	Flat	Smooth
6	Rb ₆	90.00	(++++)	Grayish White	Velvety	Raised	Smooth
7	Rb ₇	84.50	(++++)	Blackish Grey	Appressed	Flat	Wavy
8	Rb ₈	86.00	(++++)	Grayish White	Fluffy	Raised	Wavy
9	Rb ₉	89.50	(++++)	Light black	Appressed	Flat	Wavy

*Average of a triplicate set of PDA plates per isolates, Growth Rate = Fast : ++++ (81.00-90.00 mm), Medium : +++ (71.00-80.00 mm), Slow : ++ (<70.00 mm)

Based upon the colony diameter and growth rate the test isolates were grouped as fast, medium and slow growing. The fast growing isolates with colony diameter of 81.00-90.00 mm were Rb-1, Rb-2, Rb-3, Rb-5, Rb-6,

PLATE VII



Rb1



Rb2



Rb3



Rb4



Rb5



Rb6



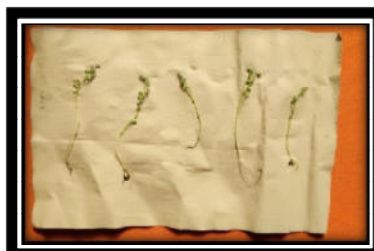
Rb7



Rb8



Rb9



Rb10

**Pathogenic variability among the isolates of *R. bataticola*
Paper Towel Technique**

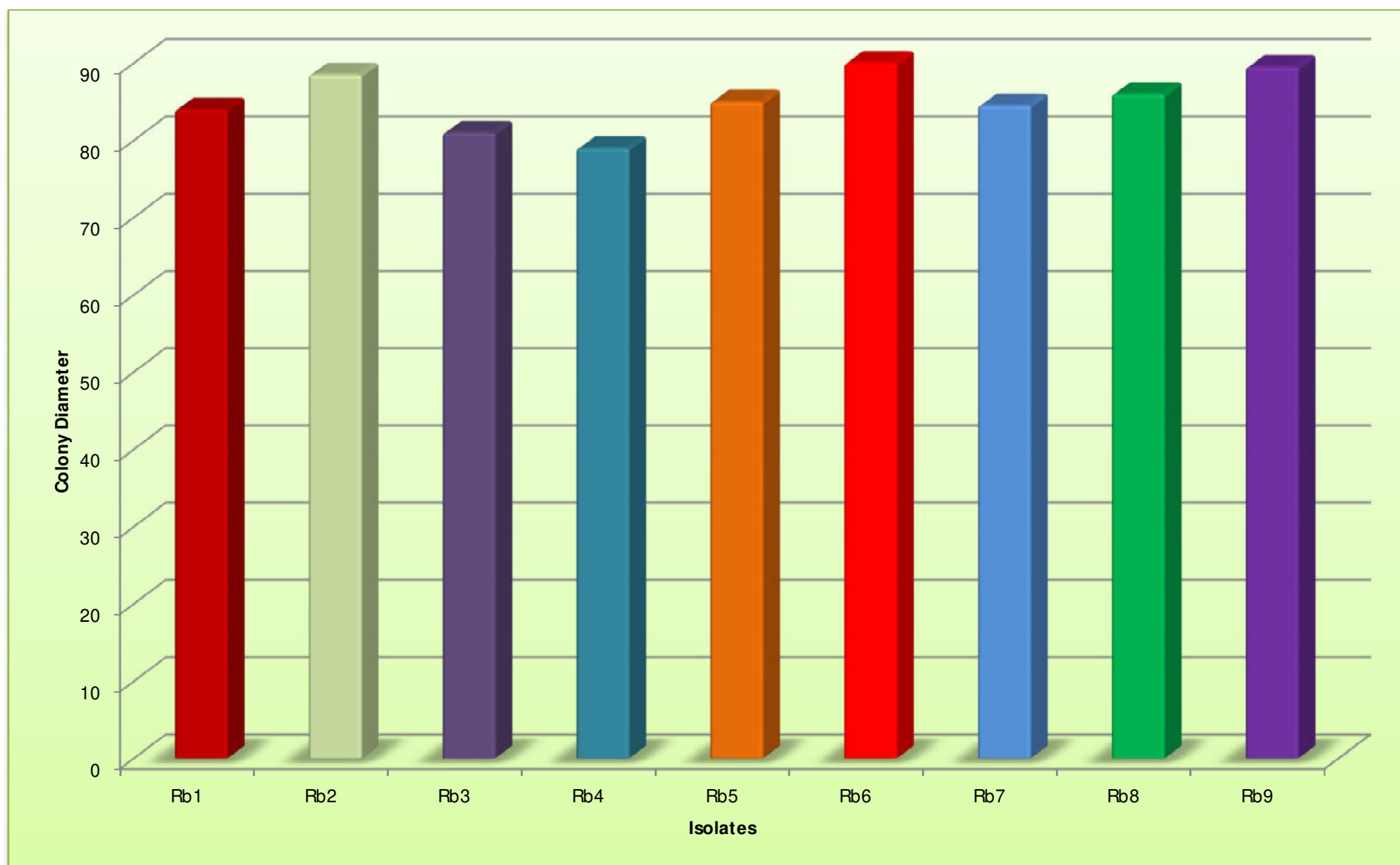


Fig. 9.Cultural variability among the isolates of *R. bataticola*

Rb-7, Rb-8 and Rb-9; medium growing isolates with colony diameter of 71.00-80.00 mm was Rb-4 and no one isolate was found be in the category of slow growing.

The colony colour varied from blackish grey (Rb-2, Rb-4 and Rb-7), charcoal black (Rb-1), grayish white (Rb-3, Rb-5, Rb-6 and Rb-8) and light black (Rb-9).

The colony textures of the test isolates were categorized in three groups such as appressed, velvety and fluffy. Out of 9 isolates, isolates (Rb-1, Rb-7 and Rb-9) showed appressed colony texture while isolates (Rb-2, Rb-4 and Rb-8) had fluffy texture (Rb-1, Rb-7 and Rb-9). Whereas, remaining isolates (Rb-3, Rb-5 and Rb-6) produced velvety type of texture.

The colony elevation was found to be raised in the isolates (Rb-1, Rb-2, Rb-4, Rb-6 and Rb-8); whereas, it was flat in the isolates *viz.*, Rb-3, Rb-5, Rb-7 and Rb-9. The colony margin was wavy in the isolates (Rb-2, Rb-4, Rb-5, Rb-7 and Rb-8); whereas, it was smooth in the isolates *viz.*, Rb-1, Rb-3, Rb-6 and Rb-9).

4.6.3. Morphological variability

The results on morphological characteristics such as hyphal width and sclerotial characters like shape, colour and size, with respect to the test isolates of *R. bataticola* are presented in (Table 10, PLATE IXA, IXB and PLATE XA, XB.)

Table 10. Morphological variability among the isolates of *R. bataticola*

Sr. No.	Isolates	Av. Hyphal Width* (µm)	Sclerotial characters		
			Av. Size* (µm) Range	Density#/ Microscopic field	Shape
1	Rb ₁	5.14	55.50	24.33	Ovoid
2	Rb ₂	6.33	90.50	51.67	Round
3	Rb ₃	4.62	50.50	18.67	Irregular
4	Rb ₄	3.80	46.50	13.67	Ovoid
5	Rb ₅	5.46	62.00	31.33	Irregular
6	Rb ₆	7.33	125.00	70.00	Irregular
7	Rb ₇	5.20	60.50	28.67	Irregular
8	Rb ₈	6.23	80.50	36.33	Irregular
9	Rb ₉	7.19	113.00	60.67	Irregular

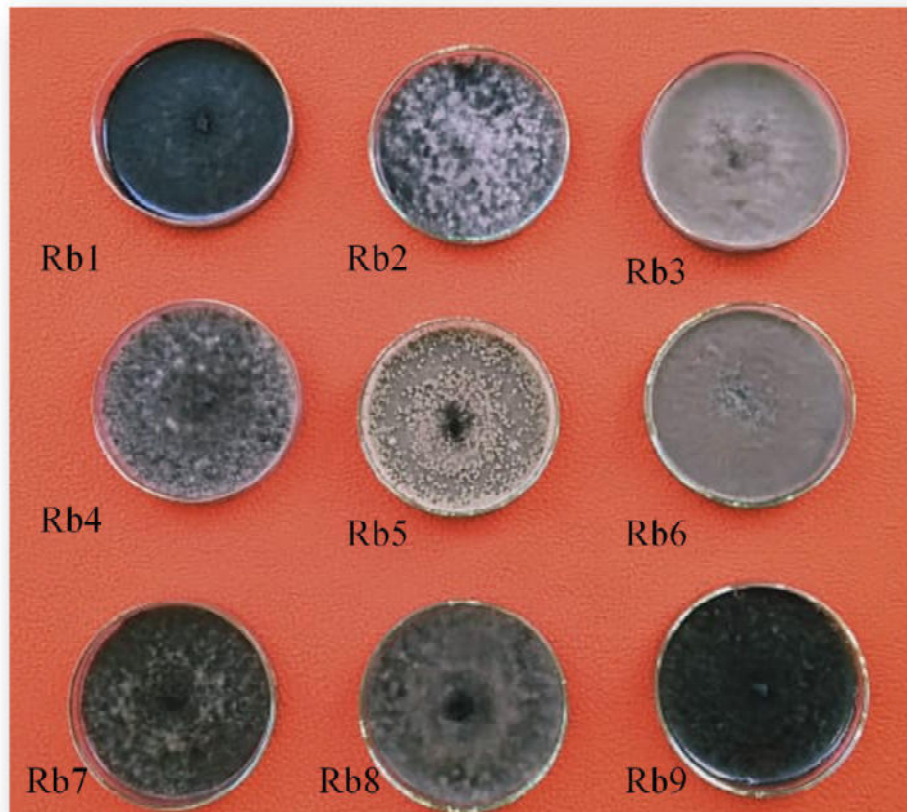
*: Mean of ten observations / isolates, #: Mean of five microscopic fields (10X).

Average hyphal width of the test isolates was to be ranged from 3.80 to 7.33 µm. Based on average hyphal width, the test isolates were classified into three categories viz., large (>5.00 µm), medium (4.00-4.99 µm) and small (<4.00 µm). Among the nine isolates, seven isolates were found with large sized hyphae (5.14-7.33 µm) viz., Rb-6, Rb-9, Rb-2, Rb-8, Rb-5, Rb-7 and Rb-1; one isolate was found with medium sized hyphae viz., Rb-3 and one isolate was found to be with small sized hyphae viz., Rb-4.

The color of sclerotia was brown to black and their shape was observed round, ovoid to irregular one. Most of the isolates produced (06) viz., Rb-3, Rb-5, Rb-6, Rb-7, Rb-8 and Rb-9, irregular shaped sclerotia and out of the rest three, two isolates viz., Rb-1 and Rb-4 produced ovoid and Rb-2 produced round shaped sclerotia.

In all test isolates found considerable variations in respect to no. of sclerotia produced (density / microscopic field) and their size. Sclerotial production among the test isolates was grouped as (Das, 1988) poor (10-20 sclerotia / microscopic field), fair (21-30 sclerotia / microscopic field), good (31-50 sclerotia / microscopic field) and excellent (>51 sclerotia / microscopic field). All-inclusive, sclerotial density was ranged from 13.67 to 70.33. However, the sclerotial density was found excellent in three isolates such as Rb-6 (70.33), Rb-9 (60.67) and Rb-2 (51.67); good in two isolates viz., Rb-8

PLATE VIII



Cultural variability among the test isolates of *R. bataticola*

(36.33) and Rb-5 (31.33); fair in two isolates viz., Rb-7 (28.67) and Rb-1 (24.33) and poor in two isolates viz., Rb-3 (18.67) and Rb-4 (13.67).

Average size produced by the sclerotia of the test isolates was ranged from 46.50 to 124.00 μm . Depending upon the size of the sclerotia, the test isolates were Classified as large sized sclerotia ($>101\ \mu\text{m}$), medium (50-100 μm) and small ($<50\ \mu\text{m}$). The large sized sclerotia were observed in two isolates viz., Rb-6 (124 μm) and Rb-9 (113.00 μm); medium sized sclerotia were observed in six isolates viz., Rb-2 (90.50 μm), Rb-8 (80.50 μm), Rb-5 (62.00 μm), Rb-7 (60.50 μm), Rb-1 (55.50 μm), and Rb-3 (50.50 μm) and small sized sclerotia was found in only one isolate viz., Rb-4 (46.50 μm).

Present results obtained on pathogenic, morphological and cultural characteristics variability among the isolates of *R. bataticola* causing dry root rot of chickpea are in consensus with those findings of several other workers. Monga and Sheo Raj (1995) and Monga *et al.*, (2004, 2007) reported morpho-cultural and pathogenic variation among *Rhizoctonia* spp. (*R. bataticola* and *R. solani*), causing root rot of cotton. The seedling mortality caused by various isolates of *R. bataticola* was ranged from 75.00-83.3 per cent. Aghakhani and Dubey (2009a) reported that all 23 isolates of *R. bataticola*, found as highly pathogenic to chickpea, which caused dry root rot incidence in the range 31.7 to 100 per cent. The test isolates were also reported to vary in respect of morpho-cultural characteristics. Sharma *et al.* (2012a, b) studied pathogenic and morpho-cultural variability among 94 isolates of *R. bataticola* causing dry root rot of chickpea and reported all test isolates as highly pathogenic to chickpea and also varied in morpho-cultural characteristics. Gade *et al.* (2018) studied pathogenic and morphological variability among 40 isolates of *R. bataticola*, causing chickpea dry root rot and reported that two isolates viz., Rb-3 and Rb-17 as highly pathogenic with 71-100 per cent mortality; the test isolates also varied in respect of density, size and colour of sclerotia produced.

4.6.4. Molecular variability of *R. bataticola* isolates

4.6.4.1. Standardization of RAPD Protocol

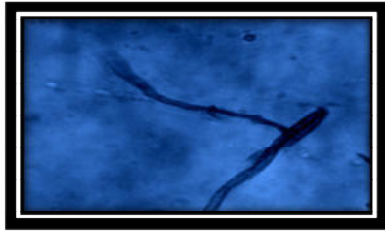
The RAPD-PCR protocol described by Williams *et. al.*, (1990) was used with some modifications, to produce RAPD fingerprinting profile of nine fungal pathogens of *Rhizoctonia bataticola*. (Table 11, PLATE XIA, XIB & XIIA and Fig. 10)

The PCR amplification reaction was optimized by varying concentration of PCR components. Amplification reaction was carried out in 25 µl reaction mixtures containing 30 ng of genomic DNA, 1X PCR buffer, 1.5 mM MgCl₂, 0.25 mM dNTPs, 0.4 µM primers and 1.25 U of *Taq*. DNA polymerase. PCR amplification was performed in master cycler gradient Eppendorf PCR Machine. The program consisted of an initial denaturation at 94°C for 4 min, followed by 39 cycles of 94°C for 1 min. 35°C for 1 min and 72°C for 1.5 min, followed by final extension at 72°C for 10 min. Finally a product was hold at 4°C. PCR amplified product was separated by electrophoresis in 1.2 % agarose gel in 1X TAE buffer stained with Ethidium bromide and visualized under gel documentation system.

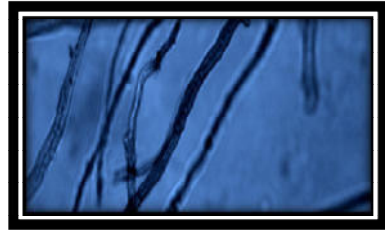
4.6.4.2. Diversity analysis using RAPD marker

The genomic DNA of all fungal pathogen was subjected for PCR amplification by using RAPD primers from Operon tech, USA. Initially 10 primers were screened with nine pathogen of *Rhizoctonia bataticola* among which five primers were produced larger no. and reproducible amplicons and further those were employed for molecular characterization of nine *Rhizoctonia bataticola* isolates.

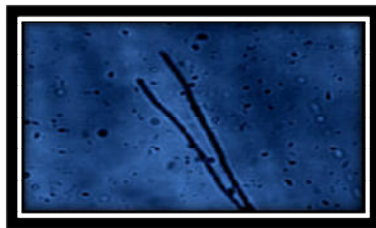
PLATE IXA



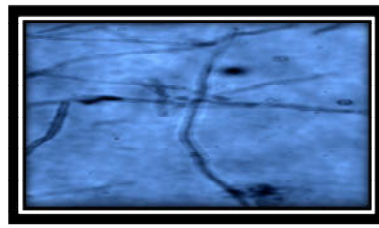
RB1



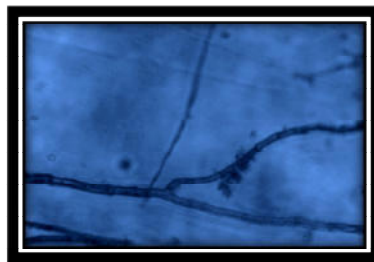
RB2



RB3



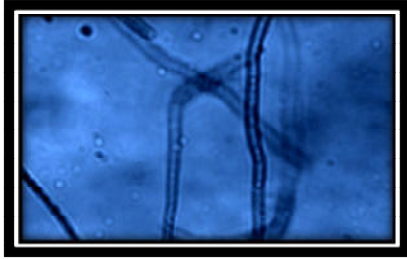
RB4



RB5

Morphological variability among the test isolates of *R. bataticola*

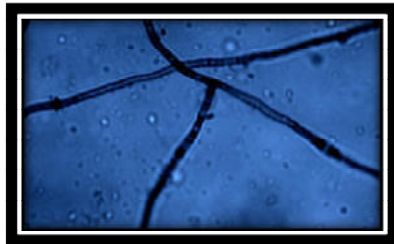
PLATE IXB



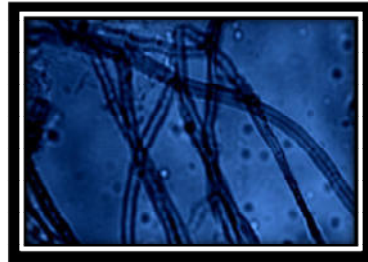
RB6



RB7



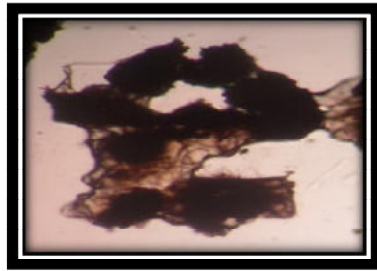
RB8



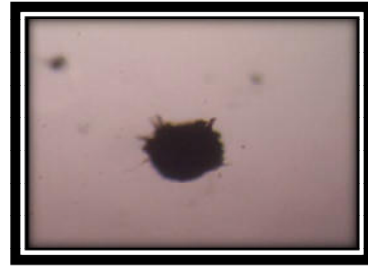
RB9

Morphological variability among the test isolates of *R. bataticola*

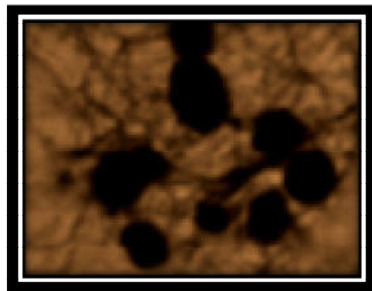
PLATE XA



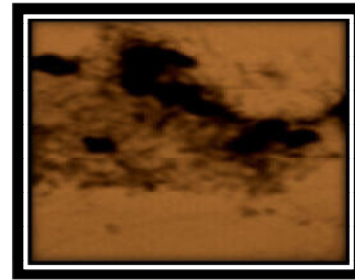
RB1



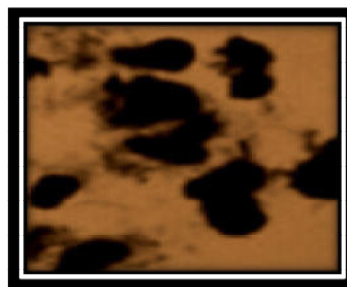
RB2



RB3



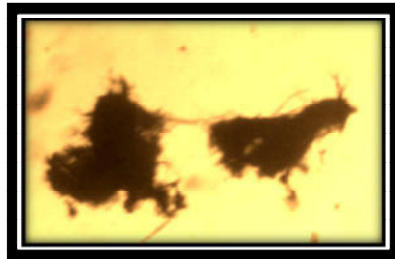
RB4



RB5

Morphological variability among the test isolates of *R. bataticola*

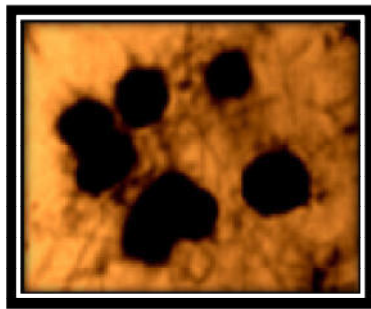
PLATE XB



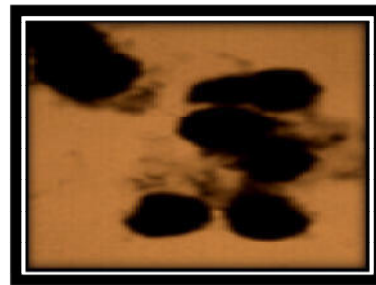
RB6



RB7



RB8



RB9

Morphological variability among the test isolates of
R. bataticola

Table 11: RAPD primers and the number of total polymorphic bands produced in 9 isolates of *Rhizoctonia bataticola*.

Primer	Sequence (5'-3')	Total No. of Bands	Total No. of Polymorphic Bands	Total No. of Monomorphic Bands	Total No. of fragment amplified	Polymorphic Percentage (%)
OPA-02	TGCCGA GCTG	15	10	5	82	66.67
OPA-03	AGTCAG CCAC	16	12	4	96	75.00
OPA-07	GAAACG GGTG	13	10	3	78	76.92
OPA-09	GGGTAA CGCC	14	11	3	62	78.57
OPA-13	CAGCAC CCAC	14	08	6	91	57.14
Total		72	51	21	409	354.30
Mean		14.40	10.20	4.20	81.80	70.86

Average polymorphism in strains of *Rhizoctonia bataticola* based on RAPD analysis was 70.86 %.

The average sizes of amplicons generated by all RAPD primer were found in the range between 100 bp to 8.0 kb. The RAPD-PCR amplification result showed that five RAPD primers generated total 72 bands, out of these, 51 bands were found polymorphic and 21 bands were reported as monomorphic with an average of 4.20 bands per primers among nine fungal pathogens of *Rhizoctonia bataticola*. The primer OPA-02, OPA-03, OPA-07, OPA-09 and OPA-13, were more found informative as they could generate total number of bands range 13 to 16. The primer OPA-07 has generated lowest number of bands *i.e.* 13 (Table 11& Fig. 11). The primer OPA-03 has produced higher number of amplicons *i.e.* 96, followed by OPA-13 and OPA-02 which has produced 91 and 82 amplicons respectively. While the primer OPA-09 has produced minimum number of amplicons *i.e.* 62.

Overall all five primers were generated total 409 amplicons with an average of 81.80 amplicons per primers (Table 11& Fig.11). Out of 409 amplicons, 220 amplicons were found polymorphic, they showed 53.78 % polymorphism. Similarly, 189 amplicons were found monomorphic. They

showed, 46.22 % monomorphism and the average number of monomorphic bands per primer were 4.20.

4.6.4.3. Genetic distance

The data obtained by RAPD markers was analyzed by NTSYS pc2.02i and dendrogram was depicted by using Jaccard's similarity coefficient. The genetic similarity matrix obtained by Jaccard's similarity coefficient of the *Rhizoctonia bataticola* isolates. (Table 12 & Figure No. 11 & 12).

The overall range of the similarity among the nine isolates was found to be very wide, which was ranged from 0.514 to 0.847 which indicated presence of variability among the isolates. Maximum similarity coefficient was observed between the isolates Rb 2 and Rb 4 followed by the isolates Rb 2 and Rb 9. Whereas minimum similarity coefficient was observed in the isolates Rb1 and Rb 8 followed by the isolates Rb 4 and Rb 8. Some isolates was observed with the cent per cent similarity coefficient on the molecular basis but could not be considered as a duplicons.

4.6.4.3. Cluster analysis of RAPD DNA fingerprint

Dendrogram generated based on UPGMA analysis of RAPD data grouped all these isolates in to two major clusters (Figure No. 12). These clusters were formed on the basis of genus as well as species level. The group I, cluster I, comprised, maximum of six fungal pathogens together which showed 64.5 % genetic similarity viz., Rb-1, Rb-2, Rb-3 Rb-4, Rb-5 and Rb-9. The cluster I comprised fungal isolates of Rb-2 and Rb-4 showed that higher similarity each other i.e. 85 %. Whereas the group II, Cluster II comprised three isolates together showing 72.2 % similarity with each other viz., Rb-6, Rb-7 and Rb-8. Theses cluster comprised fungal isolates of Rb-6 and Rb-7 showed that higher similarity each other i.e. 85 %. The cluster I and II comprised of nine isolates showed that 62 % similarity each other.

Thus, the present study on molecular variability indicated a very inadequate diversity among the population of *R. bataticola* observed in Marathwada region surveyed, of the Maharashtra state, which may be due to

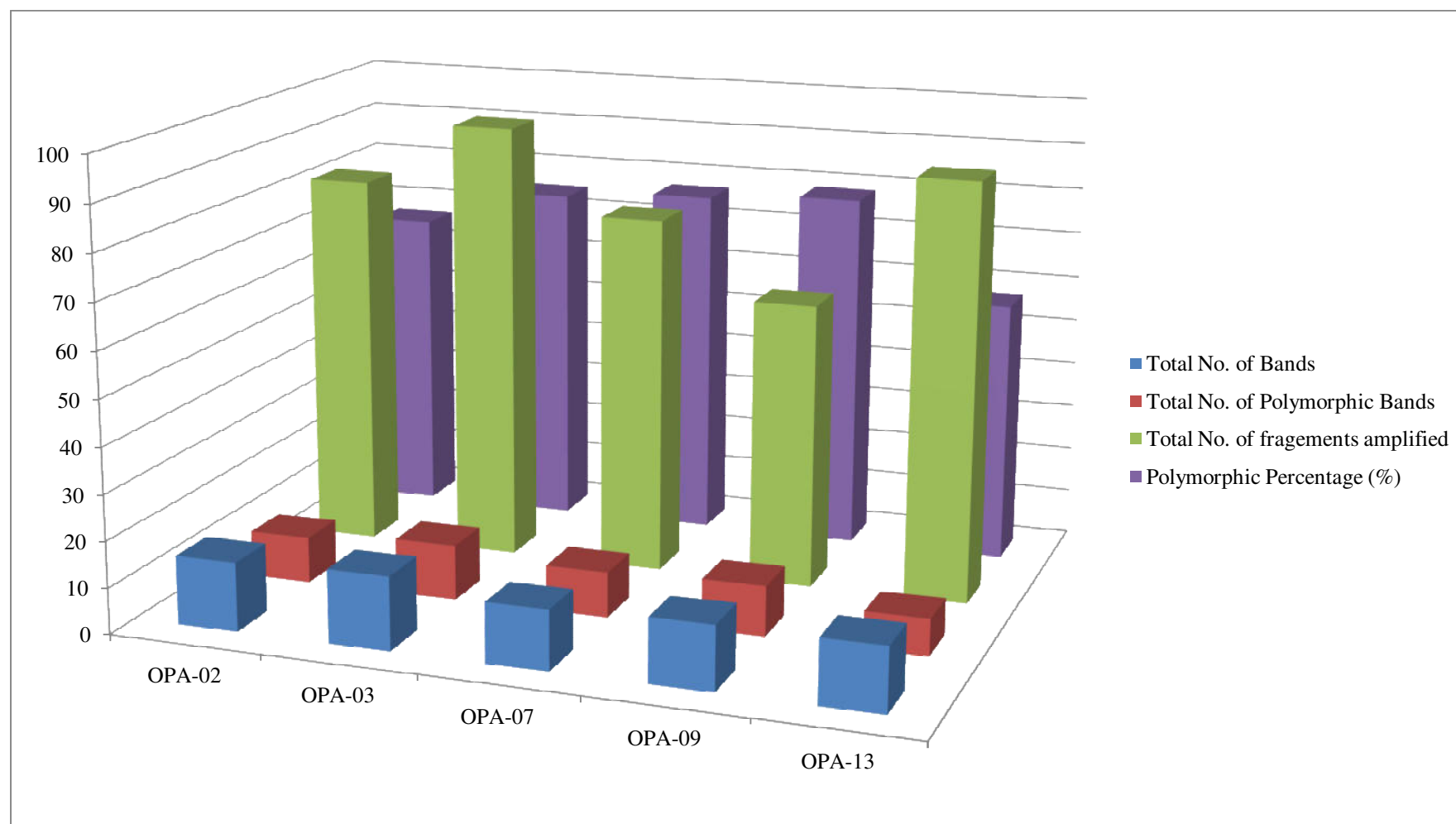


Figure 10: Graphical representation of comparative analysis of RAPD primer and polymorphic percentage

some close similarity such as the rainfall, soil type and cropping pattern found between these three agro-climatic zones surveyed, during the study.

Table 12. The similarity matrix representing coefficient of similarity among 9 isolates of *R. bataticola* based on RAPD marker

	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8	Rb-9
Rb-1	1.000								
Rb-2	0.583	1.000							
Rb-3	0.681	0.597	1.000						
Rb-4	0.625	0.847	0.639	1.000					
Rb-5	0.722	0.778	0.681	0.708	1.000				
Rb-6	0.611	0.611	0.569	0.597	0.667	1.000			
Rb-7	0.597	0.625	0.611	0.611	0.681	0.847	1.000		
Rb-8	0.514	0.653	0.583	0.556	0.569	0.708	0.722	1.000	
Rb-9	0.681	0.792	0.694	0.694	0.764	0.736	0.722	0.722	1.000
Avg. Similarity	0.668	0.721	0.673	0.698	0.730	0.705	0.713	0.670	0.756

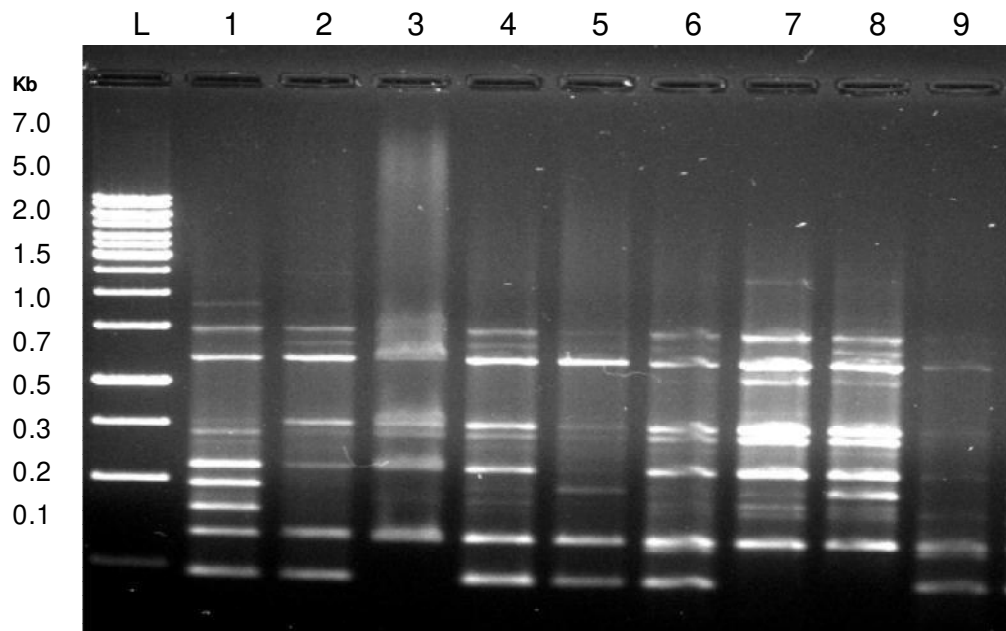
Using various molecular technique, genetic variability among and between the isolates of *R. bataticola*, causing dry root rot in chickpea and other many crops were reported by many workers (Monga, *et al.*, 2004 and 2007; Agakhani and Dubey, 2009a; Sundravandana, *et al.*, 2011; Pancheshwar, *et al.*, 2012; Belkar and Gadhe, 2016; Shriniwas P., 2016; Agle R.C., 2018).

4.7. Effect of organic and inorganic sources of carbon and nitrogen on growth of *Rhizoctonia bataticola*

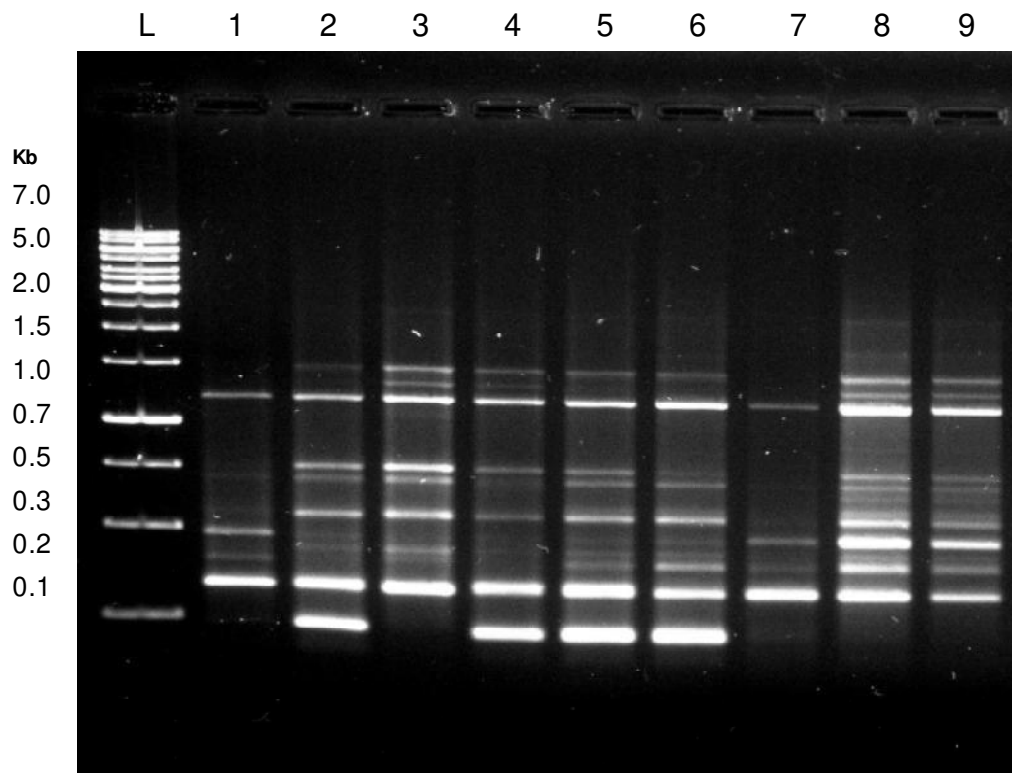
4.7.1. Effect of organic and inorganic sources of carbon

The capacity of the fungus to utilize various carbon sources is governed by its ability to produce enzymes. These enzymes convert complex carbon compounds into simpler forms, which are subsequently utilized for growth and reproduction. All of the six carbon sources (Table 13, Fig. 13, PLATE XIII) tested encouraged better growth of *R. bataticola* except carbon control.

PLATE XIA

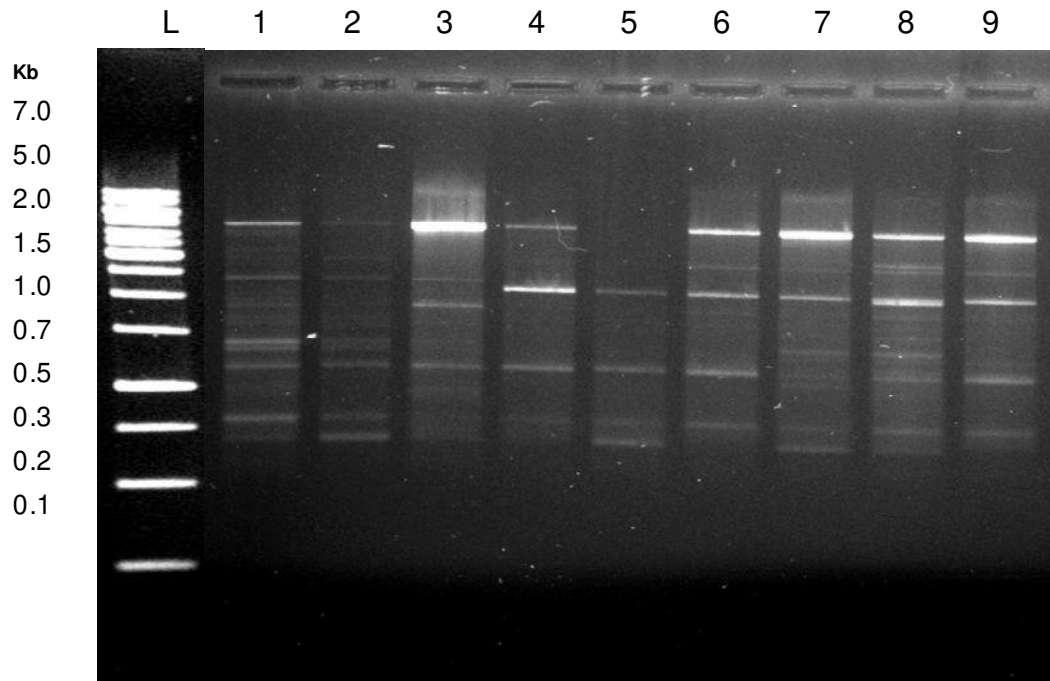


primer OPA-02: Lane L, Marker (1Kb DNA ladder);

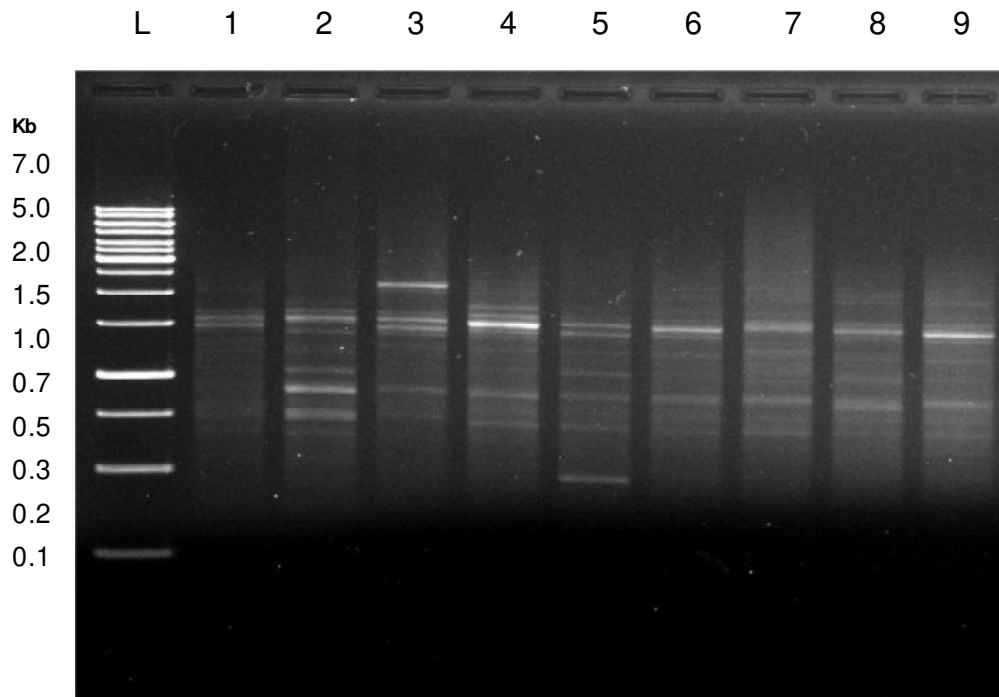


RAPD fingerprint profile of 9 pathogen isolates of *Rhizoctonia bataticola* by using primer OPA-02, OPA-03: Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of *Rhizoctonia bataticola*

PLATE XIB

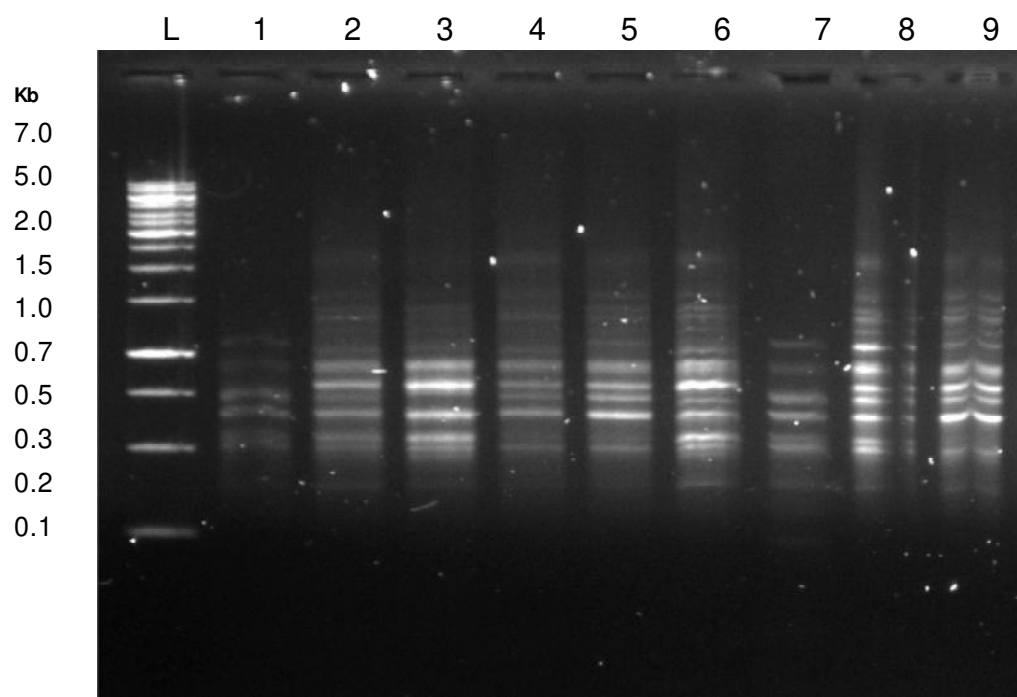


primer OPA-07: Lane L, Marker (1Kb DNA ladder)



RAPD fingerprint profile of 9 pathogen isolates of *Rhizoctonia bataticola* by using primer OPA-07, OPA-09: Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of *Rhizoctonia bataticola*

PLATE XIII



RAPD fingerprint profile of 9 pathogen isolates of *Rhizoctonia bataticola* by using primer OPA-13:
Lane L, Marker (1Kb DNA ladder); Lanes 1 –9 isolates of *Rhizoctonia bataticola*

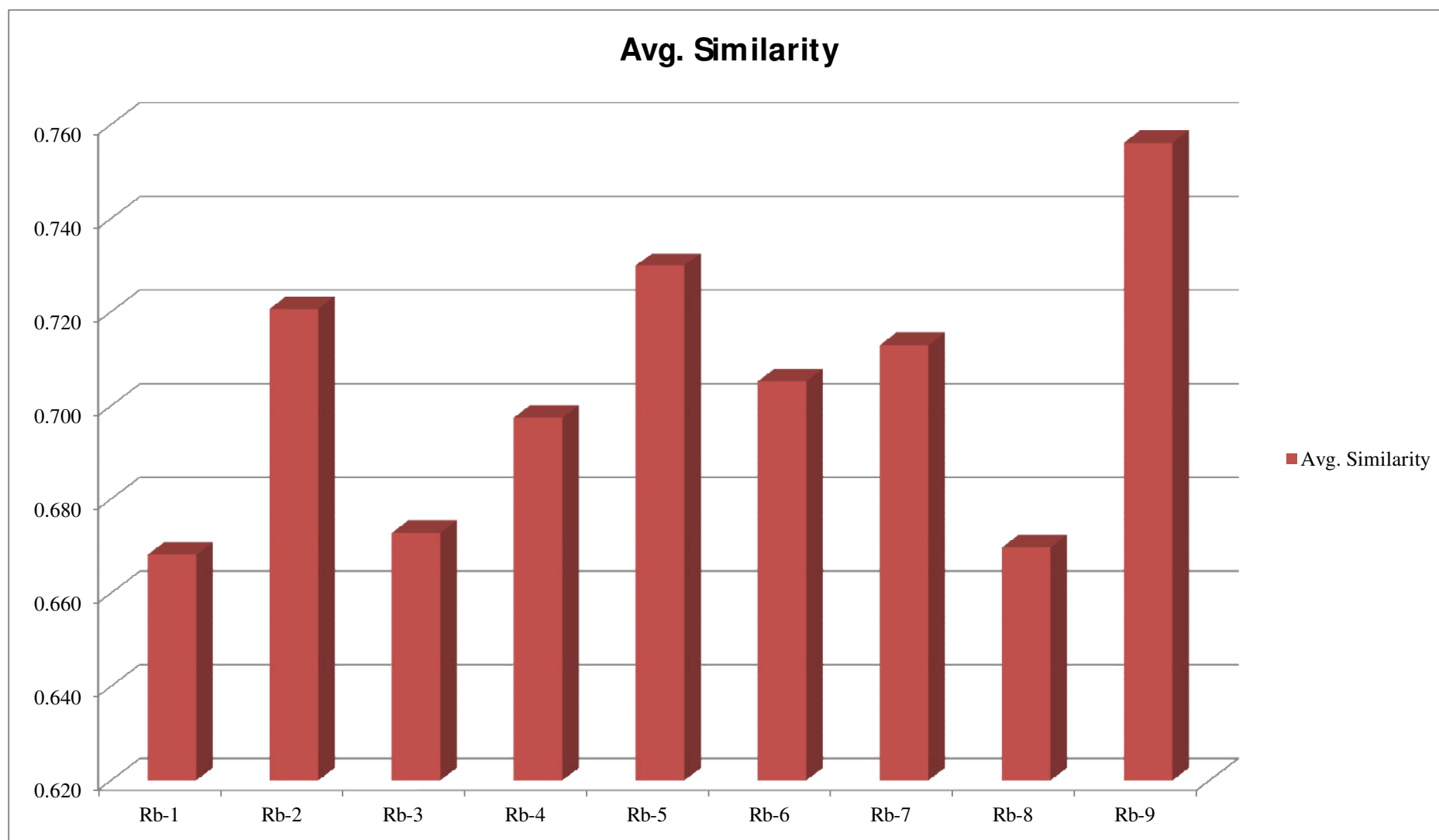


Fig. 11. Average similiarity of nine isolates of *Rizocotina Bataticola* based on *UPGMA* analysis.

Table 13. Effect of Carbon sources on growth and Sclerotial production of *R. bataticola*

Sr. No.	Carbon Sources	Mean colony diameter (mm)*	Colony colour	Sclerotial production
T ₁	Dextrose	86.58	Whitish black	++++
T ₂	Sucrose	81.28	Whitish	++++
T ₃	Mannitol	72.64	Whitish	+
T ₄	Starch	71.49	Whitish black Cottony	++
T ₅	Cellulose	51.69	Whitish black Cottony	++++
T ₆	Maltose	41.43	Black	++
T ₇	Control	26.00	Black	-
SE± (m)		0.51		
CD at 1% (m)		2.13		

++++ Excellent, +++ Good, ++ Fair, + Poor, - No.

However, significantly highest mean mycelial growth (86.58 mm) was recorded with Dextrose. The second was Sucrose (81.28 mm) and third best carbon source was Mannitol (72.64 mm). Rest of carbon sources observed mycelial growth was in the range of 41.43 (Maltose) to 71.49 mm. (Starch).

All the carbon sources tested having a wide range of sclerotial production i.e. from poor (+) to excellent (++++). However, carbon sources viz., Dextrose, Sucrose and Cellulose found excellent (++++) sporulation. Rest of carbon sources had good (+++) to Poor sporulation (+). Mukhopadhyay and Nandi (1975) reported that sucrose, fructose and glucose were best carbon sources for mycelia growth and sclerotial production.

The result in the present study is in accordance with the findings of Mukhopadhyay and Nandi (1975) and Salunke *et al.*, (2009) and Thombre & Kohire (2018). The sources of carbon viz., Sucrose, Fructose, Dextrose, Maltose, Glucose, and D - Galactose were reported to support maximum growth and sporulation in *M. phaseolinal* *R. bataticola* earlier as reported by

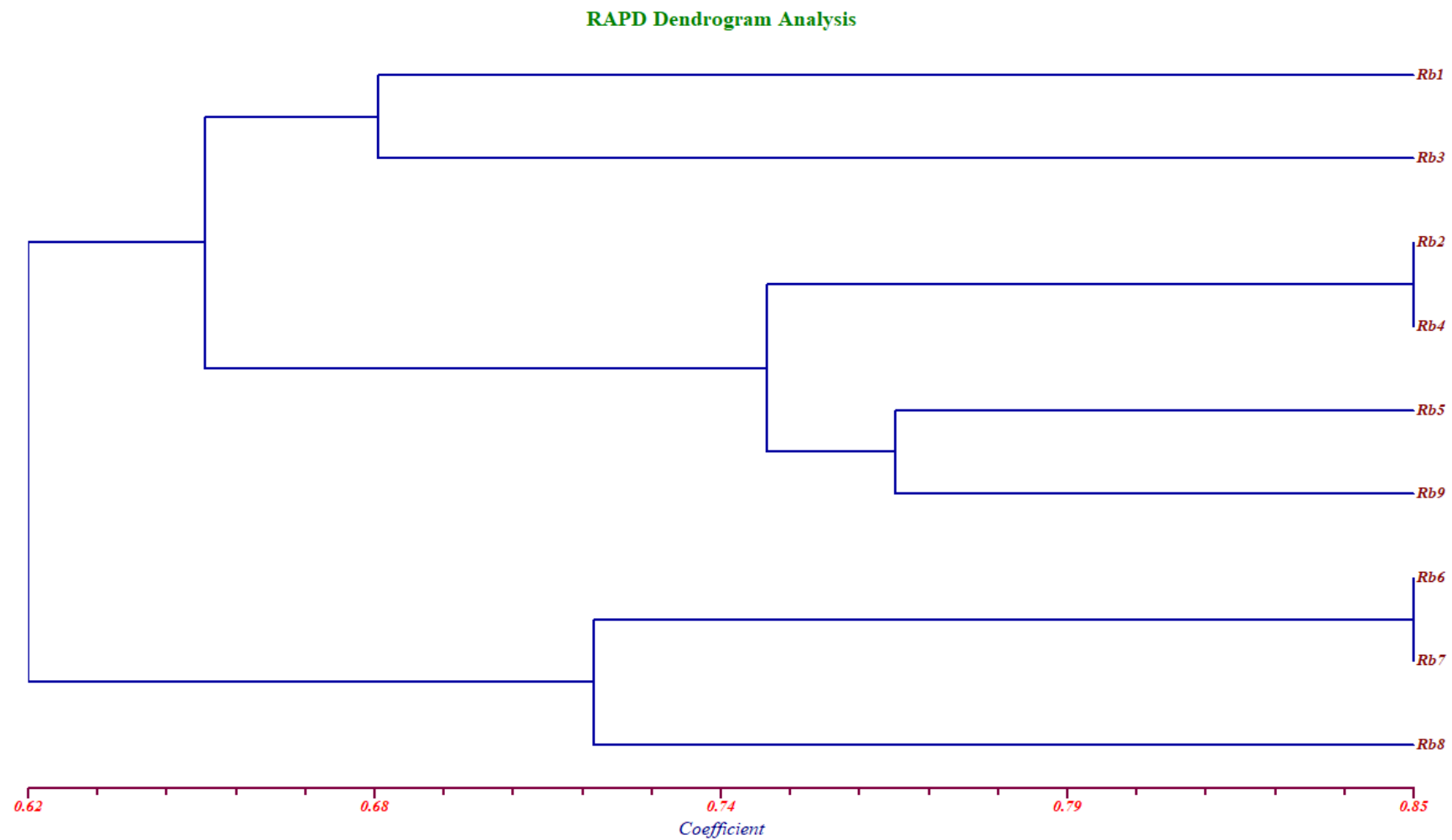


Figure 12: Dendrogram generated based on RAPD primer data through UPGMA analysis

several workers (Shanmugam and Govindaswamy, 1973; Hooda and Grover (1983); Diazfranco, 1984; Mehta and Gupta, 1992; Priya Santosh, (2008); Kumar and Gaur, 2010)

4.7.2. Effect of organic and inorganic sources of nitrogen

Table 14. Effect of various nitrogen sources on growth and Sclerotial production of *R. bataticola*

Sr. No.	Nitrogen Sources	Mean colony diameter (mm)*	Colony colour	Sclerotial production
T ₁	Calcium nitrate	62.24	Black	++
T ₂	Sodium nitrate	86.51	Whitish black	++++
T ₃	Ammonium nitrate	50.42	Whitish	++
T ₄	Mg nitrate	60.00	Grey	+++
T ₅	Potassium nitrate	84.40	Whitish black	++++
T ₆	peptone	66.60	Whitish Cottony	++++
T ₇	Control	15.33	Black	-
	SE± (m)	1.26		
	CD at 1 % (m)	5.22		

++++ Excellent, +++ Good, ++ Fair, + Poor, - No.

In the present study utilization of different nitrogen compounds (Table 14, Fig. 14 & PLATE XIV) by *R. bataticola* was recorded and it was observed that sodium nitrate was found to be most suitable one encouraged maximum radial mycelial growth (86.51 mm). The second best next nitrogen source was found to be Potassium nitrate (84.40 mm), followed by Peptone (66.60 mm), and Mg nitrate (60.00 mm).

All sources of nitrogen tested showed a wide range of sporulation from poor (+) to excellent (++++). However, nitrogen sources viz., Sodium nitrate, Potassium nitrate and Peptone showed excellent (++++) sporulation. Whereas in rest of the nitrogen sources sporulation was observed from good (+++) to Poor (+).

The result in the present study is in accordance with the finding of Salunkhe *et al.*, (2009) reported that among six carbon sources, highest growth was obtained in dextrose followed by sucrose, mannitol, starch, maltose and cellulose. Sclerotial production in cellulose was maximum followed by dextrose, sucrose, maltose, starch while, it was minimum in mannitol. Sodium

nitrate proved better nitrogen source for growth and sclerotial formation than ammonium nitrate. The sources of nitrogen viz., potassium nitrate and sodium nitrate, ammonium nitrate and peptone were reported to support maximum growth and sporulation in *M. phaseolina* earlier as reported by several workers Shanmugam and Govindaswamy (1973), Lakhpale *et al.*, (1995); Khanzada *et al.*, (2003); Sharma and Tripathi (2002); Sharma and Vishnupriya (2006); Kumar and Gaur, (2010) Thombre & Kohire (2018).

4.8. Effect of soil moisture regimes and soil types on the incidence of *Rhizoctonia bataticola*. (Polybag culture)

4.8.1. Effect of soil moisture regimes on the incidence of *Rhizoctonia bataticola*. (Polybag culture)

This disease is mainly affected by the climatic factors i.e. soil moisture depletion and soil temperature. As these factors play an important role in the development and initiation of this disease. Effect of various soil moisture levels on dry root rot incidence was tested by applying sick soil technique and sowing the seeds of susceptible cultivar of chickpea cv.JG-62, in Pot/Polybag culture under screen house conditions. The results obtained on the per cent disease incidence at 15, 30 and 45 days after sowing (DAS) are presented in the Table 15 and represented in PLATE XV and Fig.15.

4.8.1. Effect on dry root rot incidence

The results (Table15, PLATE XV and Fig.15) indicate that the disease incidence was found to be influenced by the soil moisture regimes obtained at various intervals on the effect of soil moisture regimes on the incidence of dry root rot disease (*R.bataticola*) in chickpea seed cv.JG-62.

At 15 DAS, the dry root rot disease incidence was in the range of 6.67 to 100.00 per cent, with none of the incidence at 100 per cent soil moisture level. However, the disease incidence at 35 and 40 per cent soil moisture level was significantly highest (each 100.00%), which may be due to physiological stress coupled with *R.bataticola*. These were followed by the soil moisture

PLATE XIII



Sr. no.	Carbon Sources
T ₁	Dextrose
T ₂	Sucrose
T ₃	Mannitol
T ₄	Starch
T ₅	Cellulose
T ₆	Maltose
T ₇	Control

Effect of various Carbon sources on growth and Sclerotial production of *R. bataticola*

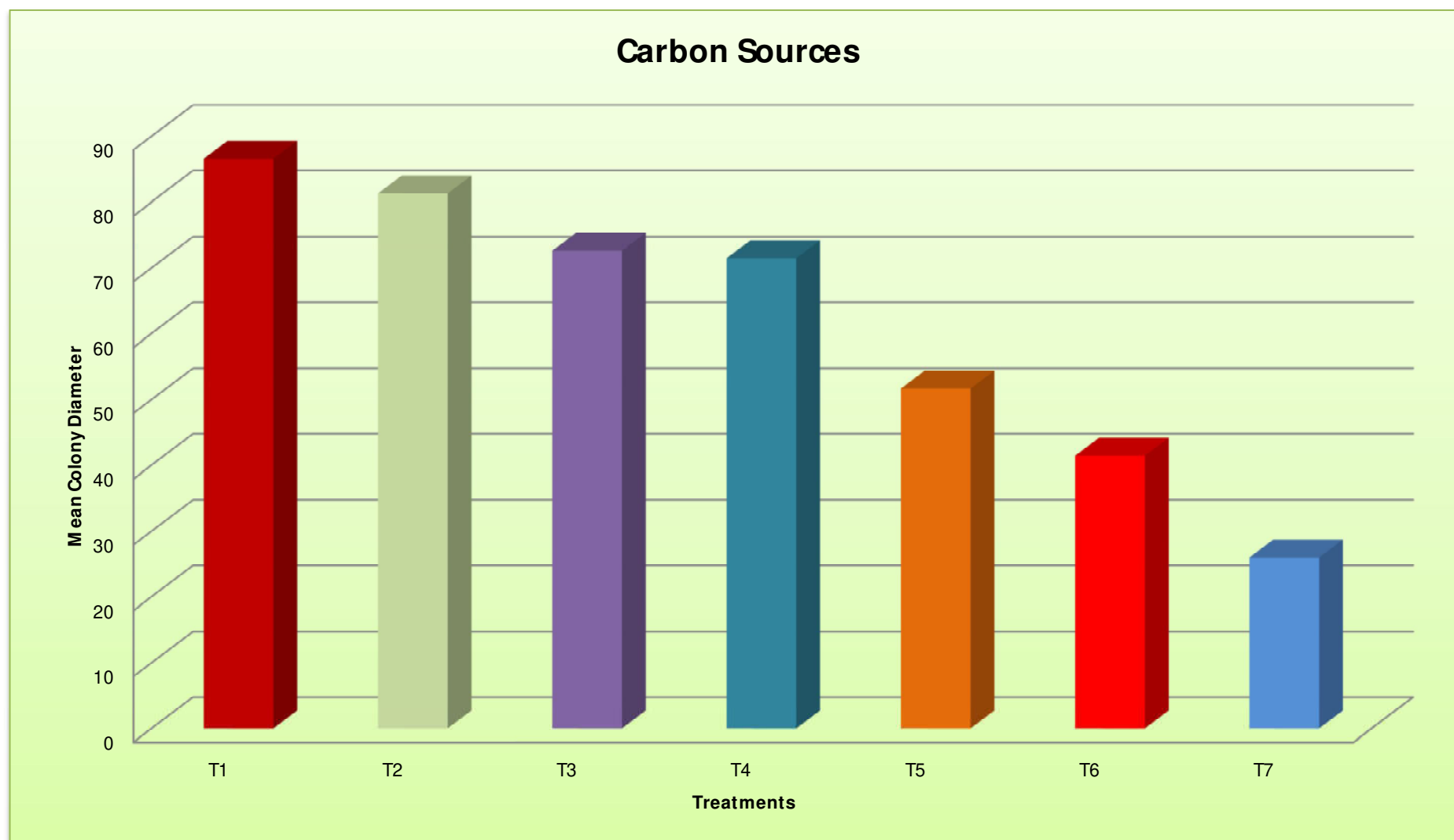


Fig. 13. Effect of organic and inorganic sources of carbon

regimes viz., 55 per cent (43.33%). At the soil moisture level above 60 per cent, the disease incidence was observed to be decreased in the range of 36.67 to 8.33 per cent.

Table 15. Effect of soil moisture regimes on dry root rot disease incidence in chickpea cv. JG-62 (Polybag culture)

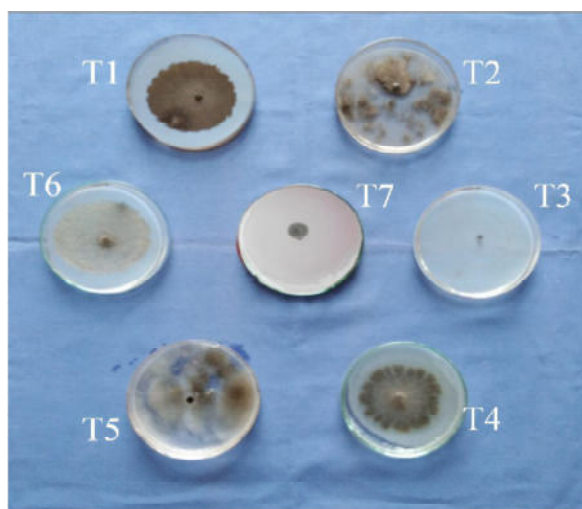
Tr. No.	Moisture level	Disease incidence (%)*			Average Disease Incidence
		15	30	45	
T1	35%	100 (90)	100 (90)	100 (90)	100 (90)
T2	40%	100 (90)	100 (90)	100 (90)	100 (90)
T3	55%	46.67 (43.08)	98.33 (85.69)	100 (90)	81.67 (71.89)
T4	60%	36.67 (37.25)	86.67 (68.66)	100 (90)	74.45 (65.28)
T5	75%	13.33 (21.33)	18.33 (25.31)	23.33 (28.78)	18.33 (25.22)
T6	80%	8.33 (16.59)	11.67 (19.89)	16.67 (24.05)	12.22 (20.28)
T7	100%	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)
S.E. $\pm$		1.03	1.86	0.96	
C.D. (P=0.05)		3.11	5.66	2.92	

*-Mean of three replications, DAS- days after sowing,
Figures in parentheses are arc sine transformed values.

At 30 DAS similar trend was observed and it was ranged from 16.67 to 100.00 per cent, as against no incidence at 100 per cent soil moisture regime. However, it was significantly highest at 35 and 40 per cent soil moisture (each 100.00%), which may be due to physiological stress coupled with *R.bataticola* infection. These were followed by the soil moistures viz., 55 per cent (96.67%). At soil moisture level above 60 per cent, the disease incidence was observed to be decreased in the range of 86.67 to 11.67 per cent.

At 45 DAS, all the soil moisture levels regimes tested showed same trend of incidence of dry root rot as that of observed at 15 and 30 DAS and it was observed to be ranged from 0.00 to 100.00 per cent, as against no incidence at 100 per cent soil moisture regime. However, the disease incidence at 35, 40, 55 and 60 per cent soil moisture was significantly highest (each

PLATE XIV



Sr. no.	Nitrogen Sources
T ₁	Calcium nitrate
T ₂	Sodium nitrate
T ₃	Ammonium nitrate
T ₄	Mg nitrate
T ₅	Potassium nitrate
T ₆	peptone
T ₇	Control

Effect of various nitrogen sources on growth and Sclerotial production of *R. bataticola*

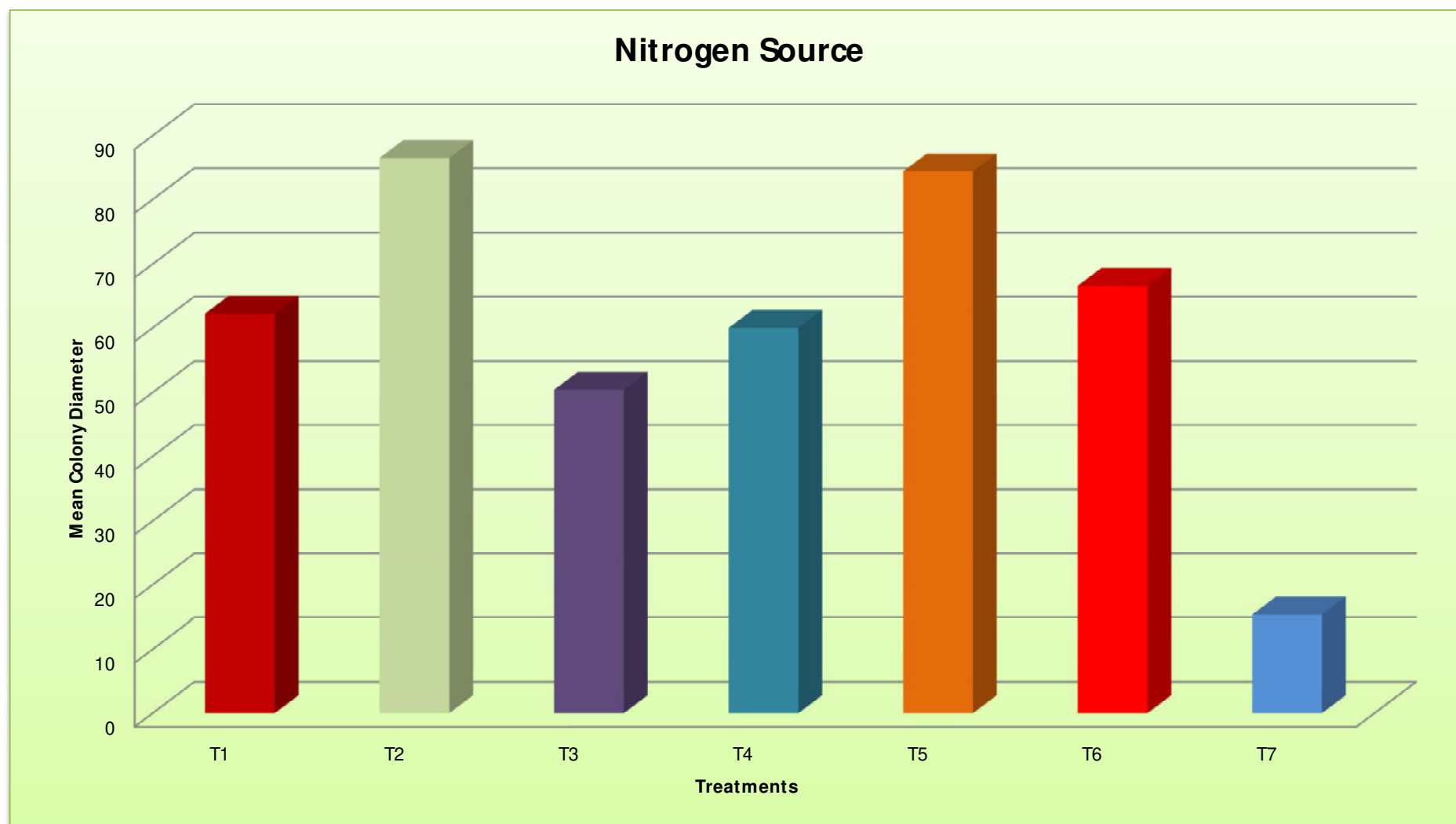


Fig. 14. Effect of organic and inorganic sources of Nitrogen

100.00%), which may be due to physiological stress coupled with *R.bataticola* infection. These were followed by the soil moisture viz., 75 per cent (23.33%). At soil moisture of 80 per cent, the disease incidence was found minimum with 16.67 per cent.

The average disease incidence noted at various soil moisture levels was ranged from 12.22 to 100.00 per cent, as against no incidence at 100 per cent soil moisture. However, average incidence at 35, and 40 per cent soil moisture was significantly highest (each 100.00%), which may be due to water stress coupled with *R.bataticola* infection. These were followed by the soil moisture level viz., 55 per cent (81.67%). At soil moisture level above 60 per cent, average incidence was found to be decreased in the range of 74.45 to 12.22 per cent.

The environmental factors such as temperature and soil moisture plays major role in growth and development of the chickpea crop to dry root rot disease incidence. Of these two major factors, moisture stress i.e.moisture depletion at the flowering stage of the crop was to be most important predisposing factor. Result obtained from this study also indicated that the average dry root rot disease incidence was also higher (range: 81.67 to 74.45 per cent) at the soil moisture level of 55 to 60 per cent.

Similar observations were recorded by Ratnoo *et al.*, (1997) observed that disease development was low in flooded soil compared to drier soil (40 to 60% moisture). Sharma and Pande (2013) also observed difference in the dry root rot incidence with change in soil moisture content. The plants grown in control showed the physiological stress at 40% moisture as compared to 60%, 80% and 100%. 40% soil moisture was insufficient for the normal growth of the plants. At 60% soil moisture, no physiological stress was found in control plants and dry root rot incidence was 100%. Symptoms of dry root rot on aerial plant parts were found to be directly related with the disease severity on roots. Blackening of the roots initiated 5 days after maintaining the

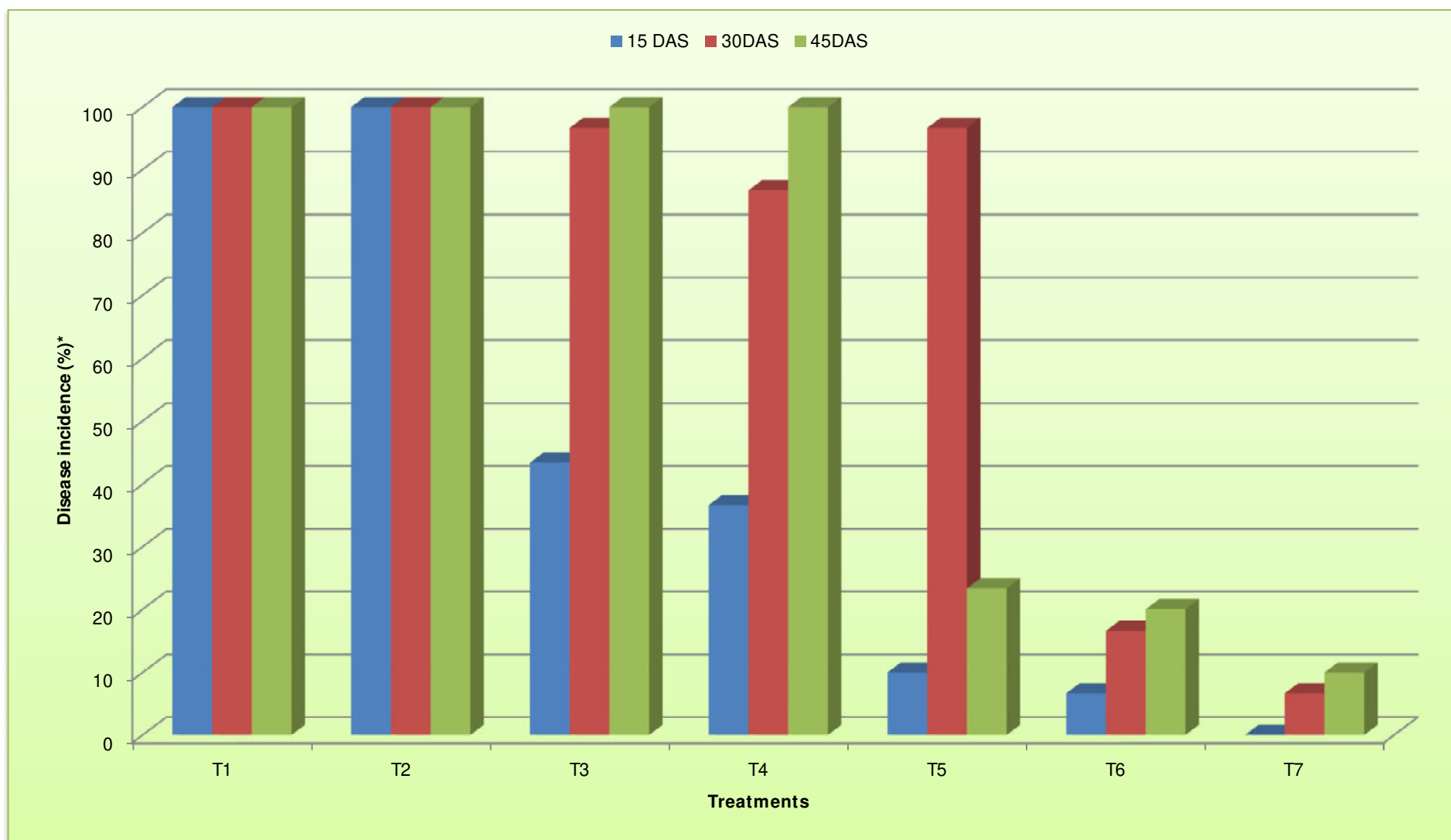


Fig. 15. Effect of soil moisture regimes on the incidence of *R. bataticola*.

moisture stress at 40% and 60%. Roots were apparently free from infection at 80% and 100%.

4.8.2. Effect of soil types on the incidence of *Rhizoctonia bataticola*. (Pot culture)

Results (Table16, PLATE XVI & Fig. 16) elucidated that, the disease was significantly maximum in sandy soil (81.85%) followed by Sandy loam soil (71.85), Red soil (66.67) and Black soil (63.33). Comparatively minimum dry root rot incidence was recorded in Clay soil (48.89%).

Table 16. Effect of Different soil types on development of dry root rot incidence of chickpea

Tr. No.	Treatments	PDI Mean
T1	Black Soil	63.33 (52.73)
T2	Red Soil	66.67 (54.73)
T3	Clay Soil	48.89 (44.36)
T4	Sandy Soil	81.85 (64.78)
T5	Sandy Loam Soil	71.85 (57.95)
	SE± (m)	2.08
	CD at 5% (m)	6.56

Results of this study are in accordance with the earlier findings of workers Sagar and Sugha (1998) who reported highest pea wilt incidence in silty loam soil (85.70 %), sandy loam soil (84.70 %) and comparatively minimum wilt incidence in clay soil (78.50 %). Chandrasekaran and Shanmugam (1984) studied the effect of soil type on wilt incidence on chickpea root rot (*Fusarium solani*) and reported highest disease incidence in sandy loam soil (96.30 %), followed by black soil (53.80 %) and red soil (35.00 %). Mallaiah and Krishna Rao (2016) reported that high incidence of dry root rot was in sandy loam (11.1%) followed by sandy clay loam (8.1%) and least in clay soils (4.3%). While studying the effect of soil types on the incidence of chickpea dry root rot diseases, it was noticed that sandy soil showed highest dry root rot disease incidence than the other soil types. Maheswari & Ramakrishnan (1999) reported that the survival of sclerotia of *R. bataticola* in

PLATE XV



35%



40%



55%



60%



75%



80%



100%

Effect of different soil moisture regimes on the incidence of
Rhizoctonia bataticola

black soil was affected by high soil moisture (80%), while the saprophytic activity of pathogen was maximum at low moisture levels (20, 30 and 40%). The adverse effect of high soil moisture on survival of sclerotia of *R. bataticola* was also reported by Olaya and Abawi (1996) indicating that *R. bataticola* cannot survive for a longer period under anaerobic conditions.

4.9. Disease management strategies

4.9.1. *In vitro* efficacy of systemic fungicides

4.9.1.1. Mycelial growth

Results (PLATE XVII, Table 17 and Fig. 17) elucidated that all six systemic fungicides tested possess a wide range of radial mycelial growth of *R. bataticola*.

Table 17. *In vitro* efficacy of systemic fungicides against *R. bataticola*

Tr. No.	Treatments	Colony Dia. *(mm) at ppm		Av. (mm)	% Inhibition* at ppm		Av. inhibition (%)
		500	1000		500	1000	
T ₁	Carbendazim 50% WP	0.00	0.00	0.00	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
T ₂	Metalaxyl	32.47	30.61	31.51	63.91 (53.07)	65.98 (54.31)	64.94 (53.69)
T ₃	Thiophanate methyl 70% WP	5.69	5.07	5.38	93.67 (75.42)	94.35 (76.24)	94.01 (75.83)
T ₄	Tebuconazole 29.9% EC	0.00	0.00	0.00	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
T ₅	Hexaconazole 5% EC	0.00	0.00	0.00	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
T ₆	Azoxystrobin	0.00	0.00	0.00	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
T ₇	Control (untreated)	90.00	90.00	90.00	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
	S.E.+	0.19	0.20		0.21	0.22	
	C.D.(P=0.01)	0.57	0.59		0.64	0.66	

*: Mean of three replications, Dia: Diameter, Av.: Average
Figures in parentheses are arcsine transformed values.

At 500 ppm, radial mycelial growth was in the range of 00.00 mm (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin) to 32.47

mm (Metalaxyl). However, no mycelial growth was observed with Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC and Azoxystrobin. The next fungicide Thiophanate methyl 70% WP has shown least mycelial growth (5.69 mm) over control followed by Metalaxyl (32.47 mm).

At 1000 ppm, similar trend was also observed as that of 500 ppm and radial mycelial growth was found in the range of 00.00 mm (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin) to 30.61 mm (Metalaxyl). However, the fungicides Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC and Azoxystrobin showed no mycelial growth. In case of the Thiophanate methyl 70 % WP the mycelial growth was found to be 5.07 mm followed by Metalaxyl (30.61 mm).

Average radial mycelial growth of the test pathogen was in the range of 00.00 mm (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin) to 31.51 mm (Metalaxyl). However, there was no average mycelial growth was found to be with Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin. On the other hand the next best fungicides were Thiophanate methyl (5.38 mm). followed by Metalaxyl (31.51 mm) over untreated control.

4.9.1.2. Mycelial inhibition

Results (Table 17 and PLATE XVII) elucidated that all the systemic fungicides tested (each @ 500 and 1000 ppm) significantly inhibit the mycelial growth of *R. bataticola*, over control (untreated). (Fig. 14).

At 500 ppm, per cent mycelial growth inhibition was in the range of 63.91 (Metalaxyl) to 100 (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin). However, Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC and Azoxystrobin recorded cent per cent (100 %) mycelial growth inhibition, followed by Thiophanate methyl 70 % WP (93.67 %).

PLATE XVI



**Effect of Different soil types on development of dry root rot
incidence of chickpea**

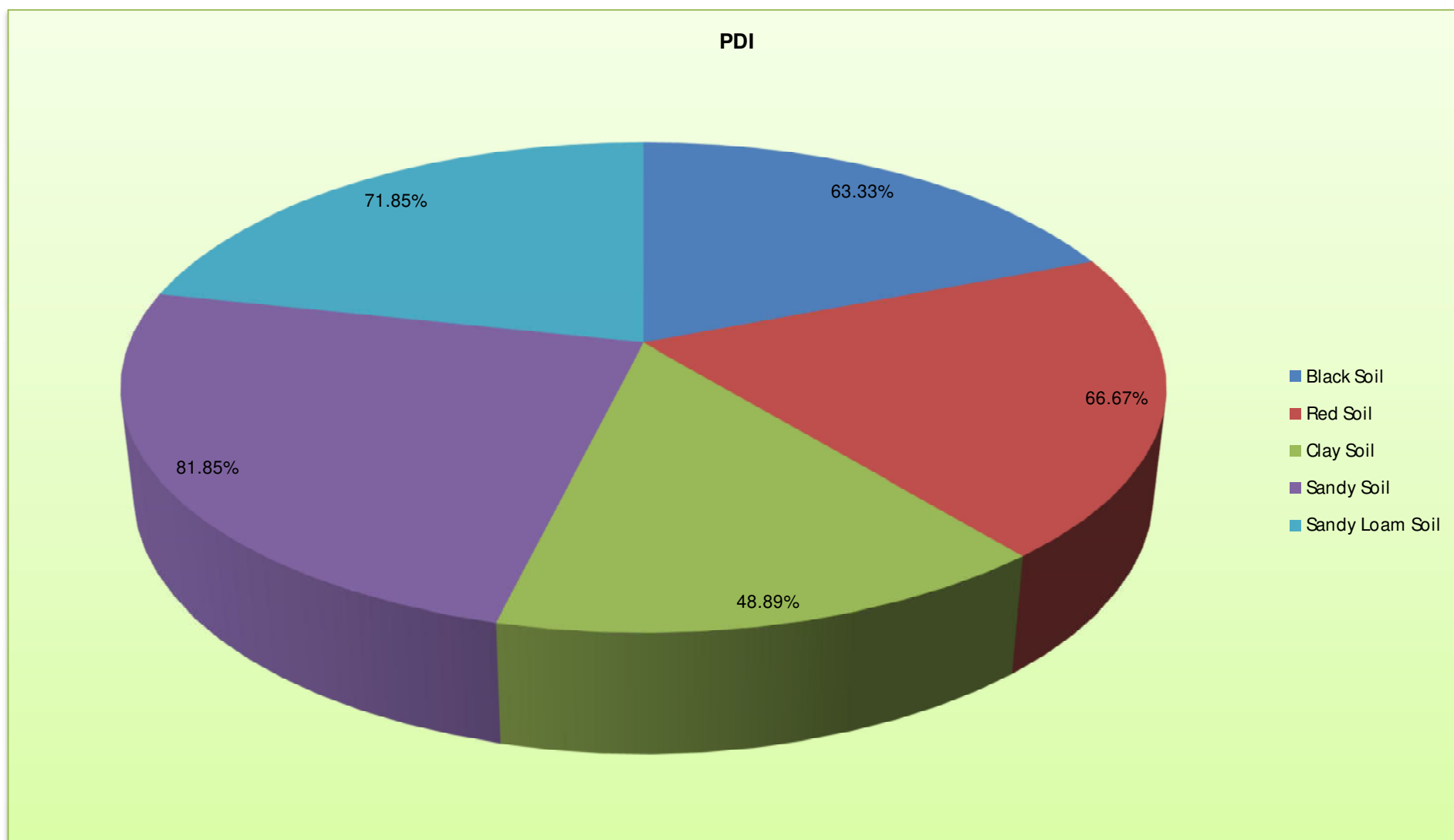


Fig. 16. Effect of Different soil types on development of dry root rot incidence of chickpea

At 1000 ppm, similar trend was observed as at 500 ppm and mycelial growth inhibition was in the range of 65.98 (Metalaxyl) to 100 (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin) per cent. However, Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC and Azoxystrobin recorded cent per cent (100 %) mycelial inhibition, followed by Thiophanate methyl (94.35 %).

Average mycelial growth inhibition noted with the test systemic fungicides was ranged from 64.94 (Metalaxyl) to 100 (Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin) per cent. However, it was found cent per cent with Carbendazim, Tebuconazole, Hexaconazole and Azoxystrobin (100 %), followed by Thiophanate methyl (94.01 %).

The general mode of action of systemic fungicides is associated with interference with the electron transport chain influencing the energy budget of the cell, reduction in the biosynthesis of new cell material required for growth and development of the organism and disruption of cell structure and permeability of cell membrane (Ravichandran and Hegde, 2017).

Thus, all of the systemic fungicides evaluated were found fungistatic against *R. bataticola* and significantly inhibited its mycelial growth, over untreated control. However, the systemic fungicides found most effective were Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC, Azoxystrobin, Thiophanate methyl 70 % WP and Metalaxyl in this order of merit.

4.9.2. *In vitro* evaluation of contact and combi- fungicides

4.9.2.1. Radial mycelial growth

Results (PLATE XVIII, Table 18 and Fig. 18) elucidated that all of the six contact and combi- fungicides tested showed a great range of radial mycelial growth of *R. bataticola* and was decreased with increase in the concentrations of the test fungicides from 2000 to 2500 ppm.

PLATE XVII



500 PPM



1000PPM

Tr. No	(Systemic)Treatments
T ₁	Carbendazim 50WP
T ₂	Metalaxyl (Apron 35 SD)
T ₃	Thiophanate methyl (Topsin M 70WP)
T ₄	Tebuconazole 29.9% EC
T ₅	Hexaconazole (Contaf 5 EC)
T ₆	Azoxystrobin (Amistar)
T ₇	Control (untreated)

In vitro* evaluation of systemic fungicide against *R. bataticola

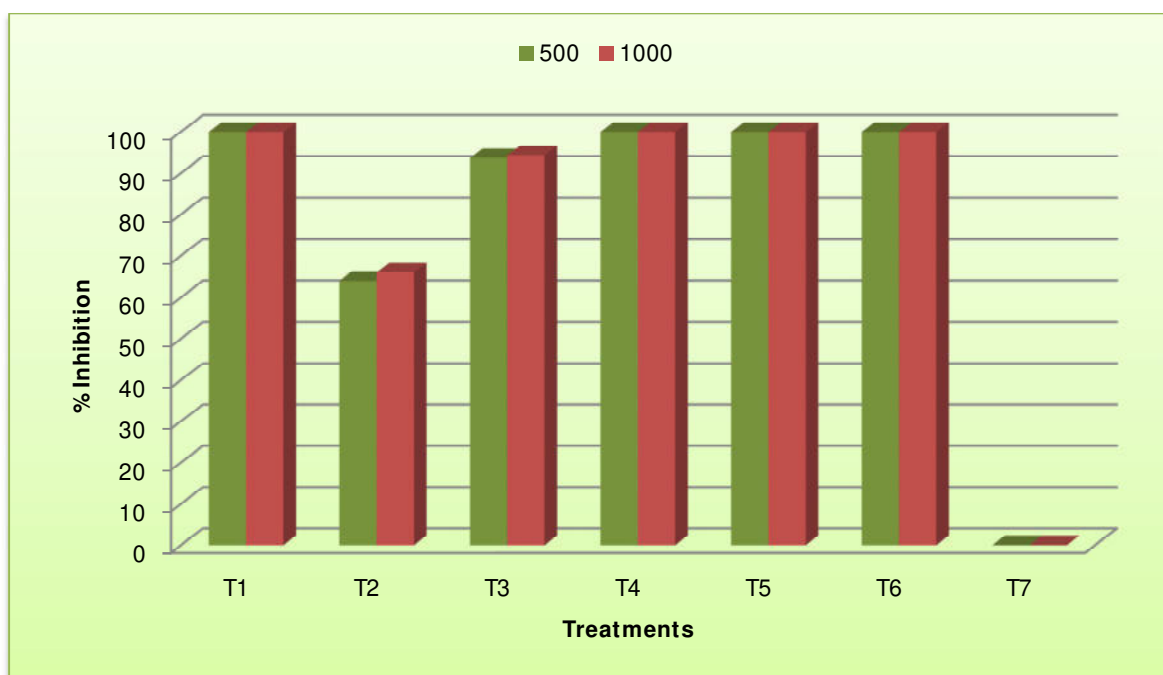
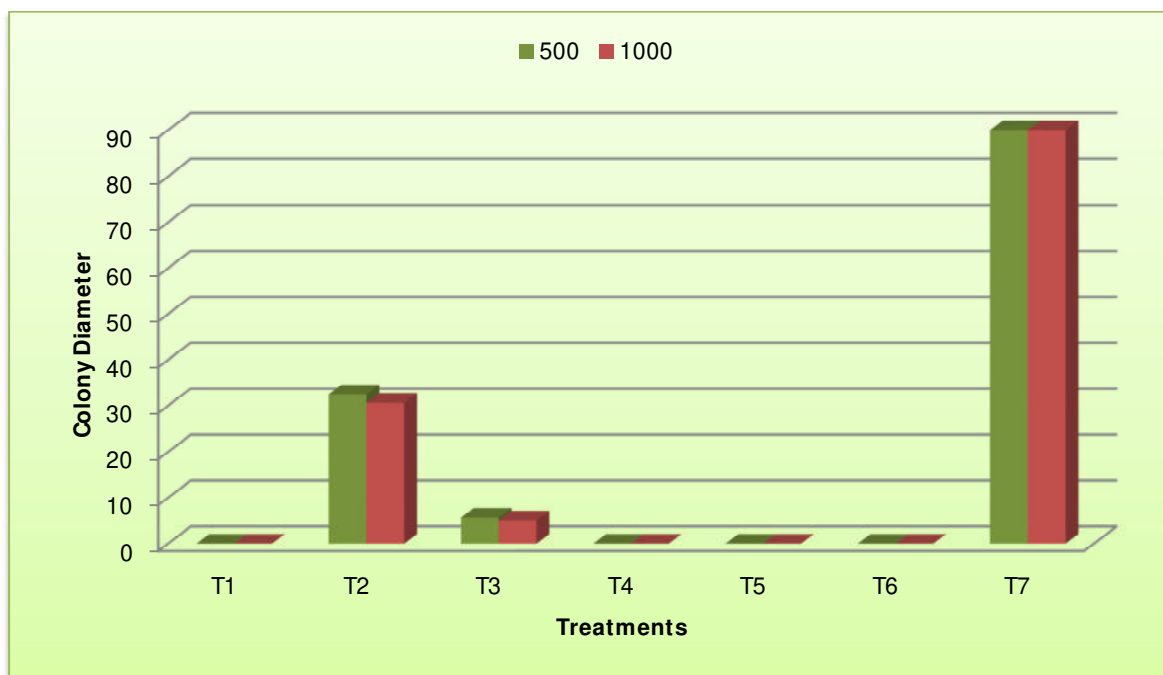


Fig. 17. *In vitro* bio-efficacy of systemic fungicides against *R. bataticola*

At 2000 ppm, no radial mycelial growth was observed with Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP. However, it was minimum with Cymoxanil 8 % + Mancozeb 64 % (5.05 mm), followed by Captan 50 % WP (6.86 mm), Copperoxy chloride 50 % WP (11.14 mm) and Mancozeb (17.79 mm) as compared to untreated control.

Table18. *In vitro* efficacy of contact fungicides against *R. bataticola*

Tr. No.	Treatments	Colony Dia. *(mm) at ppm		Av. (mm)	% Inhibition* at ppm		Av. inhibition (%)
		2000	2500		2000	2500	
T ₁	Copperoxy chloride 50% WP	11.14	6.77	8.07	87.61 (69.39)	92.47 (74.07)	90.04 (71.60)
T ₂	Carbendazim 12% + Mancozeb 63 %	0.00	0.00	0.00	100.00 (90)	100.00 (90)	100.00 (90)
T ₃	Corboxin 37.5 % + Thiram 37.5 %	0.00	0.00	0.00	100.00 (90)	100.00 (90)	100.00 (90)
T ₄	Mancozeb	17.79	15.54	16.66	80.22 (63.59)	82.73 (65.44)	81.47 (64.50)
T ₅	Captan 50% WP	6.86	5.07	5.96	92.37 (73.96)	94.36 (76.26)	93.36 (75.06)
T ₆	Cymoxanil 8 % + Mancozeb 64 %	5.05	5.00	5.02	94.38 (76.28)	94.44 (76.36)	94.41 (76.32)
T ₇	Control (untreated)	90.00	90.00	90.00	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
	S.E.±	0.53	0.67		0.59	0.75	
	C.D.(P=0.01)	1.62	2.04		1.80	2.27	

*: Mean of three replications, Dia: Diameter, Av.: Average
Figures in parentheses are arcsine transformed values

At 2500 ppm, mean mycelial growth of the test pathogen was in the range of 0.00 mm (Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP) to 15.54 mm (Mancozeb). However, it was significantly minimum with Cymoxanil 8 % + Mancozeb 64 % (5.00 mm), followed by Captan 50 % WP (5.07 mm) and Copperoxy chloride 50 % WP (6.77 mm).

Average radial mycelial growth of the test pathogen was in the range of 0.00 mm (Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP) to 16.66 mm (Mancozeb). However, it was no mycelial growth was recorded with Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP, followed by Cymoxanil 8 % + Mancozeb 64 % (5.02 mm), Captan 50 % WP (5.96 mm) and Copperoxy chloride 50 % WP (8.07 mm).

4.9.2.2. Mycelial inhibition

Results (PLATE XVIII and Table 18) elucidated that all contact and combi- fungicides evaluated (each @ 2000 and 2500 ppm) significantly inhibited mycelial growth of *R. bataticola*, over control. Further, per cent mycelial growth inhibition was increased with increase in concentrations of the fungicides tested (Fig.18).

At 2000 ppm, mycelial growth inhibition was in the range of 80.22 (Mancozeb) to 100 (Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP) per cent. However, it was cent per cent (100 %) with Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP followed by Cymoxanil 8 % + Mancozeb 64 % (94.38), Captan 50 % WP (92.37 %) and Copperoxy chloride 50 % WP (87.61 %).

At 2500 ppm, mycelial growth inhibition was in the range of 82.73 (Mancozeb) to 100 (Carbendazim 12 + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP) per cent. However, it was cent per cent (100 %) with Carbendazim 12 + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP and significantly highest with Cymoxanil 8 % + Mancozeb 64 % (94.44 %), followed by Captan 50 % WP (94.36 %) and Copperoxy chloride 50 % WP (92.47 %).

Average mycelial growth inhibition recorded with the test contact and combi- fungicides was in the range of 81.47 (Mancozeb) to 100 (Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram

37.5 % WP) per cent. However, it was cent per cent (100 %) with Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP followed by Cymoxanil 8 % + Mancozeb 64 % (94.41%), Captan 50 % WP (93.36 %) and Copperoxy chloride 50 % WP (90.04 %).

The combi product fungicides avoid the development of resistance by pathogens to systemic fungicides because these systemic fungicides interfere with only one or sometimes two functions in fungal physiology which is easily overcome by either a single mutation or by selection of resistant individuals in a population. Wherein non-systemic protectant fungicides affect too many functions in fungal physiology and to develop resistance the fungus will have to make too many gene changes. Hence the combination of both systemic and non-systemic fungicides provides better management of plant fungal disease for long duration (Ravichandran and Hegde, 2017).

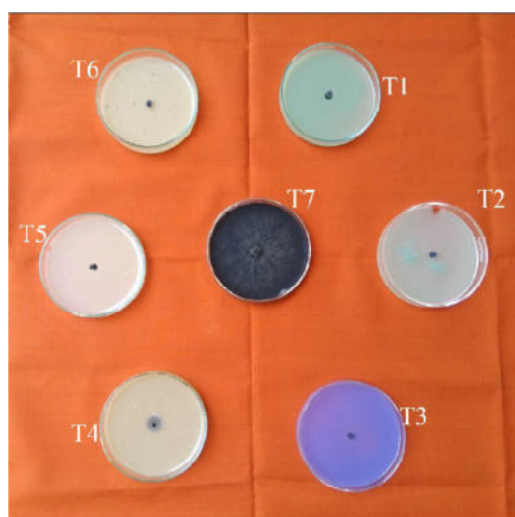
Thus, all of the non-systemic / contact and combi-fungicides evaluated were found fungistatic against *R. bataticola* and significantly inhibited mycelial growth, over untreated control. However, most effective fungicides were obtained in the order of merit viz., Carboxin 37.5 % + Thiram 37.5 % WP, Carbendazim 12 % WP + Mancozeb 63 % WP, Cymoxanil 8 % + Mancozeb 64 %, Captan 50 % WP and Copperoxy chloride 50%WP.

Results of this study on *in vitro* efficacy of systemic fungicides (Carbendazim 50 % WP, Hexanconazole 5 % EC, Azoxystrobin and Thiophanate methyl 70% WP) as well as contact (Cymoxanil 8 % + Mancozeb 64 % , Captan 50 % WP, and Copperoxy chloride 50 % WP) and combi-fungicides (Carbendazim 12 % + Mancozeb 63 % WP, Carboxin 37.5 % + Thiram 37.5 % WP) are in accordance with the earlier findings of those workers who reported these fungicides had significantly inhibited mycelial growth of *R. bataticola* / *M. phaseolina*, causing dry root rot of chickpea as well as many other crops.

PLATE XVIII



2000PPM



2500PPM

Tr. No	(Contact and Combi-fungicides)Treatments
T ₁	Copperoxy chloride 50WP
T ₂	Carbendazim 12% + Mancozeb 63% (SAFF 75WP)
T ₃	Vitavax power (Carboxin 37.55+ Thiram 37.5%)
T ₄	Mancozeb (Dithane M-45 WP)
T ₅	Captan 50WP
T ₆	Curzet (Cymoxanil 8% + Mancozeb 64%)
T ₇	Control (untreated)

***In vitro* evaluation of non systemic/ contact fungicide against
*R. bataticola***

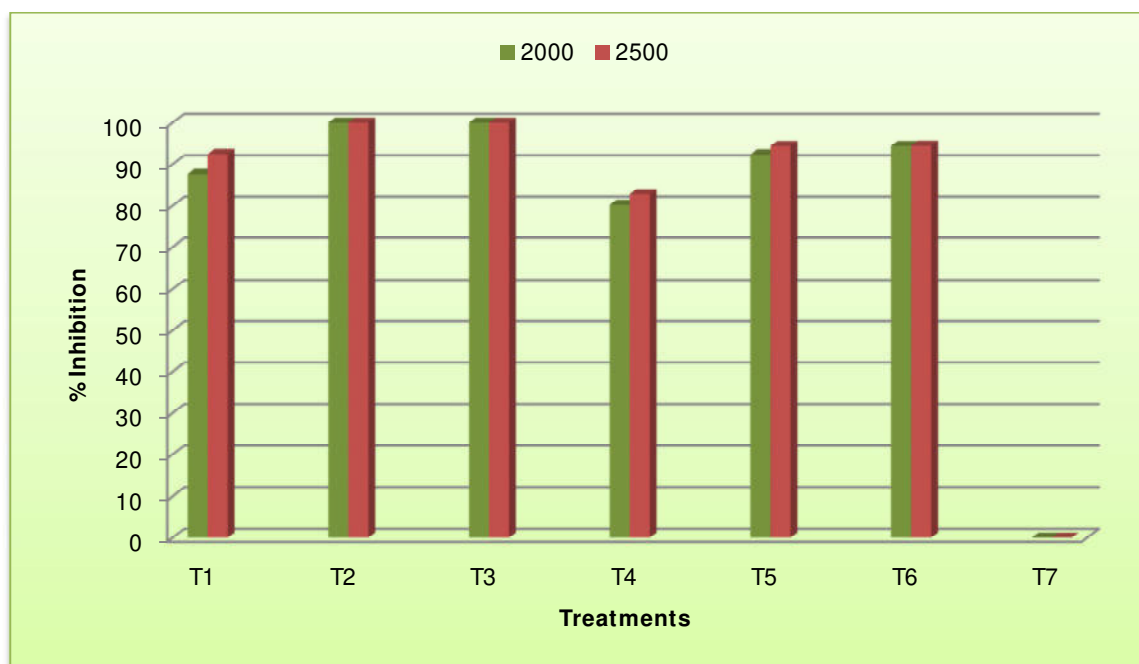
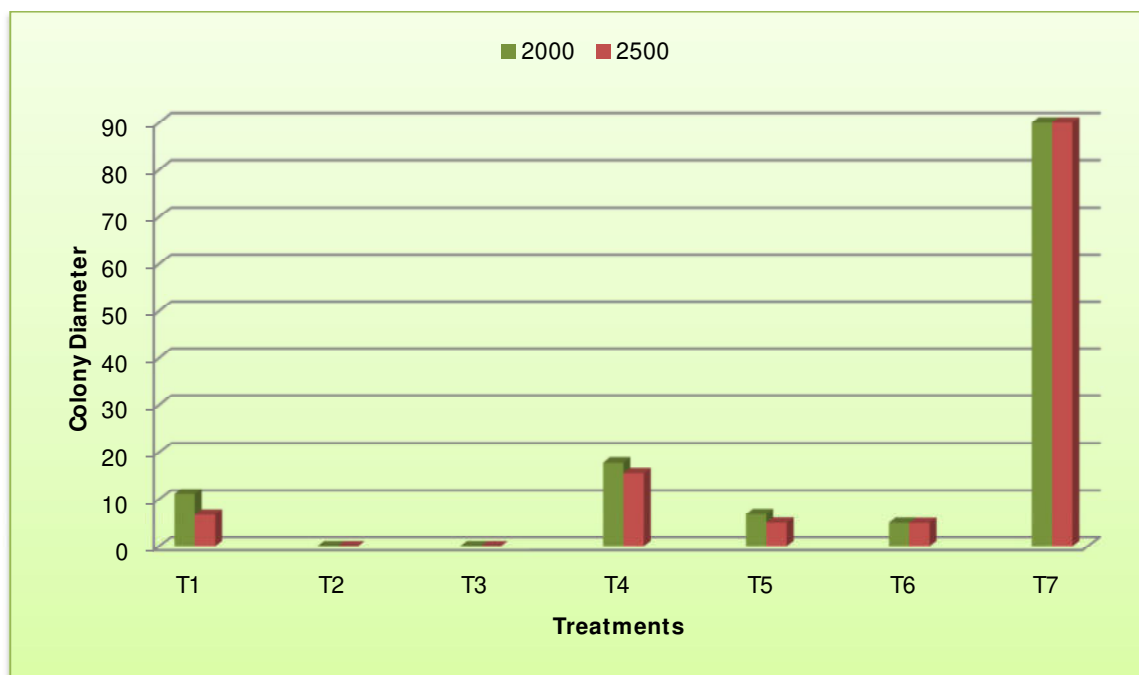


Fig. 18 *In vitro* bio-efficacy of Non systemic/Contact fungicides against *R. bataticola*

Malathi and Doraisamy (2003) found that the fungicides *viz.*, Carbendazim 50 % WP, Captan 50 % WP and Thiram 75 % WP, at various concentrations resulted with significant mycelial growth inhibition (65.5-100 %) of *M. phaseolina*. Khonde *et al.*, (2008a) observed Carbendazim 50 % WP (0.1%), Thiram 75 % WP (0.3%), Mancozeb 75 % WP (0.25%), Thiophanate methyl 70 % WP (0.1%) and Penconazole 50 % EC (0.1%) as most effective in inhibiting mycelia growth of *R. bataticola*, causing root rot of chickpea. Khan *et al.* (2012) reported that the fungicides Mancozeb 75 % WP, Carbendazim 50 % WP, Benomyl 50 % WP, Copperoxy chloride 50 % WP (each @ 0.2 %) as effective against *R. bataticola*, causing dry root rot of chickpea. Sangeetha and Jahagirdar (2013b) evaluated that the fungicides Carbendazim 50 % WP, Thiophanate methyl 70 % WP caused 100 per cent mycelial growth inhibition of *R. bataticola*, causing root rot of chickpea; whereas, Hexaconazole 5 % EC, Propiconazole 25 % EC, Carbendazim 25 % + Mancozeb 50 % WP, Carboxin 37.5 % + Thiram 37.5 % WP and Mancozeb 75 % WP caused 91.60 to 96.41 per cent mycelial growth inhibition. Veena *et al.* (2014a) reported that the fungicides Copper oxychloride, Captan, Hexaconazole and Tebuconazole were found to be highly effective (100%) in inhibiting the mycelial growth of the highly virulent pathogen *R. bataticola*, causing root rot of chickpea @ 1000, 1500, 2000 and 2500 ppm. Pawar *et al.*, (2014) reported that Carbendazim and Azoxystrobin as most effective with per cent mycelial growth were best (100.00%), tested each @ 250, 500, 1000 ppm; whereas least per cent inhibition by Copperoxy chloride was found ineffective.

Maruti *et al.*, (2017b) investigated that the systemic fungicides *viz.*, Tebuconazole 5 % EC (100 %), Propiconazole 25 % EC, Hexaconazole 5 % EC, Difenoconazole 25 % EC, Carbendazim 50 % WP and Thiophanate methyl 70 % WP (each @ 500, 1000 and 1500 ppm) resulted with 100 per cent mean mycelia growth inhibition of *R. bataticola*, dry root rot of chickpea. The non-systemic fungicides *viz.*, Mancozeb 75 % WP, Thiram 80 % WP, Chlorothalonil 75 % WP and Captan 70 % WP (each @ 1000, 2000 and 3000 ppm) and combi fungicides *viz.*, Carbendazim 12% + Mancozeb 63% WP,

Trifloxystrobin 25% + Tebuconazole 50% EC and Carboxin 37.5% + Thiram 37.5% WP (each @ 1000, 2000 and 3000 ppm) were also effective against the test pathogen.

Ravichandran and Hegde (2017) conducted lab experiment to evaluate six combiprodukt, five contact and four systemic fungicides against *Rhizoctonia bataticola* causing dry root rot in chickpea. Among combiprodukt evaluated, Carbendazim 25% + Mancozeb 50% (Sprint), Carboxin 37.5% + Thiram 37.5% (Vitavax power 75% WP) and Carbendazim 12% + Mancozeb 63% (Saaf) were found to be most effective with complete inhibition of mycelial growth of *R. bataticola* at all the concentrations tested. Among the contact fungicides tested, Chlorothalonil and Mancozeb at 0.2% were effective in inhibiting cent per cent inhibition. Among the four systemic fungicides evaluated against *R. bataticola*, Carbendazim, Difenconazole and Tebuconazole were best with cent per cent inhibition of mycelial growth at all concentrations tested.

4.9.3. *In vitro* evaluation of bioagents

The results obtained on mycelial growth of the *R. bataticola* and its inhibition with the test bioagents represented in Table 19, PLATE XIX and Fig. 19.

Results (PLATE XIX and Table 19) elucidated that all of the biocontrol agents evaluated antifungal activity against *R. bataticola* and significantly inhibited its mycelial growth, over untreated control. Among the biocontrol agents tested, *T. asperellum* was found to be most effective and recorded significantly minimum colony diameter (10.55mm) with highest mycelial growth inhibition (88.27 %) of the test pathogen, followed by *T. harzianum* (14.94mm and 83.4 %), *T. hamatum* (27.07mm and 69.92%), *T. koningii* (30.73mm and 65.85 %) and *T. longibrachiatum* (33.07mm and 63.25%) of colony diameter and its mycelial growth inhibition respectively. *Bacillus subtilis* was found to be comparatively less effective with maximum

colony diameter and minimum per cent mycelial growth inhibition i.e. 41.33mm and 54.07per cent respectively .

Table 19. *In vitro* bioefficacy of bioagents against *R. bataticola*

Tr. No.	Treatments	Colony Dia. of test pathogen * (mm)	% Inhibition
T ₁	<i>Trichoderma asperellum</i>	10.55	88.27 (69.97)
T ₂	<i>T. harzianum</i>	14.94	83.4 (65.95)
T ₃	<i>T. virens</i>	37.64	58.17 (49.70)
T ₄	<i>T. koningii</i>	30.73	65.85 (54.24)
T ₅	<i>T. hamatum</i>	27.07	69.92 (56.73)
T ₆	<i>T. longibrachiatum</i>	33.07	63.25 (52.68)
T ₇	<i>Pseudomonas fluorescens</i>	40.24	55.28 (48.03)
T ₈	<i>Bacillus subtilis</i>	41.33	54.07 (47.33)
T ₉	Control (untreated)	90.00	0.00 (0.00)
	S.E. $\pm$	0.34	0.38
	C.D. (P=0.01)	1.01	1.13

*: Mean of three replications, Dia.: Diameter,

Figures in parentheses are arcsine transformed values

T. harzianum and *T. asperellum* both suppressed the growth *R. bataticola* that might be due to mutual intermingling of antagonistic isolates with the test pathogen (Maruti a/b *et al.*, 2017, Agale R.C., 2017)

The antifungal action employed by the fungal and bacterial antagonists against *R. bataticola* causing chickpea dry root rot, which may be through the various mechanisms such as competition, lysis, antibiosis, production of volatile / non-volatile substances and production of cellulolytic / pectolytic enzymes, by the antagonistic organisms. These results are in collateral with the earlier findings of those workers who reported that bioagents viz., *T. asperellum*, *T. harzianum* and *T. hamatum* had significantly inhibited mycelial growth of *R. bataticola* infecting chickpea crop. (Lambhate a/b, *et al.*, 2002; Patil, *et al.*, 2004; Patil, *et al.*, 2007; Manjunatha and Naik, 2010; Gaur and Sharma, 2010; Khan, *et al.*, 2012; Hegde, *et al.*, 2013; Wadhawe and Navgire 2014; Wagh, *et al.*, 2015; Maruti a/b *et al.*, 2017).

4.10. Bioefficacy of organic amendments

A total 10 organic amendments were tested as pre-sowing soil application to examine their efficacy against *R. bataticola*, by sick soil method, in polybags under screen house conditions. The results obtained on per cent pre-, post-emergence mortality and average mortality and reductions, over untreated control are represented in Table 20 PLATE XX and Fig.20.

4.10.1. Pre-emergence seed rot (PRESR)

The results (PLATE XX Table 20 & Fig. 20) elucidated that the test organic amendments significantly influenced pre-emergence seed rot (PRESR) which was in the range of 16.67 to 46.67 per cent, as against 73.33 per cent in control(untreated). However, it was significantly minimum with Neem seed cake (16.67 %), followed by Compost (20.00%), Vermicompost (23.33), Castor seed cake (26.67 %), Mustard cake (27.67 %), Karanj cake (28.67 %) and Safflower seed cake (38.33%). Whereas the rest of the amendments caused comparatively maximum PRESR in the range of 40.00 to 46.67 per cent, but it was significantly highest in the control (untreated) i.e.73.33 per cent.

Table 20. Efficacy of organic amendments against *R. bataticola*, causing dry root rot of chickpea

Tr. No.	Treatments	Disease Incidence (%) *		Av. Mor. (%)	Red. (%) over control		Av. Red. (%)
		PRESR	POESM		PRESR	POESM	
T ₁	Compost	20.00 (26.57)	23.11 (28.74)	21.56 (27.367)	72.62 (58.45)	76.89 (61.27)	74.75 (59.83)
T ₂	Vermicompost	23.33 (28.88)	25.71 (30.47)	24.52 (29.68)	67.86 (55.46)	74.29 (59.33)	71.07 (57.46)
T ₃	Poultry manure	46.67 (43.09)	48.81 (44.32)	47.74 (43.70)	45.24 (42.27)	55.56 (48.19)	50.40 (41.41)
T ₄	Groundnut cake	40.00 (39.23)	44.44 (41.81)	42.22 (40.52)	36.31 (37.05)	51.19 (45.68)	43.75 (45.23)
T ₅	Cotton seed cake	43.33 (41.17)	46.94 (43.25)	45.14 (42.21)	40.77 (39.68)	53.06 (46.75)	46.91 (43.23)
T ₆	Castor seed cake	26.67 (31.09)	28.69 (32.39)	27.68 (31.71)	63.10 (52.59)	71.31 (57.61)	67.20 (55.06)
T ₇	Neem seed	16.67	18.14	17.41	77.38	81.86	79.62

	cake	(24.09)	(25.21)	(24.66)	(61.60)	(64.79)	(63.16)
T ₈	Karanj cake	28.67 (32.37)	31.49 (34.13)	30.08 (30.08)	60.83 (51.25)	68.51 (55.86)	64.67 (53.13)
T ₉	Safflower cake	38.33 (38.25)	41.51 (40.11)	39.92 (39.18)	47.32 (43.46)	58.49 (49.89)	52.91 (46.67)
T ₁₀	Mustard cake	27.67 (31.73)	31.32 (34.03)	29.49 (32.89)	61.85 (51.85)	68.68 (55.97)	66.63 (54.71)
T ₁₁	Control (untreated)	73.33 (58.91)	100.00 (90.00)	86.67 (68.59)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
	SE $\pm$	1.75	0.34		2.45	0.34	
	CD (P=0.05)	5.12	1.00		7.19	1.00	

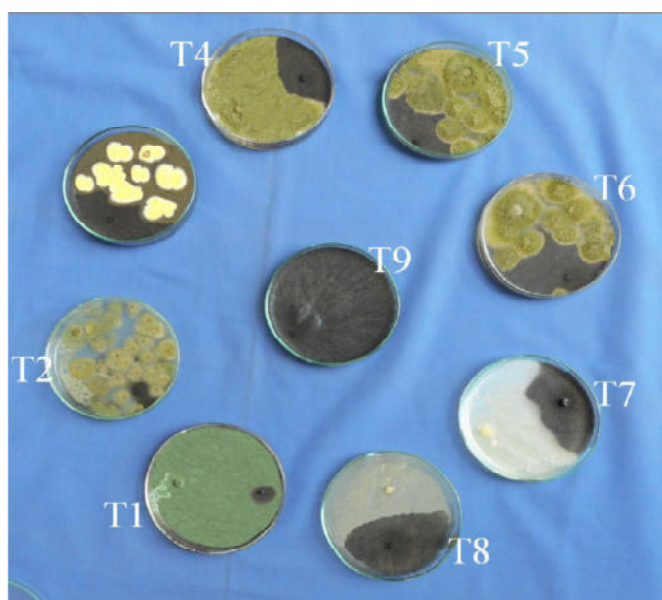
*-Mean of three replications, Figures in parentheses are arcsine transformed values, Av.: Average, Mor.: Concentration, Incr.: Increase Red.: Reduction, PRESR: Pre emergence seed rot, POESM: Post Emergence Seedling Mortality.

4.10.2. Post-emergence seedling mortality (POESM)

As that of pre-emergence mortality, similar type of trend was observed with respect to post-emergence seedling mortality (POESM), which was in the range of 18.14 to 48.81 per cent, as against 100.00 per cent in control(untreated). However, it was significantly minimum with Neem seed cake (18.17 %), followed by Compost (23.11%), Vermicompost (25.71), Castor seed cake (28.69 %), Mustard cake (31.32 %), Karanj cake (31.49 %) and Safflower seed cake (41.51%). Whereas in the rest of the organic amendments it caused comparatively maximum post emergence mortality which was in the range of 44.44 to 48.81 per cent, but it was significantly highest in control (untreated) 100.00 per cent.

Average mortality influenced by the test organic amendments was in the range of 17.41 to 47.74 per cent. However, it was significantly minimum with Neem seed cake (17.41 %), followed by Compost (21.56%), Vermicompost (24.52), Castor seed cake (27.68 %), Mustard cake (29.58 %), Karanj cake (30.08 %) and Safflower seed cake (39.92%). However in rest of the test organic amendments average mortality recorded in the range of 42.22 to 47.74 per cent, but it was significantly highest in control(untreated) 86.67 per cent.

PLATE XIX



Tr. No.	Treatments	Tr. No.	Treatments
T ₁	<i>Trichoderma asperellum</i>	T ₆	<i>T. longibrachiatum</i>
T ₂	<i>T. harzianum</i>	T ₇	<i>Pseudomonas fluorescens</i>
T ₃	<i>T. virens</i>	T ₈	<i>Bacillus subtilis</i>
T ₄	<i>T. koningii</i>	T ₉	Control (untreated)
T ₅	<i>T. hamatum</i>		

In vitro bioefficacy of bioagents against *R. bataticola*

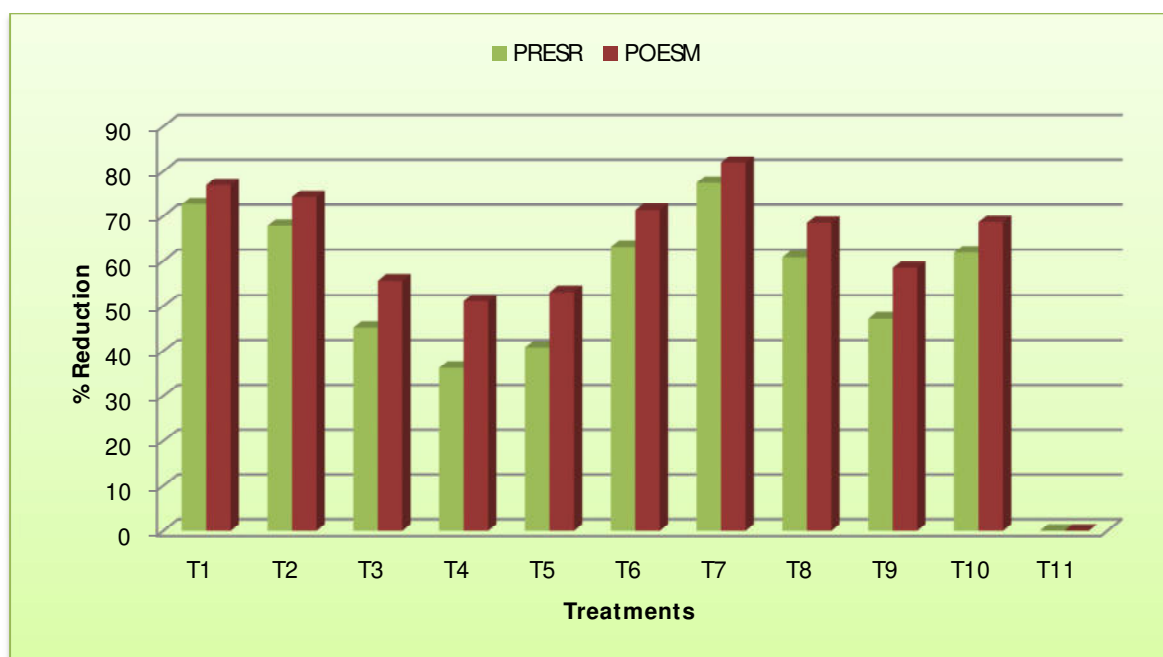
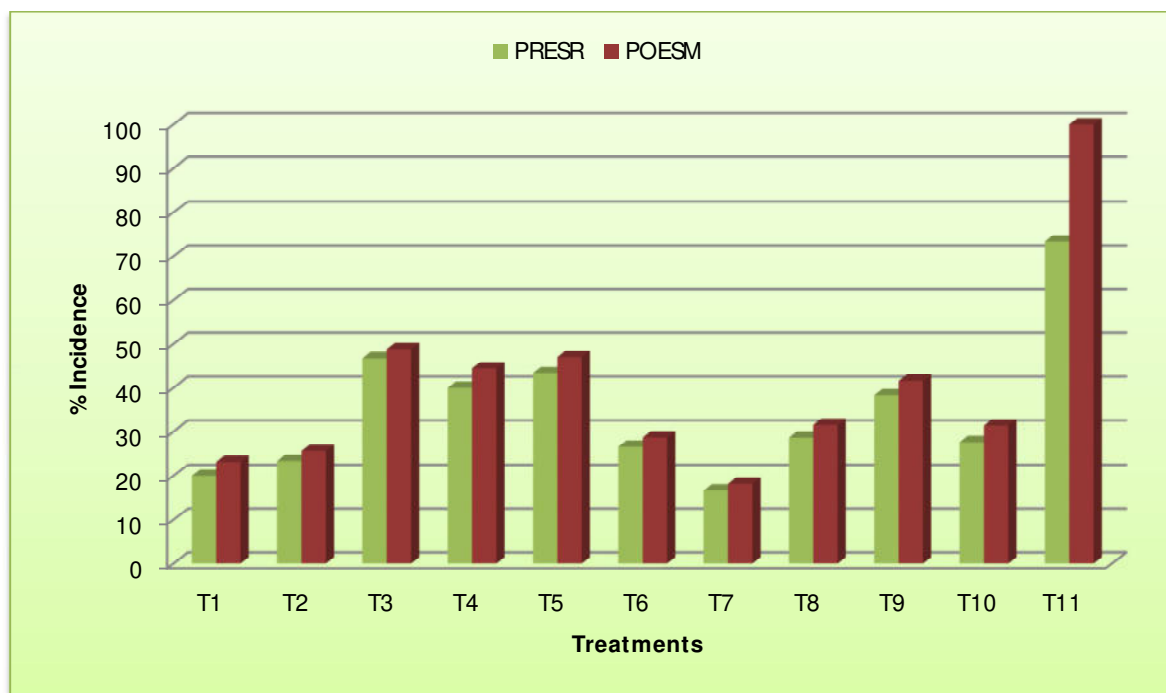


Fig. 20.Efficacy of organic amendments against *R. bataticola*

4.10.3. Reduction in mortality

All of the organic amendments evaluated (Table 20 and Fig. 20) were found to be effective in reducing the pre-emergence, post-emergence mortality and total mortality, over control (untreated).

The most effective organic amendment was Neem seed cake, with significantly highest reduction in PRESR (77.38 %), POESM (81.86 %) and their average reduction (79.62 %). This was followed by Compost (72.62%, 76.89% and 74.75%), Vermicompost (67.86%, 74.29% and 71.07%) Castor seed cake (63.10 %, 71.31 % and 67.20 %), Mustard cake (68.51 %, 64.76 % and 66.63 %), Karanj cake (68.68 %, 65.18 % and 64.67 %), and Safflower seed cake (47.32 %, 58.49 % and 52.91%). However in rest of the test organic amendments reduction was recorded in PRESR (range: 36.31 to 45.24 %), POESM (range: 51.19 to 55.56 %) and average mortality reduction was found to be in the range of 43.75 to 50.40 per cent.

In the present study organic amendments *viz.*, Neem seed cake resulted with significantly highest per cent reduction in pre-, post-emergence and total mortality induced by *R. bataticola* in chickpea. This may be due to Neem cake which contains azadirachtin, salannin, nimbin, azadiradione as the major component might be responsible to reduce the disease. It acts as a biofertilizer and helps in providing the required nutrients to plants. Neem seed cake performs the dual function, acts as a soil enricher, reduces the population of soil pest and bacteria, provides macro nutrients essential for all plant growth and these plant nutrients inhibit the growth of soil borne fungi (Pawar *et al.*, 2018).

These findings are in conformity with those reported earlier by several workers *i.e.* Hundekar *et al.*, (1998); Jaiman *et al.*, (2009); Dingani *et al.*, (2013); Pandhare *et al.*, (2015); Suryawanshi *et al.*, (2015) and Pawar *et al.*, (2018).

4.11. Disease management strategies

4.11.1. Integrated management of chickpea dry root rot, caused by *R. bataticola* (Pot/Polybag culture)

Those fungicides, bio-control agents and organic amendments found most effective during the various *in vitro* studies were integrated (alone and in combination) so as to manage chickpea dry root rot disease (*R. bataticola*), by applying sick soil method and by sowing susceptible Chickpea Cv. JG-62, pot / polybag culture, under screen house conditions. The results obtained on pre-emergence, post-emergence and average mortality are represented in the Table 21, PLATE XXI and Fig.21.

4.11.1.1. Per cent incidence of mortality (PRESR, POESM and Average)

With various treatments integrated, pre-emergence seed rot (PRESR), post-emergence seedling mortality (POESM) and average mortality incidence were observed to be in the range of 6.67 to 40.00, 9.03 to 43.24 and 7.85 to 41.62 per cent, respectively, as against 83.33, 100.00 and 91.67 per cent in control (untreated). However, the best effective treatment found was T₁₁ (Carboxin 37.5 % + Thiram 37.5 % WP (ST) @ 3.0 g + *T. asperellum* (ST) @ 10 g + NSC (SA) @ 50 g), with significantly minimum PRESR (6.67 %), POESM (9.03 %) and average mortality (7.85 %).

Table 21. Integrated efficacy of fungicides, bioagents, and organic amendments against *R. bataticola*, causing dry root rot of Chickpea (Pot/Polybag culture)

Tr. No.	Treatments	Dosages (g/kg seed or potting mixture)	Incidence (%) *		Av. Mor. (%)	Red. (%) over control		Av. Red. (%)
			PRESR	POESM		PRESR	POESM	
T ₁	Carboxin 37.5 % + Thiram 37.5 % (Vitavax power 75 % WP)	ST @ 3.0 g	28.33 (32.16)	32.14 (34.53)	30.23 (33.35)	65.97 (54.31)	67.87 (55.47)	66.92 (54.89)
T ₂	Carbendazim 50 % WP	ST @ 1.0 g	23.33 (28.88)	26.04 (30.68)	24.69 (29.79)	71.99 (58.05)	73.96 (59.32)	72.97 (58.67)
T ₃	Carbendazim 12 % + Mancozeb 63 % (SAAF 75 % WP)	ST @ 2.5 g	36.67 (37.27)	39.52 (38.95)	38.09 (38.11)	55.56 (48.79)	60.48 (51.05)	58.02 (49.62)
T ₄	<i>T. asperellum</i> (5x10 ⁷ cfu/g carrier)	ST @ 10 g	30.00 (35.26)	32.61 (37.45)	31.31 (34.02)	63.89 (53.06)	67.39 (55.18)	65.64 (54.11)
T ₅	<i>T. harzianum</i> (5x10 ⁷ cfu/g carrier)	ST @ 10 g	18.33 (25.35)	21.06 (27.32)	19.7 (26.35)	78.01 (62.03)	78.94 (62.68)	78.47 (62.35)
T ₆	<i>T. hamatum</i> (5x10 ⁷ cfu/g carrier)	ST @ 10 g	33.33 (33.21)	36.98 (34.82)	35.16 (36.37)	60.19 (50.88)	63.02 (52.55)	61.6 (51.7)
T ₇	T ₄ + T ₅	ST @ 10 g	16.67 (24.09)	19.03 (25.86)	17.85 (24.99)	79.86 (63.33)	80.97 (64.14)	80.42 (63.74)
T ₈	T ₂ + T ₄	ST @ 1.0 g + 10 g	40.00 (39.23)	43.24 (41.11)	41.62 (40.18)	51.85 (46.06)	56.76 (48.89)	54.31 (47.47)
T ₉	T ₂ + T ₄ + FYM	ST @ 1.0 g + 10 g + SA @ 50 g	10.00 (18.43)	15.1 (22.87)	12.55 (20.75)	87.96 (69.7)	84.9 (67.13)	86.43 (468.38)
T ₁₀	T ₂ + T ₄ + NSC	ST @ 1 g + 10 g + SA @ 50 g	13.33 (21.42)	20.05 (26.6)	16.69 (24.11)	84.26 (66.63)	79.95 (63.4)	82.1 (69.47)
T ₁₁	T ₁ + T ₄ + NSC	ST @ 3.0 g + 10 g + SA @ 50 g	6.67 (14.96)	9.03 (17.48)	7.85 (16.27)	92.13 (73.71)	90.97 (72.51)	91.55 (73.1)
T ₁₂	Control	--	83.33 (65.91)	100 (90)	91.67 0.00	0.00 0.00	0.00 0.00	0.00 0.00
	SE ±	--	2.58	0.35		2.76	0.35	
	CD (P=0.05)		7.53	1.01		8.07	1.01	

*-Mean of three replications, Av.: Average, Mor.: Concentration, Incr.: Increase Red.: Reduction, PRESR: Pre emergence seed rot, POESM: Post Emergence Seedling Mortality, ST: Seed Treatment, SA: Soil Application, NSC: Neem Seed Cake, Figures in parentheses are arcsine transformed values

PLATE XX



Bioefficacy of soil amendments against *R. bataticola*

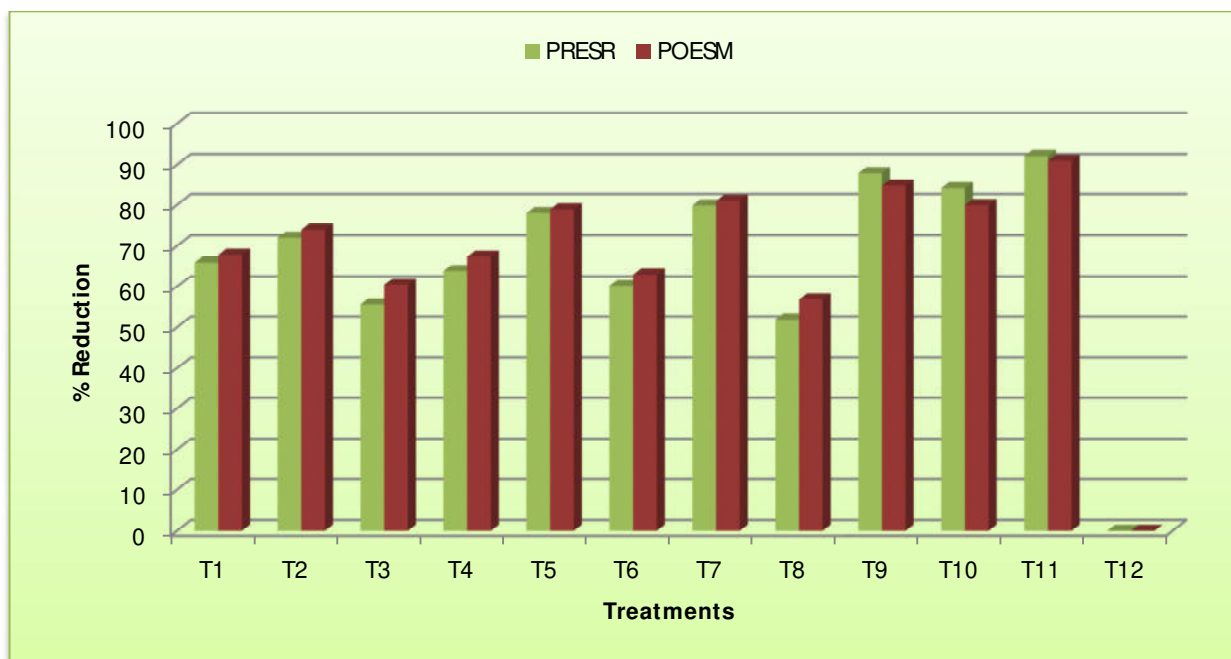
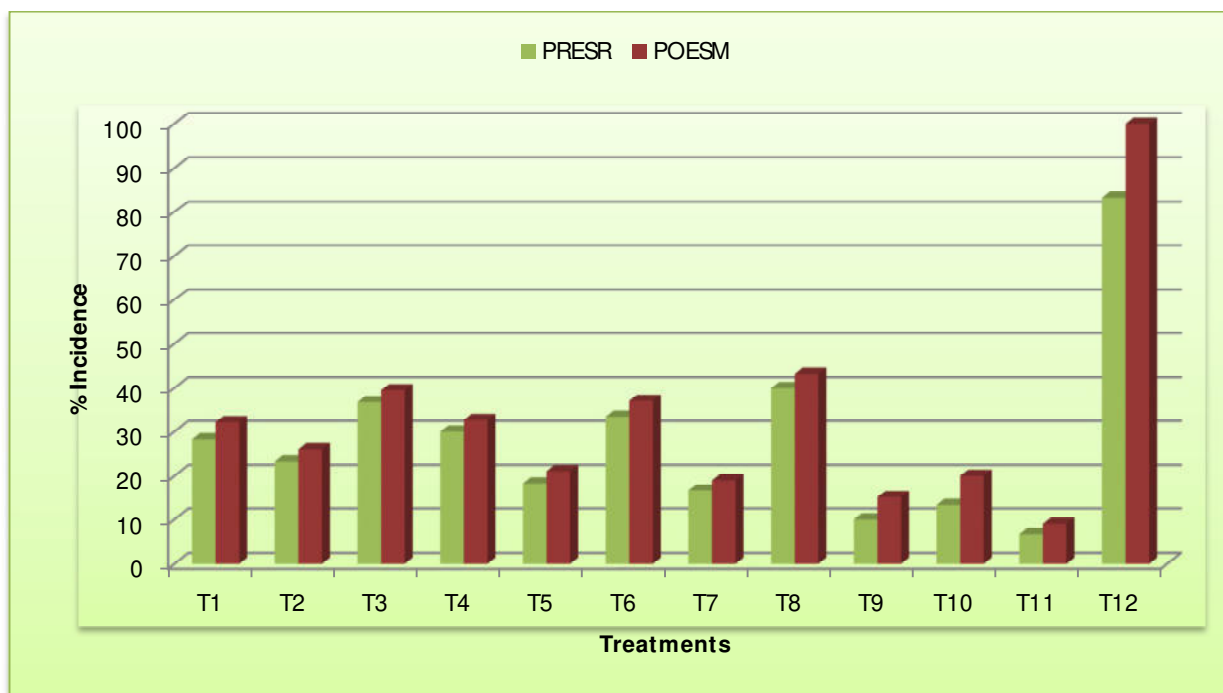


Fig. 21. Integrated efficacy of fungicides, bioagents, and organic amendments against *R. bataticola*, causing dry root rot of Chickpea

This was followed by Carbendazim 50 % WP (ST) @ 1 g + *T. asperellum* (ST) @ 10 g + FYM (SA) @ 50 g (10.00, 15.10 and 12.55 per cent), Carbendazim 50 % WP (ST) @ 1 g + *T. asperellum* (ST) @ 10 g + NSC (SA) @ 50 g (13.33, 20.05 and 16.69 per cent) and *T. asperellum* (ST) @ 10 g + *T. harzianum* (ST) @ 10 g (16.67, 19.03 and 17.85 per cent). Whereas in rest of the treatments PRESR observed in the range 18.33 to 40.00 per cent, POESM in the range of 21.06 to 43.24 per cent and average mortality was found in the range of 19.70 to 41.62 per cent as against comparatively maximum PRESR (83.33 %), POESM (100.00 %) and average mortality (91.67 %) in control (untreated).

4.11.1.2. Reduction in mortality (PRESR, POESM and Average)

Results (Table 21, PLATE XXI and Fig. 21) showed that all the treatments employed resulted in the significant reductions in per-emergence seed rot (PRESR), post-emergence seedling mortality (POESM) and total mortality, which were found in the range of 55.56 to 92.13 per cent, 66.67 to 90.97 per cent and 58.02 to 91.55 per cent, respectively, over control (untreated).

However, the treatment T₁₁ (Carboxin 37.5 % + Thiram 37.5 % WP (ST) @ 3.0 g + *T. asperellum* (ST) @ 10 g + NSC (SA) @ 50 g), was found to be best effective with significantly highest reduction in PRESR (92.13 %), POESM (90.97 %) and average mortality (91.55 %). This was followed by Carbendazim 50 % WP (ST) @ 1 g + *T. asperellum* (ST) @ 10 g + FYM (SA) @ 50 g (87.96, 84.90 and 86.43 per cent), Carbendazim 50 % WP (ST) @ 1 g + *T. asperellum* (ST) @ 10 g + NSC (SA) @ 50 g (84.26, 79.95 and 82.10 per cent) and *T. asperellum* (ST) @ 10 g + *T. harzianum* (ST) @ 10 g (79.86, 80.97 and 80.42 per cent). However in the rest of the treatments PRESR recorded in the range of 55.56 to 78.01 per cent, POESM reduction was in the range of 60.48 to 78.94 per cent and average mortality reduction was found to be in the range of 58.02 to 78.47 per cent, over control (untreated).

Results of this present study revealed that the fungicides (Carbendazim, Carboxin + Thiram, Carbendazim + Mancozeb), biocontrol agents (*T. asperellum* and *T. harzianum*) and organic amendment (Neem seed cake and FYM), applied as seed treatment and soil application (alone and in combination) were found to be effective against *R. bataticola*. Among the treatments taken, Carboxin 37.5 % + Thiram 37.5 % WP seed treatment @ 3g / kg seed + *T. asperellum* @ 10g / kg seed + Neem seed cake @ 50g / kg soil, followed by Carbendazim 50 WP seed treatment @ 1g / kg seed + *T. asperellum* @ 10g / kg seed + FYM @ 50g / kg soil and Carbendazim 50 % WP (ST) @ 1 g + *T. asperellum* (ST) @ 10 g + NSC (SA) @ 50 g were found to be most effective with significantly highest reduction in the pre-emergence seed rot, post-emergence seedling mortality and total mortality, over control (untreated).

The reason for a good control of dry root rot by fungal antagonists in integration with combi fungicides & organic amendments in this study may be due to the combi product of fungicide which performed dual action to control the plant disease *i.e.* systemic and contact. These dual action which was totally inhibit the fungal growth. Whereas *Trichoderma* is a saprophytic fungus that grows on dead organic matter and cell wall of pathogenic fungi. It secretes a range of extracellular compounds, which inhibit pathogens through antibiosis. *Trichoderma* was found effective to control the disease. Neem cake which contains azadirachtin, salannin, nimbin, azadiradione as the major component might be responsible to reduce the disease. It acts as a biofertilizer and helps in providing the required nutrients to plants. Neem seed cake performs the dual function, acts as a soil enricher, reduces the population of soil pest and bacteria, provides macro nutrients essential for all plant growth and these plant nutrients inhibits the growth of soil borne fungi (Pawar *et al.*, 2018).

Thus, these promising fungicides, bioagents and Neem seed cake could be used for integrated disease management of chickpea dry root rot disease, caused due to *R. bataticola*. These results similar with the findings of

PLATE XXI



Integrated disease management against chickpea dry root rot JG- 62 (Pot/Polybag culture)

Moradia & Khandar (2011), Vishwanathan *et al.*, (2015), Veena *et al.*, (2016) & Agale R.C. (2018). The present finding supports the results of Deshmukh *et al.*, (2016).

4.12. Reactions of chickpea varieties, cultivars and germplasm lines in polybag culture

To find out the sources of resistance in chickpea against dry root rot (*R. bataticola*), 22 chickpea entries were examined by sick soil method in pot / polybag culture, under screen house condition, during *Rabi*, 2016-17, at the Department of Plant Pathology, VNMKV, Parbhani.

The results (Table 22 and PLATE XXII) elucidated that all of the chickpea test entries found to reacted differently against *R. bataticola*, However, among the test entries, pre-emergence seed rot (PRESR), post-emergence seedling mortality (POESM) and total mortality was in the range of 6.67 to 80.00 per cent, 10.00 to 100.00 per cent and 8.33 to 90.00 per cent, respectively.

Based on these reactions (Table 23), the chickpea test entries were categorized as immune (no disease), resistant (1-10.0), moderately resistant (10.1-20), moderately susceptible (20.1-30), susceptible (30.1-50.0) and highly susceptible (>50). None of the chickpea entry was found immune to root rot disease.

Table 22. Reactions of chickpea varieties, cultivars, elite lines against *R. bataticola* (Pot / polybag culture)

Sr. No.	Genotypes	PRESM%	POESM%	Average Mortality (%)	Reactions
1	Vishal	13.33	15.55	14.44	MR
2	Vijay	23.33	25.00	24.16	S
3	Digvijay	20.00	22.22	21.11	MR
4	JAKI 9218	13.33	16.66	14.99	MR
5	SAKI 9516	8.33	10.00	9.16	R
6	AKG 1103	36.66	50.00	43.33	S
7	BDNG 13-1	10.00	11.11	10.55	R
8	BDNG 2013-2	53.33	57.14	55.23	HS
9	BDNG 2013-1	6.66	12.5	9.58	R

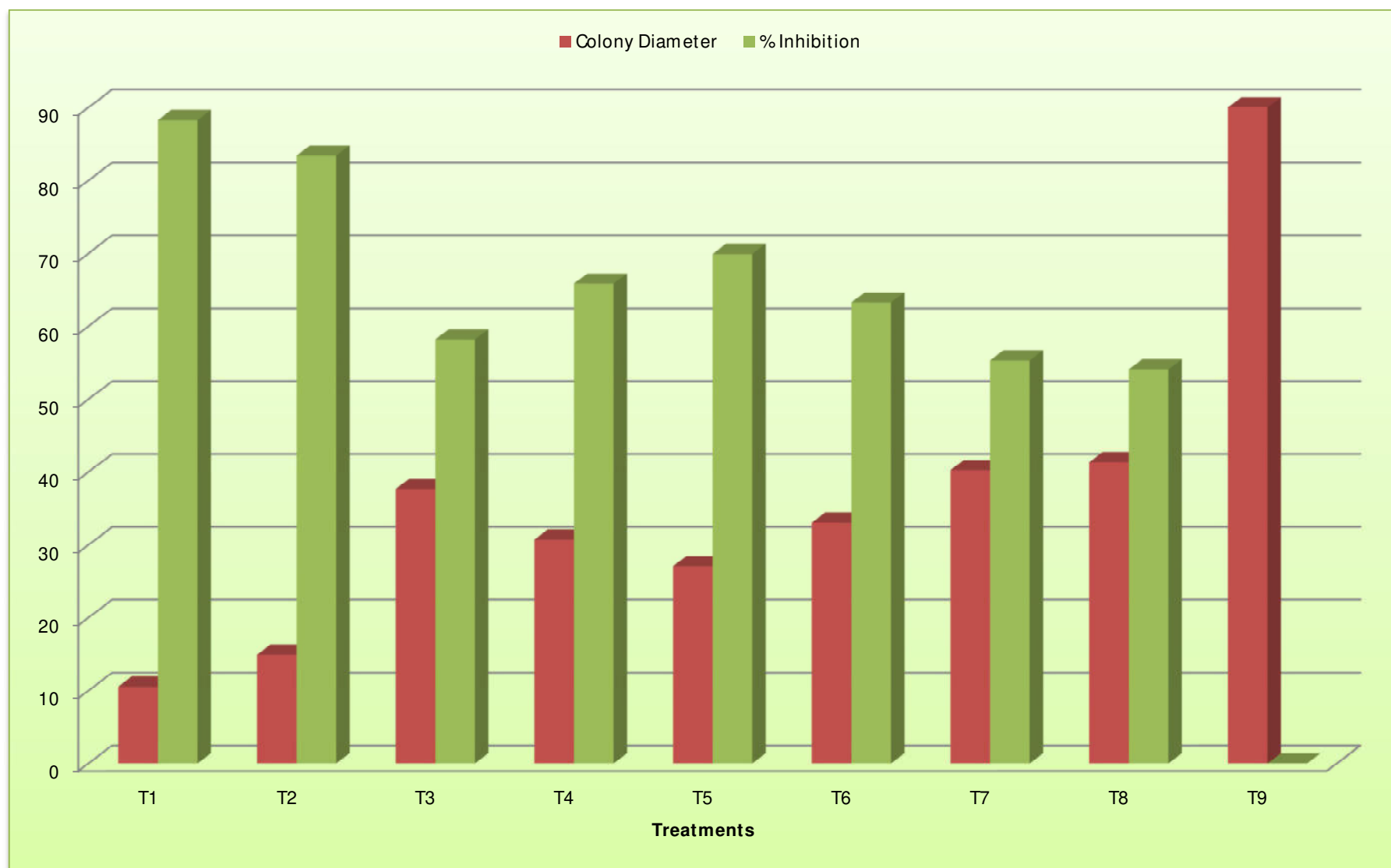


Fig. 19. *In vitro* bioefficacy of bioagents against *R. bataticola*

10	BDNG 9-3	15.00	22.5	18.75	MS
11	BDNG 797	26.67	32.83	29.75	MS
12	BCP 26	6.67	10.00	8.33	R
13	BCP 21	10.00	13.33	11.66	R
14	BCP 49	6.67	11.11	8.89	R
15	Rajas	10.00	14.44	12.22	R
16	JG 315	8.33	15.55	11.94	R
17	JG 62	80.00	100.00	90.00	HS
18	PG 0515	10.00	13.33	11.66	R
19	PG 9801	65.00	73.33	69.16	HS
20	PG - 4333	71.66	77.77	74.71	HS
21	PG 0302-10	73.33	15.55	44.44	MR
22	ICCV- 5313	23.33	25.00	24.16	MS

PRESR-Pre-emergence Seed Rot, POESM- Post Emergence Seedling Mortality, HS: Highly Susceptible, MR: Moderately Resistant, R: Resistance, S: Susceptible
Population of *Rhizoctonia bataticola* cfu/g ($\times 10^4$)

However, nine entries were found to be with resistant reactions were SAKI 9516, BDNG 13-1, BDNG 2013-1, BCP 26, BCP 21, BCP 49, Rajas, JG 315, PG 0515; four entries with moderately resistant reactions were Vishal, Digvijay, JAKI 9218, PG 0302-10; three entries with moderately susceptible reactions were BDNG 9-3, BDNG 797, ICCV- 5313; two entries with susceptible reactions were Vijay, AKG 1103; and rest four entries with highly susceptible reaction were BDNG 2013-2, JG 62, PG 9801, PG – 4333.

Use of host plant resistance is the most economical strategy for management of dry root rot of chickpea as *R. bataticola* includes a broad host range & survives in soil for long periods in the form of sclerotia and influenced by the changes in climatic condtions. The sclerotia will survive up to ten months even in the absence of the host plant & beneath prevailing dry soil conditions.

PLATE XXII



Screenhouse reactions of chickpea varieties, cultivars, germplasm lines against *R. bataticola* (Pot/Polybag culture)

Table 23. Categorization of chickpea varieties, cultivars, elite lines based on reaction against *R. bataticola* (Pot/Polybag culture)

Sr. No.	Disease Reactions	Chickpea entries
1	Immune (I)	None
2	Resistant (R)	Nine: SAKI 9516, BDNG 13-1, BDNG 2013-1, BCP 26, BCP 21, BCP 49, Rajas, JG 315, PG 0515
3	Moderately Resistant (MR)	Four: Vishal, Digvijay, JAKI 9218, PG 0302-10
4	Moderately Susceptible (MS)	Three: BDNG 9-3, BDNG 797, ICCV- 5313
5	Susceptible (S)	Two: Vijay, AKG 1103
6	Highly Susceptible (HS)	Four: BDNG 2013-2, JG 62, PG 9801, PG - 4333



Summary and Conclusion

CHAPTER - V

SUMMARY AND CONCLUSIONS

Chickpea is grown in over 50 countries of Asia, Africa, America and Oceania in rain-fed environments (Sharma *et al.*, 2015). The production of chickpea is largely constrained by *Fusarium* wilt (*Fusarium oxysporum* f. sp. *ciceris*); however, recent reports indicated that dry root rot (DRR) is emerging as a potential threat to chickpea production (Pande *et al.*, 2010; Sharma *et al.*, 2010; Ghosh *et al.*, 2013). The dry root rot is caused by *Rhizoctonia bataticola* (Taub.) Butler. (Synonym: *Macrophomina phaseolina* (Maubl.) Ashby.). *R. bataticola* is a soil-inhabiting pathogen capable of infecting chickpea at any crop stage, but most commonly infects chickpea at flowering to pod forming stage in dry and warm regions (Sharma & Pande 2013). It can affect chickpea production, causes considerable yield losses that vary from 5 to 50 % and may cause 100% losses in susceptible cultivars under favorable condition (Pande *et al.*, 2012) and considerable yield losses which may be as high as 50 to 71% (Veena *et.al.* 2014a).

Therefore, present investigation was undertaken on “Studies on variability of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea and its management”, with the defined objectives, at the Department of Plant Pathology, College of Agriculture, Vasant Rao Naik Marathwada Krishi Vidhyapeeth, Parbhani and the findings thereof are being summarized in the following paragraphs.

A survey was conducted to record chickpea dry root rot incidence during the *Rabi* seasons of 2016-17 and 2017-18, which covered 179 and 232 chickpea crop fields of eight districts, distributed under three agro-climatic zones viz., Scarcity zone (06), Assured Rainfall zone (07) and Moderate rainfall zone (08) of Marathwada region of the Maharashtra state. The results of

the survey elucidated that overall dry root rot disease incidence was observed comparatively maximum during *Rabi*, 2017-18 season compared to *Rabi*, 2016-17. During *Rabi*, 2016-17 and 2017-18 maximum dry root rot incidence was 24.23 and 30.55 per cent, respectively, with maximum pooled mean incidence of 27.39 per cent in Scarcity zone, followed by Assured rainfall zone (22.17 %, 26.63 % and 24.4 %) and Moderate rainfall zone (16.18 %, 20.29 % and 18.23 %). These variations may occurred due to the some close similarity in rainfall pattern, soil type, temperature and cropping pattern etc. and a rangy chickpea cultivars cultivate by the farmers. The pooled mean (*Rabi*, 2016-17 and 2017-18) results showed that the maximum average dry root rot incidence of 26.85 per cent and 30.91 per cent observed in Jalna district, respectively, during *Rabi* 2016-17 and 2017-18, with maximum pooled incidence of 28.88 %. This was followed by the districts of viz., Parbhani (27.01%), Aurangabad (24.45 %), Osmanabad (24.88 %), Beed (24.00 %), Latur (21.12 %), Hingoli (21.79 %) and Nanded (19.68). Of various chickpea varieties cultivated in these three agro climatic zones, maximum mean dry root rot incidence was found in Vijay (32.90 %) followed by the varieties viz., Digvijay (29.37 %), Local (Deshi) (25.93 %) and Vishal (22.99 %) In rest of the varieties, the root rot incidence was in the ranged of 16.56 % to 22.99 %.

The pathogen (*R. bataticola*) can induce various diseases viz., seedling blight, seedling decay, dry root rot, charcoal rot, wilt, stalk rot, stem blight, fruit rot, leaf blight etc. in many crop plants. During the studies, viz., survey, pathogenicity test, Pot/Polybag culture experiments, typical symptoms of the disease observed were: At seedling stage, infected seedlings were appeared stunted without any lateral roots and when uprooted the tap root was observed black in colour. In the vegetative to flowering stage, symptoms on affected plants were observed as bronzing of the leaves on one or more of the lower branches, leaves became yellow to brown in plants which shows advanced disease symptoms. The terminal part of the tap root and lateral roots

became brown to black. The tap root without any lateral roots was also observed which characteristic symptom of dry root rot is.

The pathogen (*R. bataticola*) isolated successfully on basal culture medium Potato dextrose agar and purified. Naturally dry root rot diseased plant roots samples collected during survey were subjected to the tissue isolation on PDA and 9 isolates of *R. bataticola* were obtained, purified and maintained on PDA slant tubes for further studies. Pathogenicity of the test pathogen isolates was proved successfully on susceptible chickpea Cv. JG-62, by sick soil method in polybag culture/pot culture, under screen house conditions, which showed that all 9 isolates as moderate to highly pathogenic to chickpea.

The test pathogen was identified as *Rhizoctonia bataticola* (Tabb.) Butler based on symptomatology (naturally and artificially diseased chickpea plants), morpho-cultural characteristics, microscopic observations and the pathogenicity test. Further, its identity was made confirmed by comparing its cultural and morphological characteristics with the descriptions of *R. bataticola* (Butler, 1918, Sneh *et al.*, 1991).

All 9 isolates of *R. bataticola* exhibited a wide range of pathogenic variability, employed by sick soil and paper towel methods, of which both methods were found to be most efficient. On the basis of which, all of the 9 isolates of *R. bataticola* were categorized into various groups (Pawar, 2010). Based on results of both methods, the isolate Rb-6 (Loha, Nanded district) was found as highly virulent and or pathogenic to chickpea, which was selected and used for further *in vitro* studies.

Studies on cultural characteristics of *R. bataticola* 9 isolates indicated a inadequate variability amongst them. However, the isolates Rb-6, showed maximum mycelial growth of (90.00 mm), followed by Rb-9 (89.50

mm) and Rb-2 (88.50). The colony colour exhibited varied from blackish grey (Rb-2, Rb-4 and Rb-7), charcoal black (Rb-1), grayish white (Rb-3, Rb-5, Rb-6 and Rb-8) and light black (Rb-9). The growth rate was fast (++++) of the isolates viz., Rb-1, Rb-2, Rb-3, Rb-5, Rb-6, Rb-7, Rb-8 and Rb-9; whereas, it was medium (+++) of the isolates viz., Rb-4.

The studies on morphological characteristics of *R. bataticola* isolates indicated great variability with respect to the hyphal width and sclerotial characters like shape, colour and size. Seven isolates were with large sized hyphae (5.14-7.33 μm) viz., Rb-6, Rb-9, Rb-2, Rb-8, Rb-5, Rb-7 and Rb-1; one isolate was with medium sized hyphae viz., Rb-3 and one isolate was with small sized hyphae viz., Rb-4. Majority of the isolates (06) viz., Rb-3, Rb-5, Rb-6, Rb-7, Rb-8 and Rb-9, produced irregular shaped type of sclerotia. Overall, sclerotial density was ranged from 13.67 to 70.33. However, the sclerotial density was found excellent in three isolates viz., Rb-6 (70.33), Rb-9 (60.67) and Rb-2 (51.67). The large sized sclerotia were found in two isolates viz., Rb-6 (124 μm) and Rb-9 (113.00 μm); medium sized sclerotia were produced by six isolates viz., Rb-2 (90.50 μm), Rb-8 (80.50 μm), Rb-5 (62.00 μm), Rb-7 (60.50 μm), Rb-1 (55.50 μm), and Rb-3 (50.50 μm) and small sized sclerotia was found in only one isolate viz., Rb-4 (46.50 μm).

Dendrogram generated based on UPGMA analysis of RAPD data grouped all of the 9 test isolates (*R. bataticola*) into two major clusters. The Cluster I included 6 isolates viz., Rb-1, Rb-2, Rb-3, Rb-4, Rb-5, and Rb-9, collected from three zone of Marathwada region which showed 64.5 per cent genetic similarity. Cluster II comprised of only two isolates Rb-6 and Rb-7, with similarity coefficient of 85 per cent.

The studies on effect of organic and inorganic sources of carbon and nitrogen on *R. bataticola* indicated that all the carbon and nitrogen sources tested showed a wide range of sclerotial production from poor (+) to excellent

(++++). However, carbon sources viz., Dextrose, Sucrose and Cellulose recorded excellent (++++) sporulation and nitrogen sources viz., Sodium nitrate, Potassium nitrate and Peptone recorded excellent (++++) sporulation.

The studies on Effect of different soil moisture regimes and soil types on dry root rot disease indicated that the average dry root rot disease incidence was found to be higher (range: 81.67 to 74.45 per cent) at the soil moisture level of 55 to 60 per cent. While studying the effect of soil types on the incidence of chickpea dry root rot diseases, it was noticed that sandy soil showed highest dry root rot disease incidence than the other soil types.

All of the 12 fungicides (six systemic, three contact and three combi- fungicides) tested *in vitro* at various concentrations were found to be effective against *R. bataticola* and however, the per cent mycelial growth inhibition was observed to be increased with increase in concentrations of the test fungicides. Of the systemic fungicides tested, However, Carbendazim 50 % WP, Tebuconazole 29.9% EC, Hexaconazole 5% EC and Azoxystrobin gave cent per cent (100 %) mycelial inhibition. Among non-systemic / contact and combi-fungicides tested, mycelial growth inhibition was cent per cent (100 %) was observed with the Carbendazim 12 % + Mancozeb 63 % WP and Carboxin 37.5 % + Thiram 37.5 % WP followed by (Cymoxil + Mancozeb) (94.38), Captan 50 % WP (92.37 %) and Copperoxy chloride 50 % WP (87.61 %).

All of the eight biocontrol agents examined *in vitro* were found to be antifungal to *R. bataticola*. However, *T. asperellum* and *T. harzianum* were found most effective with significantly maximum mycelial growth inhibition (88.27 and 83.4 %, respectively), followed by *T.hamatum*, *T. koningii*, *T. longibrachiatum*, *T.Virens*, *P. fluorescens*, *B. subtilis*, respectively with 69.92, 65.85, 63.25, 58.17, 55.28 and 54.07 per cent mycelial growth inhibition.

All 10 soil amendments examined were found to be fungistatic to *R. bataticola*. However, Neem seed cake, was observed with significantly highest reduction in PRESR (77.38 %), POESM (81.86 %) and their average reduction (79.62 %). This was followed by Compost (72.62%, 76.89% and 74.75%), Vermicompost (67.86%, 74.29% and 71.07%) Castor seed cake (63.10 %, 71.31 % and 67.20 %), Mustard cake (68.51 %, 64.76 % and 66.63 %), Karanj cake (68.68 %, 65.18 % and 64.67 %), and Safflower seed cake (47.32 %, 58.49 % and 52.91%). Rest of the test amendment recorded reductions in PRESR (range: 36.31 to 45.24 %), POESM (range: 51.19 to 55.56 %) and average mortality reduction in the range of 43.75 to 50.40 per cent.

All the treatments (fungicides, bioagents and organic amendments) carried out (alone or in combination) in pot / polybag culture were found effective in reduction of pre- / post- emergence and total mortality, induced by *R. bataticola* chickpea Cv. JG-62. However, significantly maximum reduction in average mortality was recorded with Carboxin 37.5 % + Thiram 37.5 % WP (ST) @ 3.0 g + *T. viride* (ST) @ 10 g + NSC (SA) @ 50 g, (91.55 %). This was followed by Carbendazim 50 % WP (ST) @ 1 g + *T. viride* (ST) @ 10 g + FYM (SA) @ 50 g (86.43 per cent), Carbendazim 50 % WP (ST) @ 1 g + *T. viride* (ST) @ 10 g + NSC (SA) @ 50 g (82.10 per cent) and *T. viride* (ST) @ 10 g + *T. harzianum* (ST) @ 10 g (80.42 per cent).

Under artificial situation of *R. bataticola* (pot / polybag culture), all of the 22 chickpea entries had different reactions, during 2016-17. However, nine entries were found resistant (1-10.0%) reactions SAKI 9516, BDNG 13-1, BDNG 2013-1, BCP 26, BCP 21, BCP 49, Rajas, JG 315, PG 0515; four entries found with moderately resistant (10.1-20%) reactions were Vishal, Digvijay, JAKI 9218, PG 0302-10; three entries found with moderately susceptible (20.1-30%) reactions were BDNG 9-3, BDNG 797, ICCV- 5313;

two entries found with susceptible (30.1-50.0%) reactions were Vijay, AKG 1103; and rest four entries was found with highly susceptible (>50%) reaction were BDNG 2013-2, JG 62, PG 9801, PG - 4333.

CONCLUSIONS

Thus, from the results obtained on various aspects during present investigations on chickpea dry root rot (*R. bataticola*), following conclusions are being drawn:

- *Rhizoctonia bataticola* (Taub.) Butler is one of the major constraints in the cultivation and production of chickpea, causes both quantitative and qualitative losses in the crop.
- Survey studies indicated that chickpea dry root rot (*R. bataticola*) was distributed widely in the Marathwada region and it was found more severe in *Rabi*, 2017-18 than *Rabi*, 2016-17 season, on all chickpea cultivars / varieties cultivates in the region. The disease was noticed to be more severe in Scarcity and Assured rainfall zones, with 500 and 500-700 mm rainfall and may be due to climatic conditions found favourable compared to Moderate rainfall zone.
- The pathogen, *R. bataticola* was successfully isolated on PDA, from naturally infected dry root rot diseased chickpea plant roots of the samples and about 9 isolates from eight districts surveyed were obtained, purified and maintained for further studies.
- Pathogenicity of all representative 9 isolates of *R. bataticola* proved on susceptible chickpea Cv. JG-62, by sick soil method in polybag culture, elucidated that the test isolates was moderately to highly virulent to chickpea and the most highly virulent isolate Rb-6 was selected for further *in vitro* studies.

- Based on the symptomatology, morpho-cultural characteristics, microscopic observations and pathogenicity test conducted, the test pathogen was identified as *Rhizoctonia bataticola* (Tabb.) Butler, cause of chickpea dry root rot.
- All of the 9 isolates of *R. bataticola* had a wide range of variability with respect to pathological, cultural, morphological and molecular but it needs to be confirmed further.
- All those organic and inorganic sources of carbon and nitrogen shows a wide range of sclerotial production. carbon sources viz., Dextrose, Sucrose and Cellulose and nitrogen sources Sodium nitrate, Potassium nitrate and Peptone recorded excellent sporulation.
- Soil moisture level 55 to 60% and sandy soil, sandy loam soil favourable for the pathogen growth and disease development.
- All those fungicides, bioagents and organic amendments examined *in vitro* (plate and polybag culture) were found fungistatic against *R. bataticola*, causes significant mycelial growth inhibition as well as reduction in mortality. However, the fungicides Carbendazim 50% WP, Hexaconazole 5% EC, Carboxin 37.5% + Thiram 37.5% WP, Carbendazim 12%+ Mancozeb 63% WP and Captan 50% WP; bioagents viz., *T. viride*, *T. harzianum*, *T. hamatum*, *T. koningii*, *T. longibrachiatum* and organic amendments viz., Neem seed cake, Compost, Vermicompost, Castor seed cake and Karanj cake were found most effective against *R. bataticola*, which resulted in significant mycelial growth inhibition as well as significant reduction in the pre and post-emergence mortality and average mortality, over untreated control.

- The studies on integrated disease management of chickpea dry root rot (*R. bataticola*) disease in polybag culture (*Rabi*, 2016-17) showed that the treatments Carboxin 37.5% + Thiram 37.5% WP @ 3 g + *T. viride* (ST) @ 10 g + NSC (SA) @ 50 g, Carbendazim 50 % WP (ST) @ 1 g + *T. viride* (ST) @ 10 g + FYM (SA) @ 50 g, and *T. viride* (ST) @ 10 g + *T. harzianum* (ST) @ 10 g noted as most effective and economical for the efficient management of dry root rot disease of chickpea.
- The studies on screening of chickpea entries, (*Rabi*, 2016-17) in polybags culture revealed nine entries as resistant, four entries as moderately resistant, three entries as moderately susceptible, two entries as susceptible and four entries as highly susceptible to dry root rot disease. Though nine entries were found resistant but as the disease influenced by environmental conditions the resistance was constant. May these entries found susceptible to dry root rot when experiment performed in different climatic situation.

***Literature
Cited***

CHAPTER – VI

LITERATURE CITED

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***Thesis
Abstract***

Studies on variability of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea and its management

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ABSTRACT

Dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler, is a widely distributed and destructive disease of Chickpea [*Cicer arietinum* L.] causes accountable losses. The pathogen can induce various diseases viz., dry root rot, charcoal rot, wilt, stalk rot, stem blight, fruit rot, leaf blight etc. in many crop plants. Therefore, present investigations was planned on *R. bataticola* with the objectives viz., survey, symptomatology, pathogenicity, variability (morpho-cultural, pathogenic and molecular), effect of carbon, nitrogen sources, soil moisture, soil types, *in vitro* disease management strategies and screening of chickpea entries were undertaken during 2016-17 to 2017-18 at the Department of Plant Pathology, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani (M.S.), India.

The survey studies indicated that dry root rot (*R. bataticola*) disease in chickpea crop widely observed throughout the Marathwada region of the Maharashtra state. The disease was most severe in *Rabi*, 2017-18 as compared to *Rabi*, 2016-17. Almost all chickpea varieties grown in Marathwada region were found to be affected by the disease. The disease was observed more severe in Scarcity zone (27.39%), followed by Assured rainfall zone (24.4%) and Moderate rainfall zone (18.23%).

The test pathogen produced varied symptoms such as seedlings were appeared stunted without any lateral roots when uprooted and the tap root was black in colour. Affected plants were observed as bronzing of the leaves on one or more of the lower branches, leaves became yellow to brown in plants

showing advanced disease symptoms. The terminal part of the tap root and lateral roots became brown to black and shriveled. The tap root without any lateral roots was also observed. A grayish mycelium can sometimes be seen on tap root.

The test pathogen (*R. bataticola*) was grown well on basal culture medium of Potato dextrose agar. The dry root diseased chickpea plant samples collected during survey, when isolated on PDA, 9 isolates of *R. bataticola* were yielded. In pathogenicity test, *R. bataticola* was found to be highly pathogenic to chickpea (cv. JG-62).

All of the 9 isolates of *R. bataticola* shows a wide range of variability with respect to the virulence, morpho-cultural characteristics and molecular variability. Out of the 9 isolates, a single most virulent isolate Rb-6 (Loha, Dist. Nanded) was selected for further *in vitro* studies.

All those organic and inorganic sources of carbon and nitrogen had a wide range of sporulation. Carbon sources viz., Dextrose, Sucrose and Cellulose recorded excellent sporulation whereas among nitrogen sources viz., Sodium nitrate, Potassium nitrate and Peptone recorded excellent sporulation.

Among different soil moisture regimes average dry root rot disease incidence was also higher (range: 81.67 to 74.45 per cent) at the soil moisture level of 55 to 60 per cent. While studying the effect of soil types on the incidence of dry root rot disease of chickpea, it was noted that sandy soil showed highest dry root rot disease incidence followed by sandy loam soil and then the other soil types.

All of the 12 fungicides (six systemic, three contact and three combi- fungicides) treated *in vitro* were found to be most effective with significant mycelial growth inhibition of the test pathogen, over control. However, resulted (100%) inhibition in the Carbendazim 50% WP, Carboxin

37.5% + Thiram 37.5% WP and Carbendazim 12% WP + Mancozeb 63% WP, Hexanconazole 5% EC, Tebuconazole and Azoxystrobin (100 %).

Among the biocontrol agents tested, the most effective bioagent with significantly highest mycelial growth was *T. asperellum* (88.27%), followed by *T. harzianum* (83.4%), *T. hamatum* (69.92%), *koningii* (65.85%) and *T. longibrachiatum* (63.25%).

Among all organic soil amendements examined, Neem seed cake was found most effective with significantly highest reduction in mortality (79.62%), followed by Compost (74.75), Vermicompost (71.07), Castor cake (67.20%) and Karanj seed cake (64.76%).

Among various treatments integrated in polybag culture under artificial condition in screen house to manage chickpea dry root rot, the most effective treatment was found with significantly highest disease reduction is Carboxin 37.5 % + Thiram 37.5 % WP (ST) @ 3.0 g/kg + *T. asperellum* (ST) @ 10 g/kg + NSC (SA) @ 50 g/kg (91.55%), followed by Carbendazim 50 % WP (ST) @ 1.0 g/kg + *T. asperellum* (ST) @ 10 g/kg + FYM (SA) @ 50 g/kg (86.43%) Carbendazim 50% WP (ST) @ 1 g/kg + *T. asperellum* (ST) @ 10 g/kg + NSC (SA) @ 50 g/kg (82.10%).

Of the 22 entries of chickpea evaluated in polybag culture (*R. bataticola* sick soil), nine entries were resistant, 4 entries moderately resistant, 3 entries moderately susceptible, two entries susceptible and four entries highly susceptible to dry root rot disease.

APPENDIX -I

Composition of various culture media used for *in vitro* studies

1) Potato dextrose agar

Peeled potato	: 200 g
Dextrose	: 20 g
Agar agar	: 20 g
Distilled water	: 1000 ml

2) Potato dextrose broth

Peeled potato	: 200 g
Dextrose	: 20 g
Distilled water	: 1000 ml

APPENDIX -II

Composition of various stains, buffer, solutions and reagents used for molecular work

1 Composition of Lactophenol cotton blue:

Phenol (crystal)	:	20 g
Lactic acid	:	20 ml
Glycerol	:	40 ml
Distilled water	:	20 ml

2 Stocks solution for DNA isolation

Tris HCl (1M, pH 8.5)	:	Tris base 15.76 gm in 100 ml distilled water.
NaCl (5M)	:	NaCl 29.22 gm in 100 ml distilled water.
EDTA (0.5M, pH 8.0)	:	EDTA 18.61 gm in 100 ml distilled water.
P:C:I (25:24:1)	:	Add Phenol 25 ml, Chloroform 24 ml and Isoamyl alcohol 1 ml.
Sodium acetate (3M)	:	Sodium acetate 24.60 g in 100 ml distilled water.
Ammonium acetate (7.5M)	:	Ammonium acetate 57.75 gm in 100 ml distilled water

3 DNA extraction buffer (50 ml)

Tris HCl (1M)	:	5 ml
NaCl (5M)	:	5 ml
EDTA (0.5M)	:	2 ml
SDS (10%)	:	10 ml
Distilled water	:	28 ml

4. Reagents for agarose gel electrophoresis

a. 50X TAE buffer composition (1L)

Tris base	:	242 g
Acetic acid	:	57.10 ml
Na ₂ EDTA	:	37.20 g

Dissolve above components into distilled water and make up vol. 1
lit

b. TE buffer composition (100 ml)

Tris HCL	:	10 ml
Na ₂ EDTA	:	200 µl

c. 6X DNA gel loading dye

Bromophenol blue (BPB)	:	0.25 %
Xylene cyanol (optional)	:	0.25 %
Glycerol in water	:	30 %

d. Ethidium bromide (5 µl/ml)

Dissolve 100 mg Ethidium bromide in 10 ml of distilled water.

e. Recipe for 1.2% agarose gel (30 ml)

Agarose	:	360 mg
1x TAE	:	30 ml
(5 µl /ml)	:	5 µl

5. RNase stock (10mg/ml)

Dissolve 10 mg RNase in 1ml of sterile distilled water and store at
– 20 °C.

Vitae



CURRICULUM VITAE

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SUBJECT OF SPECILIZATION: PLANT PATHOLOGY

(Mycology)

Master's Degree: "STUDIES ON LEAF SPOT OF CABBAGE CAUSED BY
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